

Robert P. Taylor
Editor

THE COMPUTER IN THE SCHOOL: TUTOR, TOOL, TUTEE

Pioneiros na Informática Educacional

Robert Taylor

Alfred Bork

Thomas Dwyer

Arthur Luehrmann

Seymour Papert

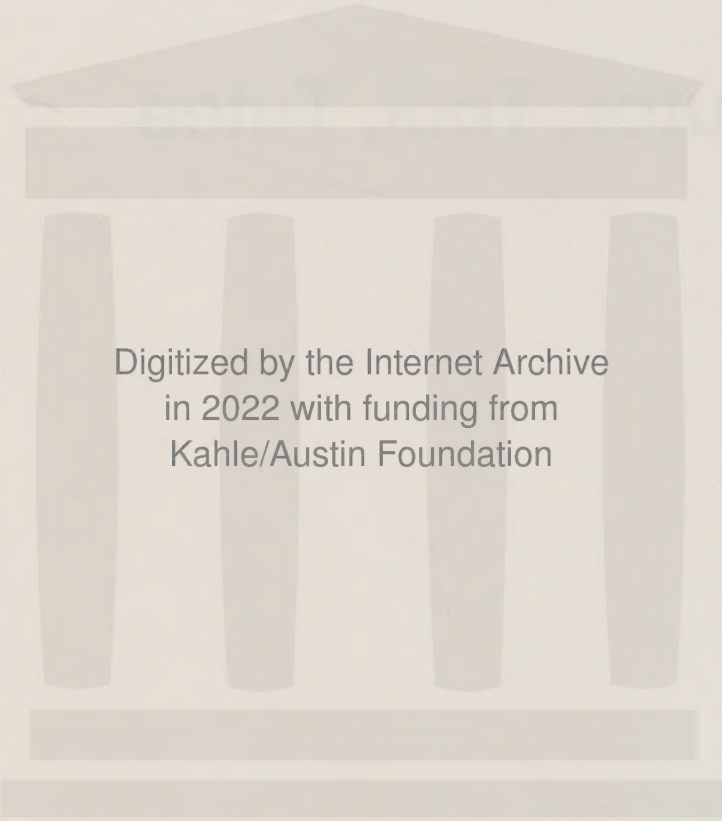
Patrick Suppes



THE COMPUTER IN THE SCHOOL

Tutor, Tool, Tutee

O computador na escola: tutor, ferramenta, aluno



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Tutor, Tool, Tutee

Edited by Robert Taylor

*DIRECTOR, PROGRAM IN COMPUTING IN EDUCATION
TEACHERS COLLEGE, COLUMBIA UNIVERSITY*



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Preface

MY FORMAL INVOLVEMENT in the field of computers in education goes back to the summer of 1965 when I taught my first course for in-service secondary school teachers. By that time some people had already been involved in the computers in education field for a decade or more, but these people were few and far between.

The past 15 years have seen marvelous progress in computer hardware, software, and courseware. The number of people involved has been multiplied manifold. The computer in education field has gained in stature—indeed, it is now a “respectable” field of endeavor.

Much of the progress can be traced to a relatively small number of pioneers who have done the seminal thinking, research, and implementation. It is not possible for a single volume to recapture all of the work of all of these pioneers. What is provided here is a carefully selected subset of the work of just five of them.

The current rapid proliferation of microcomputers in education serves as a strong motivation for the production of this book. Many thousands of educators are just beginning to think about computers in education. They are apt to spend much time “rediscovering the wheel.” This book can save such educators an immense amount of time and effort. It provides an excellent summary of much that has been done in the past, and points to what needs to be done now or in the future.

As chairman of the Association for Computing Machinery’s Elementary and Secondary Schools Subcommittee, I am especially happy to see this book produced. It is a good response to a major concern of that subcommittee, one which manifested itself in the creation of a subgroup, the taskgroup on “Ar-

*Chairman, Association for Computing Machinery’s Elementary and Secondary Schools Subcommittee; Professor, Dept. of Computer Science, University of Oregon

guments Supporting Use of Computers in Education.” As its head, Professor Taylor persuasively suggested that such arguments had already been made by a number of articulate experts and needed only to be brought together from the scattered and often obscure sources where they had originally been published. This book demonstrates that he was right, bringing a good representation of the best of such arguments together in one volume, thus making them readily accessible to anyone who needs to become familiar with them. As such, it certainly becomes a welcome output from both that taskgroup and the subcommittee as a whole. Anyone wondering either what role computers can play in education or why their incorporation into the curriculum should receive the highest priority should certainly read this book from cover to cover, as soon as possible.

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them anew. The articles are therefore a key to the future as well as a record of the past.

Such writings should be read by anyone interested in computing and education because they suggest what has already been accomplished. To know what has already been accomplished is the first step, whether one merely wishes to find out what the field is like or whether one wishes to determine the point from which to begin one's own work. However, simply plunging into the field and attempting to assimilate the ideas may not work. Initially some conceptual help may be needed.

Approaching the Diverse, Technically Foreign Area of Computing in Education

The application of computing to education encompasses a range of complex activity, formidable in its apparent diversity even for those who are simultaneously both computer specialists and educators. Approaching such a complex area for the first time, especially as a computing novice, can be very confusing. This book attempts to minimize the unnecessary confusion three ways. First, by limiting the number of authors included, it arbitrarily limits the diversity of what is presented. Second, by presenting only articulate spokesmen, the issues and the work discussed are presented in an intelligible fashion. Third, by introducing a succinct framework (tutor/tool/tutee) for classifying all educational computing, the book provides the reader with a simple scheme for intellectually grasping a somewhat chaotic range of activities.

The *major* role of this introductory essay is to present the tutor/tool/tutee framework. The basic framework and a summary set of comments on each of the five authors are presented. Then, the application of the framework is demonstrated, in terms of the work of those five.

To assist the reader interested in understanding more of the context of the author's work, a brief biographical sketch precedes the presentation of that author's articles. To assist the reader interested in reading more of a particular author's work, a short selected biography for that author can be found at the end of the book.

Tutor, Tool, Tutee — The Three Modes of Using Computing in Education

The framework suggested for understanding the application of computing in education depends upon seeing all computer use in such application as in one of three modes. In the first, the computer functions as a *tutor*. In the second, the computer functions as a *tool*. In the third, the computer functions as a *tutee* or student.

THE COMPUTER AS TUTOR

To function as a *tutor* in some subject, the computer must be programmed by “experts” in programming and in that subject. The student is then tutored by the computer executing the program(s). The computer presents some subject material, the student responds, the computer evaluates the response, and, from the results of the evaluation, determines what to present next. At its best, the computer tutor keeps complete records on each student being tutored; it has at its disposal a wide range of subject detail it can present; and it has an extensive and flexible way to test and then lead the student through the material. With appropriately well-designed software, the computer tutor can easily and swiftly tailor its presentation to accommodate a wide range of student differences.

Tutor mode typically requires many hours of expert work to produce one hour of good tutoring, for any or all of several reasons. (a) As intuitive beings, humans are much more flexible than any machine, even a computer. (b) Creating a lesson to be delivered by a human tutor requires less time because it omits much of the detail, relying upon the spontaneous improvisation and performance of the instructor to fill in both strategy and substance at the time of delivery. (c) Computers are still relatively crude devices and the only means we have of programming them are awkward and time-consuming. (d) Human instruction rarely aims to accommodate individual differences because the normal classroom situation prohibits such accommodation; hence lesson preparation and design are simpler and swifter. Because such accommodation is possible with the computer as tutor, the substantive and strategic details needed to individualize the lesson tend to get included, thus often greatly lengthening lesson design and preparation time.

THE COMPUTER AS TOOL

To function as a *tool*, the classroom computer need only have some useful capability programmed into it such as statistical analysis, super calculation, or word processing. Students can then use it to help them in a variety of subjects. For example, they might use it as a calculator in math and various science assignments, as a map-making tool in geography, as a facile, tireless performer in music, or as a text editor and copyist in English.

Because of their immediate and practical utility, many such tools have been developed for business, science, industry, government, and other application areas, such as higher education. Their use can pay off handsomely in saving time and preserving intellectual energy by transferring necessary but routine clerical tasks of a tedious, mechanical kind to the computer. For example, the burdensome process of producing hundreds or even thousands of employee paychecks can be largely transferred to the computer through the use of accounting software; the tedious recopying of edited manuscripts of texts or

even music can be relegated to the computer through word or musical notation processing software; the laborious drawing of numerous intermediate frames for animated cartoons can be turned over to the computer through graphics software; or the fitting of a curve to experimental data can be done by the computer through statistical software.

To use the computer as tutor and tool can both improve and enrich classroom learning, and neither requires student or teacher to learn much about computers. By the same measure, however, neither tutor nor tool mode confers upon the user much of the general educational benefit associated with using the computer in the third mode, as *tutee*.

THE COMPUTER AS TUTEE

To use the computer as *tutee* is to tutor the computer; for that, the student or teacher doing the tutoring must learn to program, to talk to the computer in a language it understands. The benefits are several. First, because you can't teach what you don't understand, the human tutor will learn what he or she is trying to teach the computer. Second, by trying to realize broad teaching goals through software constructed from the narrow capabilities of computer logic, the human tutor of the computer will learn something both about how computers work and how his or her own thinking works. Third, because no expensive predesigned tutor software is necessary, no time is lost searching for such software and no money spent acquiring it.

The computer makes a good "tutee" because of its dumbness, its patience, its rigidity, and its capacity for being initialized and started over from scratch. Students "teach" it how to tutor and how to be a tool. For example, they have taught it to tutor younger students in arithmetic operations, to drill students on French verb endings, to play monopoly, to calculate loan interest, to "speak" another computer language, to draw maps, to generate animated pictures, and to invert melodies.

Learners gain new insights into their own thinking through learning to program, and teachers have their understanding of education enriched and broadened as they see how their students can benefit from treating the computer as a tutee. As a result, extended use of the computer as tutee can shift the focus of education in the classroom from end product to process, from acquiring facts to manipulating and understanding them.

Five Pioneers of the Application of Computing to Education

Cinco pioneiros da
aplicação
da computação à educação

Though many computer scientists have broad general interests, most have only a few really dominant specific interests and those few are typically

shaped and informed by the individual scientist's particular point of view. Before looking at the work of our five authors in terms of the tutor/tool/tutee framework, therefore, it may be helpful to summarize the dominant interests and point of view of each.

ALFRED BORK

Bork is a physics professor at the University of California at Irvine where he has directed the Physics Computer Development Project for a number of years. That project produces computer-based material that can serve as the primary source from which first year physics is learned at Irvine. As this implies, Bork's major interest is the application of computing to physics instruction. His work strongly emphasizes concept mastery, self-paced instruction, and computer-resident testing. Though his work beautifully demonstrates how computer/student dialogs can function and how graphics can be carefully and integrally used to enhance these dialogs, he does not argue that all instruction should be computerized, even in a subject like physics. Bork sees stand-alone computers as the major vehicle in the new generation of computer-assisted learning. He is also careful to point out repeatedly that good software in any reasonable quantity is more likely to be developed by software factories or institutes than by individual professors, teachers, or researchers.

THOMAS DWYER

Dwyer is a computer scientist and educator at the University of Pittsburgh, who for a number of years ran a series of projects involving high school and junior high school teachers and students. The projects were characterized by an exploratory approach to using computing, one which tended to depend upon and generate a new way of looking at learning in the school, one which Dwyer himself dubbed "Solo-Mode" learning. In such learning, the framework is provided by the teacher but the pupil must work autonomously, learning to "fly solo." This mode Dwyer contrasts with the more usual classroom situation that keeps the teacher in complete control and has the student "flying dual."

Dwyer's work stresses an heuristic, exploratory approach based on principles rather than a closed one based upon a formula of what to do. He places a heavy dependence upon the teacher as a supportive human being, stresses that the teacher is crucial, and addresses teacher education as a major concern of any attempt to use computing broadly and creatively in the schools. Though much of the solo work dealt with math and physical science, Dwyer's work through solo and elsewhere also applied computing to other subjects, including music.

ARTHUR LUEHRMANN

Luehrmann is now associated with the Lawrence Hall of Science at Berkeley, where he is directing projects to integrate computing into museum science exhibits to make them interactive, and projects to teach computing to a broad, general public served by the museum. Prior to going to Berkeley, he was a professor at Dartmouth and was involved in many successful projects there, applying computing to instruction.

As several of his article titles suggest, his strongest emphasis is upon the computer as a new and fundamental technology worthy of study on its own. He sees the mass impact of this new technology as very substantial and stresses the need for popular literacy, the need for everyone to acquire programming skills, and the need for a good stand-alone personal computer. Though trained as a physicist, Luehrmann's work has dealt with applying computing in many instructional areas, not simply those related to the physical sciences.

SEYMOUR PAPERT

A professor of mathematics and an educator at M.I.T., Papert is best known for his development of the LOGO language and its application to teaching computing and mathematics to young children.

His major thrust definitely is to teach a way of mathematical thinking that young children can intuitively master. By encouraging anthropomorphizing, play, and intuitive guesswork he tries to capitalize upon the existing insights and mental frameworks of children. His strong attention to how and what children are thinking is in part based upon his extended association with Piaget in Switzerland. Papert's work has been exploratory, centering on children's use of computing, emphasizing almost exclusively the child learning to program. It has included imaginative use of robots, graphics, and sound as a child-attractive alternative to traditional textual output. Throughout, the computer tends to be used to create a problem-rich environment, presenting the child with interesting, challenging problems that require a computer for solution.

PATRICK SUPPES

Suppes is a philosophy and mathematics professor at Stanford, where he pioneered the development of computer-assisted instruction.

His work stresses the applicability of the computer to skill areas such as mathematics, logic, and language. It aims to produce complete courses of in-

struction to be delivered by the computer. He has always stressed how little we know about learning but has carefully used what is available to design a considerable quantity of computer-assisted instruction. Much of this instruction is on the market and in wide use on minicomputers. For example, whole systems stressing mathematics and language arts skills are commercially available through the Suppes-founded Computer Curriculum Corporation. His work stresses individualized learning and increased educational productivity.

Using The Tutor/Tool/Tutee Framework

Now that the framework and the five pioneers have been introduced, let's look at some of their writings in terms of the three modes of that framework. We will only cite a few of the articles included in this book and will discuss none in detail; our aim will be to merely suggest the framework's utility. The authors speak well, even brilliantly, for themselves and do so clearly enough to need no explanation. The framework must be accurate enough for the reader to make the associations between it and the articles for himself or herself.

EXAMPLES OF THE TUTOR MODE

Historically, this mode has its roots in programmed instruction. However, when properly deployed it is far more flexible than any book- or material-based programmed instruction. For one thing, in tutor mode, the material can be presented interactively, and dynamic graphics and other sophisticated teaching aids can be integrally used. For another thing (as pointed out earlier), in tutor mode the performance history of one or more pupils can be collected and stored, then subsequently used for evaluating the material and as a basis for routing a particular pupil through the material. At the same time, this mode can be designed to move the student at a wide range of speeds and to be interruptible more or less at the student's convenience. Though the label has been applied to broader applications than just this one of using the computer as a tutor, this mode has often been called CAI (Computer-Assisted Instruction), probably because the ancillary tasks it performs are similar to those that could be performed by ideally competent teaching assistants.

The work of Bork and the work of Suppes both exemplify the tutor mode at its best. All of their included articles deal with this model. Bork has concentrated much of his thinking on how best to develop good tutor material for physics instruction, while Suppes has developed material for a wide range of subjects. Both have used the computer to store, analyze, and act upon student

results, and both have used such sophisticated peripheral devices as audio or graphics to maintain student involvement and enrich the nature of the tutoring.

The time period encompassed by Suppes' work alone reveals how *many* years have already gone into work on tutor mode computing applications. The breadth of those applications is suggested by his prepared statement for the Congressional hearings on Computers and the Learning Society, "The Future of Computers in Education." No one who reads either that or his Marseilles conference article, "Impact of Computers on Curriculum in the Schools and Universities," will mistakenly believe that CAI on microprocessors is completely new. One gets a sense of how much work goes into producing good material in this mode by reading Bork's detailed, "Preparing Student-Computer Dialogs: Advice to Teachers," or by reading his very thoughtful "Learning about Graphics." Anyone who would produce good tutor mode material should certainly be thoroughly familiar with both these pieces.

This mode has had both its advocates and its critics. Criticism from those who are deeply involved in computing and education is usually directed at those making extreme claims about the positive benefits to be derived from tutor mode computing. Good criticism of this kind is exemplified by Luehrmann's "Should the Computer Teach the Student or Vice-Versa?" Neither he nor any other pioneer, however, would argue that tutor mode computing should not have a significant place in education.

EXAMPLES OF THE TOOL MODE

Tool mode is probably seen to be the major mode of computer use by most people outside computing and education. Because it receives considerable attention and encompasses such a wide range of activities, tool mode computing is popularly seen as synonymous with computer use, period. For example, most business data processing, whether routine accounting or word processing and office automation, uses the computer as a tool. Thus the school's administrative activities use the computer in a tool mode, from payroll and inventory to pupil scheduling and grade reporting. Even library automation involves the computer merely as a tool.

In tool mode, the computer provides a service that the user needs and more or less knows how to use. It is not primarily a teacher or tutor as in the tutor mode; it is not user-alterable and not a set of raw building components as might be provided under tutee mode. Use of the computer in tool mode may teach the user something during use, but any such teaching is most likely accidental and not the result of any design to teach. In tool mode computing, the user can only explore activities and ideas for which the tool at hand is appropriate; one can explore musical inversion with a composition and playback

tool, but not a word processing tool or a regression analysis tool.

Most people in computing and education frequently and creatively use the computer in a tool mode, because of their everyday familiarity with computing capabilities. However, possibly because they are familiar with such use, possibly for other reasons, most computing-and-education people do not see this mode as something they want to focus primary energy upon. All five pioneers whose work is included in this book fit this general position. All assume heavy use must be made in education of tool mode computing; none advocates it as most important or focuses his own major interest upon it.

All five have advocated the use of the computer as a calculator and a word processor, and all have advocated various other tool mode uses as well. Bork and Suppes, for example, suggest that the computer as a calculator and record keeper should be available simultaneously to anyone using the computer in the tutor mode. Bork further suggests that various graphic tool capabilities should be similarly available. Dwyer, Luehrmann, and Papert all argue that various tool capabilities should be available to be utilized by anyone who needs them in exploratory problem-solving, writing, or anything else. Dwyer, for example, argues that the student flying solo on the computer would freely utilize many of its tool capabilities as he pursued his overall project.

EXAMPLES OF THE TUTEE MODE

The tutee mode is the one upon which Dwyer, Luehrmann, and Papert focus their energies. This book reflects that.

One of the early and still one of the best arguments for this mode of computer use is Luehrmann's "Should the Computer Teach the Student, or Vice-Versa?" In it, he argues that in teaching the computer, the child learns more deeply and learns more about the process of learning than he or she does from being tutored by software written by others. Papert extends the argument by suggesting how children using the computer as a tutee may learn more of what they should be learning of mathematics than they can in classrooms without computers. This position is clearly articulated in both "Teaching Children Thinking" and "Teaching Children to Be Mathematicians vs. Teaching about Mathematics." Dwyer extends the concept in a different way in "Some Principles for the Human Use of Computers in Education," defining his now well-known concept of solo-mode computing and showing how it relates to the total curriculum question.

In these essays and others, all three suggest that in using the computer as tutee, the learning the child experiences is qualitatively different than he or she might otherwise experience in any school setting. None downgrades the role of the teacher in a tutee mode environment, but all see it as different from the teacher's typical role now. Dwyer states the case very well for all three in

his "Some Thoughts on Computers and Greatness in Teaching."

Papert suggests that the computer as tutee can, with appropriate graphic and robotic capabilities, serve as a means to enable the child to link his or her experience to the deep, fundamental mathematical ideas we most want children to learn. In his "Personal Computing and Its Impact on Education," he suggests that this may be the only way to avoid having most children spend most of their time struggling within the dismal reality of dissociate learning. Both in that essay and in "Computer-based Microworlds as Incubators for Powerful Ideas," Papert contrasts dissociate learning, the attempt to somehow internalize great quantities of information apparently of no use in the child's world, with a more natural learning that resonates with the child's experience.

Using the Framework Without Becoming Blinded by It

The tutor-tutee-tool framework has been presented to help those who would like to get an organized initial grasp on an apparently complex field. It will serve to overcome hesitation and initial trauma. Of course it can and should be used later, so long as it conveniently provides insight. It is a reasonably broad framework and suffers from no more shortcomings than any other schema or typology. As such a schema though, it can divert attention from relevant insights when used too slavishly.

Reasonable alternatives to this framework certainly can be advanced. They might be entirely different or be simple extensions of it. For example, I seriously considered extending the tutor, tool, tutee framework to include a fourth mode, making it tutor, tool, tutee, toy. There are numerous games, simulations, and models of many sorts that one spontaneously classifies as toys—they appear to have been created above all to play with and enjoy, whatever other merits they might possess. I finally decided against that extension, though, believing that such software is just as well subsumed under one or more of the earlier three modes. The point is, one need not be bound by this framework. If modifying it or replacing it by an alternative framework helps with the process of internalizing the ideas advanced in the various articles, then such modification or replacement is in order.

The articles are the main thing. They provide a good introduction to computing and education. If you prefer a different order, follow it. If the framework gets in your way, disregard it. But do read the articles—all 19.

Editorial Notes

In editing a collection such as this one, the editor faces many temptations. Three that were successfully resisted are particularly worthy of mention.

(1) Textual changes other than correction of typographical errors were not made, despite the temptation to impose more stylistic uniformity by making at least minor changes in footnoting, bibliographical presentation, and so forth.

(2) No editorial asides were inserted anywhere, despite the temptation to qualify or clarify. Where they do appear, parenthetical asides are strictly those that were present in the original article.

(3) No footnotes or other additions were made to bring out-of-date details up-to-date. Writing about high technology always loses the battle to keep the product current, and these articles are not exceptions. In any case, the important contribution of these articles is their articulation of issues that transcend the immediate detail spawning them, issues that will remain significant as the technology continues to change.

ROBERT P. TAYLOR
February 3, 1980



ALFRED BORK

ALFRED BORK is widely known in computing circles as an expert on applying the computer to the teaching of physics. For more than 10 years he has been professor of physics and information and computer sciences at the University of California at Irvine, where he directs the Physics Computer Development Project. As founder and director of that project, Bork has pioneered the use of the computer as a creative component in the teaching of undergraduate physics. He has authored numerous computer-based dialogs to teach various aspects of physics; these dialogs employed advanced technology, especially computer graphics, with computers of all sizes. Bork has written a great number of articles and several books, addressed many learned conferences, served both as officer and member of various societies and special interest groups relating to computers in education, and directed or helped to direct a wide range of funded projects to improve instruction through appropriate use of computing.

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é amplamente conhecido nos círculos da computação como um especialista em aplicação do computador ao ensino de física. Por mais de 10 anos, ele foi professor de física e de ciência da informação e computação na Universidade da Califórnia em Irvine, onde dirige o Projeto de Desenvolvimento de Computadores para Física.

Como fundador e diretor desse projeto, Bork foi pioneiro no uso do computador como um componente criativo no ensino de física para alunos de graduação. Ele é autor de inúmeros diálogos baseados em computador para ensinar vários aspectos da física; esses diálogos empregaram tecnologia avançada, especialmente computação gráfica, com computadores de todos os tamanhos.

Bork escreveu um grande número de artigos e vários livros, palestrou em muitas conferências acadêmicas, atuou como dirigente e membro de várias sociedades e grupos de interesse especial relacionados a computadores na educação e dirigiu ou ajudou a dirigir uma ampla gama de projetos financiados para melhorar o ensino por meio do uso apropriado da computação.

2. Preparing Student-Computer Dialogs: Advice to Teachers

BY A DIALOG I mean in this document a “conversation” between a student and a teacher, where the student is at a computer display and the teacher is conducting the dialog through a medium of a computer program. Typically such a dialog follows a pattern like this. First something will be typed to the student, possibly some information. Then the student will be asked to reply, or perhaps to ask a question. Several modes of reply are available. Depending on what the student put in, a number of things might be displayed next.

I CAN'T READ YOU . DO YOU KNOW HOW TO ADD COM-
PLEX NUMBERS?

?YES

THEN TRY AGAIN PLEASE BUT THIS TIME ADD THE
TWO COMPLEX NUMBERS WITH PAPER AND PENCIL
BEFORE TYPING THE ANSWER .

WHAT IS

$$(2+3*I) + (7-8*I)$$

EQUAL TO?

? ■

If you have access to dialogs, you should examine a variety of such dialogs. Experience in running good dialogs is the best way to develop a quick appreciation of what is possible.

In this paper I offer advice to those first attempting to prepare such student-computer dialogs. Writing a dialog is only a partially understood process at present, so *any* advice is partially subjective, and should not be taken too seriously! We all have much to learn about the educational use of computers.

Nevertheless some tentative experience can be brought to the attention of the teacher who is designing such material. This discussion attempts to do this, using the experience in the Educational Technology Center and the Physics Computer Development Project at the University of California, Irvine, as the basis.

Getting Started

The critical decision the dialog writer must face is *what* to write dialogs about. At present the use of computer-based dialogs is experimental, and in many areas, untested. Although some evidence exists to show that dialogs *can* be highly effective in assisting students in learning, as compared to other methods, much usage is untested and judged only subjectively.

Hence, the burden of educational proof is upon the dialog writer, who cannot assume that simply putting standard existing material into interactive dialog form in trivial ways will improve the learning environment. Furthermore, the preparation of extensive dialogs can be a lengthy job, putting a premium on a wise choice of subject and approach for a dialog.

One way to approach the problem of preparing dialogs is to ask how you could, within the teaching of a particular subject area, gain some unusual leverage with computer dialogs. Where can the computer have maximum learning impact? The answer to this question might be different for different areas and could only be given by someone with an extensive knowledge of both subject and pedagogy. It is very important to *ask* the question and to concentrate dialog efforts in areas and approaches that look promising for this technique.

It is reasonable to debate with colleagues as to where the effort should be concentrated. Often such a question leads to a fundamental rethinking of the area involved, a reconsideration of deep educational objectives. This direction often involves a *fundamental* rethinking of the goals and objectives desired in the unit of course under development. Given this powerful new learning device, it may be possible to learn important ideas at an earlier stage in the student's education; examination of the topics and technique stressed in more advanced courses may be useful at this stage. Some tasks previously considered very difficult or almost impossible can be learned using the computer.

Ideally this process should also consider the other educational media available for learning the specified objective. It is beyond the scope of this paper to discuss the full problem of course design. To take an entirely new approach to a course rather than simply an approach that adds on existing material is desirable, but it is a difficult and expensive process. The instructional design activity is an extensive science in itself, and one that many teachers may be unwilling to approach simply because there is much to learn there. For a full curriculum development, it is essential.

Everyone will have different views about the best approaches for course design and instructional development. My own prejudices in this direction are pointed toward the work that The Open University has been doing in designing their courses for the last few years. While The Open University courses have used computers only in minimal ways, their general approach to course design combines both the research information and practical considerations in a very desirable blend. I will not describe that process in more detail in this discussion.

Goals

Perhaps some brief comments should be offered concerning short-term versus long-term learning goals. In preparing any activities for learning we should decide what we are trying to do, specify our learning objectives.

If we consider a science course, there may be a factual piece of information at a point in the course that the student needs to learn—the standard theories that already exist in the area, the mathematical techniques that go along with these theories, etc. But our interest in learning science is not all archival, intended to persuade people to look admiringly at these lovely mental structures of past science. Rather we hope to prepare people who can proceed to *use* this information, modestly or in great creative leaps beyond the present status quo. Learning factual material is one task, but being able to use it is often a different matter. The moment of truth for students in a science course comes when they must work difficult problems, problems which demand that they obtain new information from the information and techniques presented. The long-range goal of most science courses is to teach people to make these developments themselves.

Learning to tackle difficult problems successfully is a hard task. The heuristic strategies involved in problem solving, for example, are seldom discussed with students (a glowing exception is George Polya's book, *How to Solve It*, highly recommended for both teachers and students).

The dialog writer should keep in mind long-range teaching goals, and stress them in whatever way possible. They are very easily ignored because they pose far greater difficulties than the mere presentation of information.

Although these comments on goals are directed toward science courses, the consideration is important in curriculum development in all areas.

Types of Dialogs

A dialog which recreated a book, or a printed program text, or some other teaching method, is not likely to have long survival value at the present time. Perhaps when the cost of computer usage is considerably less than it is today, and when we become more knowledgeable in the use of computers, the computer *may* replace the text, in many situations, but this is a little in the future.

A corollary is that it is too early to prepare a "complete" computerized course; we should concentrate on small segments and study their effectiveness. Innovation in many directions is still essential, and trial and error is still important. Using the computer does not imply abandoning traditional technologies for learning such as the book or film. But experiments already underway on text display indicates that the situation may soon be different.

I will not attempt a full classification of the many types of possible dialogs. Several types lend themselves to the beginner in the area, but the developer need not use any of the types described.

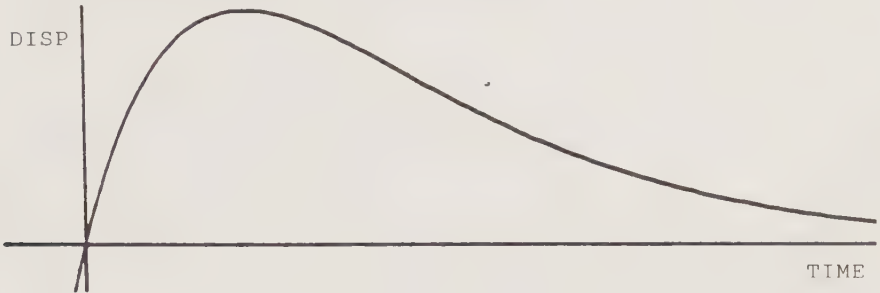
In recent years we have discovered that perhaps the most powerful of our computer applications, in terms of both our judgment and reactions of students, have been the on-line tests. The advantage of the on-line tests are that they can not only offer immediate feedback to the student as to whether the student has answered things correctly but can also offer immediate aid. Thus, we can diagnose in many cases exactly what problem the student had and immediately show what went wrong with his or her attempt to solve a problem. Typically this use has taken place in a personalized system of instruction environment, but it can be done anywhere.

Thus, this material serves both a teaching and testing function, and when coupled with a full course management system often can be a driving force for the entire course.¹

A class of powerful dialogs are the *remedial dialogs* which determine the student's weaknesses in an area, and give assistance just where needed. These can occur *within* an on-line test or as a separate program. One useful procedure is to begin by assuming that the student *does* know the area, giving a series of questions which selectively test the student's knowledge, perhaps by working examples. These problems need not be difficult; if they are to be repeated an infinite set can be "generated" by means of a problem generating sequence. Students will only be sent into the remedial parts of the dialog if they cannot handle these problems after several attempts (to allow for typing errors). This approach has the advantage that students receive assistance only where they are weak, so the program can be highly responsive to individual needs.

One variant is a dialog which assists students who have trouble working a problem, finding where the trouble occurs and giving help. Figures 1 to 5 illustrate the process; the pictures are from the help sequence in an on-line quiz called **SLOPE**.

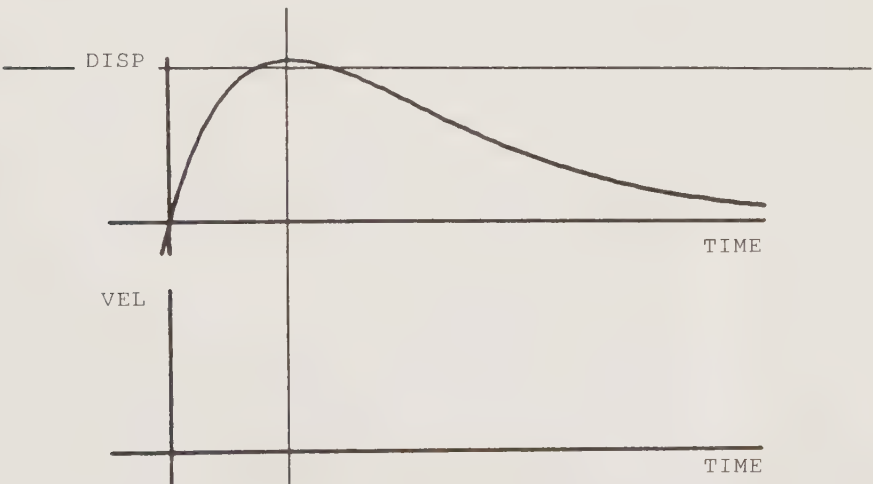
Suppose we have a particle moving on a straight line. Let's plot where it is as time goes on.



Our problem is to construct the velocity-time curve for this same motion.

PRESS RETURN TO CONTINUE . . . ■

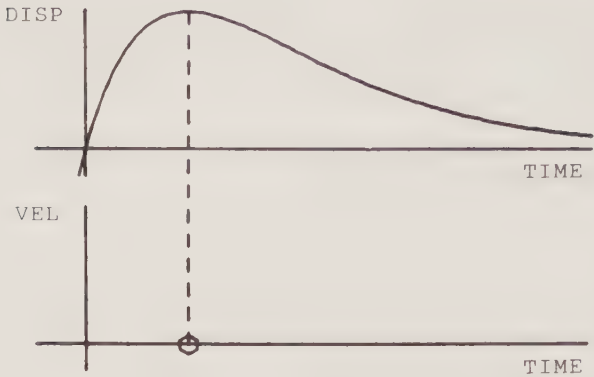
Figure 1



Is there a time when the velocity is zero? YES

The velocity is zero at one time.
Point to it with the cursor.

Figure 2

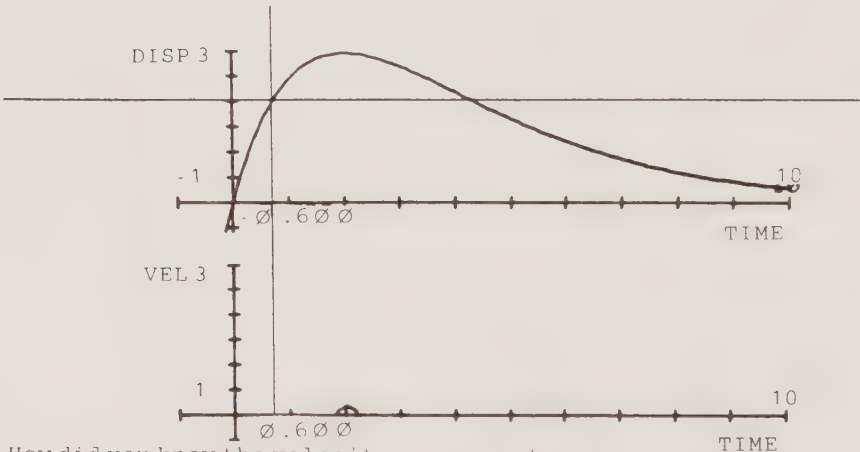


Is there a time when the velocity is zero? YES

The velocity is zero at one time.
Point to it with the cursor.
GOOD! Let's indicate that fact on the graph.

PRESS RETURN TO CONTINUE . . . ■

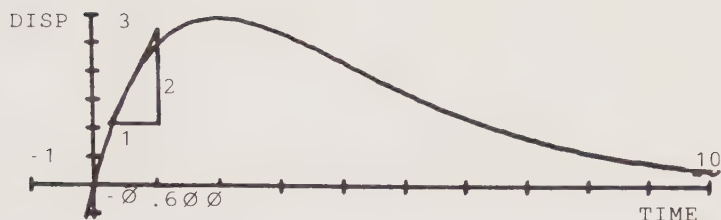
Figure 3



How did you know the velocity was zero at that time? BECAUSE THE SLOPE IS ZERO

Yes, the tangent is horizontal.
Find a place where velocity is positive.

Figure 4



We'll draw the tangent at one location.

How can we determine the slope at this point? GET X AND Y

You have only the curve to work with, not an explicit function. So you need a geometrical way to take the derivative.

We'll proceed this way.

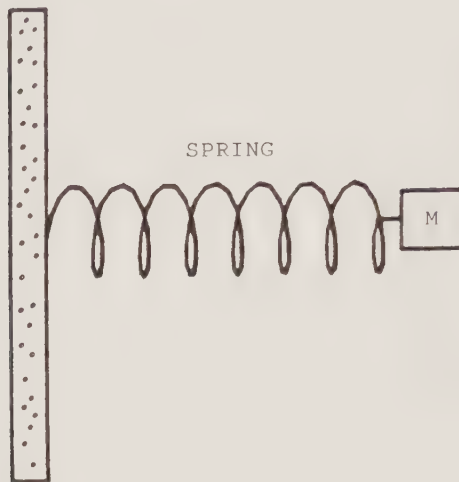
Let's draw a right angle triangle with the tangent as the hypotenuse with sides parallel to the axes.

The slope is the ratio of the vertical side to the horizontal side.

The slope is 2.

Figure 5

Another area of the sciences where we think that dialog material is usually effective is that of the *interactive proof* or problem. The idea is to allow students to attempt to prove some of the important results in the development partially on their own, making choices and guesses along the way, perhaps in response to suggestions in the program. The process of developing difficult proofs can be made an active process rather than the passive one (for most students) of listening to a lecture or reading a text. The students attempt to make the decisions rather than just seeing the decisions the instructor makes, and so hopefully learning something of the art of proving substantial results. Figures 6, 7, 8 and 9 are from a dialog on the harmonic oscillator.



WE FIX A SPRING AT ONE END , AND DISPLACE
THE OTHER END A DISTANCE x TO THE RIGHT ,
SO AS TO STRETCH THE SPRING THAT AMOUNT .

WE HAVE TO PULL ON THE SPRING
TO STRETCH IT .

IN WHICH DIRECTION SHOULD WE PULL?
? ■

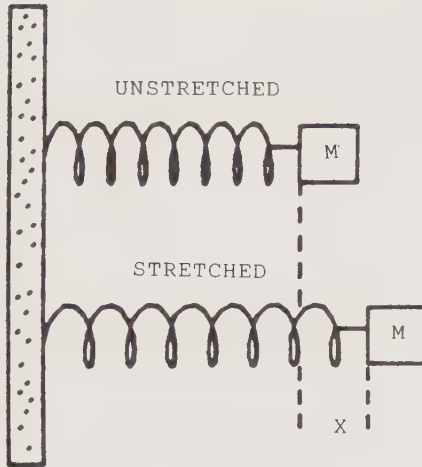
Figure 6

O.K. SO A SPRING STRETCHED TO THE RIGHT EXERTS A
FORCE TO THE LEFT. BUT WHAT ABOUT THE MAGNITUDE
OF THE FORCE? DOES IT DEPEND ON THE AMOUNT OF
STRETCH?
?SURE

IT DOES. THE FORCE IS DIRECTLY PROPORTIONAL TO
THE CHANGE IN LENGTH BUT OPPOSED TO THE DIS-
PLACEMENT. SPRINGS OBEY HOOKE'S LAW.

JUST PRESS RETURN TO GO ON . . . ■■■

Figure 7



THE FORCE EXERTED BY A STRETCHED SPRING
COULD BE USED TO ACCELERATE SOMETHING .
WE COULD IMAGINE ONE END OF THE SPRING
CLAMPED TO A RIGID WALL AND A MASS M
ATTACHED TO THE OTHER . SUPPOSE WE DISPLACE
THE MASS A DISTANCE X TO THE RIGHT , AND
RELEASE IT .

WRITE AN EQUATION THAT ALLOWS US TO
PREDICT THE SUBSEQUENT MOTION OF THE MASS .
?MA - KX ■

Figure 8

BUT LOOK HERE: X MEASURES BOTH THE CHANGE IN
LENGTH OF THE SPRING AND THE DISPLACEMENT OF THE
MASS . SO FOR THE VELOCITY WE COULD WRITE

$$v = \frac{dx}{dt} = x'$$

WRITE AN EQUATION FOR THE ACCELERATION :

$$A = \frac{d^2x}{dt^2} = x''$$

Figure 9

Similarly, a problem at the computer has advantages over a textbook problem; you can, for example, make students *ask* for relevant information, rather than giving it all in advance as in the typical textbook problem. So they must decide what information is relevant, necessary in the “real world” outside the textbook environment.

Another possibility is to present an investigative facility, or *interactive world* for student use. In such an environment, students can explore freely, gaining experiences that may not normally be available in any other way. Such a world can be used either freely or as part of an author-structured experience, perhaps a controlled experiment with the computer showing what the student is doing and offering advice where appropriate.

One of the principal values of the interactive world dialog is that it provides the possibility of building up student intuition and insight into the way things behave, often by providing a wide range of experiences to the student. Such intuition is an important aspect of learning, and one that is often difficult to do anything about within the traditional courses. We are often criticized for producing students who know how to carry out the mathematical manipulations in science courses, but who are unable to intuit beyond the formal derivations. However, to use such program in this fashion demands various auxiliary material we have found.²

Many other types are also possible.

Mechanism

How should the dialog writer work? As with matters of style, this is very individualistic, and further will be heavily dependent on the facilities available. If you *must* work in one way, because that is all that is available, you may want to skip this section. All that is attempted is to mention and comment on some of the possibilities. A fuller description is available from the Educational Technology Center at Irvine.³

Should you work alone or with others? Our experience is that it is highly desirable to work in groups of two or three.⁴ A single person will miss many possibilities, while a large group tends to spend all its time in debate, accomplishing little. Work with someone with similar views of the learning process. It is desirable that one member of the group have previous experience in preparing dialogs.

After the basic area has been chosen, the authors should prepare at least a brief outline of their “mainline” approach, showing the material to be covered. A one-page outline is often useful. A more detailed treatment is often a useful beginning. If mathematical derivations are involved, these should be fully worked out in advance, allowing for all the reasonable variants likely to occur with the students who will use the dialog.

Some authors will develop the mainline material almost fully in preparing the dialog, developing the messages that will be typed to the student who is answering correctly in full detail. Then they will go back and fill in bad branches and loops, or other mainlines. However, some teachers prefer to work on a "frame-by-frame" basis, outlining the principal development briefly, and then going sequentially through the program branches along the way.

My own preference is very much for the second style, although with very complex dialogs the sequential approach may present a problem in remembering where one is along the process. The psychological advantage of this approach is that it continually obliges the teacher to think how to respond to students who are confused or who do not know what they are doing, while the mainline approach may lead one to be impatient in filling in the details. Dealing with the troubled student is the most important aspect of the dialog; the very good student does not need much assistance in learning.

Another aspect of the mechanism of dialog preparation is the author's relation to the computer programming language or languages involved in the final preparation of the material. A good many variants are possible. First there is what might be termed the "Coursewriter approach," the one the original developers of the Coursewriter language in IBM seemed to have in mind. This involves the author, the teacher, in using the language itself in writing the dialog, writing statements directly in that language as he or she thinks through the learning sequence.

Many such languages intended for computer-aided learning exist, but most have seen little usage in developing large collections of student-usable materials; Coursewriter and Tutor are the two major exceptions. The languages developed for this purpose have often claimed to be easy to use for the novice, but the non-programmer does not necessarily agree with this. Further, these languages often restrict the resources available to the author. The most versatile recent language used in this direction is Tutor, developed at the PLATO project. Most of the languages, including Tutor and Coursewriter, are old and so do *not* conform to the best standards of modern programming activities.

A second approach is for teachers to work in developing the dialog in a (modified) flowchart form, in a way that does not depend on the details of computer mechanism to be used. (A variant on this approach is to use decision tables.) The teacher sketches out the conversation by a series of boxes, lines, and other graphic aids, showing what to "say" to the student, what responses to handle, the message typed or displayed for each response, the record keeping desired. The following flowchart, a fragment, illustrates the process.

Note that this flowchart shows a variety of things. First it shows the text, including formulae, that are to be written out to students. It also shows sketches of what is to be drawn at a particular point. You will see too the occasional use of a "delay" between text and graphics and vice versa. It also

* * New Page * *

The proportionality constant depends on units chosen,

In one common set of units, MKSA or System International-

$$F = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{r^2}$$

Where charge is in COULOMBS, distance in METERS, and $\epsilon_0 =$ _____

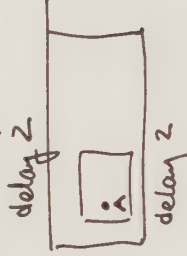


RESET BADZ

* * New Page * *

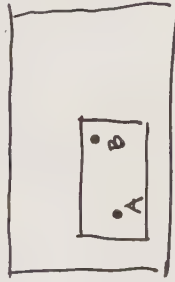
Next on our agenda is the direction of the force
delay 1

Here's a positively charged particle, A



and another B also positive

Delay 1



A and B not on same horizontal line!

Delay 2

The force on B is a vector quantity. Then out in time B. Point with cross hairs to top of array +

1
This doesn't seem
any +R4 agency

CURSOR

* other cases bump BADZ
Save his cursor position

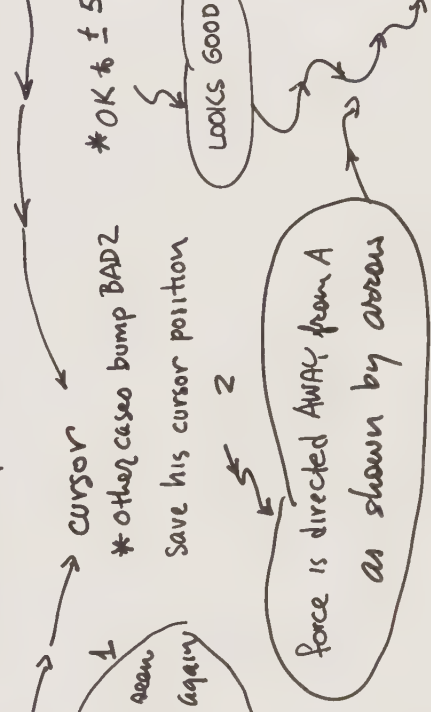
* OK to $\pm 5^\circ$

* Toward A $\pm 15^\circ$

No, the force is repulsive,
not attractive.
Try again

LOOKS GOOD

Force is directed AWAY from A
as shown by arrows



shows the logic involved in responding to student input. In this case the student is using the built-in pointing device, the cursor, on the display. Note that there are a variety of possibilities examined as to what the student does. It is not important to read the individual details; this flowchart is clearly rather messy, typical of those that are produced. Our purpose is to show how the process goes. Sometimes additional extensive instructions will need to be given to the programmer about how the material is to be displayed, particularly when there is elaborate interweaving of text and graphics as is often the case.

A third possibility is the use of a facility that prompts the instructor, sitting at a terminal, for the various pieces of the dialog that will be necessary, like the Scholar-Teach system, the DITRAN system developed by Noah Sherman at the Lawrence Hall of Science at Berkeley, and PLANET.

Again I state that my own strong preference is for the second way, using teacher-oriented flowcharts. This would remove teachers from the computer details and allows them maximum freedom to do *what* they want to do within the program without worrying about *how* to do it. This approach also enables the instructor to interact with a variety of specialized helpers, secretaries, graphic designers, and programmers, who will be responsible for the computer-related and other details.

The learning program, the dialog, must be entered in the machine. Either typing at a display or key punching is possible, depending on the computer system employed. But key punching is fast vanishing. Since dialogs are to be used on interactive systems, either timesharing or stand-alone, it is advisable to use the often powerful editing capabilities of such systems to assist in the preparation of the dialog material; so on-line entering and correcting of programs is very desirable except where it is ruled out by financial considerations or by system unreliability.

A good individual for typing or keypunching is the trained secretary, rather than a programmer, a student, or a teacher. Programmers and teachers in general are *not* good typists. Anyone who is acquainted with dialogs knows that a vast amount of time must be spent in typing the material, since much of it is the text to be shown to the student. It is not difficult to train secretaries to use computers and to work in well-designed programming languages and editing systems.

Recently in our work at Irvine we have moved away from secretarial entry. This is because of the increasing importance in our situation of tackling the entire problem of screen design, how the material is to appear on the screen. This includes the placement of text and the placement of pictorial material as well as such things as timing considerations. Another type of individual is often best to do this, a person who has training in design in the sense of industrial or applied design. The process resembles that of magazine layout, but

the tools are somewhat different. If this procedure is to be followed, a collection of on-line interactive graphic aid tools must be developed and maintained.⁵

Style

To be doctrinaire about style is unwise, even more so than in the rest of this discussion, because style is so individualistic. It seems reasonable that dialogs should *not* always be in the same style; different people have different ways of learning.

One tendency is to approach the problem of preparing student-computer dialogs as if writing a text or a paper or giving a lecture. But the difficulties and the opportunities are greater with dialog material; the style of the dialog should reflect these differences, both advantages and difficulties. Most of the concern with a text is with the “main line,” the right way of handling the developing material. Usually only one main line is considered, although occasionally alternate proofs may be given. A dialog may include multiple main lines, to react to different ways the students may proceed.

The dialog writer should also expend great energy and effort in sections that never appear in texts—wrong approaches, mistakes to which you should respond in some reasonable way, remedial assistance for a student who is having mathematical difficulty. If a computer-student dialog is to prove valuable, it will need to be much more responsive to student needs than a static book. This means that the non-main line sequences are extremely important for the dialog; typical students are likely to be in these areas of the program much of the time, so they deserve utmost care.

Dialogs should resemble *conversations*. The name “dialog” suggests the model of the student conversing with the teacher in the office; the teacher asks questions designed to help students learn the material or responds to the students’ questions. Clearly we cannot fully realize with the computer the model of the office conversation; some people have even objected to *trying* to make a computer dialog resemble a student-teacher discussion. However, it seems possible to follow this model. I believe this is a viable approach; we are slowly becoming more skillful at creating such dialogs.

The model of human dialog suggests that computer dialog style, in a literary sense, should be more like a conversation and less like a book. Talking is less formal than writing, and often more redundant. Humor and light touches are desirable and welcome, although experience shows that not everybody agrees as to what is humorous! Informal vocabulary and structure, as opposed to elaborately structured and carefully formulated sentences, is desirable. While some people talk in the type of long and involved sentences we see in learned articles, this is rare even among college instructors! But it is probably

wise to avoid the "in" phrases of the moment, because they date rapidly. Other style considerations are like those employed in good prose of any type.

Another disputed issue, with little *empirical* evidence, is the question of the use of a first person style. Should the computer say "I" or "we"? Most dialogs developed at Irvine have used the first person style, while many on other systems have not. The Irvine students, when queried about the first person style, supported its use. More information is needed, perhaps through psychological studies, as to whether the computer should be saying "I." We used one dialog at Irvine with two branches, randomly chosen, one of which uses the first person, one not; little difference could be seen in student performance.

Students have a number of ways of interpreting such an "I" in a computer dialog. They may think of it as the authors of the program rather than the computer itself. Dialogs should identify the author or authors, and this might make "I" more natural. We prefer that authors' names appear at the beginning of a dialog. Or you can, if you prefer, choose not to use personal pronouns.

More generally the question of how style influences student response is undetermined. It has been suggested that relatively small changes in style, in, for example, technical vocabulary, may have considerable influence on student output, but little evidence exists. Again a randomly chosen two-branch dialog to explore this question showed little difference in student performance with altered vocabulary.

Perhaps one of the hardest things for teachers to keep in mind in preparing a dialog is that they have at present limited facilities for analyzing the student response. A carefully organized and prepared dialog will sometimes miss the meaning of what the student is typing, even though the dialog has already been improved from past student usage. The computer is *not* a person, and does not have all the resources of an actual teacher for dealing with the students' comments.

Largely we identify responses by string matches, looking for key words or letters in the student input. Even with elaborate care for different types of string matches, we cannot react accurately to everything the student says, and we cannot currently approach the capabilities of human beings except in restricted situations and with large amounts of computer time. Care in how the questions are stated is valuable, but does not do the entire job. The authors try to find all the reasonable alternatives the student may use, but will not always succeed.

This weakness indicates that a degree of humility and modesty is required in the response to student comments, particularly the comments which we have been unable to analyze and which we presume to be wrong. To *tell* the student unequivocally that he or she is wrong can be dangerous, except in environments where the response is carefully controlled by the situation, or

where extremely detailed analysis of input is made. Your program may be missing an unusual variant of a right answer.

Along with the previous suggestions a pedagogical point seems reasonable. A tendency exists, particularly with impatient individuals, to be scornful of the students' lack of success in a particular place in the program. Abusive language, or language which questions the student's intelligence, is seldom desirable in a learning situation, or dialog. Thus, it is not reasonable to call students "stupid" because they do not put in the response you were looking for at a particular place.

A tendency to allow technology control is inherent in employing technological aids to education. This trend is a mistake; teaching aims and teaching purposes should always be in the forefront in developing dialogs. Thus, in applying computers to learning, you should resist the temptation to be guided by the facilities available. Rather the primary emphasis should be on what students are to learn and how you want to teach it, the pedagogical aspects. Ideally authors should develop the dialogs without much regard to the details of how they will be put on the machine, although they need to have some background of what is *possible* with the computer. Pedagogy should take precedence over technology in all cases.

A stylistic tendency often noticeable in new writers of computer-based teaching materials is to spend too much time "talking" to the students, accepting only trivial responses and presenting long messages. We might call this the "textbook malady" or the "lecture complaint." There are places where one does want to type long messages, or interact only minimally, but a dialog which does no more than give lengthy written responses may not be worth putting on the computer, since it becomes a book typed to the student. A dialog writer should *involve* the student in a different way, stimulating meaningful responses which contribute toward learning. The fact that we can have students play an *active* role rather than the passive role often seen in the lecture environment is one of the most important aspects of the computer as a learning mechanism.

Interesting sidelines involving much display without instruction can be made optional; thus, in a science dialog, historical discussions of the issues may not interest all students, but may interest some. Letting students choose in such situations is reasonable and increases the flexibility of the material. Similarly, a review might be optional for students who have done well, but automatic for students who have not; you can save information about performance to use in making such decisions. Auxiliary printed information can be available and students can be queried about its presence.

In many instances it is reasonable to allow students several tries at the answer, perhaps even a large number of attempts. There should, however, be an eventual exit in all cases, to avoid having students trapped at some point, not

knowing how to get out of the trap. Setting of counters and testing on counters can allow flexible looping for additional tries. You can have a series of hints, successive pieces of advice, which can be given to students not putting in the expected results or full diagnostic sequences to allow for individual differences in student background.

You can give advice particularly tailored to wrong responses. For example, if you are expecting an equation and the student is not entering an equal sign or some equivalent word, then you could state that you are looking for an equation and not identifying it in the input. You may expect a formula or equation that contains certain symbols, which do not appear, you can reply with the information about what is missing. It may be useful for the student to try the question again. If students have only *part* of the answer, they can be asked to enter only the aspects missing; don't require more typing than necessary. Wrong responses may also be given special attention, particularly if the student can be led to understand *why* the answer is incorrect.

In looking for verbal input, a good policy is to look for only *part* of each key word, thus bypassing some problems of poor spelling or typing. You might also look for likely misspellings or mistyping; this is much easier to do when you are revising the dialog. Thus, one of our dialogs recognized PLOY for PLOT, a common typing error.

The quantity of retries and specialized advice can vary from place to place within the dialog. For some important results students may need many, many attempts, but in others more than one may be unreasonable. The dialog author can spend an infinite amount of time preparing any one question in the program in an attempt to analyze the student response. But the teacher should use judgment as to where a point of diminishing return is reached, usually a pedagogical decision. Usage with students will provide information on improving program responsiveness. The author should also be prepared for programming errors to become more of a problem as dialog complexity grows.

An analysis of the student's input is not always necessary. In some situations the program can simply accept the input and go on. Thus, you may want the student to *think* about the material, or to have some pause in between sections of material. Or you may want students to make an input, intending to say the same thing no matter what is entered. Or you will at most give a pat on the back for a correct response.

Another situation in which the nonanalyzed input is of value is with student comments. Dialogs should usually invite long multiline comments from the student at the end, and presumably these may be too complex to allow any immediate reply. Another related device useful in providing feedback as to the teaching success of the program is to ask for a long verbal description or summary of the situation being studied. Thus, in a dialog involving standing waves on a string, we ask students to describe a standing wave and tell what

types of standing waves are possible; if the program has worked students through the first normal mode, they can describe the second normal mode. Such a long entry, involving many lines, could not presently be analyzed in a very meaningful way, although one might still choose to respond to some key words. But the teacher can examine these detailed comments, stored in a disk file, and determine if they *do* indicate that most students understand the material covered. This mechanism can also be used for getting feedback to the students. A student can be asked to sign his questions or queries with the promise that the reply will be coming soon.

One stylistic question on which there is not universal agreement is the necessity for what the behavioral psychologists call "positive reinforcement." A view supported by many psychologists and teachers is that when students make the right response they should *always* be told they are right. However, others argue that we do not do this in normal conversation and so are not willing to do it at all times. One can have compromise positions, sometimes responding favorably to correct answers and sometimes not. I believe that it should be done frequently but not all the time.

A place in dialog writing where imagination tends to be limited is the constant need to say the same basic thing over and over, but in different ways. The typical situation is "try again," the response that the student should attempt the question at least one more time. Congratulating the student on a right answer is another similar situation, reinforcing the response. It is convenient to have built-in facilities to vary the response in these situations.

One of the most important aspects of the dialog is the ability to respond reasonably to the *wrong* answers. If students say something which is wrong and you can tell them just *why* it is incorrect, and perhaps allow another try, then the dialog is serving an important interactive function. In thinking about the possible responses for every question the teacher needs to consider what the student can say that is not right, and what response is reasonable. This is not necessarily easy, and some discussion with others may help; lectures and textbooks seldom take this approach. Groups of two or three in the writing process are useful in coming up with the many possibilities of incorrect responses. Good dialogs often devote more of the program to responding to wrong answers than to the main line material, sometimes dramatically more. Don't worry about how the professionals can slip by; the dialogs are written for *students*, and need to be responsive to *them*.

No matter how detailed the analysis is on a student reply, you must be prepared for the possibility that none of your tests have succeeded; you have not been able to identify anything in his input. So you must always be prepared to say something in such a case. Often students are allowed to try again after a hint. How many times they are allowed to repeat the question requires a pedagogical decision in each case. The number of tries may depend on the

importance of the question for the material being taught, or it may be at least partially determined by your resourcefulness in thinking of genuinely useful help sequences. And you must allow for the case that all the hints fail, so that the student must be told the response you have been seeking. In some situations you might want to note such a failure by setting a counter, so that you can return to this point later for this student. Or you might suggest that the student go to a text, to the instructor, or to another dialog.

Graphics

In early educational use the computer had available only typewriter-like devices, so communication to and from students could only proceed through alphanumeric information. Recently, display units with good graphic capability have become reasonable for large numbers of students. Hence, modern preparation of dialogs can make use of visual methods of presentation as well as alphanumeric information.

We need only examine the blackboards around the universities to see the widespread use of pictorial or iconic representation of information. In addition to the more obvious graphs and curves other devices are in common use, such as underlining, putting boxes around important information, drawing arrows to show logical connection, drawing more elaborate flowcharts to indicate processes, and employing varying size type faces for emphasis. Lectures or discussions also employ a simple graphic "input" device, simply pointing to places on the screen. All of these, and many others, are possible within a computer-student dialog, and if at all possible, given the existing facilities, should be taken into account by the dialog writer.

We believe strongly at Irvine that the educational material that will survive in the period ahead is almost certainly graphical. The importance of pictorial information as an alternate form to alphanumeric information is great. However, teachers are not necessarily skillful in using iconic representations. On the contrary, because teachers are often the successful products of a highly verbal education, they may be less effective in using pictorial means of communication than people in other professions. Thinking about and studying these possibilities is important. The best reference in the available literature but not oriented toward interactive computer graphics is the book by William Huggins and Doris Entwisle.⁶

The point has already been made that alternate ways exist of representing the same information or of representing slightly different aspects of the same basic situation. The use of these alternate ways enhances the educational value of the material. Students think in different ways, and an approach which will be successful with one student may not be as successful with a different student. Hence, the chances of success of the material will be greater if

more modes of communication are used. Often we can display different aspects of what we are trying to get by these two different modes, even if the *informational* content is the same. One example often used is to compare a set of numbers with a curve.

When the curve appears on the screen, it can appear dynamically developing in time if this is important. The informational content of these two representations is really the same: Given the set of number pairs, a person who knows about graphing could produce the curve. Nevertheless, the effect on the student is very different. If our objective is to build insight into the way planets move around the sun, the visual information is probably much more effective. On the other hand, if we need detailed information about a particular position, for astronomical or other purposes, the table of numbers might be more valuable. This is a simple example, but the ideas are repeated in a great many different situations. The fact that the visual information is more useful in developing insight into the nature of the physical world is no accident, but is a common property of pictorial ways of representing either realizable or unrealizable physical phenomena.

Another important aspect of graphics in learning situations is the motivational impact of pictorial material, particularly if this material is well designed. If a large corporation develops an advertisement for a magazine, that advertisement is almost always some very carefully prepared combination of graphic and alphanumeric material. It would be cheaper to produce such ads without any pictorial material and cheaper not to use colors; but the manufacturers are well aware of the motivational aspects of well-designed graphic material and so rely heavily on these factors. In education we are often not so wise in dealing with the motivational issues: we have much to learn from this situation.

In thinking about graphics with the computer it is important to realize the dynamic nature of what is happening, the control over time that we do not have in a printed page. In a sense, the position of someone developing computer dialogs is more like that of a filmmaker than that of a textbook illustrator. The textbook picture is there all at once, seen fully as soon as the student looks at it. The information referred to in the text must be keyed to the picture. For example, in geometrical proofs in textbooks, typically letters are attached to points on the diagram and the text refers to these letters. These techniques are not needed in a dynamic environment where there can be an interplay between alphanumeric and graphic information, with suitable delays. Hence, when a line is mentioned, that line, after a brief pause to allow the eye to shift from alphanumerics to graphics, is drawn. Elements of the picture already on the screen can be reinforced by flashing them or by encircling them or using color or with some other graphic device. The teacher can reflect the student's input, so that new pieces of the picture can depend on how the student is responding either graphically or alphanumerically.

Some devices not normally thought of as graphic are possible on graphic displays, but not on alphanumeric displays. Some of these have already been suggested. One is the occasional use of larger letters for titling or to emphasize a particular point. Graphic letters of any size, form, or orientation can be drawn. Figures 11, 12 and 13 all make some use of large letters.

PROJECTILES

a problem aid

UNIVERSITY OF CALIFORNIA
IRVINE

alfred bork ■

Figure 11
Help Sequence from PROJECTILE,
an On-line Quiz

POTENTIAL



A University of California
Physics Module

Alfred Bork

U.C. Irvine ■



Figure 12
Title Page from POTENTIAL

Suppose that as we examine the objects
one of them moves slightly

The two bodies are then stationary again.

Will the potential change? YES



Yes, the distance is slightly less.

BUT

when does the change occur?

return please ■

Figure 13
Sequence in POTENTIAL

Another "graphics" idea is the careful arrangement of the alphanumeric information, avoiding the tyranny of the left-hand margin and seeking more interesting layout on the paper; the text can be broken with spaces and other appropriate information to assist the student in understanding what is happening. Figures 14 through 22, taken from two different dialogs, indicate the wide variety that is possible.

Here is the problem.

A batter hits a ball over a 25 foot fence
300 feet away.

The ball is hit 5 feet above the ground
at an angle of 45 degrees.

We want the speed at which the ball is hit.

How would you get started with a problem like this?? ■

Figure 14
PROJECTILE

I'd always start this way.

R E A D

the problem

carefully! ■

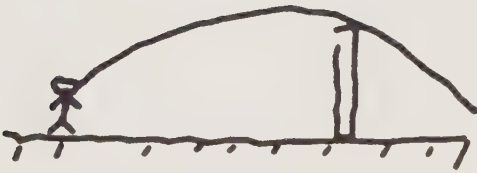
Figure 15
PROJECTILE

You don't seem to have mentioned
the importance of a

PICTURE.

at the beginning stages of problem solving

Figure 16
PROJECTILE Help Sequence



Now let's talk about
the PHYSICS
of the problem.

After the ball is hit, what force acts on it?

Figure 17
PROJECTILE Help Sequence



Now let's talk about
the PHYSICS
of the problem.

After the ball is hit, what force acts on it? GRAVITY

YES

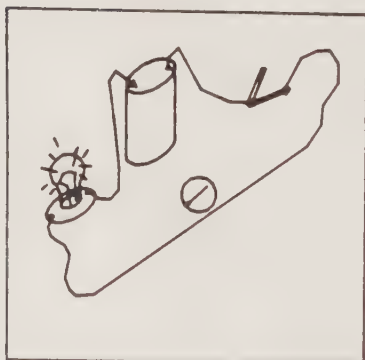
GRAVITY. ■

Figure 18
PROJECTILE Help Sequence

Suppose a small compass is placed along the wire.

The wire is parallel to the compass needle
when no current is flowing.

Now we close the switch . . .



The lighted bulb indicates current

Are you aware
of what else happens in this system
? ■

Figure 19
MFIELD

This physical effect
was first observed
by the Danish physicist
Hans Christian Oersted
in 1820.

The discovery created a sensation
and precipitated a line
of investigation
similar to the one we follow
in this dialog.

Previously
magnetic effects were known
to be associated
ONLY
with permanent magnets
and not with electrical phenomena.

Figure 20
MFIELD

Next we reverse the connections of the battery terminals.
Would you expect the compass needle to alter its behavior?

? . . . SURE

In what way?

? . . . OPPOSITE

Yes, we observe a deflection in the OPPOSITE direction!

We say that
a current flows in the wire
when it is connected to the battery.

Figure 21
MFIELD



Figure 22
MFIELD

A further graphic device is the overwriting of text making it appear to flash, or the putting of text in bold by overwriting with slight displacement from its previous position.

Special characters are also much more easily available on a graphic display, although at some loss in communication speeds. Newer devices are taking a more flexible attitude toward this problem, allowing many choices.

$$\vec{\nabla} \cdot \vec{D} = \rho$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \times \vec{H} = \frac{\partial \vec{D}}{\partial t} + \vec{J}$$

Figure 23
MAXWELL

Thus, if the Greek alphabet is needed, and the display has only the standard ASCII set, these characters can be drawn. The use of this capability from the standpoint of students may depend upon what special software and other facilities have been developed to aid the student.

Although time control, the insertion of delays in the presentation is not strictly a graphic feature, it is almost essential in the graphic environment, to allow time for eye movement. It can also play an important role in the alphanumeric aspects of what's happening. Thus, the computer can stop at the end of a sentence to allow the student to read that sentence, it can even pause briefly at places where a speaker would pause briefly to emphasize the meaning of the sentences, and can follow such natural rhythms as pausing after an equal sign before writing the rest of the equation.

One can sometimes increase the readability of the material by the way the alphanumeric information is handled on the paper or on the display screen. Although some of this breaking up can be left to the programmer, some of it must be carefully supervised by the instructor. One aspect is the way lines are divided on the screen, and the use of possible delays after natural phrases in the lines. For example, if we write:

This has problems. Please

Try again

This is a much less natural way of stating it than the following:

This has problems.

Please try again.

This can extend not only to the sentence level, but to phrases within a sentence, as in the "bad" and "good" pair of examples which follow.

The kinetic energy before and
after collision is
conserved.

The kinetic energy
before and after collision
is
conserved.

One may not want to write all material in this broken fashion, but particularly important items that need to be emphasized, or where it is essential that the student read it, might well be put into this form.

A related aspect of this is *where* the graphic material appears on the screen. All older computer-aided learning material typically was driven by the tyranny of the left-hand margin. Everything began at the left-hand margin. Newer systems allow the user to move material on the screen in convenient forms, and thus make a more interesting and visually exciting display. This too serves to emphasize the verbal messages contained in the materials.

So far our discussions referred primarily to the ability of computer displays to draw pictures as part of a learning dialog. A second important aspect is graphic input, entering graphic information. This facility provides an alternate mode for student input.

Typically, capabilities here are still limited, but nevertheless valuable. You should inquire about what capabilities are available in your situation and see examples of running programs which use graphic input. With graphics, perhaps more than in any other area of developing dialogs, the examination of a series of well-working dialogs employing graphics will be very useful in providing ideas.

The typical inexpensive graphic input available is a pointing device, allowing students to point to a given area on the screen. At the simplest level the student might point to a location on a curve and the computer might respond with the coordinates of that point, or might enlarge the region around the point, or some similar approach. Students might be asked to locate an object.

Figures 24, 25 and 26 show a pointing example in one of the Irvine dialogs, LUNA. The student is asked to point to where the sun would be so that the observer on earth — the little box — would see a full moon. You can see a variety of possibilities. In each case the sun is drawn at the location specified, and the program uses this information to make internal decisions.

Suppose we take a position far out in space, looking down on the earth-moon-sun system from above the north pole of the earth.

The observer on earth is located at the small box.

To make the circles visible, we've made the moon and earth much larger than they would appear on a photograph.

Suppose the observer, in the location indicated, sees a full moon. Use the crosshairs to point to a possible location of the sun. Type S when you want to proceed.

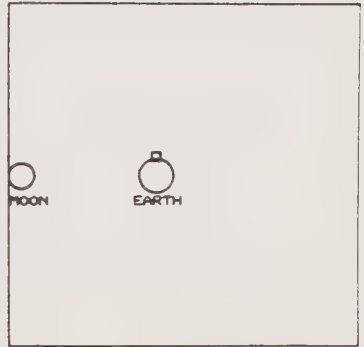


Figure 24

Type S when you've selected your position.

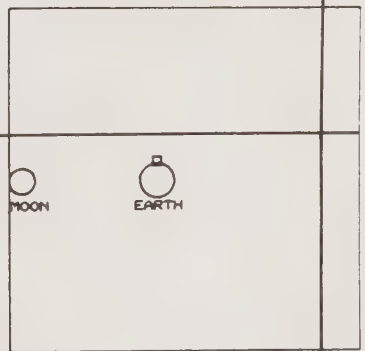


Figure 25

Type S when you've selected your position.

For your choice of the sun position light would travel from the sun to the moon along these lines.

The situation you have indicated does indeed occur. But consider what the observer sees under such circumstances. The earth blocks out some of the light.

Is the face of the moon fully illuminated under these circumstances.
?NO

That is correct.
This is the relatively rare occurrence where the earth's shadow does fall on the moon, a lunar eclipse.

To continue push return ■

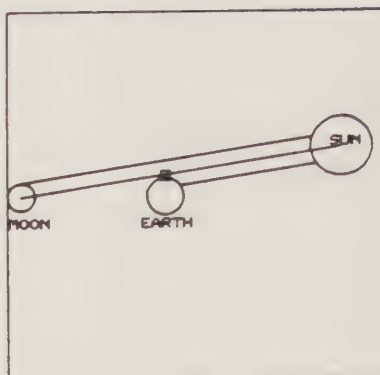


Figure 26

More elaborate graphic capabilities may be available, such as a light pen or table, which allow free-form input of curves in a convenient manner. However, these are still rare with large-scale student use.

One of the advantages of graphic input is simply the mode of input; the student is so often typing information that the possibility of an alternate way of communicating with the computer provides some relief and variability in the program.

How is the teacher, the developer of the dialog, to convey graphic intents to the computer or to some intermediary such as a programmer or graphic designer? Clearly the system will vary from place to place. The coding of graphic material is even more of a specialist's task than the coding of alphanumeric material, for a variety of reasons. One difficulty is the task of carefully interweaving graphic and alphanumeric information on the same screen, so that the graphics do not overwrite alphanumeric information, and the total effect is pleasing to the eye of the student.

If the facilities are available for working through programmers, the author can proceed by means of drawing pictures directly within his flowcharts,

showing how the screen is to look for graphic output, with some verbal information on the chart describing these pictures. If parts of the pictures are to come on at different times, interwoven with textual information, this can be indicated in at least two ways. One possibility is to number the components of alphanumeric and numeric information, showing the programmer the order in which they are to appear; another possibility is to make use of pens of different colors and color code the drawing along with the alphanumeric information. Both of these techniques have been successfully used at Irvine. With graphic input the job is to describe what decisions should be taken in the program for different choices of graphic input. These can be described in the instructor's own language, with the task of implementing the details left to the programmer. All this assumes that programmers are available to assist the teacher. If not, the author is dependent upon the graphic capabilities of the particular language and has no recourse but to learn how to use these effectively.

A recent development is the use of on-line programs for authors or programmers where the graphic and alphanumeric information can be described by pointing to the places on the screen desired, entering the information in convenient ways, and rearranging the components, again by pointing or by changing size or orientation, until the display takes the desired form. These programs then write the code necessary to produce the display just formulated.

Such a facility can be used by the programmer in implementing an instructor's needs as indicated by diagrams or can be used directly by the instructor. The possibility of involving a graphic designer in the process is also one that should be considered; as mentioned, this mode of working is not too different from that normally employed by graphic designers.

The software available for this in Irvine is available in APL and is known as DPL; a description is available.⁷

Let us follow an example of the use of this software to give something of the flavor. First, the problem here is to design a title page that identifies the material that is associated with the Educational Technology Center at Irvine. First, in Figure 27 the designer specifies that he wants some large letters. Note that this object has a name, A.

```
DI A←XBIG
RING:IRVINE
```

IRVINE

Figure 27

Figure 28 shows the next stage in the process, putting into the diagram a thick, vertical line identified by the variable B.

DI
B ← XTHI



Figure 28

The designer now wishes to put in some text saying “Educational Technology Center” and wishes this text to be right justified. In Figure 29 we see the text being entered. It is initially entered left justified, since right justification is not possible until the entire message is in. Note that there was a typing error and that one letter was erased.

DI B ← XTHI
← XLONGR

EDUCATIONAL
TECN/HNOLOGY

CENTER



Figure 29

In Figure 30 the Educational Technology Center has been moved to its correct position, and we are now experimenting with drawing a polygon around the entire figure. The initial polygon, as can be seen, is too small.

```
DI A,B
DI C←C RIGHT 10
DI D←XPOLY
RADIUS (X,Y):400 399|-|00
SIDES: S
```

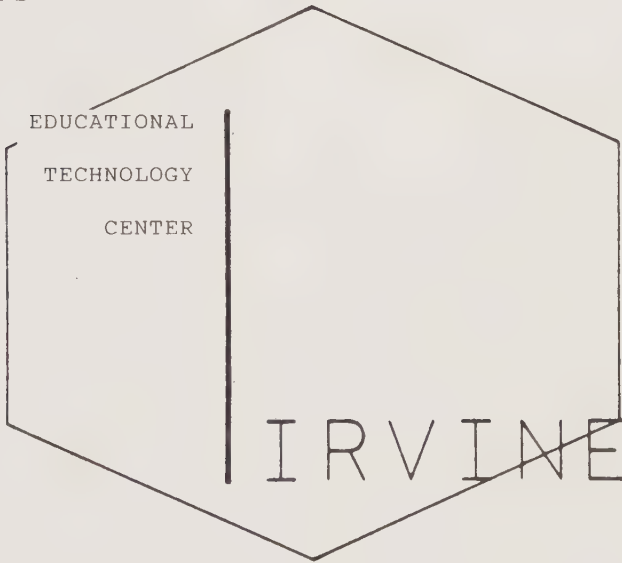


Figure 30

Then in Figure 31 we see some attempts at expanding the polygon until it is visually in the desired form as seen finally in Figure 32.

DI A, B, C

DI D←D EXPAND 1.5.1

DI D←D EXPAND 1.1.2

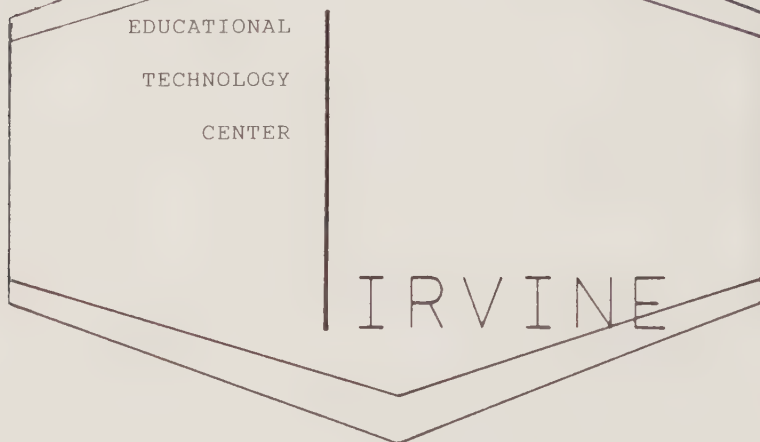


Figure 31

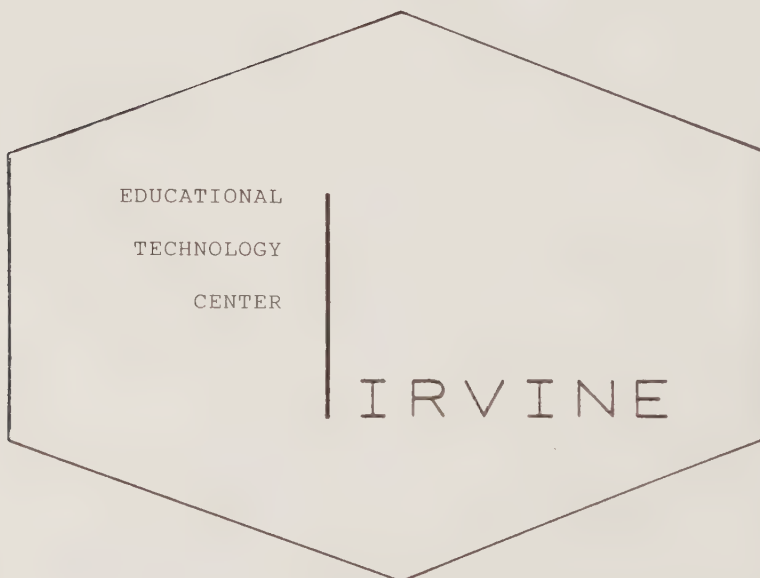


Figure 32

Feedback

It has been suggested that feedback from student use of dialogs can be important in improving dialogs for later groups of students. This is indeed one of the main hopes for producing dialogs which will be effective learning devices. Dialogs as initially written, even the best available today, have only limited success in dealing with student response, so feedback and rewriting are critical in producing programs adaptable to individual student performance.

The question of *what* feedback is wanted from student use of the dialog, and how the feedback is to be used, should be carefully considered in advance. The dialog should be consciously planned to generate the kind of information useful for analyzing students' responses, so that this information can serve to improve the next version. You should be careful not to bury yourself under too much information, for example, but to select information that is *relevant* to improving the dialog.

In the physics conversations developed at Irvine we have found it reasonable not to save *all* student responses, because with large student usage too much would be obtained to be analyzed successfully. What is usually helpful in improving the conversation are responses that the program was unable to analyze.

Some of these responses may be right answers, but answers that your matching program was too crude to find. Others may give further suggestions as to what students are likely to say that is wrong, and that should be immediately commented on in the program. The saved responses may also indicate areas which are extremely weak and which need to be extended, because very few students are able to respond reasonably; or they may indicate ambiguous terminology in the question being put to the student, or a poor stylistic approach. They can even indicate that the student's use of the English language is at variance with the teacher's! In saving responses it is valuable to store also information which allows identifying the student who entered them. Thus, some insight into the problems of the dialog may be obtained by watching the progress of individual students.

Authors should also consider whether they want to keep a numerical record of performance during the dialog — how many things the student got right, which areas were employed. Again, some thought in advance as to what information should be gathered during the student performance, and how that information is going to be analyzed, is important.

The author needs access to convenient sorting programs in handling these responses, sorting both on the location within the program at which the input was obtained and on the inputs associated with each student. The result with a large class will be extensive; with sorted output the author can effectively set to work on the next, and better, generation of the dialog.

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3. Interactive Learning

WE ARE AT the onset of a major revolution in education, a revolution unparalleled since the invention of the printing press. The computer will be the instrument of this revolution. While we are at the very beginning—the computer as a learning device in current classes is, compared with all other learning modes, almost nonexistent—the pace will pick up rapidly over the next 15 years. By the year 2000 the *major* way of learning at all levels, and in almost all subject areas will be through the interactive use of computers.

I do not intend to offer any full discussion of this impending revolution. Many of the factors involved go far beyond the question of computing. I will review those factors briefly.

Our colleges and universities will be drawing from 15% fewer high school graduates in 1985 than they did in 1975. Because of the development of a variety of alternate modes of education, the formal institutions of higher education—*our* institutions—will predictably be enrolling a *smaller percentage* of the high school graduates in 1985 than they are today. So enrollments will drop significantly. The demographic factors are not uniform across the country. In the U.S. the northeast and north central regions will sustain the greatest impact.

Coupled with decreasing numbers of high school graduates will be a shift in the nature of our student population. Hodgkinson, at a conference on general education held by the University of California and the California State University and Colleges systems, commented that our students will soon be “older, poorer, and blacker” than our present students. Thus, many of the students will not be our “traditional” students, but will be students that Cross

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designates as “new” students. These students are often very poorly served by our current strategies in higher education. Their presence will demand changes. Finally, universities all over the country are in trouble with the public, with increasing pressure to reduce the cost of education. Particularly in time of economic difficulty, universities are increasingly vulnerable to this unfavorable political-social climate.

These factors will combine with the rise of the computer as an inexpensive and effective teaching device, to bring about tremendous changes in our instructional institutions. These changes will not always be ones that we like. I will discuss briefly the types of change at the end of my comments. We can look forward to an exciting and demanding period.

Any view of radical changes in education should begin by asking some fundamental questions, with the hope of influencing these changes. What kinds of learning opportunities *should* we ideally provide to students in our university environment, particularly within our physics courses? I will begin with such considerations. From this basis I review the problems with our current courses. Then I discuss the role of the computer in the learning process with emphasis on the future.

What Should Education Be?

All of us come into our teaching experiences as the highly successful products of the existing education system. But I do not wish to start with what we do now but rather with an ideal view of the situation.

One way to begin is to consider education historically. If we go back very far in human experience, we see that most learning must have taken place as individuals interacted with the environment, generalizing on the basis of that interaction, or with individuals interacting with other people. The emphasis should be on the word *individual*. The notion of *formal* learning situations is relatively new in human history.

This early learning was rich in experiences. We can get some view of it by noting a place in our society where it still happens, early childhood education. There education is often a natural outgrowth of play. Eble, in *The Perfect Education*, queries this way: “Where does education begin? Surely it begins in play and continues in play for all of our lives.” Play is a way of gathering, under highly motivating circumstances, a variety of experiences, possibly even focused experiences. An experiential base is a vital ingredient in the learning process.

Much of this information is visual. Nelson, in *How to See*, discusses visual literacy, asking “. . . how does it happen that young children, all of whom quite naturally absorb great quantities of visual information, grow up to be visually illiterate? The answer, as far as I can make out, is that this early cap-

ability is simply beaten out of them by the educational process.”

One critical aspect of play is that it is always an *active* process. The child, or the adult, engaging in games for learning or for pleasure, is playing an active role, constantly interacting with the environment and manipulating the environment. We cannot play passively!

The experiences of play are unique experiences for each individual. Each child does different things in the environment and obtains different results, even in group situations. Individuals do not play at a single fixed pace.

We do not have to think long about play to see the instructional usefulness of those features just enumerated: experience, the active role of the student, and the individualization of the experience for each student. Most contemporary learning theorists, coming from a variety of approaches, would see these as critical components. We see the same factors in the learning activities of primitive societies.

Although we often look upon the development of written language as a very positive step in human history, and in education, not everyone agrees. Plato, for example, in a story in one of the dialogs, describes the invention of writing as a catastrophe in human history, killing the oral tradition and killing mental training in remembering. In Plato's letter to the relatives and friends of Dion, he comments: “. . . every man of worth, when dealing with matters of worth, will be far from exposing them to ill feeling and misunderstanding among men by committing them to writing . . . if one sees written treatises composed by anyone . . . these are not for that man the things of most worth, if he is a man of worth . . .” However, most of us would feel that writing was a major advance.

A new mode of learning came into prominence in Greek times, seen in its best form in the Socratic dialogs in the works of Plato. To some extent this interactive process, the dialog, was a formalization of early learning by interacting with others. The “teacher” played a special role. The teacher did not *tell* things but rather tried through a series of carefully formulated questions to lead the student to understanding. Eble comments about this process as follows: “The Socratic method, dialectic, is one of the inheritances from the classical past that is essential to maintaining the dynamics of learning. Basically it is a method of arriving at a firm answer to a series of focusing questions that rests on an even more basic assumption that thought must be exercised in order to develop. It also implies that answers to questions are best arrived at through this strenuous kind of questioning.”

In Greek times the lecture also became prominent. Plato did not think highly of this idea, but it has become almost a dominant educational delivery mode.

The invention of the textbook was also a critical moment in education. Here, we go back to the introduction of printing in the west. It is significant,

in thinking about our current scale of development, that almost 200 years elapsed between the invention of the printing press and the extensive use of textbooks within formal institutions of education.

Many of these “recent” educational innovations occurred not so much because of their educational desirability as much but rather as responses to the increasing problems of educating very large numbers of people. The notion of one to one student-teacher interaction still is held in great esteem, but this type of interaction, Socratic or otherwise, is seldom realizable given the millions of students in schools and universities.

We can approach the question of what education should be in many directions other than historical. For example, we can examine courses to determine their deep objectives. I am not thinking of the objectives in the sense of specifying carefully for each individual unit just what the student is to learn — a necessary development in any rational curriculum organization — but rather our *hidden agenda*.

One way to seek deep objectives is to see what is tested. The vast majority of the testing in science courses is based on problem solving. The ability to increase the student’s problem solving skills is the major hidden agenda in science teaching, one of the abilities we hope students will retain from our courses long after they have forgotten particular statements of physics.

Another underlying objective guides many courses in the sciences. We are very much concerned in teaching not simply the details but the *methodologies* of how we arrived at these details. Knowledge for most of us is not archival but rather evolving. We hope to foster the growth of people who will be able to further this evolution in the next generation. We hope to show students too that these methods are relevant outside the formal study of science.

Problems with Our Current Situation

I suspect that many of you are beginning to feel uncomfortable by the ideal description I have just given of what is “good” in education. The difficulty is clear. Most of our courses are far from the ideal learning situation.

The vast majority of physics courses at the present time are lecture-based courses, often lectures to very large groups. Physics is not unique; this situation prevails in most academic areas.

There seems to be reasonable experimental evidence that the lecture, while having advantages with regard to motivation and role modeling, conveys information poorly. But our courses do not seem to take this evidence into account. Given scientists’ commitment to the empirical process, it seems ironic that they are so unlikely, in a statistical sense, to pay attention to research on how learning takes place. I do not want to claim that this research is always marvelous. As a whole it is of poor quality. Nevertheless we cannot as scientists dismiss all of it just because some is poor.

Another characteristic feature of the contemporary situation is the dominance of the published textbook. Only a few books account for most of the courses in each discipline. This by itself is not an objectionable idea. Certainly reading is one of the major ways to learn, although many of our students have difficulty learning this way. But the textbook situation in the last several years has taken on an unhealthy complexion, one that has been little noted within the academic community. Standard publishers tend to refuse to publish a textbook that departs even in minor ways from the existing successful textbooks. How many new textbooks do you see published that do not use SI units? How many new textbooks introduce significantly different ways of approaching a particular subject area? How many textbooks recognize the existence of the computer? How many textbooks discuss topics different from all the others? In all these cases the answer is “very few.”

Conservatism among textbook publishers is a recent development. In the past they have published books which were very radical for their own times, such as Sears and Zemansky and Halliday and Resnick. How ironic it is that these previously radical books are now a major deterrent to new ideas in the teaching of physics! The difficulty arises from several factors. The declining economic situation in the textbook industry a few years ago made publishers much more cautious about producing books. They moved into a “marketing” strategy with study of “successful” books and with extensive surveys sent out in advance asking what the content should be. I am sure many of you have received these surveys from publishers. Such surveys yield a least common denominator; any idea not accepted by most current teachers will be excised from the eventual book. We cannot expect books generated out of surveys to develop radically new ideas.

Hence, the current method of publishing textbooks is antithetical to cultivating exciting new approaches. We need to bring this problem to the attention of the commercial textbook industry. Perhaps we will have to resort to models such as the Teachers Insurance Annuity Association (TIAA) in insurance, with universities banding together to form profit or nonprofit corporations to fill the void left by the commercial publishers’ abdication.

Another difficulty in our current situation is the meager resources we provide for students to learn problem solving. I have already noted that problem solving, as judged by how we test students, is a major component of our courses.

Our method of teaching problem solving in most courses is a “follow me” approach with examples of solved problems in books or similar examples displayed by instructors or teaching assistants on the blackboard. In most cases students do *not* see the solution of problems, but only the polished product of the solution, the building with the scaffolding removed. Everything happens magically, in exactly the right progression. Students working at home later

discover that *their* problems do not work out in this neat logical fashion! This is no surprise to us. We know that the solutions to problems are not typically so neat and logical at first. No clear presentation of the *process* of solving problems exists in most of our courses.

This type of information *does* exist. In a variety of books, such as Polya's *How to Solve It*, explicit heuristic strategies for problem solving are developed. But very few of our courses expose students to these strategies.

Many of our courses use primarily passive learning methods through lectures and books, most often keeping students within a fixed pace rather than letting them proceed at their own rate. Physics has pioneered in the use of variable-paced techniques. But these techniques are still in relatively little use in spite of research supporting them, summarized in recent article in the *American Journal of Physics*. Again, the problem is not with our "good" students, who can be counted on to go beyond the required material, but rather with the majority of students.

Another problem with our courses is that they are too verbal, too restricted to teaching people to repeat material. While this is a successful mode with some students, it is a barrier to others. Many faculty members have difficulty recognizing this problem, because they are highly verbal. But many students require more visual or other aids in the learning process, aids that are not stressed in our learning materials. The new students of the future will require still more nonverbal learning modes.

This situation is particularly bad in courses such as the physical science survey. This course, often taught to prospective teachers, gives a false impression about what science is all about; rather it perpetrates the notion that science is a miscellaneous collection of odd pieces of information, "revealed" information not to be questioned. Given the number of elementary teachers who have had such courses it is not surprising that they misrepresent science when they teach it in the elementary schools. This is a major national problem, demanding our attention.

The notion of where terms in a scientific theory come from, how they are connected with experience, the notion of overall structure of scientific theory, the notion of scientific models and how they are used, and the notion of operational definitions and the relations of terms to experience, all these are missing from many courses.

Many of us are aware of a newly discovered problem which plagues our current courses. Typically we have assumed that our students are all capable of abstract or formal reasoning. But recent extensive experimentation, reported in the *American Journal of Physics*, and elsewhere, indicates that many of our students are still deficient in important characteristics of formal reasoning. Yet almost all of our courses *assume* such capability on the part of the students. Thus, we are in the unfortunate position of teaching courses

which are inherently unreachable by perhaps a third of our students in those courses.

I could go on; I have not covered all the ills of American education. I refer everybody interested to the many insightful papers of Arons for a fuller discussion.

What Can We Do with Computers?

Does the computer represent learning modes which will overcome some of the kinds of problems that I have emphasized? How is the computer to offer us anything new? One way to approach the problem is to talk about all the different ways in which the computer *can* be employed within the learning process. There is not a single unique way, but a whole variety of ways, bearing on different aspects of some of the problems just identified.

I should like to make it clear immediately that I do not consider all current uses of computers in education as progressive or even as pointing to useful future directions. Much of the existing computer material is less than impressive. As with any new and powerful learning medium, we are at an early stage of learning how to use the medium.

Yet we have made and continue to make progress in a variety of modes of computer use. We are beginning to understand both the process of how to use computers effectively in education and the process of producing effective course materials.

I do not consider that the *current* equipment in our schools is the equipment that large-scale delivery of computer-based learning materials will use. Although most of us are now running on time-sharing systems, the new generation of stand-alone equipment, the elegant products of the large-scale integration technology, will become more and more the dominant delivery mode. While time-sharing systems will still continue in modified forms to have some use in education, most future use will be primarily with systems in which all or almost all of the processing takes place at the device itself.

I will review briefly some of the modes in which computers can be used and some of the advantages of the computer as a learning device.

a. Interactive learning. You might guess from the title chosen for this speech that the most valuable aspect of the computer in education is that it allows us to make learning interactive, with students constantly cast as *participants* in the process rather than as *spectators*. Psychologists agree that the best feedback is that which comes immediately after the event.

Anglin comments on Bruner's view of education: "... The acquisition of knowledge, be it the recognition of a pattern, the attainment of a concept, the solution of a problem, or the development of a scientific theory, is an active

process. The individual . . . should be regarded as an active participant in the knowledge getting process . . .”

In many lecture situations the students *are* passive. In some reading of textbooks, particularly with the less brilliant students, the process is also one of passively letting the textbook information “flow” into the individual. Good interactive computer programs can provide a very different environment. As soon as a small amount of information is given to the student, the program can begin to ask questions. The process can be, with skillfully written programs, a *dialogue* in the full sense of the word. The student is not conversing with the computer but rather with the authors of the material. The authors of the material are creating not a single dialog but a whole collection of such dialogs, conversation with each student. Visual information should play a critical role in this “conversation” if we are to serve the needs of all students.

b. Individualization. In the computer dialogs just described each student response can be analyzed. Different actions can be taken depending on the exact student input. Cumulative records of student performance in that session and even previous sessions, can be maintained and used to affect the flow of the learning sequences. A student who does not learn with a particular approach can be presented with alternate learning materials. The learning experience for each student can be unique, tailored to the needs, desires, and moods of that student.

I see this individualization as a *humanization* of education, particularly compared to what typically happens in the large lecture situation. There the process is fundamentally the same for every student, a mass production system. With the computer each student can have a unique learning experience.

c. Experience. I stressed earlier the role of experience in initial learning. But when a student enters the university, experiences directly relevant to the learning situation are not typically available. Here a new mode of use for the computer becomes important. It can amplify everyday experiences. The computer can create worlds which are not available in convenient form for the students to play with and explore the possibilities. Thus, we can create realms of experience with the hope of enriching the formal learning environment to follow and with the hope of building up student insight and intuition about the physical processes that are later to be described by mathematical details.

A typical example of such material is the use of electric field plotting to give insight into the way static electric fields, or later changing electric fields, behave. Visual representations of field lines and of equipotential surfaces can give students, through a structured set of experiences, a view of the way charges lead to fields that is not obtainable by working with formulae or differential equations.

As with all play material, something is needed beyond the play. Our early experiences with this material at Irvine indicated that while faculty were of-

ten extremely interested in our controllable worlds, only the brighter students, a small fraction of the total class, made active use of them. It was a classic example of material that was attractive to the teacher but not that attractive to many of the students. We have better success if we provide a structured learning situation, in which students have at least some guidance as to what kind of play is reasonable—the kind of thing you might say happens in nursery school. This can be provided either directly within the dialog or in separate written material. We also work to give students some explicit way of seeing whether they have indeed understood what we hoped they would understand with this play. For example, with electric field lines we have available an on-line quiz, trying to find out whether students can “read” the field diagrams and have picked up the ideas we would like to be understood.

d. Intellectual tool. So far the type of computer usage we have been observing is one in which students interact with programs prepared by others for some specific pedagogical purpose. But programming itself is increasingly a fundamental skill in modern society.

Even more importantly, programming can often lead to new and powerful ways of approaching a subject matter. For example, if students are in a position to write programs, they can be brought much sooner to an understanding of the laws of motion as different equations. Many of our beginning courses present the laws of motion as purely algebraic structures, in terms of the kinds of problems that the students can work. Yet we know from intermediate and advanced courses that the real power in these laws is in their use as differential equations. The fact that they are not discussed in the beginning level is due to the mathematical difficulty involved. But the computer, through simple numerical treatment, allows an “end run” around these difficulties. This particular subject has been very well discussed in the literature, in noncomputer forms and in the Feynman lectures in physics and by many of us (Luehrmann, Peckham, Merrill, and myself to mention only a few). Materials are available through CONDUIT.

One of the advantages of this approach is that students grow up feeling that the computer is a natural tool to use in a variety of different areas. Such a tool will become as important as reading, writing, and arithmetic in the future. As with any learning tool we want to introduce it to students in such a way that they will use it in reasonable and proper ways.

e. Student control of pacing. Do all students learn at the same rate? Do all students spend the same amount of time in the learning process? These are not entirely settled research questions. But there is evidence that the answer to both questions is “No.” The typical courses of today force everyone to move at essentially the same pace, not allowing for individual differences. The mid-quarter exam comes for everyone at exactly the same time. No allowance is made for students to move at differing rates through the material, perhaps re-

viewing individual learning sequences where necessary.

The computer is not a necessary component for variable student pacing. Indeed most variable paced systems such as the Personalized System of Instruction (PSI) have not used computers. But the computer makes individualized pacing convenient and commercially practical.

One of the advantages of variable pacing is that it allows for students with very different backgrounds. Thus, if remedial materials are available, students may be referred to those materials and may spend several weeks with such materials before continuing with the mainline materials. Such students are typically lost in our current courses, although some help possibilities may be available such as in the learning resources centers. The computer can provide a much more flexible set of alternatives.

f. Time and sequence control. Related to the student's ability to be able to control movement through the material is the ability to control the sequence of flow of material within a learning sequence and to control timing of presentation.

In a film the filmmaker constantly decides on timing. Pauses will be inserted to allow students to absorb a particular idea. This is in contrast with what happens when one turns the page of a book, where the entire material is spread out in front of the reader. There is no control over the order in which the material on the page is seen by the student and no control over the timing within that order.

The computer more closely resembles the film, or with another analogy, the development of material on the blackboard in a good lecture. The sequence and timing are under program control, and can be modified on student request. The dialog can move back and forth between alphanumeric material and graphic material.

Delays between such material can allow time for human concentration and reaction. Within alphanumeric material we can stop at the end of a sentence, allowing time for reading. We can come back and reinforce ideas by various graphical approaches, such as underlining, making words flash, and encircling them.

g. Student control over content. In a typical education situation every student sees almost the same content in a course. The only difference is in such things as term papers or special projects. Thus, the main learning outlines fail to take into account any individual preferences or any differences in background. Because the computer can provide a great variety of interactive learning experiences, and can provide management capabilities to be discussed shortly, there is no reason why students cannot be allowed a variety of choices.

The implication is not that the student could simply do *anything*. The instructor can still control the types of sequencing that are possible. But within these constraints students can be allowed considerable flexibility. In one of

our courses we specified six different tracks through the quarter, based on two sets of materials.

h. Testing as a learning mode. One of the newest and most exciting roles of the computer is in an intimate combination of testing and learning. This kind of use is of particular interest in a self-paced environment based on mastery learning, such as the PSI environment. Cross in *Accent on Learning* comments that "Mastery learning is a revolutionary concept that lies at the heart of the new teaching strategies. Simply stated, mastery learning permits all students to learn to the same high degree of achievement regardless of the time period. Traditional education, on the other hand, permits the level of *attainment* to vary while the amount of *time* is perceived as a constant across the group of learners."

In a traditional course tests play several roles. They furnish an input to the grading system, typically unfortunately a norm-based grading system that compares students with other students rather than determining just what they have learned. In a PSI environment the tests are in a competency-based direction, with students obtaining feedback as to just which of the objectives in the unit needs further study before they can progress to the next unit. Thus, the emphasis moves away from the evaluation to aiding students in learning.

The computer as a medium for tests allows us to go much further than this. Students can be reinforced immediately when an answer is correct not only by being told that the answer is good but by auxiliary construction or by review of the problem. A wrong answer can not only be identified immediately as wrong, but in many cases it is possible to determine just *why* the answer was wrong and to offer immediate learning sequences to the student dealing with that precise problem. It will be seen that such an experience is an intimate blend of testing and teaching. In terms of student learning this mode of computer use, although new, is one of the most powerful and promising. We can provide immediate and precisely formulated feedback, offering direct aid to students.

A variety of techniques can be used to make each exam unique, yet each testing for the same objective. Our physics tests, developed primarily by Franklin, Marasco, and myself, illustrate the tremendous power of such procedures.

i. Management. If students are taking tests on line, this leads naturally to the notion that the computer can also maintain the class records. In a large class environment, keeping accurate records and allowing students to check that no errors occur is a nontrivial problem. In our own department we have a half-time secretary whose major occupation is to maintain records, primarily for the big beginning courses. Faculty members too put considerable amounts of time into this process, and no one would claim that this is the most productive use of their time.

The computer is a very powerful information gathering and handling de-

vice. The exams write directly into an online database; other items not covered with on-line exams can be entered into the same database. Feedback can be provided to both students and instructors as to what is happening, what problems are developing, and which students need aid.

j. Communication. Another use of the computer in classes is as an additional mode of communication, beyond the traditional ones usually available. The computer can drive an electronic mail system, with the instructor broadcasting messages to the students, with the students sending queries to the instructor, and with the instructor replying to such queries. In my own large courses this mode has assumed increasing importance. Typically I will answer about fifteen computer letters per night from my home terminal.

Any way of increasing communication between the teacher and the student is desirable. Electronic mail does not *replace* personal contact — presumably the instructor will still have office hours and will meet with individual students — but it does offer another significant channel of communication.

k. Personal factors. The computer has no prejudices. All of us are subject to inherent prejudices, often at the subconscious level. These unconscious personal feelings present in all of us may well affect our students. Many of us, for example, are consciously or unconsciously supporters of the better students, those most “like” us, and so tending to have our sympathy. The problems of students struggling to learn a particular piece of material are difficult for many instructors. Even the best teachers with the most devotion to the vast majority of students may occasionally tire of such problems.

Some students may *prefer* to deal with the learning material in an impersonal way, rather than to come into the faculty member’s office.

Production and Distribution

I only touch on important issues concerning widespread availability. At the present time production of computer learning materials is almost at the cottage industry stage, with individuals producing and sending a few copies to friends. Journals, such as the *American Journal of Physics*, have aided in wider distribution. New organizations, such as CONDUIT, have been set up by the NSF particularly for this purpose. But so far these activities have only scratched the surface.

Fundamental to widespread use of computer learning materials are more structured ways of producing and distributing the materials. Efficient production demands that we examine carefully the process for production and distribution, setting up centers with particular expertise in this direction. These centers may be within universities or may be within commercial organizations. National centers were first proposed in the Carnegie Commission report of about five years ago, *The Fourth Revolution — Instructional Technology in Higher Education*. In the report seven such centers are suggested.

No such centers have yet developed, but they still seem to be an attractive possibility.

What types of firms will be involved? Perhaps they will be the traditional book publishers. Perhaps they will be computer vendors. Perhaps they will be special companies, profit or nonprofit, formed particularly for the distribution of such materials. Within the elementary level a very successful company already exists, Computer Curriculum Corporation steered by Suppes. Perhaps universities will become the producers and distributors of these new types of curriculum materials.

Production methods for these materials must take into account the variety of tasks. The pedagogical specification of the materials is a different task from that of programming them to run on a particular machine and requires different talents. Graphic and instructional designers must also play an important role in the process. Programs must be easily modifiable over a considerable period of time, as experience is gained with direct student usage.

Institutional Changes

I have already suggested that institutional changes are likely to be drastic because of the demographic factor, the changing nature of our student body, increasing legislative and public control, occasional economic depressions. I realize the inertia of the system, but I believe these factors will be sufficient to overcome this inertia. The computer will offer us our best approach to coping during this difficult period.

We already see signs of this change in our current institutions. Cross (in *Individualizing the System*, edited by Dyckman W. Vermilye) comments, "By the year 2000 an instructional revolution will have changed higher education in fundamental ways. Signs of that revolution have already appeared."

We can expect more self-paced courses, more emphasis on mastery learning. These changes are likely independent of the computer. Furthermore, we can expect more emphasis on *self-paced curricula* in which the curriculum is not tied in with fixed time constraints with courses beginning only at the beginning of the semester, but rather is adapted to individual students. There is no reason, given the computer environment discussed in this paper, why a student cannot start a course at any time and finish a course at any time, subject to any desired constraints. The self-paced curriculum may lead to the final destruction of perhaps the single most sacred feature of American universities, the four year degree. New attitudes will be generated with regard to grading, with more emphasis on competency-based grading. The question of credit, and similarly degrees or other marks of achievement, may well be brought into question also. In all of these developments the computer will play an important role in suggesting solutions, with computer use steadily becoming larger.

Perhaps the most exciting development in institutional change will be the rise of entirely new kinds of institutions, ones that depart from our traditional patterns of education. They will compete with our traditional institutions for the limited number of students available.

The most exciting example, although with little use of computers so far, is The Open University in the British Isles. In its first six years of operation The Open University received over a quarter of a million applicants and registered about 74,000 students. In 1975 the total cost per student including governmental funds and student fees was about 369 pounds per year. The Open University is effective in terms of its learning procedures and *very* cost effective as compared with our traditional universities. The emphasis is very much on the development of course materials, and much less expensive delivery systems are featured. Thus, the university has no student "campus" with many expensive buildings. Rather students work primarily in their homes and in scattered learning centers all over the country. Some aspects of The Open University are probably not appropriate in our hemisphere, but we have much to learn from this exciting educational experience.

Central to learning is the creation of an environment particularly conducive to the area involved. Papert, at MIT, likes to compare two situations for learning French. One is that of the typical high school course in French in our schools. Learning to speak French in this environment is often a difficult process; not all students are successful. Yet in France all young children learn to speak French! Can we create such an environment for physics, a Physics Land? That is the challenge that is before us!

What Can We Do?

What is our role, as educators, in this process? First, we are, among all the faculty in higher education because of our own technical training, the group in perhaps the best position to advise and help administrators and faculty colleagues in understanding the pressures for change. We can aid in creating changes which lead to healthy educational environments rather than to unhealthy environments.

Many of us hope to play major roles in the development of computer-based learning materials, the new and exciting materials the computer will make possible. We can work with the companies that may be involved in either the production or the sale of these new types of materials, particularly in the early uncertain period. We can exhibit materials to our faculty colleagues to show that the computer is not a frightening device at all, but rather is one that has great potential for improving learning.

Like many uncertain futures this future can be both terrifying and exciting. It is terrifying because we cannot see all the possibilities that will develop in the troubled period of the 1980's. It is exciting because we stand at one of the great moments in the history of education.

4. Learning through Graphics

GRAPHICS PLAY AN extremely important role in the learning process.¹ A look at existing teaching methods, textbooks, blackboards, overhead projectors, or other aids would quickly show the widespread use of graphic devices, not only in the sciences but in many other areas. Textbooks in many areas average a picture per page, in spite of the increased production costs implied by this use of visual material.

Examination of blackboards in any school or university also shows a wide range of graphic devices in active use. These include varieties of graphs, actual sketches, flowcharts for summary or for showing the logical structure of the area, underlining or boxing of important information or ideas, arrows to indicate connections, variable size letters for stresses, and careful location of visual and alphanumerical information. Even a simple graphic “input” mode can be observed when instructors lecture at the blackboard, pointing to critical or relevant previous material still on the board. Other visual media—films, video, slides, overhead transparencies—are also often used, although the logistics of the present technologies in these areas often present considerable barriers.

This wide use of pictorial material can be seen in almost all areas. Some aspects are more common in certain areas, while others are universal. Thus, the explicit use of graphical information in the narrower sense—curves and surfaces corresponding to known functions or experimental or calculated data—is more common in the sciences, and even there more used in physics and mathematics than in biology. But the various learning devices mentioned above in connection with the blackboard are widely applicable.

¹ This article originally appeared in: *Computers and Communications: Implications for Education*. New York: Academic Press © 1977.

Graphics in Education

The importance of visual information in early childhood education is extreme. At that stage the child has very little language and so depends mostly on visual and auditory data. That children learn some complex activities as speaking, writing, and arithmetic at very early ages indicate the intrinsic power of these channels of communication.

Graphic and other communication modes provide for the older student an alternate to alphanumeric modes. Most of our ways of educating at the moment are very much oriented toward the use of language. For students who are linguistically apt, verbally sophisticated students who learn well in this way, this emphasis on words does not present any problems. But many students, at all levels of education, are not this sophisticated in assimilating verbal information, and would be greatly aided by the use of other channels of communication. Nonverbal modes are particularly important with students from minority backgrounds. One difficulty these students have is understanding what is important and what is peripheral. Visual devices of the type indicated in the discussion of blackboard can be very helpful with such students.

Even the "same" information, presented in different forms, can have different meaning to the student, conveying, different impressions. Thus, the data showing the motion of a comet can be given with numbers indicating position, or can be shown by means of a curve developing dynamically in time. But the effect on the student is entirely different in the two situations.

Even alternate *graphical* modes may have different impact. The concept of a vector field in physics is a difficult one for beginning students—an infinite collection of triplicates of numbers must be viewed as a single entity, with different collections corresponding to different physical situations. Further, the field may be time-dependent. Textbooks use a variety of graphical techniques in enabling students to comprehend this vital concept. The most common is the field line which follows the direction of the field at each point; another employs many small lines with arrows, pictorial representation of vectors. Both forms are partial, and neither is an ideal visual presentation, but experience in teaching this area indicates that both are useful in assisting students.

Another example of alternate graphical modes can be found in the practice of research in science and engineering. The following examples are discussed in a recent book² on data analysis: horizontal bar charts, vertical bar charts, pie charts, area diagrams, volumetric diagrams, correlation graphs, silhouette charts, line charts, scatter diagrams, graphs relating more than two variables, perspective, projections, contour plots, log and semilog plots, calculating charts, and nomographs. These same alternate modes are also valuable in learning.

Thinking about these alternate modes, and their uses, leads us to one of the most valuable contributions of iconic representation in learning, the building

of student intuition or insight. This is an extremely important goal of traditional school and university education, and one that is difficult to reach by most of our current modes of learning. A new Ph.D. in the sciences is good at carrying out the manipulative skills required for problem solving, either theoretical or experimental, but is often less successful in understanding intuitively the critical problems or needed directions of advance. The intuition we develop in everyday life comes from the rich collection of phenomena we experience, as indicated above about children. But as we move to more and more abstract areas, everyday experience becomes less and less relevant. With computer graphics we can extend these experiences.

We have many examples of this development of insight in the work of the Physics Computer Development Project at Irvine.³ Our most widely used student-computer dialog is MOTION, an " $F = ma$ " world for the student to explore freely. Students control, in a highly interactive manner, the force laws, equation constants, and initial conditions. Thus, they can examine many more situations than they can in the "real" world, with much more control. Further, they need not view the systems only in configuration (x - y) space, but can plot any two or three physically meaningful variables, thus moving toward viewing the system as existing in a wide variety of spaces normally unavailable. The very wide use of this dialog by students *not* enrolled in physics, classes testifies to its success.⁴ Another dialog, SPACE, allows students to explore Lorentz transformation of space-time, again not a possibility without computer graphics.

On a simpler level the ability of mathematics students to examine curves as parameters are varied under their control, is similarly useful in building up intuition. Many areas of mathematics or mathematical physics present similar opportunities. The ability to generate many examples of conformal maps from one complex plane to another, or to see the results of a Fourier Transform (available in APL workspaces at Irvine), again leads to understanding difficult to obtain by the usual analytic treatment. In all these interactive graphics is the critical component.

Another extremely important educational role of graphics is to motivate the student. In many areas, learning is not an easy task; students need some help to stay with the difficult parts. Often this factor is ignored in current classroom practice, but the recent changes in student attitudes make it clear that, for many students whose self-motivation is not high, we cannot ignore motivational issues. The main evidence for the motivational accomplishments of well-done graphics comes from the media and advertising industry. But we can also find many clues in education itself. At U. C. Irvine we have an extensive collection of both graphic and nongraphic dialogs available for student use. Both types are used in classroom-associated modes; however free use, outside of classes, is almost entirely graphic.

In thinking about the role of graphics in learning, it is important not to

overlook the obvious, the situations in which the fundamental information is basically pictorial in content. Perhaps the best known PLATO lesson, involving the breeding of fruitflies, is a good example. The object we examine here, and the characteristics we are interested in, *are* visual. An Irvine dialog, LUNA, uses the visual model of the phases of the moon as a vehicle for introducing the notion of a scientific model.

As a case study, I would like to consider another very important objective of almost all science and engineering courses, and an increasingly important objective in other areas of high education, the development of problem-solving skills. We want students to be able to use their knowledge to create new knowledge! Again existing modes of teaching are inadequate; the old approach is to show the student (in lecture and textbook) many polished examples of worked problems, giving little insight into *how* problems are practically solved. Interactive computing can make a major contribution to this objective. A central problem is to show students how to organize an attack on the problem.⁵ Here graphical visualization is important; the flowchart of any successful attempt must begin with what is given, must end with what is to be found, and must exhibit logically plausible intermediate steps. A common student difficulty is a failure to generate such a "map," or series of maps, of possible processes and directions; a student who does not is in danger of wandering into areas having little reference to the problem at hand, as shown by work in the Sesame program at the University of California, Berkeley.⁶ We also encourage students to represent the situation, including the given information, through a diagram. If the student is working the problem at a terminal, with the computer dialog assisting with this problem and also approaching the general heuristics of problem solving, these visual modes are important. Also useful is the ability to show the interconnections of the mathematical structure while the details are underway. And finally, as problem-solving is not easy, the motivational aspects of graphics are important too.

It must be admitted that we are not skillful in the use of visual images in education. Humans were more skillful in this direction before the development of writing, because learning depended more at that time on graphics than it does now. Most of us are the successes of a highly verbal educational system, so it is not surprising that the main abilities of most present-day teachers lie in verbal directions. We tend to be clumsy in employing iconic information in educational environments, and can expect a long, slow learning cycle before we can develop good capabilities. In many areas the best graphical realizations are not obtainable with present equipment.

Experience at Irvine indicates that prospective authors in *any* area developing computer-based educational material, given a choice between graphics and nongraphics, will almost inevitably choose to use graphics. Concerning situations elsewhere, one can see two factors which will generally preclude the

use of graphics. First, it is obvious that graphics will not be used if the terminals widely available to students in that environment, or the languages used, do not support line-drawing capability. Less obvious is the fact that authors will not use graphics if it increases considerably their exposure to the computer. Thus, if instructors must learn not only a computer language that handles branching, but must also learn about graphic capabilities and learn the codes which will produce them, this will discourage such usage. Neither factor is present at Irvine; the vast majority of the Irvine material is produced by instructors who do *not* program the dialogs, but only describe them pedagogically; graphics is specified by rough sketches and descriptions.

Research in Graphics for Learning

I would like to make the case for graphics and other media in learning environments on the basis of very careful research. Although competent research exists, I do not regard this research as definitive at the present time. Good research in educational areas is extremely difficult. I often find myself not impressed with the results that are obtained, because the cases are too specialized or because the numbers involved are not great enough, or because of factors not taken into account. Although the following quote refers particularly to typography, it is applicable to educational research.

“Even when he starts with a relevant problem, a researcher may still be misled by the ‘classical experimental paradigm.’ A researcher starts off with, say, the relation of line-lengths to legibility. He then tries to control all other variables. This is not only impossible (as we shall show) but actually *mistaken*, for any real situation will bring in dozens of other decisions, and those variables will, indeed, interact. Perhaps this very real interaction is what should be the focus of the study

“We attribute such mistakes not to the incompetence of researchers, but to their mistaken use of the classical experimental paradigm drawn from the physical sciences. No one, in our opinion, can possibly predict all the variables that might interact with the one chosen for experimentation. And even if a comprehensive list of variables could (by some magic) be drawn up, the permutations would generate untold millions of experiments — far more than could ever be executed.

“The few results that can be reliably applied to practical situations are further debilitated by technical questions about the way the experiments were conducted or, in some cases, by the way they have been reported in the journals”

This quote comes from a paper written at The Open University in Britain, where a great need exists to find suitable technology that might lead in a very practical way to improved learning situations. The Open University goes to

great expense to produce courses, and has very competent educational technology advice. Often, however, as indicated in this particular passage, existing research turns out to be inadequate for the practical day-to-day problems. Unfortunately the same is true with regard to media use in education, but some useful work does exist.⁸

Current Educational Graphic Systems

The vast majority of the current educational graphic systems in student environments depends on the storage tube technology. Tektronix terminals are relatively inexpensive, as compared with most other graphic devices, reliable, and offer reasonable software support. A school can afford large numbers of them, and thus they can be employed in very large classes. The situation at the University of California, Irvine, shows the possibilities. We currently have about 30 Tektronix 4013s for student use, and another dozen or so similar Tektronix terminals in use by faculty, most of those in developing student dialogs. A similar situation exists in the two community colleges in our area, Orange Coast College and Golden West College, and Dartmouth has similar numbers of Tektronix terminals also. With this number of terminals it is possible to use graphic learning materials in our beginning physics classes; during the coming academic year, we expect to employ our computer dialogs with about 600 students at Irvine.

Limitations of the Tektronix device are well known. It does not have intensity control, it does not have color, its baud rate is too slow in most typical environments, it provides only limited control over the character set, graphic input capabilities are somewhat limited. Nevertheless this terminal has had a tremendous impact in showing the practicality of graphic interactive environments with large numbers of students. Another "problem" is that it has so little competition at the moment.

The plasma display terminal, developed by Donald Bitzer in the PLATO project, is also in widespread student use at the University of Illinois and other PLATO environments. Its advantages over the Tektronix technology are the ability to use selective erase and nonstore capability and the availability of terminal-defined characters. Its disadvantages are relatively high cost, cruder resolution, power requirements, even poorer graphic input than Tektronix for careful use, the current lack of color and the physical engineering of the present terminal.

Other terminals based on TV or related technologies are in limited educational use. The Princeton Electronics and Hughes terminals use a TV monitor refreshed from an internal silicon digitally, but has cruder resolution than the other terminals mentioned.

Much more elaborate display systems than either of these are in use.⁹ These

systems are more expensive, and so are much more limited as to the number of students that can access them. None has been designed primarily for educational use, so they often have many features that are not of prime importance in use with very large student groups. But they show the potentialities for the future for graphic displays. Prominent vendors of such advanced graphic systems are Vector General, Adage, Information Displays, Evans and Sutherland, Imlac, and Digital Equipment Corporation.

Requirements for Future Educational Graphics Systems

In this section I specify a series of requirements for the graphic components of future computer-based learning systems, both in terms of hardware and backup. These are important features for future systems. The first part considers physical characteristics, the second software and courseware, and the third mentions other *possibly* desirable features.

PHYSICAL CHARACTERISTICS

1. *Color.* It is essential that future computer-based systems have full color capability, the ability to display *any* color in any graphic manner. At least two aspects are important. The first is that color can be a conveyor of information, a new dimension in the kind of information that can be carried. Thus, in a complex physical situation, where many different variables are to be represented, only a few of them can be represented by traditional spatial graphic modes, and color allows additional representation. Furthermore, it allows stresses and other kinds of graphic devices which enhance alphanumeric presentations. We need only to look at the way students use color markers in textbooks to appreciate this last item.

Another very important aspect of color concerns the affective component of education. Color is very useful in motivating student learning. Students are bombarded daily by very high quality design employing color — television, magazines, etc. These materials, prepared by extremely competent industrial designers, constitute the “background” world the student lives in. If the educational world presents a monochromatic view, it fails to compete for students’ attention. Too often in educational situations we neglect these motivational aspects of teaching, but they should be considered in systems of the future, particularly in a time when education will be more and more pressed by its external situation; I am referring particularly to the expected difficulties in higher education in the 1980s, with the decline of student population and with increasing numbers of students seeking presently nonstandard modes of learning.

2. *Wide Bandwidth.* Currently most graphic uses of computers in education are taking place on 1200 baud communication lines. The amount of graphic information which can be communicated with systems in this range of speeds is limited. In drawing a function on a screen it is plotted somewhat faster than you could draw it by hand. Full dynamic evolution in time, such as we are familiar with in films, is clearly impossible on-line in this situation, although limited dynamics is feasible. Most computer-generated films, where full dynamic animation is used, demand either off-line equipment or much fancier and more elaborate graphic displays than currently possible with large numbers of students, displays operating at baud rates much greater than 1200 baud.

Full dynamic capability is essential for systems of the future, so something approaching TV bandwidth is needed. The teacher should *not* be restricted in preparing materials by the limitations of the speed of communication lines. This increase can be accomplished in three ways: 1) vastly increased transmission speeds, much cheaper (the cost being the present limitation on the widespread student use of such high speeds); 2) intelligence in the display; and 3) the putting of *all* intelligence directly within the display device, so that no transmission lines are required. Those possibilities will be discussed later.

3. *Integration of Other Media.* Graphic computer systems should not function alone, but they should be fully interfaced with all the other media. Thus, a computer display device should be in a position to show, at any point within the learning program, a video sequence of arbitrary length. The decisions should be controllable in the learning program, dependent on the student's performance. It should be prepared to display large numbers of static pictures. It should be prepared for audio input and audio output, given the very important communication virtues of the use of the spoken language. Although there are systems which do one or more of these at the moment, I believe that none of these systems handles this problem adequately with a fully integrated modern technology. This technology potentially exists, in the form of the videodisc systems.

4. *Computational Capability.* It is sometimes said that educational uses of the computer do not demand much computation. While this is true of certain types of usage, other types, extremely valuable in learning environments, demand considerable computational capability. My own experience is primarily with use in undergraduate physics classes. There our *current* teaching materials often demand more in computation than we can obtain from our sizable time-sharing system. At times we have avoided approaches we believe to be pedagogically viable because even greater computational demands would be made. My experience also indicates that these restrictions are even greater for typical current minibased time-sharing systems. Physics, and other areas of science and engineering, may be special cases here, because models in these

areas often depend on ordinary or partial differential equations. But I have heard similar comments from those in other areas.

Graphics increases these computational demands. First, the graphic facilities themselves imply additional hardware to handle the graphic aspects of the display. Further, the preparation of graphic data internally in the program often involves computation. It is difficult to quantify this computational need. A leading exponent of stand-alone systems likes to say that the home computer of the future will have roughly the power of a CDC 6600. While not everyone would agree with the need for this much power, it serves to emphasize that the computational capability of the contemporary minicomputer is inadequate for the future learning systems employing graphics.

5. *Character Capability.* Computer dialogs should be able to call on a wide variety of symbols, not just a standard ASCII set. In every human endeavor man develops special symbols, and these symbols are widely used by the practitioners in the area. A physics student needs different symbols than a Russian student.

Two aspects must be considered. First, the computer should be able to *display* any conceivable symbol. On a graphic terminal any symbol can be drawn, but such symbols require computer time and input-output time. The sensible procedure is to allow the program to redefine symbols, and instruct the terminal that this has been done. Only PLATO currently allows this capability. The second aspect is the *entering* of special symbols. No current displays have a desirable way to enter symbols. Ideally, what is displayed on the keys should change, showing the new symbols. This might be done by LEDs on the keys, or by projection method as in a recent Bell Labs telephone operator console.

SOFTWARE AND COURSEWARE

1. *Graphic Software Facilities.* In the first items, the type of teaching material discussed has not been mentioned. It is essential that graphic capabilities be available in a wide variety of forms, not only in the programs which we prepare for student use, but also for students to use themselves within the common programming languages. Thus, a full graphic development for education should be certain that graphics is present in all the languages students use to program. Efficiency and transportability demand that these facilities become an intrinsic part of the language, rather than an "add-on." Comparisons at Irvine of APL graphic capabilities, either building facilities directly within APL, or the use of APL functions, indicates the first is more efficient by an order of magnitude.

2. *Production Centers for Graphic Material.* The use of all these additional facilities described will complicate the preparation of effective computer-

based teaching material. The "cottage industry system," with each instructor working alone, will be inadequate to prepare effective materials. The type of organization suggested in a study¹⁰ by the Carnegie Commission on Higher Education, the establishment of large regional centers for production and dissemination of such materials, is one possibility. This question cannot be considered independently of the technological needs, but is part of the same problem, so it must be raised together with the technological requirements.

Other Possibilities

Existing experience with graphic-based systems suggests that these six requirements will be important for future systems. But other graphic facilities may or may not be critical for such systems. The ability to do realistic drawings might be desirable in some situations, but some research indicates that sketches are to be preferred to realistic drawings, at least in some pedagogical situations. And the educational case for surfaces and textures also seems more ambiguous to me, although undoubtedly useful in some situations. The situation is similar for hardware curve-drawing facilities.

Models for the Future

Given the usefulness of graphics in learning, and given the requirements as just developed, it is not difficult to picture a number of ways the current technology could evolve for such facilities.

One natural extension of some of the current developments, including such diverse activities as the ARPANET and the PLATO project, is the use of very large networks, either based on a single CPU with terminals widely spread or based on very large collections of CPUs. A number of types of networks are being actively developed, and network software is receiving great prominence from several major computer vendors.

The most important limitation, in terms of our above requirements, for such systems is transmission speed, and the cost associated with this. While it is possible to transmit at higher speeds, so far it is not clear that the costs will come down so dramatically that it will be possible to have very large numbers of students using machines this way. Full color animation will require, as suggested, TV bandwidths for each terminal. It is difficult to visualize hundreds of thousands or even millions of such terminals in use of those bandwidths, although there *are* optimists! The advantages of such systems would be that the transportability of materials would be relatively easy. Development of the graphic devices could include even in this case some intelligence within the displays themselves and so lose some of their requirement for line speed.

Here, as with all the other developments, there is some interplay between the bandwidth necessary for transmission of pictures and the amount of intelligence involved in the display.

The second model suggested by current technology, and one that is currently flourishing, is the use of small timesharing systems, such as those based on minicomputers. The number of terminals per system is smaller, but more such systems are contemplated. The advantage is greater reliability — demise of a system affects fewer users. The disadvantages are like those above, with transmission speed an important consideration. Another disadvantage is that materials must be physically transported from one machine to another, and small variations in the machines will lead to problems. I can testify from experience with my own project that the transferring of complex educational materials even to machines with identical hardware and very similar software will often present problems. Another disadvantage of such systems for many current applications is the limited computer power available. In visits to educational projects based on such systems, I often see materials limited by the hardware capabilities. While future systems of this kind will undoubtedly be less limited, because of increasing speed and decreasing cost of hardware, it will I think still be a factor even ten years from now. A recent visit to Britain, where a sizable educational development is going ahead on minibased timesharing systems, reinforces my belief that such development tends to restrict the materials developed. This handicap may be overcome with the increasing use of chip technology.

Another major disadvantage of the small-scale timesharing systems in their present format is their restriction to one on-line language. Students function in a variety of ways in using the computer, and one important way involves the standard programming languages. It seems reasonable that they should have access to a wide variety of languages, to employ in courses, to give a more accurate view of what the computer is like, and to prepare students for future computer uses. Furthermore, certain languages are superior to others for particular types of educational use. A system that considers the student only as a user of existing programs is, I believe, an unfortunate choice educationally, just as a system that restricts people only to writing their own programs is unfortunate. I am impatient with arguments that suggest that one or the other of these is the desirable way for students to proceed; both have their uses in educational environments and there is no reason why we should “choose” between them. So a single-language system is too restrictive for student use.

The third future model is that of primarily stand-alone systems, systems which contain within them not only the graphic capability but full computing capability also. While these systems might occasionally be used on-line to one

or the other type of systems described, they would function with students *primarily* in a stand-alone mode. On-line use would probably be for accessing extremely large or rapidly changing national or international databases for nationwide record keeping activities, or for massive computational access. Reliability of such systems would be greater as a malfunctioning system would affect only the individual user. System and software design could be much simpler, because it would not be necessary to support a multi-language timesharing system or even a simple timesharing system as usually available in the second model above.

Stand-alone systems have fewer limitations with regard to speed of display. Since the computational facilities and the display unit are at the same physical location, very fast display techniques are possible and feasible even in a relatively inexpensive environment. The stand-alone system is the natural final evolution of the LSI technology, the building of computer components in smaller and cheaper packages.

Distribution of materials — providing access to a wide range of languages and educational materials — is a problem with stand-alone systems. The most likely solution to this problem from current technology is the videodisc system, such as those being developed by Phillips, MCA, and other major companies. These systems allow the storage of vast amounts of data in a relatively inexpensive, hard-to-pirate, and sellable form. The technology of production is related to that for present musical records. The storage techniques are *not*, as with other computer storage techniques, magnetic.

Although the developers of videodisc systems have thought primarily of using the videodisc for bringing movies into the home, such videodiscs, used in stand-alone systems, could have a combination of video sequences, audio sequences, slides, and computer code, all on the *same* inexpensive disc. Thus, a videodisc might hold a complete elaborate teaching dialog that the student employs at a stand-alone display, complete with slides, video sequences, and the necessary computer code. Or it might contain the code for a full language, such as APL, along with a full set of materials, including video and slides, for learning that language. Thus, the videodisc would allow a full multimedia approach, combining the computer with other media.

Given the capacity of such discs, roughly storing an hour's worth of TV on currently projected systems, the digital information could be stored redundantly for greater reliability. Systems with videodisc capability will probably need local inexpensive storage such as a floppy disc or a cassette. But the storage requirements may be lower than might be expected, because such devices would be functioning only as an extension of main memory or for record keeping purposes.

The only software-based, stand-alone system currently being marketed is the Digital Equipment Corporation Classic. While this is a step in the right

direction, it does not have graphic capabilities, and it has far more limited processing capability than essential to support fully an educational stand-alone environment. Program input is through floppy discs. Hardwired stand-alone systems are also available from Wang, Hewlett-Packard, and other vendors; they are unfortunately restricted to a single language. It is doubtful that the current materials we are using with students at Irvine, in physics teaching, could be made to run in existing stand-alone environments; the fact that the educational materials currently being provided with the Classic system are much more limited also supports this point of view. But nevertheless their potentialities for the future are interesting.

Other stand-alone systems exist in experimental form. The work of Alan Kay with the Alto system, using Smalltalk with young children, is particularly interesting. Seymour Papert has developed a two-display system, and hopes to market such a device. The Papert system is not intended primarily as a stand-alone system, but it is built around an internal CPU, and could be configured in versions that would allow it to function primarily stand-alone.

With regard to the color display aspects, again we can see a number of projects beginning to develop experimental systems in color display. The CHARGE system being developed in HumRRO is one such interesting example.

Given these three models of the educational future, I believe the most promising is the stand-alone system. Hence, I will confine the remainder of the paper to that possibility.

Additional Research Needed

The development of stand-alone systems such as described here is primarily a development effort as opposed to a research effort, as most of the ingredients already exist. But some additional research will also be needed.

Perhaps one of the major problems is to learn how to use the decreasing cost of computational power. The CPU on a chip is now extremely inexpensive, leading to the possibility that a stand-alone system could have many CPUs in it, perhaps individualized for different types of usage or to different languages. Redundancy is easily possible too. But the critical organization of such systems, including the necessary software and firmware in such an environment, is still a difficult issue. I would expect very different types of architecture to evolve than the typical multiprocessor systems seen with large computers.

Reorganization of the user software, the programs that the students are using, might also need considerable rethinking within such an environment, so research is needed here also. It might be possible in a system using the video-disc to get by with less fast storage because of the large static storage capabil-

ities of the videodiscs. Part of the information stored on the videodisc might be firmware, to emulate various types of computer systems on which the material was initially developed. Here too some thought about the type of organization that will make this practical would be needed. Although much talk occurs about the use of microprocessing to emulate other machines, and one sees a few examples, this technique is employed relatively little in a really dynamic mode.

With regard to research needed on the aspects of the display itself, some study comparing raster scan techniques and random access techniques, with regard to their relative efficiency, particularly in a color environment, is necessary. Standard color TV sets are a likely possibility, but they will require some work in improving the resolution of these sets, by going to a greater number of lines or by other techniques. Some study of the relative advantages of generating raster scan storage from other kinds of pictorial storage, by means of hardware or hardware plus software, are also useful. Such systems as the HumRRO CHARGE system are already investigating the pure hardware possibilities here, but additional work is needed.

Since much of the work on videodiscs has been done behind closed doors in major companies, it is difficult to say what additional research for interfacing such systems within computer-based systems would be required. Present work is oriented toward the home TV market. As the developers have not thought of education as their major market, some effort in this direction will be necessary. Additional research on audio input will also be required to make this practical.

We do not currently know enough about the educational usefulness of "advanced" graphical hardware, such as the ability to represent surfaces and shading. Research comparing such drawing with less expensive line-drawing capabilities would therefore be useful.

Development

To what extent can we depend on the conventional computer manufacturers to develop the type of stand-alone system suggested here? Although vendors have some interest in this direction, as indicated by the existence of the Classic, nevertheless the prospects of having such systems developed entirely through the conventional computer manufacturers at the moment is not great. Support from the government will probably be necessary if the time scale is to be reasonable.

Although computer vendors can see that in principle the educational market, involving very widespread use of computers consider everything from kindergarten to postgraduate school, it is also clear to them that this is not an immediate market. It is difficult for the average vendor to see how to meet the

demands of that market. The fact of our current economic life is that most terminal and mainframe vendors are dominated primarily by the commercial market and its requirements, since this represents their largest selling area. That situation is not likely to change dramatically in the next two or three years.

What form should government support take to make such projects most effective? I should go on record as *not* favoring any “blockbuster” projects in this area, projects with very large infusions of money over very brief periods of time. A slower development period, perhaps covering at least five years, would be more reasonable, both in terms of effective use of the funds within projects and in terms of the rapidly evolving technology at the moment. This development might be undertaken by a combination of universities, with particular interest in using the computer in learning environments, and computer vendors. Further, it seems *highly* desirable that there be support for not just a single project, but *several* projects all working toward the same goal.

While the projects might focus primarily on hardware development, the hardware development should be oriented toward first running existing, educational materials, at least as a minimum requirement. While these materials do not in general use all the facilities that have been suggested as essential for our graphic environment, they nevertheless represent a considerable body of useful teaching materials.

Some additional developments could also take place as part of the projects, beginning to exploit how to use the newer facilities in learning environments. This information would then be used to improve the displays. But large-scale development should probably wait until such systems are in full operation and some prospect exists for very wide-scale distribution of the systems.

Market Costs

My expertise goes in different directions from that of estimating the costs of such systems. Such estimations clearly depend on projections about current hardware capabilities.

The final product must be inexpensive if it is to be widely used. If we assume a university of 10,000 students, each using the learning systems 3 hours a week, about 500 systems will be required. So we must be talking about equipment not too much more expensive than the current home TV.

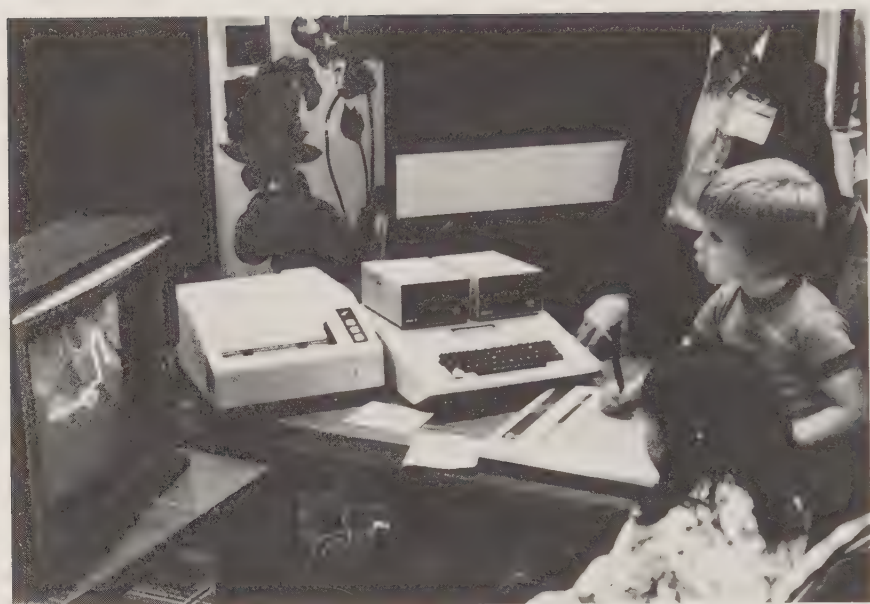
Costs of preparation of suitable materials that exploit all the capabilities indicated will be nontrivial. As indicated, this can be done only by moving away from the current cottage industry situation and by setting up groups having this preparation as their principal mission. The type of organization found in the Open University in England seems to be one that could well be followed. Typical course production in the Open University costs about

\$200,000 and involves a group effort involving both subject-matter and media specialists. Similar expenses for course development in this area would be reasonable.

Finally I want to state strongly my belief that meetings such as the current one are absolutely essential. Educational computer use is still, I believe, in a very early stage. We must look very carefully into the future, seeing the technological possibilities and trying to mold developments in directions conducive to use in learning environments. Often pedagogical uses of technological devices have followed the technology; for effective educational use this is the *wrong* order. It is important to examine desirable futures from the standpoint of improving learning environments and to influence the development of the technology to create these environments. I am particularly suspicious of arguments *from* technology, arguments which go along the lines of "I have a very powerful piece of hardware," or "I have a very powerful software idea." The sensible approach is to *start* from actual learning and teaching problems; these then should influence the hardware, languages, and computer techniques.

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THOMAS DWYER

Thomas Dwyer is well known for his work in applying computing to the teaching of mathematics at the secondary school level and for his dual/solo conceptualization of learning in that context. For more than 10 years, he has been a professor of computing science at the University of Pittsburgh and for much of that period was director of a series of NSF-funded projects concerned with integrating computing into the high school curriculum. While those “solo” projects did explore computer applications in traditional subjects, such as mathematics, they also explored the use of computing in a number of other ways in the school curriculum and raised some fundamental questions about the role of the teacher in computer-based learning. Dwyer has authored and coauthored numerous articles and books relating to computing and instruction, has addressed numerous learned conferences, and has produced several interesting multimedia displays about computing in education.

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5. Heuristic Strategies for Using Computers to Enrich Education

Introduction

Recent advances in technology, especially those related to the ideas of computer science and human problem-solving, offer fascinating potential as agents for implementing a rich and quite deep view of education. Yet many educational programs involved with computer technology studiously avoid entanglement in such issues, preferring instead to accept the safe but shallow waters of drill and practice, frame-oriented tutoring, computer-aided testing, and other traditional management-of-student applications. There has undoubtedly been a tendency to associate technology with some of the more mechanistic aspects of instruction.

The purpose of this paper is to describe an educational viewpoint — both a philosophy and a methodology — based on the assumption that the complexity inherent in human nature should “drive” the relationship between technology and education; that deep technology is of little value without a deep view of education. It will also be argued (paradoxically perhaps), that the workability of this philosophy in the real school world is very much dependent on advanced technology.

This viewpoint is derived from three years of experimentation at the secondary school level in Pittsburgh (Project Solo), and a study of similar work in several other cities (Critchfield, Dwyer, Fennell & Taulbee, 1973). Some excerpts from interviews conducted during this study are given in Appendix A. Earlier examples of student work at the secondary school level are given in Dwyer (1971); additional examples are shown in the film *My Computer Understands Me* (1972), and in the multi-media report *Midnight to Midnight*

(1973). An example of how the viewpoint described here is being applied to a new math education program (Soloworks) will be described at the end of the present paper.

The Goals of Education

Defining the goals of education is a basically ungrateful task; the results will seldom meet with concurrence. Yet it is an essential task if one is to make progress in applying technology at a non-superficial level; at a level which focuses on more than instruction in facts or skills. I will therefore first attempt such a definition, trying to be time-independent, that is, as independent as possible of the strengths, weaknesses, attitudes, or politics of a particular point in history. I shall then describe a heuristic methodology for applying this definition to the real world of education and I shall also try to do this in a fairly time-independent manner. In this case, time independence will mean developing a methodology that is independent of the technology of a particular era. However, I will also argue — and this will be my principal point — that the computer-related technologies and concepts of the '70s have the unique distinction of being the first to match the requirements of the methodology being proposed.

First, let me attempt to define education and its goals. I will base my definition on ideas from a wide range of educational theorists, including Aristotle, Aquinas, Whitehead, Dewey, and Piaget, on the writings of psychologists like Rogers and Maslow, and on some of the contemporary insights of people like Illich and Papert. I think that the common thread one can find in these writers is the view that education should be linked to the ideas of liberty, and fulfillment of human potential. The following shall be used as my definition: *Education is that which liberates human potential, and thus the person.*

The first step in translating this definition into more practical terms is to note that it implies helping people achieve certain kinds of control over their lives. We could thus characterize education as that which shows people how to achieve “liberating control” — control over both their environment, and themselves. Control over environment would include control of social, physical, and economic factors. More important, education should teach internal control. After three decades of work with progressive education, Dewey (1938) could still write, “The ideal aim of education should be self-control.”

When addressing the question of control, a deep view of education recognizes the importance of understanding “the human condition”; it recognizes that one more often deals with failure (or partial failure) than with success. It is thus important for the educational process to work at equipping people to deal with partial failures, and to suggest ways of doing something about this — to try alternatives, to “debug” the procedure that resulted in failure, to

find challenge and satisfaction in working out a personal “can do/can’t do” mix. Such a view can have a very positive long-term value, as has been described quite directly by Maslow (1970): “I’m hopeless with numbers; I’ve always had trouble with numbers. I can’t remember them; I’m not interested in them. It’s something constitutional or genetic, from my earliest days. So what? I forgive myself for it. I learned the minimum I had to. I did not become a mathematician; I’m a poor statistician. But I’m good at something else, and I’ve built on my own capabilities rather than trying to overcome defects.”

It should be noted that the general definition of education proposed here passes the test of specializing to more specific definitions. For example, control over one’s economic environment yields the idea of career education for a decent job. Control over physical environment suggests education in science and engineering. Control over self includes education toward social responsibility through acceptance of limits on one’s own behavior in respect of the rights of others. Control over self also implies that it is in the domain of education to address the problems of self-knowledge and self-determination, somewhat along the lines advocated by Carl Rogers (1969). It also endorses educational goals concerned with fostering life-long artistic, intellectual, and emotional growth; with developing the satisfactions found in an active “life of the mind.”

Translating Goals into Practice

The preceding adds up to an admittedly theoretical, and perhaps somewhat utopian, view of education. Let me now try to relate it to the real world of schools, teachers, and students.

The insights of other disciplines (including, in particular, those of computer science) suggest that there are two systematic approaches for translating educational theory into large-scale practice.

The first might be called an algorithmic approach. This approach is oriented toward defining predetermined sequences of activities for achieving predetermined measurable goals. The method suggests starting out by specifying output (often in the form of what educators call “behavioral objectives”), and then designing a procedure for producing this output from a wildly varying, quite mysterious input made up of children and “curricula.” This approach is associated with what is sometimes called educational technology (as distinguished from computer technology); educational technology, in turn, draws many of its ideas from the field of “systems analysis” (Hinst, 1971).

The algorithmic approach has the merit of giving structure and precision to what might otherwise be a fuzzy process, and of encapsulating that precision

in a reproducible form. Furthermore, the reproducible form theoretically possesses the logic and detail needed if it is desirable that people (or computers) can be trained (or programmed) to replicate the system exactly.

This last characteristic of the algorithmic approach explains one of the reasons it is also invoked in educational circles. Much educational research is predicated on a "pipeline" model, where the end product of that research is shipped out to users after development. This model views the teacher as a transmitter of these products; it must therefore package the products in reproducible form. From an intellectual viewpoint, the teacher has little prestige in such a system. As Piaget (1970) notes, "What the schoolteacher lacks, in contrast to all these (professionals), is a comparable intellectual prestige. And the reason for this lack is an extraordinary and rather disturbing combination of circumstances. The general reason is, for the most part, that the schoolteacher is not thought of, either by others or, what is worse, by himself, as a specialist from the double point of view of techniques and scientific creativeness, but rather as the mere transmitter of a kind of knowledge that is within everyone's grasp."

The second method, which I find much more attractive and very powerful in both theory and practice, is to employ open-ended procedures, by which I mean that neither output nor algorithm are completely specified. Direction and control are achieved instead by applying "heuristic strategies." I'll use the phrase "heuristic strategy" to mean a local procedure for pointing to global goals; a principle or guideline that helps one make both decisions and discoveries, but leaves open the question of the universe in which they are to operate. A heuristic strategy assumes that there are many procedures available; its function is sometimes to suggest new directions, sometimes to suggest backtracking. Its real power, however, lies in suggesting ways of redefining the present state of the procedure being used as the initial state of a new one. This second method is related to the theory of Dynamic Programming (Bellman, 1961); it also bears some resemblance to the goal-oriented strategies employed in artificial intelligence (Newell & Simon, 1961).

Further insight into the heuristic approach is given by associating it with the word "learn." Heuristic devices do not tell one what to do; they tell one how to learn what to do. Thus a heuristic system takes a very different view of the teacher. It insists that the teacher become more and more proficient, eventually arriving at an expertise that transcends that of the original designer of the heuristic system. This is in contrast to a teacher functioning in an algorithmic system where the instructor is often viewed as a technician for implementing predetermined aspects of the system. When critics of educational innovation say that it will not work because teachers will reject it, they are really talking about innovation implemented under an algorithmic, not a heuristic system.

I shall give examples of four such heuristic strategies, and simultaneously

say something about how computer-related technology can make it possible to employ them in a manner that is unprecedented in both its workability and its profundity.

The Authoritarian View vs. Respect for the Learner

The first strategy is a principle for dealing with the problem of “authoritarian” vs. “liberal” views of education. An authoritarian view of education is not an evil one. It is, in fact, valid and essential, since it holds that each civilization presents its young with a large history of accomplishment, that there is always a significant heritage worth examining, that there is invaluable information about theories that did and did not work, and that these can (in fact must) be passed on for the profit of succeeding generations.

The difficulty with this view lies not in stressing the value of accumulated know-how. The problem is with the tendency of human teachers (and even more so, authors of CAI) to embed the information they transmit within their own personal and unalterable interpretations (models) of how to use this information. Piaget (1970) tells us that, indeed, the “lesson is always bound to conform to the tendencies of the teacher, since that is by far the easiest solution.”

My first heuristic device would be to insist on educational procedures that continually reassess how well they recognize (and do something about) the distinction between transmitting accumulated experience, and transmitting models of that experience which are not valid for the new learner.

Let me give a simple example (which also explains our use of the word “solo”). It has to do with flight instruction where a distinction is made between dual and solo mode training. Dual mode involves an instructor, and much of it is authoritarian. The student must use a certificated aircraft, he must obey air traffic control, he must use the right airspeed to optimize climb, and so on. Yet he knows that he is moving to a solo flight, where he can only succeed if he develops his own models of how to use all this “past” heritage. The student alone can build the “right” model for solving a given problem, say, making a landing. The instructor knows how to land an airplane in the sense that he can do it; he can also theorize about how he does it. But he will never know much about the student’s internalization of this information. So the instructor’s primary task is really not to tell the student how to land the aircraft, but to help the student build his own model of the process.

The importance of helping students build their own models of the world comes home in a striking way when one observes experienced teachers of blind children at work. These teachers know that the manner in which these children “see” the world will forever be a mystery to the instructor. Such instructors become educators only when they respect this mystery, and organize their instruction accordingly. As one teacher put it, “you don’t help them do

it — you help them do it for themselves.” I submit that adding the sense of sight does not change the essential rightness of education based on this important distinction.

What has technology to do with these ideas? I think the answer is that we are at a point in history where we have an unprecedented technology upon which to build educational systems that distinguish between the transmission of past heritage, and the eliciting of new understanding. It is called student-controlled computing. Student-controlled computing is to solo mode, as CAI* is to dual mode. Student-controlled computing means that the student uses technology to develop and test his own models; that the student learns to deal with failures; that it is in his power to debug the procedures that caused these failures. He also develops a powerful store of ideas (“ways of thinking”), from his contact with real computer science (Minsky, 1970). The user of CAI is, as Arthur Luehrmann (1972) points out, denied all of these riches.

Expectation of the Unexpected

The second heuristic strategy I would propose hinges around words like “trust,” “expectation,” and “recognition.” The words assume that a human teacher and human relationships are an indispensable part of education. A human teacher is important for clarifying, inspiring, guiding, and propelling. Formal educational systems do not build habits of lasting self-propulsion. But trust, and expectation of the unexpected, are grade A propellants in any setting. They are especially valuable if they initiate a history of success. This means that expectation should match student capabilities.

The role of technology in supporting this heuristic device is quite direct. When a teacher hands over control of the sophisticated tools of computing to a student (including the bells and whistles of real hardware), this is an act of trust and expectation. It does not need to be verbalized, and it does not require a teacher with a Ph.D. in psychology to be effective. Moreover, the general nature of computing, and the natural way in which it invites problem segmentation, allows for success on many levels. We have seen numerous examples in our work that verify the power of technology in this role.

There are some words of caution that should be added at this point. Teachers who encourage solo mode learning in the manner I have advocated must face up to a problem that is not found in dual mode instruction. They will have to develop an approach to handling that aspect of maturation in students we sometimes call “moral judgment.” They will have to anticipate that at

*Computer-assisted Instruction, used in the sense of computer-managed drill, tutoring, and testing.

times it will seem as though their trust has been betrayed. This can be a devastating experience; the experience can be even more traumatic when the teacher must also face the wrath of those who feel obliged to fix blame on one person. I should also warn that even when there are no “failures,” solo mode can be rougher on the instructor than the student; airport tower operators have learned to spot a solo flight more by the apprehensive nail-biting figure of the instructor on the ground than by the performance of a student flying in their traffic pattern.

Lest these remarks seem overly pessimistic, I should hasten to add that “betrayals” of trust are exceptions, not the rule, in a student-controlled computing environment. Even when they do occur, they are usually traceable to a combination of “unthinkingness” about the consequences of certain actions, and a desire to do things (often of a quite advanced technical nature) which had been declared off-limits for students. Thus one is sometimes faced with the strange dilemma of having to classify a desire to learn as an offense. I think the solution to this dilemma lies in the idea of “control over self” that was part of my original statement of purpose for education; in stressing the development of a “can do/can’t do” mix; in the pursuit of legitimate ways of changing (debugging) the system that produced the dilemma in the first place.

Many other things can be said on this subject. Some interesting work in the area of developing responsibility through control of “dangerous” technology is going on in other areas. The film *Choosing to Learn* (1971) describes one such project for very young children. Such experiments are interesting because the problem of teaching “moral judgment” has usually been skirted in public education. The reason is the traditional association of moral education with religious beliefs. Strange as it may seem, technology (in the form of student-controlled computing) may provide the beginnings of an answer to this impasse.

Environmental Learning

The third heuristic strategy is based on the principle that education, in the sense defined earlier, takes place best in a rich and joyful environment. Learning and play go hand in hand. Artifacts, both simple and sophisticated, when used correctly, can make complex ideas transparent. The work of Seymour Papert (1971) at MIT in teaching young children advanced mathematical ideas (through the use of sophisticated artifacts) illustrates this idea. The success of certain community learning centers (e.g., the People’s Computer Company), and imaginative museum settings (e.g., Frank Oppenheimer’s

“Exploratorium”) is greatly attributable to the spirit of fun environment they have developed.*

One *caveat* should be given: environmental learning, devoid of purpose and guidance, can be inefficient, and at times harmful. A person who lives in the wrong part of France can learn bad French; South American babies who learn to swim by falling off their family house-raft don't always survive. Like most good ideas, environmental solo learning needs to be coupled with other factors — with a supporting structure — before it reaches its full potential. A study of the factors which support solo mode flight instruction (which is an excellent example of environmental learning) gives particularly good insights into the role of supporting structures. Some of these factors are discussed in Dwyer (1970).

The relation to computing is this: if the student learns to make a computer (and related technology) sit up and do tricks [to be a “symbiotic partner” in the sense described by John Kemeny (1972)], this student is working in one of the richest intellectual environments yet devised by man — and it is fun. A proof of this last statement is not needed by any teacher who has tried it — the stories about students staying very late after school or getting up at 5 a.m. to be the first “on-line” are legion.

The richness of computer environments is presently limited by student-machine interface technology. I suspect that we will see fundamental growth in our understanding of the potential of environmental learning as these limits are gradually removed. Some of our own work that is moving in this direction will be described later in this paper.

There are two pitfalls to be avoided in building technology-based environments. The first is an administrative one that comes about when local control of equipment is sacrificed to the god of centralized efficiency. The folly of such an approach can be seen even in the use of simple technology. For example, a teacher who has exclusive control of a projector and suitable screening room is a much more effective user of films than one who must share a central resource. Central resources look great on paper, but they can lead to the reversal of important educational priorities in practice.**

The second pitfall is to develop environments that are too rigid or too simple. The prepackaged lab experiment that is built around a cookbook-type curriculum commits exactly the same kind of educational error as the instructor who insists that there is only one way to attack a problem. The environ-

*Artifacts need not always be expensive. For example, the Nuffield math materials contain a rich store of ideas for constructing low-cost devices.

** There are notable exceptions to this rule; the Dartmouth computer center and our own experience with a commercial time-sharing service (Comm-Share, Inc.) come to mind as examples of educationally superior systems that are partly centralized.

ment that is simplified to ensure foolproof operation assumes (literally) that students are fools. The first heuristic device, with its insistence on the importance of learning to debug unsuccessful procedures, tells us that this is wrong. Educational environments need to have a complexity which invites alternate explorations, and (possibly) alternate solutions.

The Teacher Rediscovered

The final heuristic strategy I will suggest is to organize educational change around a system that encourages teachers to become expert in, and enthusiastic about, an area that students admire. I think it is no secret that schools of education do not in general produce the kind of expertise that is regarded highly by students, or even by the teachers themselves. Yet it becomes clear in dealing with young people, that despite an apparently cynical attitude toward authority, they are still very much influenced by qualities of expertise in teachers. They are even more impressed by such qualities as a zest for learning, and enthusiasm for discovering new answers.

What has technology to do with supporting such a principle? The answer is simple, and circularly (recursively?) ingenious. The way to apply this fourth heuristic strategy is to reshape a good segment of teacher education along the lines suggested by the first three heuristic strategies. I have used the phrase teacher education rather than teacher training, to emphasize (using my original definition) that I am really proposing a liberation of the human potential of those who will work at liberating the human potential of others.

Actually, the role of technology in supporting teachers goes well beyond teacher education. One of the most difficult things for any teacher to handle is a prudent student-teacher relationship; one that recognizes that such relationships are essential, but also that they can be taken advantage of. The "objects" of technology can serve as intermediaries for resolving this conflict. They do this by acting as the common ground to which both students and teachers can relate (usually in quite different ways). They can thus serve as the link for establishing such relationships, as the buffer for safeguarding them, and as later-life reminders of the values discovered through them.

The Role of CAI

The preceding discussion has emphasized the value of student-controlled computing as a technology for implementing heuristic strategies derived from a humanistically oriented view of education. This does not imply that CAI, used in the sense of author-controlled technology, cannot make important contributions to an educational system based on this same view. On the con-

trary; CAI can be an important subsystem for the “transmission of past heritage” (just as student-controlled computing is an important subsystem for eliciting new understanding of, and contributions to, that heritage).

More precisely, I view CAI as having unique value as a subsystem of dual-mode instruction, especially when it frees the human instructor for the global aspects of teaching. A good instructor learns that many things are best left unsaid, and that his primary function is to focus the student’s attention on the forest, not the trees. But without trees, there are no forests. CAI can be an excellent tool for planting and nurturing educational trees.

CAI becomes especially valuable when dual and solo modes build on each other, and when there is planned transition between modes. Some of the technical consequences of insisting on transitional ease between CAI and student-controlled computing are discussed in Dwyer (1972).

An Illustrative Example

An application of the preceding ideas to a math education project for beginning secondary school students (aged 13-17) is presently in progress at the University of Pittsburgh.* Extensive use of technology is involved, but the role of this technology has been primarily determined by the heuristic educational principles just described. The math curriculum developed will in fact be deliberately structured so that its replication elsewhere will be dependent on the conscientious application of these same principles. This is a weakness of the curriculum in the sense that it assumes teachers and administrators who understand and agree with the heuristic approach. It is a strength in the sense that it gives the curriculum the ability “to learn and adapt.” In the description that follows, square brackets will be used to point out the heuristic strategies being applied.

The Soloworks math program will be organized around four labs called Computer Lab, Dynamics Lab, Synthesis Lab, and Modelling/Simulation Lab. Each lab will focus on both the achievement of major skills, and on the pursuit of research projects. A research project is defined as any endeavour in which the outcomes are not completely predetermined [expecting the unexpected, trust]. The instructional plan behind the labs is shown in Fig. 1.

The role of the teacher might be described as one of “knowledgeable facilitator.” This means that our teacher-training program must focus on having teachers live with, experience, and become good at some (not all) of the skills connected with the labs [encourage teachers to become expert in areas that students admire]. The supporting structure box for each lab consists of both

*Sponsored in part by NSF grant EC-38063. The informal name of the project is “Soloworks.”

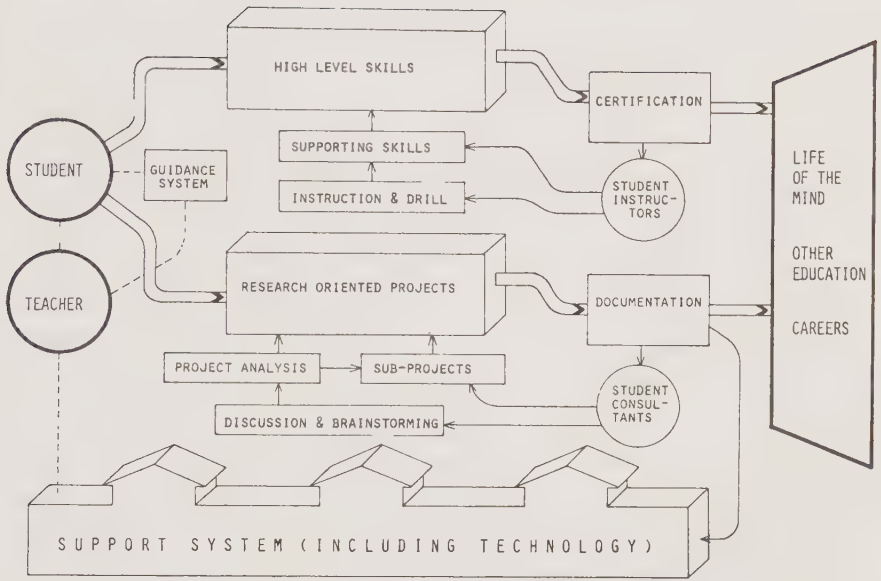


FIG. 1. Organization of the Soloworks Labs.

curriculum material and exciting technology [a rich and joyful environment].

(1) The Computer Lab will focus on those aspects of mathematics that are well described by algorithms, and encourage student access of a local computer and terminals. A major skill students will achieve in this lab is that of computer programmer. Examples of some projects that have already been done by students in this lab are developing programs to do accounting and inventory for a small store, programs to manage computer dating, generation of random ballet dances, programs to plot all kinds of mathematical curves, programs to predict and plot their intersections, etc. The significance of the project approach is that although students have made available to them past "heritage" appropriate to attacking the project, they are also expected to develop new and unique extensions of that heritage [respect for the learner's need to develop his own models]. We have found that this is difficult for students at first; most of them exhibit a high reliance on an instructor's explicit orders. But we have also found that patience is rewarded; students can and do make the transition.

(2) The Dynamics Lab focuses on mathematics that describes processes that take place in time. One form of technology being used in this lab is a flight simulator [rich environment]. The skill acquired in using this particular equipment is that of making a full instrument landing, or becoming good at

instructing a fellow student to do the same [trust, recognition]. An example of a project would be to sample analogue readings of heading, time, and speed from the flight simulator, translate these into digital data, and then write a program that plots the path of the flight simulator. Another type of artifact being used in the Dynamics Lab is the "Rube Goldberg" machine, a gadget designed by the student to do nothing useful, but to be a mind-stretching exercise in imagination.

(3) The Synthesis Lab is concerned with mathematical methods that make use of the principle of superimposition, producing complex effects by adding together simple ones. Two special pieces of technology we will use are the "Music Monster" (a kind of programmable band-organ), and a multi-skills associated with these devices are composer and media-designer. Projects will focus on the design, debugging, and performing of original works.

(4) The Modelling/Simulation Lab uses mathematics as a tool for creating new models of reality that can be studied and manipulated [new models vs. old heritage]. Some of the models are physical (e.g., bridges, elevators, lunar landing modules, etc.), some are abstract (e.g., an ecology). The skill developed here is really that of applied mathematician, while the gamut of possible projects is open-ended. This is because the computer available to students is general purpose, allowing them to simulate systems not heretofore dealt with [expecting the unexpected].

The sketches in Fig. 2 give some idea of the form of the technology we are or will be using [fun environment!]. We are just concluding our first year of what will be a three-year program to develop and test these ideas.*

Conclusion

The methodology I have proposed has difficulties of course. The source of these difficulties is pinpointed in the words of Piaget: "The heart-breaking difficulty in pedagogy, as indeed in medicine and in many other branches of knowledge that partake at the same time of art and science, is in fact, that the best methods are also the most difficult ones." I would argue that this is precisely the challenge that those engaged in developing technological innovation for education ought to embrace as their own. Just as the complexities of technology cannot balance out inadequate educational philosophy, thoughtful philosophy cannot survive without a methodology attuned to the complexities of human nature. This is the primary standard against which technological innovation in education, present or future, ought to be judged.

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A study of similar work was supported in part by NSF grant GJ-33221.

FIG. 2. Some Examples of the Soloworks Lab Components.



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*Progress reports in the form of a newsletter are available by writing to the author.

Appendix A

A problem that sometimes confronts an educational system based on student-controlled technology is the request to “prove” its worth through evaluation based on standardized tests. Using such evaluation can be a misleading exercise, since standardized tests (and the test-makers who devise them) favor the outcomes of dual mode instruction; both questions and answers are influenced by the models of the test-makers. Papert suggests that using such tests to evaluate an innovative instructional system concerned with eliciting new models is like evaluating the potential of a precision electric motor by measuring its ability to drive a cart meant to be drawn by horses.

There are several kinds of useful evaluation that can be applied to education based on a dual/solo model. One is based on an analysis of new student output and the structure of the models that the student built to produce such output. Such evaluation is very impressive when done well; some work along these lines is currently in progress as part of the Soloworks project.

Another kind of valid evaluation is based on the argument that it is more important to fill in glaring gaps in a system than to effect small percentage gains in areas that have already accomplished the goals for which they were designed. An illustration of one such gap is given by Begle (1973) in an article that discusses the experience of the MSG “new” math program:

... student attitudes towards mathematics seem to be rather favorable at the beginning of fourth grade and improve slightly during the remainder of elementary school. However, at the beginning of junior high school, student attitudes towards mathematics begin a slow but steady drop that continues to the end of high school. These attitude changes seem not to be affected by the nature of the curriculum to which the student has been exposed.

The ability of student-controlled computing to fill this kind of gap appears to be one of its strong points. The following excerpts from some interviews* with administrators, teachers, and students who have used student-controlled computing for a few years will give some insight into the nature of this phenomenon.

“In a computer math course you have the freedom to allow students to become very expert, which a lot of students want to do, while in a regular course you don’t have that freedom.” (Teacher)

*Conducted at George Washington High School in Denver, and Fox Chapel High School in Pittsburgh.

"... it's exciting to use computing in other classes. That portion of the class becomes quite exciting." (Teacher)

"... Every time I wrote a program I always wondered what I could add to make it better; the only thing wrong with that approach is I would still be working on my first program." (Student)

"I came running out [after seeing my plotter program work]. I found [I had discovered] a very important principle in calculus." (Student)

"It's really amazing how [students] will go ahead on their own; they go beyond my scope. My best students are much better than I am." (Teacher)

"I am completely bewildered at times by our big computer. . . . I haven't learned it all. But I have students who are very interested and I have *them* present that part of the program. . . ." (Teacher)

"I have absolute confidence that, in the sphere of high school math, these kids can write algorithms in whatever areas you wish." (Teacher)

"We hope that next semester we'll be able to expand. Right now we're operating in what may be termed a converted broom closet. We need more space, . . . we're trying to get it, and I think we will." (Principal)

"They found that . . . students seemed to not only learn the subject taught, but were using the generalizations produced with the computers to enhance their skills in related subjects." (Teacher)

"Finally, when we finished the survey, the job of analyzing the [social studies] data was given to another group of people, of which I happened to be a part. We decided that the only obvious solution would be to use the computer." (Student)

"My private study project is . . . travel in space involving relative time conversion using the Lorentz transformation equations developed by Einstein . . . I have taken all the physics courses offered at G.W. . . . the way I increased my knowledge in physics was through the computer math course." (Student)

"They've written programs for the language classes. . . . And then they've written programs for refresher arithmetic." (Teacher)

"They also solve problems for other teachers . . . a girl modelled the political growth of a city." (Teacher)

"[Steve] has been writing practice programs for basic math students . . . games with the computer so that they have to add and subtract and give the right answer." (Teacher)

"I see a teacher in there for whom we just can't get enough money and terminals and computer time. . . . In my case I've found all I have to do is get out of the way and let her [the teacher] run." (Administrator)

"We immediately rushed down to the principal's office and showed [the project] to him. He was very interested in it, and he encouraged us to [go ahead]." (Student)

"We couldn't be more excited about the program. . . . We've had very, very favorable community reaction and continue to have." (Principal)

"It was challenging [to introduce computing], but I always liked a challenge . . . I'm more enthusiastic about this than about any other area I have ever taught. . . ." (Teacher)

"I see young people in there till five o'clock in the evening . . . not only do they understand the computers but they seem to understand the why of the mathematics behind it. . . ." (Administrator)

" . . . This is the hard part in senior high school, trying to get the kids the least bit interested in learning math . . . I heard one little girl who was really animated about describing what had happened . . . she was talking . . . in a quite sophisticated way, yet she never had a history of achievement in mathematics before." (Administrator)

"They go out and try [new things] — we have several projects going on right now . . . such as modelling where they actually construct mathematical models." (Teacher)

"We've even had trouble because they were staying after school was closed. . . . They would hide . . . until everybody disappeared and then they would come here [computer room] . . . They found them at 8:00 at sometimes. . . . They want to be on these machines from midnight to midnight. We have to throw them out at night. It's really fantastic." (Teacher)

" . . . I became so engrossed in computer programming that I would work on programs at night by flashlight or wake up at 5:30 to get to the computer lab when it opened. I would work all weekend on various programs to do my homework, or to solve problems in chemistry, math, and even carpentry. After a while I realized that there is more to life than computers and I slowed down a bit." (Student)

6. The Significance of Solo-Mode Computing for Curriculum Design

Introduction

There is growing evidence that education may have caught a tiger by the tail. It comes in the form of an activity called "student-controlled," or more accurately, "solo-mode"¹ computing. Such use of computers (as the tiger image suggests) often exhibits an unexpected raw power for eliciting complex learning behaviors in all kinds of students. The untamed animal image also reflects the anxiety many feel about an educational activity that isn't (by definition) under direct control; an activity that often seems to lack "purposeful" or "accountable" direction.

Whether lack of direction is a potential problem, or a strength in disguise, depends a great deal on the orientation of the educational system making the judgment. If the system puts all of its emphasis on attainment of measurable competencies through application of pre-packaged materials, then there can be a conflict. Solo-mode computing's most interesting characteristic is its ability to support explorations that are endlessly unique. The sure way to lose this uniqueness is to try to specify or package it.

Careful packaging is, of course, a useful technique. But it has its price. Anyone who has had the opportunity to compare solo exploration of a new country with a packaged tour of the same area, will understand why. They will also understand that the concept of learning by means of solo exploration is an idea which arises not by design, but by necessity, necessity born of the human spirit's need to probe in ways that are not fully understood by even the best of instructional paradigms.

It is this relationship to the complexity of human nature that provides the

¹ This article originally appeared in: *EDU* Sept., 1974. Published by Digital Equipment Corporation, Maynard, Mass.

key to making the most of solo computing in the real school world. The secret is to recognize that the deeper issues in human learning cannot be accommodated by totally explicit instructional models. There are some aspects of personal growth that, in effect, vaporize the moment they come into contact with strong control structures imposed by others. There seems to be a kind of educational "Heisenberg" principle at work; the more we sharpen the explicitness of instructional control, the less likely are we to pinpoint the wellsprings of deep understanding.

Does all of this argue for abandonment of control, or the neglect of disciplined competency? Certainly not. Rather it argues for a much broader understanding of the idea of educational structure. What needs to be abandoned is the notion that managerial techniques designed to increase productivity of material things, will do the same for education.² The work of education is far more intricate, partaking as Piaget notes, of "both art and science."³ It requires a model of considerably greater sophistication, than that usually implied by words like "management," "efficiency," or "accountability."

This significance of solo-mode computing for education is that it has opened the door on a world of insights about exactly such a model. Not all of these insights are clearly perceived as yet, but the focus is sharpening. It is also becoming clear that this is a door that must be kept open; it would be a mistake to freeze (or package) any one structure based on even the best of this experience. It is for this reason that it is better to formulate these new insights in terms of "style" rather than doctrine. The moment we make everything explicit will be exactly the moment in which we lose contact with the best aspects of the solo idea.

The purpose of this paper is to set down some thoughts about a generalized view, not of a complete model for education, but of an important component in any such model, namely the curriculum. I will do this by first offering a generalized view of the idea of a curriculum, and then analyzing the style of curriculum that seems to surface in an environment committed to the importance of solo-mode learning.⁴ As will be seen, this style is the antithesis of the mechanistic image usually generated when the words "technology" and "education" are encountered together. Our experience has been that the support of solo-mode computing inevitably induces a view of curriculum that is not only faithful to the ideals of a liberal education, but one which is remarkably effective in suggesting plans of action for implementing the liberal ideal.

A Generalized View of Curriculum

To get at a generalized view of the idea of curriculum, I think that the right question to ask is this: how do you combine the interesting things that happen when students explore their own interests in their own ways (doing what is

sometimes called “just messing around”), with the undeniable need of these same students for intellectual and emotional growth, a growth that must feed on the ideas and guidance of others? I believe that the answer to this question can be found by recognizing that people grow in two ways. They grow by assimilating good ideas from the past (what I’ll later call a growth in repertoire), and they grow by moving into the arena of creativity as early as possible. It’s this second aspect of growth that can be short-changed by an overzealous management approach to education.

The obvious (and only) way to grow through creativity is to be frequently creative. But how do you get started? It’s clear from history that a small number of people find the answer to this on their own, with very little help from formal education. But that shouldn’t prevent formal education from asking how it might evoke this “natural” capability in more students, and at higher levels than normally anticipated.

Part of the answer to this last question is that a direct approach is unproductive. It’s very doubtful that what results after one person tells another person how to be creative, can ever be really creative stuff. To help people grow in creativity, indirection (multi-direction may be a better word) is called for.

I think that a powerful choice of indirection to take, in order to stimulate growth in creativity, is found in the idea of problem-solving, using this phrase in a general sense. Our experience at Project Solo, and that of others,⁵ indicates that it does make sense to talk about helping students become habitual, enthusiastic, catholic, and fluent problem-solvers. As this fluency develops, creativity will also start to appear. Just how this works is currently something of a mystery; neither psychology nor the new disciple of artificial intelligence would pretend to have models for generating the world’s next captivating melody, influential book, or deep theorem.

But the “habitual problem-solving” approach to starting on the road to creativity works, and that’s what is important. Knowing that this particular form of indirection works makes it fair to propose the following general definition of curriculum, even in the most formal of educational systems:

A curriculum is a structure that brings about and nurtures two kinds of growth in students: growth in repertoire through assimilation of (and practice with) good ideas from the past, and growth in creativity through developing a habit of creative problem solving.

The generalized nature of this definition leaves open the question of what the “structure” alluded to looks like. It could, for example, be a conventional arrangement of a standard number of courses spread over so many years. It could be a very loosely structured but rich environment of people interacting with “things,” ideas, and with each other. It could be an alternating work-study-discussion program. Or it could be some combination of these and other elements. What’s important, is that no matter what form of structure is elect-

ed, there must be close attention to developing both growth in repertoire, and growth in creativity.

Of course some forms of structure are probably better suited to circumstances than others. We could ask, however, whether there are any especially powerful techniques that would make a big difference in any structure. I believe that the answer is yes, and that an outstanding example of such a technique is solo-mode, general-purpose computing.

To see why this is so, let me now discuss some specific ways in which an environment based on the idea of solo-mode computing makes a fundamental difference in the way one thinks about curriculum.

Curriculum Design Factors Suggested by Solo-Mode Computing

If a curriculum is viewed as a structure that helps students grow, it becomes obvious (by contemplating the great variety in people and their cultures) that such a structure must be dynamic and flexible. It cannot be, for example, little more than a set of courses to be taken, or a list of topics to be covered. This is particularly obvious when solo-mode computing is introduced. A curriculum that is completely prescriptive in nature will tend to cut off (or more often, never permit to arise) the best aspects of solo-mode exploration.

For this reason, it is not too surprising that the most important things one learns about curriculum design in a setting that relies heavily on solo-mode computing are related to style, rather than to content. But how does one describe style, or share insights about it with others? I guess the answer is "with great difficulty." But let me try to set down a few ideas anyway, by trying to answer the following question: What key phrases summarize the kind of advice we are currently giving ourselves as we work at the problem of developing curricula that match the full potential of solo-mode learning? Here are six candidates for such a collection.

THINK TOP-DOWN

This phrase suggests designing a curriculum by starting with an end-idea; something substantial, challenging, and really representative of the field. Then, through a process of backtracking, one develops a structure that leads to this end-idea. This structure should encompass enough detail so that students who master it will be able to make creative attempts at extending the end-idea.⁶

Now I would suppose every curriculum designer believes that he really proceeds in this way, but what comes out in practice usually misses the idea behind the top-down approach. There's very little you can really do with the

typical textbook end-idea. It's the presence of real computing, with its ability to assemble component ideas into a large system with real output, that makes "text-book" ideas come alive. Working with general purpose computers reveals that the theory found in modern school books (e.g., the better "news" math books) is an excellent source of component ideas. But it also reveals that one can (in fact must) go beyond the applications anticipated by the authors of these books, if the power of solo computing is to be fully exploited.'

It's important to note that the top-down approach carries over to non-computer activities, but it does this best in a computer "atmosphere," where the students are used to assembling basic ideas into complete and functional end-systems. Such an atmosphere (and such students) are an extremely good antidote to the uninspirational mediocrity of most "text-book" end-ideas.

ENCOURAGE CONJECTURE

This is a piece of advice that dawns on you after you learn to sit back and quietly observe the working habits of students who have broken the creative barrier. Given such a student, a new challenge (say figuring out a new machine), and a carefully constructed sequentially arranged instruction book on how to run that machine, I have never seen such a student attack the challenge by first opening the instruction book (and *certainly* never to page 1). The human learner has an ingrained inclination to focus on just about anything *except* a linear sequence of logical steps. I think this characteristic of people needs to be recognized and taken advantage of; the conjecture habit should be encouraged, not discouraged.

It should, of course, be obvious that a curriculum structure has to embrace much more than written materials if the conjecture (plus validate) habit is to be supported. One good mechanism, for example, is to infiltrate a group with a few inveterate "conjecturers." It is also important never to call any student's conjecture a poor idea; just the opposite, the teacher and others must learn to explore the potential of all ideas. There's an important difference between calling a conjecture "poor" and examining exactly where its problems or virtues) may lie. Just how these two mechanisms are to be built into the curriculum will be left as an "exercise for the reader." However it should be clear that the conjecture atmosphere could be a natural part of the problem-solving component of a curriculum. And as observed earlier, the achievement of a habitual and fluent problem-solving capability is the prelude to genuine creativity.

PREP THE AUDIENCE

An extremely important consequence of designing a curriculum so as to encourage the conjecture habit, is that the students so encouraged will become a

very different kind of “audience” for more conventional instructional methods. These will be students who are looking for ways to prove or disprove their latest theory. I can’t think of a better prepared group for a good old-fashioned lecture, or a good old-fashioned book. It is a sound principle that the best reader (or listener) is one with lots of questions. The curriculum designer who manages to take advantage of this phenomenon automatically multiplies the effectiveness of the descriptive and explanatory components in his curriculum many times.

ILLUMINATE WITH INDIRECT LIGHT

The importance of indirection was discussed earlier in connection with the idea of creativity. Indirection is also important in other areas of curriculum design. Allowing for it is more difficult than one may realize, because indirection implies pursuing paths other than those the teacher (or curriculum designer) had thought about. It takes special teachers and special structures to follow for this kind of “heresy.”

Indirection does not mean aimless wandering. It means pursuing alternate paths, and comparing them. Thus indirection really needs lots of direction! For example, we are currently working on a project to bring students to a point of fluency with the concept of mathematical integration. We are encouraging students to arrive at their understanding through many paths (multi-direction is again part of our concept of indirection).

We do this by *direction* of the sort that says: see if you can design a machine that solves this second order differential equation using hydraulic mechanisms; now do it with ball-disc integrators; now with operational amplifiers; now digitally; now with the formal techniques of calculus, etc. The end result is a very deep, multi-dimensional view of the subject, something like one gets by walking around (or flying over) an object. Further, we are continually obliged to tie these various views together (students ask!). This is frequently an embarrassing exercise, for one soon realizes that these various approaches all have different vocabularies, each with its own unique brand of fuzziness. You end up with an appreciation of how really difficult it is to produce good instructional material, and a bit shocked at finding out how provincial instructional materials can be.

EVOKE IMAGINATION

Sometime back we produced a computer game that asked the student to find four hidden players in a rectangular grid. He did this by guessing at the positions of the players, and then receiving feedback on the distances between his guess and the actual positions. The folks at PCC (People’s Computer Company) renamed the game MUGWUMP, and added a few other imagina-

tive interpretations to the exact same situation. The result was much more effective. Kids are able to generate a fantasy world of considerable complexity, and hanging cognitive explorations onto such a world is a very good idea.

Attaching cognitive processes to an imaginative framework is also a sophisticated idea, having roots in the "little man" (frame-of-reference observer) and "Maxwell Demon" techniques of classical physics. It is really an application of the fact that learners attach new ideas to existing internal structures. A corollary of this fact is that it *is* worthwhile for education to help children build a repertoire of images and fantasy structures, and these can be a lot more complex than most adults seem to remember.

SILENCE CAN BE GOLDEN

A few years ago I started to do some flight instruction. When I first took students out for their initial ride, I spent a lot of time briefing them on everything from how the airplane worked to what all the correct values of airspeed were for different maneuvers. It didn't take me long to realize that this was a complete waste of time. All these students wanted to know was whether they were going to get back alive. The "language" of the airplane itself, not English, was the right one for these circumstances.

Perhaps the anxiety about life and limb is not as pressing an issue in the typical classroom, but the principle that some things are best left unsaid for a while is a universal one. Of course something has to replace English, and this is where both the kinesthetic and artificial-language aspects of computing have fundamental value. English words are mixed in, but in a crisp, precise way that seems to ring sympathetic chimes in most children (and adults).

A non-computer example of a really effective piece of instructional material that has come our way is a brochure on how to build some mechanisms with a model building set that comes from Sweden. The brochure is in French, but it has very good photos. Students seem to get a lot more out of unraveling these instructions than they do from corresponding, rather verbose manuals in English. I think that two reasons explain why the French version is so effective. One is the fact that the student *knows* he won't be able to follow the instructions. This puts him in a creative mood, one where he plays with options in interpreting the photos, not being afraid to conjecture, try, re-conjecture, etc. The second is that the photo sequence in this book is top-down! You first see the whole machine, and then watch various pieces "explode" away. By the time you get to the starting point where it's time to assemble the mechanism, you have a good idea of where you are headed. (I wish to parenthetically remark that I am not at all advocating the kind of super-simplicity found in books today that are produced by publishers who argue "kids can't read." On the contrary; I'm advocating high level, complex explanatory mate-

rial, but in a variety of “languages,” some of which are quite different, and quite superior to English.)

Conclusion

Although the curriculum design techniques suggested here have been phrased in general terms, they all come from specific work done over the past few years in a solo-mode computing environment. This has certainly been a tremendous advantage. In fact, it would probably be a bad idea for anyone to try to adopt these techniques without the presence of some kind of equivalent environment. A more general way to summarize the pros and cons of the philosophy of curriculum design presented here is to admit that most innovative thinking about education is partial in nature, and it may have little practical worth if implemented in isolation.

Conversely, good ideas flourish when they interact with other good ideas. As just mentioned, the ideas about curriculum design described here are linked to another good idea, namely the availability of solo-mode technology. A second complementary “good idea” (one that may in fact be essential), is to work hard at getting students involved in thinking about their own learning. This implies an openness on the part of curriculum designers to student/teacher input. All the curriculum design techniques discussed here certainly work much more effectively in a setting where both teachers and students are viewed as part of the total curriculum design team. This, in fact may be the most important design secret of all.

References

1. The term solo-mode is derived from flight instruction, where a student first practices under the guidance of an instructor (dual-mode), in order to eventually learn by managing all the problems of flight alone (solo-mode). Solo learning and dual instruction work best when they alternate. It should be noted that while solo-mode implies student-control, the converse is not true. For example, there is a great deal of student-control in a dual flight, but it's exercised in the context of the instructor's model of how things should be done. Thus solo-mode is a much deeper idea than “student-controlled.” It implies that the student must develop his own model of how things work.
2. The November 1973 issue of *Educational Technology* contains several papers that discuss the limitations of “managerial” education.
3. J. Piaget (1970), “The Science of Education and the Psychology of the Child,” Orion Press, N.Y.
4. Most of these ideas are derived from the experience gained with Project Solo, an experiment in computing for secondary schools, and a successor project called Soloworks. A newsletter describing Soloworks is available from the author. Soloworks is supported in part by NSF Grant EC-38063.
5. See, for example, the comments of educators quoted in T. Dwyer, “Heuristic Strategies for Using Computers to Enrich Education,” *International Journal of Man-Machine Studies*, Vol. 6, 1974.

6. Lest there be any doubt, let me hasten to point out that an end-idea is in no way, shape, or form related to a Behavioral Objective. An insightful essay on the limitations of BO's is found in Leo Ruth, "Mismanagement by Objectives: A Learning Module for Planners in English," *Educational Technology*, Vol. 13, No. 11, pg. 37.
7. Numerous examples of end-ideas based on a modern first year algebra book are given in T. Dwyer, "Computer Resource Book for Algebra," Houghton Mifflin Co., Boston (1974).

7. Some Thoughts on Computers and Greatness in Teaching

Introduction

In a paper entitled “The Computer and the Fourth Revolution,” [1], Molnar gives a panoramic view of recent efforts to harness the potential of the computer for education. He concludes that “computing is so compelling a tool that it cannot be stopped.” Our own experiences, both at Project Solo and through numerous contacts with others, confirm this conclusion, and in fact suggest an even stronger one. We have found that computing, placed in the hands of well-supported teachers and students, can be an agent for catalyzing educational accomplishment of a kind that is without precedent. We believe that there has simply been no other tool like it in the history of education.

Our enthusiasm is tempered, however, by a growing respect for the importance of human relationships, particularly in a technology-based environment. The more advanced the accomplishment of students, the more keen seems to be the need for the recognition and guidance that only a dedicated teacher can give. Developers of computer-related instructional systems seldom make explicitly negative statements about the role of teachers. There have been, however, unfortunate implications that computers are important because they can make education “teacher-proof.” I think that this is not only an incorrect attitude, but one that cuts off the real power of the computing tool. And the most important users are not research and development people, but “teacher/student people.” Put simply, I believe that computers in education are revolutionary because they make possible great teaching in a system dedicated to mass education. But they make this possible by supporting person-to-person educational influence, not by replacing it.

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Some Background

To see the consequence of this last statement, let's back up a bit. It is a truism that there have been, and always will be, inspirational teachers. The know-how with which they have practiced their art has remained pretty much an untapped resource, however. The classical "pipeline" model of educational R&D (educational product shipped to end users *after* development) makes no attempt to tap this resource. How, it would be argued, could a concept as ill-defined as "great teaching" ever be encapsulated and shared in a mass education system? More to the point, how could a word like "greatness" ever characterize the efforts of the majority of teachers who are (by definition) "average" educators?

It has been tempting to respond to these questions by proposing that the art of the great educator be captured and disseminated through the medium of computer-assisted instruction (CAI). One early worker in CAI went so far as to predict that "millions of school children will have access to what Philip of Macedon's son, Alexander, enjoyed as a royal prerogative: the personal services of a tutor as well informed and responsive as Aristotle." [2] But CAI is also based on a pipeline model. The years have verified that what comes out of the pipe, even when a computer is at the other end, bears little resemblance to the charisma of an Aristotle. CAI has its virtues, but reproducing person-to-person teaching is not one of them.

There is another use of computer-related technology, however, that is triggering originality and depth in student work, as well as such vital qualities as enthusiasm and zest, particularly in the area of "math" education.¹ It is doing this, not so much by reproducing pedagogical greatness, but rather by *inducing* such greatness.² It is use of technology that works its magic by freeing teacher/student groups from the restricted format of drill in facts and formulas. Teacher and student become, instead, co-discoverers of truths. The methods they learn to use, and the results they obtain, display a freshness that suggests that they have personally discovered a secret that transcends the art of any one great teacher. Their secret, put simply, is to use computer technology to build an environment in which learning mathematics is both natural and exciting. Because their work is predicated on the belief that the student alone must (ultimately) control what is done with the technology, such use has been called "solo mode" computing. [3] Because the technology provides a world of experiences to be lived in by both teacher and student, the resulting

¹The word "math" is used here in a very extended sense. See references [3] and [4].

²In a subtle way, clever technology can also "reproduce" some of the talent of the inventor (e.g., Papert's turtle), and in this way transmit certain aspects of great teaching.

environment has been affectionately dubbed “math-land.”³ [4]

On the negative side, it is clear that environmental solo learning can easily flounder if it does not build on regular injections of disciplined study. Our experiences at Project Solo have repeatedly shown the value of intermixing “dual instruction”⁴ with “solo learning.” To use an even more evocative terminology, we believe that it is as important to support growth in repertoire as it is to encourage creativity. Works of artistry are founded on solid technique; the outcomes of education are no exception.

The problem then is to build an educational setting that supports creative solo learning while simultaneously addressing the importance of a standard repertoire. Teacher-student controlled computing provides exactly such a setting. In addition to bringing out the creative spark at many levels, such computing demands no-nonsense technique. This is because it is essentially a self-validating process. Trivial output is as obvious (and painful) as an off-key operatic aria. The prescriptions are clear in both cases: hard work to eradicate some clearly pin-pointed difficulties. It is also clear that the prescriptions have special meaning only to those who first venture into the solo spotlight.

Providing easy and enjoyable access to such on-stage experiences is what solo-mode computing is all about. But — and this is the important point — a real teacher who cares and worries is essential if the most is to be made of such experiences. It is difficult to be one’s own critic; it is impossible to be one’s own encouraging audience. Only the real-life teacher who cares enough to agonize in the wings will do. Any claim that CAI will ever fill this role must be labeled as a senseless hoax.

From a technological point of view, the future of solo-mode computing is very bright. Some recent work we have been doing with extended solo-computing environments indicates that the surface has just been scratched.⁵ But all of our experiences continue to highlight the importance of master teaching in technological settings. It is important, then, to ask how this ideal can be related to reality.

The Ideal and Reality

A realistic view of education admits that human teaching can be uneven, but it also understands that the relationships involved between students and

³The reference to “math-land” was made by Seymour Papert in a session entitled “The Computer as Super-Toy” given at SJCC, 1972.

⁴We use the word “dual” to mean a two-person process: teacher instructing real student, CAI author directing imagined student, student teacher explaining to fellow student, etc. Reference [3] discusses the significance of this terminology in greater detail.

⁵These experiments are part of a project called “Soloworks” which is concerned with developing computer-based math lab environments. This work is supported in part by NSF grant EC-38063. A project newsletter is available from the author.

teacher transcend the “logic” of even the fanciest instructional flowchart. A realistic view sees students and teachers as deserving of all the help and encouragement they can get as they wrestle with some very tough problems. They are people working with little more than their “bare hands” in a system whose priorities do not recognize the crucial importance of environments that support rather than restrain the work of teachers. There is an innate “genius” to teacher/student groups, but the flourishing of this genius is next to impossible without sophisticated backup. Jerome Bruner states the need emphatically:

... in 1965 I took a leave of absence from Harvard and taught fifth grade for three months . . . I taught every day until 1:15 and came back to my office exhausted. Like the other teachers I had to prepare most everything myself. Now I know that after you teach for a while, you can't expend energy that hard and lay your ego that constantly on the line. How can a teacher cope without some kind of supporting backup? [5]

Thus, supporting backup is the essence of reality; when it is backup that also induces unique accomplishments in teacher/student work (as does locally controlled computing), then it deserves being called a “compelling” educational phenomenon.

The Problem of Measurement

One of the important facts of life that seems to be by-passed in most educational research is the distinction between success in growth as a human being, and success in scoring high on tests. Educational evaluation doesn't attempt to look at the first kind of success on the grounds that it is too fuzzy a concept. What really should be said is that it is too complex a concept for ordinary evaluation instruments. The solution to this difficulty is to switch to a different kind of instrument, one based on the ability of experienced, practicing educators to detect extraordinary educational outcomes. If they tell us that something is working unusually well, we should listen.

Such detection is done on the basis of hundreds of clues, comparisons, and judgments, all assembled into an intricate pattern of great subtlety. It is a pattern that cannot be translated into words or numbers any more than can the emotional impact of a Bach cantata. But the phenomenon is no less real or no less important because of this.

The closest thing to a written form of evaluation that focuses on the importance of human teachers is found in the diary style of the type used by James Herndon in *The Way it Spozed to Be*. [6] Two shorter and more scientifically constructed evaluations that also include anecdotal information, giving a good

feel for the specific teacher/student relationships made possible by solo mode computing are given by Milner [7] and Critchfield [8].

Teacher Education

A consequence of viewing computing as a support system that can induce greatness in teaching is that one gets a new view of teacher education. One understands, for example, that it is very important to prepare teachers to use such supportive systems *creatively*, and to help them think about the importance of getting their students to do the same. I use the word “creatively” to suggest that teachers should be educated with the view that they are life-long researchers, not technicians who implement other people’s ideas. Another difference would be to immerse teachers in very rich technological environments as part of their education, again with the objective of making them fluent at extending (rather than just imitating) ways of using similar supporting systems.

The overall goal of teacher education should be to give teachers a strength and confidence based on expertise; to help them master the art of molding themselves and their students into very strong personal units within the large impersonal systems that arise in mass education. Since such strength is earned rather than granted, this suggests that teacher education programs ought to give much more attention to developing subject matter expertise than has been the case in the past. Simultaneously (and this is not a contradiction), teachers must learn to build on the expertise that each of their students can bring to a classroom.

Another thing that teacher support programs must do is to provide the ammunition needed in fighting “city hall” to get the right facilities. Such facilities provide more than an educational setting. They take on a very important protective function, insulating teacher/student groups from the erosive forces of non-educational concerns found in a big school system. Any teacher who has worked in a school with good departmental facilities will understand the significance of this remark.

Summary

A good way to summarize these remarks is to say that I believe computer technology has brought us the capability of making a very fundamental change in education. The change is a return to greatness in teaching. It is a change that eschews the use of technology to automate pedagogy. On the contrary, it’s a view that starts out by recognizing the extraordinary power of human relationships. It believes that teachers are important because they are the proper focus of such relationships in education. It sees technology as a

compelling force in education because it has the power to melt away the real-world constraints that manacle even the best of teachers. In a very real sense, it is a view that says we are now able to back off from the frightening dead-end of school systems with no belief in their own power to teach, and explore the possibility that greatness in teaching is a goal that is no less realizable than it is reasonable.

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8. The Fundamental Problem of Computer-Enhanced Education and Some Ideas About a Solution

Some time in the late 1940's, a small exploratory movement started to take form, mostly in the workshops and homes of individual experimenters. A sprinkling of amateurs, working at times in local groups, at times independently, began to assemble chunks of technology (much of it originally developed for professional use in the motion picture "palaces" of the day) into personal high fidelity sound systems. The results were, for those who had the opportunity to observe and listen, quite sensational. There was no need for formal evaluation; what came out of these strange looking boxes and mysterious concoctions of electronics was clearly a new, and tremendously exciting phenomenon. An underground of know-how soon spread, and within a short time even higher quality components became available for a widening group of knowledgeable users. The names of some of these early pioneers eventually became synonymous with quality in the components themselves; names like Brooks, Scott, Hartley, Williamson, Bozak, Villchur, Cook, Klipsch, Kloss, and many others.

Two observations about this phenomenon are relevant to a discussion of the future of computers in education. Although these early breakthroughs have since evolved into some of the most sophisticated technology that laymen can acquire and use, hardly anyone thinks of these complex and mysterious products of science as threatening or demeaning. They are a personally desired and appreciated technology, and therefore, to use Illich's terms, "convivial" rather than "manipulative" vis-a-vis the aspirations of people.

The second observation is that the development of the "software" for this particular technology (records and tapes) has ranged from the problem of

payola-influenced trash, to the richness of an unprecedented catalog of both contemporary and classical works. This contrast sheds much light on the difficulty (but more so the possibility) of ennobling uses of technology, and the importance of supporting diverse creative users in a potentially manipulable market. We can also see that a listing of those applications which different people would classify as "great" will always show healthy variety. It is a truism that even highly complex technology must span human cultures if it is to serve people well. As with the much earlier technology of books, greatness in the new technological media is dependent on creative use by many people with many insights, in an overall setting that fosters freedom, thoughtfulness, professionalism, and open-ended growth.

This brief excursion into the history of one recent technological development (and the generalizations about good use of technology it suggests) has been made to give some flavor of the spirit that biases our* view of the future. It is a spirit that sees great power in combining the enthusiasm and generosity of professionals, all in a dynamic setting that insists on continual improvement. It sees ways of enriching this setting by utilizing the incredible advances of certain new technologies (especially personal computers), while also contributing to this advance.

Spirit is one thing. What can be done with it is quite another. Before offering some thoughts on the challenge of converting spirit into action, it will be helpful to say more about the source of the biases behind our thinking.

Background; the Solo Saga

A correspondent recently complained (we think) that one of our newsletters repeated something we had written elsewhere. Well, at least we were consistent. Actually, there has been change, but there has also been a consistency. The consistency is derived from seven years of work pursuing the idea of solo-mode computing. The word "solo" describes a pedagogy based on the intensity of involvement and accomplishment that occurs when something is a personal quest: the learning of a student-pilot on a solo flight; the learning of a language student getting along in a foreign country; the learning of a blind person who must internalize the world in a completely unique way. Additional information about application of the solo idea to computing is given in references [1] and [2].

Our first formal investigation of this potential took place in 1969-1972 under the name Project Solo. We worked with three large public schools in Pitts-

*The plural pronoun is used to credit the invaluable thinking of the Project Solo staff. Any errors in reporting our biases are due to the biases of the author.

burgh, with the goal of enhancing standard courses through “solo” (as opposed to “dual” or CAI) use of computers. We also shared our findings with other educators through a series of 33 newsletters, through circulation of a large collection of curriculum modules, and through descriptive articles, workshops, and three 16 mm. films. The pros and cons we found to in-school solo computing are summarized in [3].

In 1972 we supplemented our experience at Project Solo by visiting other computer projects, analyzing the common elements that seemed to characterize their work and preparing a multi-media report on what we found. The appendix to [2] gives additional information.

Based on the Project Solo experience, and on our study of the work of others, a new project was initiated in 1973 under the informal name of the “Soloworks Lab” [4]. Our purpose here was to develop the components for a computer based math-lab in conjunction with working teachers and regular students, but in an environment freed of the technical and pedagogical limitations we had encountered as follows: “To develop powerfully different educational tools using artifacts and technology from the computer age in a setting defined by the possible more than the habitual.” It is this Soloworks experience, both good and problematic, that most directly influences our thinking about “What next?”

The Solo Learning Environment — What’s Right About It

There are two ways to summarize the positive features of a learning environment based on the use of solo-mode pedagogy and technology. The first is to compare the atmosphere in the “place” that results when these ideas are put into practice, with the atmosphere found in conventional classrooms. In the classroom setting, it is very difficult to escape the tradition that puts a class and its teacher both literally and figuratively on opposite sides of the room. The tensions that can result are not really the fault of one side or the other; lock-step schedules, constrained classroom settings and limited opportunities for intensive thinking about challenging problems, all conspire to let the students know the limitations of their role in conventional institutions. The problem is also not the fault of the teacher, who is required to be literally “on stage” during four, five, or six 45-minute slots per day, 180 days per year. The view that schools are “factories in which children are to be shaped into products”¹ is still an influential one.

¹This quotation and the origins of the factory model are discussed in Chapter 5 of Callahan, *Education and the Cult of Efficiency*, University of Chicago Press, 1969.

The atmosphere in a structured solo learning environment is quite different. There is still pain in hard work, there are still differences in personalities to be understood and lived with, and there are still exhausting demands on teachers and administrators. But there *is* deep thinking, and it becomes an almost continuous occurrence once the ball gets rolling. There is also a clear shift away from the tradition of opposing roles for teacher and student, and a natural movement toward a mutual zest for learning, discovery, and best of all, invention. One teacher's observation that students "want to be on these machines from midnight to midnight" sums up the situation in simple but profound terms.

The second way to describe the advantages of a solo environment is in terms of specific accomplishments.² These include such things as:

- (1) student projects in areas currently thought to be limited to higher education, e.g., the solution of coupled systems of differential equations to solve multi-body orbital problems.
- (2) student projects in areas at the outposts of knowledge, e.g., a program that does better than people in solving puzzles such as the Soma cube.
- (3) student projects that break through the disciplinary lines usually imposed on school subjects, e.g., the mathematics of art, dance, or orchestration.
- (4) the emergence and flowering of extraordinary talents in students formerly classified as "F" in school, e.g., self-study of a complex electronics course, or the design, building, and testing of several new loudspeaker configurations.
- (5) the recognition, challenging, and *correcting* (by students) of wrong or misleading statements printed in "eminent" textbooks, e.g., detection of a wrong solution to a problem in electric field theory, and the plotting of fields to show what really goes on; or the case of incomplete and misleading statements about non-Euclidean geometries in a college book that led to both theoretical and computer-based explorations in differential geometry.
- (6) perserverance with long-term problems requiring the elimination of very difficult "bugs," e.g., the extension of a comprehensive swimming meet simulation to include relay events in a real-time system.
- (7) *really* long-term projects, e.g., a two-year development of a program to grade, analyze, and 'comment' on geometry proofs (this one by a teacher).
most remarkable of all, the management of all the above by "ordinary," working high school teachers.

²It should be clear from these examples that "solo" is *not* synonymous with "unstructured" or simple "discovery" learning. A solo airplane flight is highly dependent on previous instruction, support from controllers, careful design of equipment and facilities, and many other "dual" factors; generalized solo learning needs the same level of back-up.

Solo Mode Learning — What’s Wrong with It

The main difficulty with the solo-learning philosophy is that while the basic idea possesses the virtue of compelling correctness (that is, it is an accurate encapsulation of what most people know as the basis for their own most memorable learning experiences), it is nevertheless a very complex thing to orchestrate. It is also an idea with more subtlety than one might expect. It takes a more comprehensive model than that implied by just the word “solo” if the idea is to bear real fruit.

The second difficulty is that even with the right model, the solo idea has a fragility that makes it an easy prey to the natural tendency of people, and particularly educators, to restructure things to their way of understanding. Just as it is hard for a flight instructor to resist taking over the controls from a student pilot, it is difficult for a teacher to guide learning in ways different from his own. Some form of highly stable, impartial mechanism for resisting this tendency to regress back to mostly “dual” instruction is needed.

The third difficulty is that while the preceding difficulties can be solved, and in a very impressive manner, the solution will probably be viewed by many as being too avant-garde,³ too expensive, or too esoteric.

This then leads to the following dilemma: if an idea is too complex, too subtle, or too fragile for its time, should it be scrapped until the time is right? We think that the answer is obviously no, for otherwise the “right time” will never occur.

But more than a “let’s forge ahead” response is needed. In particular, very careful attention needs to be given to another quandary. If an educational idea is complex, should it therefore be developed only by specialists, with “education” in its proper use to take place at some later date when the design is finished? We think that this is the most searching question of all. In fact, we believe that this issue and the problem of its resolution, encompasses all the other difficulties we have cited.

For this reason, we think it should be called the “fundamental problem” facing serious computers-in-education people today.

Toward a Solution

There are at least three parts to a good solution to a tough problem. First there needs to be a core-idea; a conviction founded on a gut reaction about the

³As the vice president of one publishing house told us, “Keep me informed about all that ‘kooky’ stuff you’re doing, but in the meantime, how about writing a nice, down-to-earth basic math book.” However, this is a different view from that of most teachers who write us. One correspondent who requested as many curriculum modules as possible after seeing the titles in one of our newsletters, wrote, “It’s like looking in on paradise from the outside.”

right way to go. Then there needs to be a more exact model (sometimes called a paradigm) that helps give precision and direction to this conviction. Finally, there has to be a plan of action that spells out ways to apply, evaluate, and improve the model. I'll conclude this paper by describing only the core-idea that seems to be settling out of our work with students and teachers at Solo-works.

It's an idea that is partly captured by the words "frontier" and "pioneer," together with the notion of a place that serves as a "base-camp" for the exploration of new educational worlds. The base would have strong communication and supply links to the present world of schools, and it would draw upon the expertise of members of the local community. But its primary function would be to forge new paths, feeding on the imagination and excitement of people with the courage (and opportunity) to follow their convictions and aspirations. Most of these people would not be professional researchers.

Our "value-judgement" is that teachers and students have much to offer. They really *do* share the adventurous spirit they have seen in considering the past. They are even more intrigued by the future, where both a philosophical point of view (e.g., the visions of Teilhard de Chardin), and the exercise of fantasy (e.g., the Star Trek phenomenon) dare one "to go where no person has gone before."

While the outpost "place" we propose would have elements of both imagination and daring in its makeup, it would be based on realistic prototype experiences. It is thus not a complete unknown. However the rate at which we have seen ideas multiply in our own prototype situation convinces us that there will be discoveries and outcomes which none of us can envision at present. This is why we consider outpost-based exploration influenced by practicing teachers and their students an educational need of the highest priority. It is a need as fundamental and inescapable as human nature itself, and therefore a need intrinsic to education at all levels.

Further Steps

In addition to settling on a core-idea, we have finished most of the work on the second phase of the solution strategy just outlined, namely the development of a model. We call it the 'Community of Learning' model, and will describe it in [5]. As to the third phase, namely applying, evaluating, and refining the model, that's still at the planning stage. We will certainly share any future developments in this direction with the membership of SIGCUE, and as always we invite comment.

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Further information about the solo idea can be found in these other papers by the author.

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ARTHUR LUEHRMANN

Associate Director of the Lawrence Hall of Science, a public outreach organization within the University of California's Berkeley Campus, Arthur Luehrmann was previously a prominent, long-time member of the group of computer enthusiasts at Dartmouth who revolutionized that institution's incorporation of computing into the academic mainstream. Though the application of computing to physics instruction has long been a special interest for Luehrmann, his extensive work with school computing, first at Dartmouth and more recently at LHS, has been characterized by efforts to bring computing into instruction at all levels, in all subjects of the curriculum. Luehrmann has published sparingly but has addressed numerous learned societies and has directed a number of funded projects directly designed to expand the appropriate use of computing in formal education and in informal education, such as that increasingly carried on by museums.

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9. Should the Computer Teach the Student, or Vice-Versa?

This sermon begins with a parable.

Once upon a time in the ancient past there was a nation in which writing and reading had not yet been invented. Society was as advanced as possible, considering that it had no mechanism for recording the letter of the law or of writing agreements, contracts, or debts. Nor was there a way of recording the heritage of information and knowledge that had to be passed on from generation to generation.

As a result, a great fraction of the total effort of the society was spent in oral transmission of information. Master teachers, who themselves had been taught by older master teachers, lectured before children and young people of the society. Training a master teacher was a long and expensive process, and so the society could not afford many. For reasons of economy the curriculum was quite rigid and lectures were on a fixed schedule. Teaching, obviously, was a labor-intensive industry based on skilled, expensive talent. Education, per force, was a luxury that could be afforded by the elite classes only.

Then, one day, writing and reading were invented. Not surprisingly, the first application of this new technology was to business and government. Money was printed; laws were encoded; treaties were signed. In response to these needs, a reading and writing industry grew up. Within a few years it was able to offer a broad range of reading and writing services to its customers. The customers found this to be a convenient arrangement, since hiring readers and writers from service vendors eliminated the need for each customer to invest in an expensive R & D effort of its own. The customers remained illiterate.

At first the situation was somewhat chaotic. Each vendor of reading and writing service tended to develop its own favorite language and its own technique for encoding information, leading to incompatibilities that impeded the spread of the new technology. After a winnowing-out period, however, the number of competing systems settled down to a few and major difficulties were handled by translators—though inevitably something seemed to be lost in the process.

Always looking for new markets, the vendors of reading and writing service began to examine the area of education. In view of its elitist role in the society it had been dismissed at first as too limited a market. A few, more imaginative people, however, argued that the application of reading and writing technology could turn education into a mass market. They proposed the following plan of attack. Reading and writing specialists and master teachers would work as a team. The master teachers would deliver their best, most carefully prepared lectures to the reading and writing experts, who would write them carefully *verbatim* into books. The books would then be copied many times, and each copy would be made available to a new type of educational functionary—the *reader*. His only job would be to assemble groups of students and to read aloud to them the recorded lectures of the master teachers. In view of the fact that training such a reader would be far less expensive than the education of a master teacher, the on-going cost of such a program would be far less than that of the conventional lecture method. The new method came to be called Writing Assisted Instruction, frequently abbreviated to WAI.

Needless to say, WAI had its opponents. Established master teachers expressed doubt whether a less skilled reader would be able to communicate subtleties of inflection, and they were certain that a mere reader could not process student responses with skill or intelligence. WAI proponents counter-charged that the master teachers were merely expressing their vested interest in the present educational establishment, and, indeed that they ought to be fearful because the superiority of WAI would ultimately drive out the conventional practitioners. Even within the education establishment some younger members became WAI supporters on the grounds that the new method was a boon to education research. Until then, teaching had been something of a black art, shrouded in the privacy of the classroom. To compare one teacher with another was impossible. But in the future, they said, the written record of the lectures of master teachers would make the teaching experience explicit and subject to analysis, comparison and improvement. It was high time, the young Turks exclaimed, that the teaching profession act with accountability to the public it served.

Unfortunately, such controversy remained for many years on a hypothetical plane. The number of actual WAI efforts was very small and their results were not striking. There was also a credibility problem. Many of the most outspoken advocates of WAI, especially in the legislature and in business and on

local school boards, were themselves almost totally illiterate in the new reading and writing skills. How could they evaluate a new technology if they had not mastered it themselves?

Finally, government, business and some members of the education establishment decided to mount two or three large-scale demonstrations of WAI in order to show publicly the advantages of the new educational technology. For a period of several years curriculum experts collected information on a few key courses of lectures by assorted master teachers. The reading and writing experts wrote down the best series and read them aloud to the curriculum experts, who would criticize them and make improvements. The reading and writing experts would then incorporate the improvements in the next draft. Then came the field test. Readers began to read the drafts aloud to actual classes of students, and this led to further revision by the curriculum experts and rewriting by the reading and writing experts. At the end of a few more years a summative evaluation of the projects was undertaken by an independent, reputable educational testing organization, whose mission was to compare the cost and effectiveness of WAI with conventional education.

The parable is nearing its conclusion now. Actually it has two alternate endings, one happy and one sad. The sad ending, which follows now, is brief.

The educational testing organization reported that the projects were a complete vindication of Writing Assisted Instruction. It found that students taught by WAI performed even better on standardized tests than students taught by the average master teacher, that the student liked WAI better, and that the total cost of WAI was about a fourth that of conventional instruction. These pilot projects were imitated on a grand scale and education was revolutionized. Special institutes turned out vast numbers of readers and within ten years they were reading courses of lectures aloud to masses of people who could never have been educated before the new instructional technology arrived. The nation grew and prospered and thanked the day that the reading and writing industry was founded.

That is the sad ending. The happy ending is somewhat longer and more complicated. Here it is:

The educational testing organization found that WAI was neither measurably worse than conventional instruction, nor better. It found that costs were somewhat higher than anticipated, mainly because market demand for people with reading and writing skills had driven their wages up near those of master teachers.

But this lukewarm finding was anticlimactic when it came, for the impact of reading and writing on education had taken a new turn during the intervening years. Here is how it happened.

At first a few master teachers had themselves found it necessary in pursuing their own research to spend the enormous effort required to master the skills of reading and writing. As they became more and more competent read-

ers and writers, they began to see clearly the power of the written word within their own disciplines. Naturally enough the humanists were the first to apply this new intellectual tool to their fields of interest. Literature specialists collected stories, wrote them down, exchanged them with each other and began to develop literary criticism to a new height. Language specialists compiled lists of grammatical rules, which became writing manuals. Scientists were slower in becoming literate, with mathematicians leading the way, since they grasped the possibility of writing mathematical concepts in abstract notation. Nevertheless, for many years scientists continued to remain in verbal darkness.

While reading and writing had its primary impact on scholarly research, at the same time many master teachers across the land began to wonder whether it might not be beneficial to introduce elementary uses of reading and writing to students in their courses. A few language teachers began to show students how to write phrases and sentences, and the more venturesome teachers even asked students to write sentences of their own. Such experience, they claimed, greatly enhanced a student's understanding of syntax and rules of grammar. Even in subject areas far removed from language, to which reading and writing have a natural affinity, teachers began to report pedagogical gains due to having students carry out elementary reading and writing tasks as an adjunct to conventional instruction.

One obstacle to student use of reading and writing was the awkwardness of the main systems of notation, which had been developed mainly for research and business applications. The most popular such system was particularly difficult to format, since its characters all had to be positioned accurately in a fixed number of columns. Occasionally there were rumors that a group of teachers in a remote province near the northern frontier had developed a simpler writing system and all their students were using it daily. Such rumors were hard to verify; only a few people ever voyaged that far north, and, in any case, experts in the reading and writing industry seemed confident that anything that made the current system simpler would also take away its power and elegance. So most teachers adhered to it.

Within a few years teachers began to hold national meetings to tell one another how their students used reading and writing within their courses. Advocates of this type of use, which came to be called *adjunctive*, insisted that it be distinguished clearly from WAI. Writing Assisted Instruction, they charged, was nothing more than an improvement in the technology of delivering instruction. Adjunctive use of reading and writing by the student, on the other hand, represented a change in the intellectual content of instruction. They argued from the following philosophical premise:

Reading and writing constitute a new and fundamental intellectual resource. To use that resource as a mere delivery system for instruction, but not to give a stu-

dent instruction in how he might use the resource himself, was the chief failure of the WAI effort, they said. What a loss of opportunity, they exclaimed, if the skill of reading and writing were to be harnessed for the purpose of turning out masses of students who were unable to read and write!

WAI advocates responded that it was well and good that a few elitist schools teach their students the difficult skill of reading and writing; it was enough that WAI teach lesser skills to masses that might otherwise remain uneducated and unemployable.

How much longer, asked the WAI opponents in rebuttal, will an illiterate person be considered educated? How long will he be employable and for what jobs if elitist schools are turning out competent readers and writers by the hundreds?

The more visionary advocates of mass literacy told of foreseeing the day when students would spend more hours of the day reading and writing than listening to lectures. Small research libraries had indeed sprung up at some schools, but they were expensive operations limited to a few specialists who had to raise funds to pay for their use. Such people were particularly incredulous at the suggestion that every school ought to adopt as an educational goal the establishment of a significant library open freely to all students. School administrators were at first appalled at the idea that the library should not be on a pay-as-you-go basis but should be budgeted as part of the general institutional overhead costs.

But as time went on and even school administrators became competent and imaginative users of the skill of reading and writing, all schools gradually accepted as a mission the bringing of literacy to all students. Accreditation agencies examined the quality of libraries before approving schools. Books began to appear all over and finally even in people's homes. WAI did not die out altogether, but continued as a cost-effective alternative to the lecture. But as books reduced dependence on lectures, students made less use of both WAI and lectures and spent more time on their own reading and writing projects. The nation grew and prospered and wrote poems in praise of the day that reading and writing were discovered and made available to all people.

End of parable.

It is a perilous strategy, bordering on bad taste, to tell a joke and then for several pages explain why it was supposed to be funny. However, this allegorical tale has been told here not merely for entertainment but mainly for the moral lesson it carries. To compare reading and writing with computing might be dismissed as an amusing frivolity; but that would be wrong. Our fundamental philosophical premise here is that, like reading and writing,

“[computing] constitutes a new and fundamental intellectual resource. To use that resource as a mere delivery system for instruction, but not to give a student instruction in how he might use the resource himself, has been the chief failure of

the [C]AI effort. What a loss of opportunity if the skill of [computing] were to be harnessed for the purpose of turning out masses of students who were unable to [use computing]!"

As this example shows, it is a trivial editing task to go through the entire reading and writing fable and turn it into a story about computing and its uses in education. In fairness, the author admits that the story really *is* about computing and that reverse editing was done in the original telling so that it would seem to be about reading and writing. Yet, as a story about reading and writing it has considerable plausibility, doesn't it? The Writing Assisted Instruction program outlined in the story is not a totally absurd idea for putting reading and writing to use in education. One cannot argue against claims that committing lectures to writing would make education available to more people, would invite critical comparisons and a consequent improvement in subsequent revisions of written materials, and would be an asset to the study of the learning process itself. What does appear absurd, however, is the failure of these mythical WAI proponents to recognize that the best educational use of reading and writing is the teaching of reading and writing itself to everyone. Mass literacy is an educational mission about which few of us have doubts today.

Yet that consensus among us seems to vanish when one substitutes "computing" for "reading and writing" and "CAI" for "WAI." Mass computing literacy is not an agreed-upon educational goal. Today very few courses at any educational level show students how to use computing as an intellectual tool with applications to the subject matter being taught. Oh, there are a few isolated, subject-matter-free courses in computer programming; but their market is largely restricted to vocational-education students, at one end of the spectrum, and future computer professionals at the other. It is true that most schools consider it prestigious to have a large and powerful computer facility; but the fact of the matter is that such computers are usually the captives of research and administrative interests and operate on a pay-as-you-go basis. Ironically, it is in the most prestigious universities that students are least likely to be permitted to use those prestigious computers. It is a rare secondary school, college, or university that budgets and operates its computer facility in the same way that it budgets and operates its library. (There is a persistent rumor of an exceptional example in some remote province near the northern frontier, but so few people ever travel that way that the report is hard to verify.) In the main, literacy in computing simply is not an educational goal at many schools. Most educators seem to find bizarre the suggestion that accreditation agencies examine schools for the quality of their educational computing facilities, just as they now do with libraries.

The distressing truth today is that educators, local school boards and federal policy-makers are far more receptive to the plans of CAI proponents for

using the technology of computing as a cost-effective delivery system for instruction in math or remedial English than they are to making computing itself a part of education. This statement should not be taken as a blast against CAI. On the contrary, CAI advocates are to be commended for their desire to reduce the cost of instruction, to tailor it to the different learning styles of students, to develop systems that encourage closer examination of what is being taught and systems for improving instruction, and to hold teachers and schools accountable to their clientele. With enough developmental work on CAI, it is likely that students will perceive the computer as a very superior teacher. Above all, CAI promises to make education a less labor-intensive industry and so to enable masses of people to become better educated. This is certainly a goal worth working for.

But there is a higher goal. If the computer is so powerful a resource that it can be programmed to simulate the instructional process, shouldn't we be teaching our students mastery of this powerful intellectual tool? Is it enough that a student be the subject of computer administered instruction — the end-user of a new technology? Or should his education also include learning to use the computer (1) to get information in the social sciences from a large database inquiry system, or (2) to simulate an ecological system, or (3) to solve problems by using algorithms, or (4) to acquire laboratory data and analyze it, or (5) to represent textual information for editing and analysis, or (6) to represent musical information for analysis, or (7) to create and process graphical information? These uses of computers in education cause students to become masters of computing, not merely its subjects.

It will be countered that such an educational mission is well and good for a few elitist schools, where students are willing to learn the difficult skills of computing; but it is enough that CAI teach lesser skills to masses of students that might otherwise remain uneducated and unemployable.

In response we ask, how much longer will a computer illiterate be considered educated? How long will he be employable and for what jobs if elitist schools are turning out competent computer users by the thousands?

The true story about computing and education is at its midpoint. Like the reading and writing parable, it has a sad ending. Which one actually occurs will be determined by you — teachers, school administrators, computer professionals, and government policy-makers.

10. Prepared Statement on Research, Development and Planning for Computers and the Learning Society

Introduction

The bare bones of my testimony today can be put in five sentences. I will first enumerate those sentences. Then I will amplify each one and attempt to convince you that it is factually accurate. Finally, I will point out the research, development, and planning implications that follow from these facts. Here are the statements.

1. The United States is the world leader in manufacturing microelectronic hardware.
2. That technology makes it possible today to deliver powerful personal computers to millions of individuals.
3. The biggest single impediment to the further growth of this industry is the fact that the vast majority of Americans are uneducated in the use of a computer.
4. To carry out the educational task ahead will require a substantial investment in research, development, planning, and delivery.
5. The private sector is not organized to make significant investments in the education of the public.

This was a prepared statement for Computers and the Learning Society. (Hearings before the Subcommittee on Domestic and International Scientific Planning, Analysis and Cooperation, of the Committee on Science and Technology, U.S. House of Representatives, Ninety-fifth Congress, First Session, October 4, 6, 12, 13, 18 and 27, 1977 [No. 47].) Washington: U.S. Government Printing Office, 1978. Pp. 548-569.

What Is “Education in Computer Use”?

The key idea in these statements is contained in the phrase “education in computer use.” Before going on, I want to clarify what I mean by it and to distinguish my concerns clearly from the concerns of the witnesses who appeared here during the first two days of these hearings. They were concerned mainly with the *form* of education. I will be emphasizing the *contents*.

They focussed during the first day on computer *management* of instructional resources and, the second day, on computer *delivery* of instruction. In both cases the instruction thus managed and thus delivered was in traditional subjects, such as reading and mathematics. Their focus, therefore, was on the computer as a *medium* on instruction. My focus today will be on the computer as an *object* of instruction — not a thing to be taught with, but a thing to be taught *about*. Not a *means*, but an *end*.

Proper questions for the Subcommittee to ask about the computer as a *medium* of instruction are (1) whether it is better and (2) whether it is cheaper than conventional ways of managing and delivering instruction. But the appropriate questions to ask about the computer as an *object* of study are quite different — (1) Is it more important to learn to use a computer than to learn subjects now being taught? (2) Can computer use be taught effectively to a broad spectrum of students? Negative answers to one set of questions must not prejudice the answers to the other set. The cost effectiveness of a CAI math course tells us nothing about whether children can or should be taught to write computer programs.

I am not quarrelling with CAI or CMI; both will improve instruction in traditional subjects. My point here today is that neither CAI nor CMI will teach people to use the powerful personal computers which American technology is now capable of delivering to our citizenry. While it is true that a person taking a CAI lesson will learn rudimentary typing skills and rules about interacting with a computer, he or she will *not* learn how to write interesting computer programs, or how to structure a problem for computer solutions or how to evaluate other people’s programs. When I use the phrase “education in computer use,” I am referring to the development of these latter skills and abilities — programming, structural thinking, and critical evaluation of computer applications. It is these skills that are presently lacking in the public and are needed if the public is to benefit from the personal computer.

Back to the Main Argument

Now I want to return to those five bare bones of my testimony and to flesh them out.

1. “The United States is the world leader in manufacturing microelectronic hardware.” Basic research in solid state physics in the 1950s led to the in-

vention of the transistor at Bell Laboratories. Since that time the U.S. has led the world in the technology of miniaturizing electronic circuits on silicon crystals. A 1977 study by Arthur D. Little, Inc., forecasts a \$4 billion U.S. microelectronic industry in 1980, growing to \$10 billion in 1985. This growth is occurring at a time when other "native industries," such as automobile manufacture, are losing ground against world competition.

2. "Microelectronic technology makes it possible today to deliver powerful personal computers to millions of individuals." Already two companies with national distribution are marketing \$600 personal computers complete with TV screens, keyboards, and magnetic tape units for recording programs and data. These are the Tandy TRS-80, marketed through Radio Shack, and the Commodore P.E.T., marketed by independent stores. Vastly more powerful machines, now available for several thousand dollars, are expected to follow the steep cost-decline curve well known in the microelectronic field. In a few years they too will be about as costly as a color TV set. Texas Instruments, Timex, and Sears are reported to be entering the field. *Time* magazine (September 5, 1977) reports a Dallas microcomputer retailer as saying, "Some day soon every home will have a computer. It will be as standard as a toilet."
3. "The biggest single impediment to the further growth of this industry is the fact that the vast majority of Americans are uneducated in the use of a computer." The technology, in short, is far ahead of our ability to make use of it. The same *Time* article points out that "even though prices have dropped, microcomputers remain complicated devices that require long hours of study to use properly."

There is a striking analogy between the situation today with regard to computers and the situation in Germany in 1492 when the Gutenberg printing press was developed. A new technology had vastly reduced the cost of delivering books directly to masses of people. The people, however, were illiterate. To them the book was also a "complicated device that requires long hours of study to use properly." Computer technology today is entering a world populated by computer illiterates — those who have not yet learned to use a computer. Our Dallas dealer, therefore, should restrain his enthusiasm. Most of us have already mastered the flush toilet. Not so, the computer.

4. "To carry out the educational task ahead will require a substantial investment in research, development, planning and delivery." Preliminary R&D projects, such as the ones reported on today by Drs. Kay, Papert, and Brown, give strong evidence that a knowledge of computer use can be a powerful aid to the intellectual development of children. My twelve-year experience at Dartmouth College and now at the Lawrence Hall of Sci-

ence convinces me that many if not most people enjoy learning to use a computer and that they succeed at it. More important, I have observed that a student who is able to structure a problem for solution by computer nearly always has a better understanding of the problem than he or she got from a purely verbal or purely mathematical description. In a very real sense, there is an intellectual dimension to computer problem solving. Some research projects report that computer users develop what amounts to a new language for expressing their ideas. I will not elaborate this point because the testimony you have already heard today has dealt almost exclusively with research into the educational value of learning to use a computer.

A great deal of additional work needs to be done before mass computer literacy is achieved, however. Several fundamental research questions need to be answered. For example, what sorts of logical structures should be contained in the computer languages we teach? What grade level is best for classroom instruction in computer use? How should emphasis be balanced between teaching the grammatical rules of programming and teaching meaningful problem solving by computer?

While that research is in progress, several pilot courses in computer use need to be developed, tested, evaluated, and improved. Teachers need to be trained, or retrained, to conduct the courses.

As computer courses become viable and are recognized as valuable additions to the curriculum, the schools must engage in planning activities. The schools must find ways and means of introducing the new courses, and of supporting their continuing costs in the future.

If these research, development, and planning activities do not occur, then the majority of the public will remain illiterate in computer use. Furthermore, those who do become skilled users are likely to come from affluent families who can afford a home computer and can teach themselves to use it.

The analogy with books comes to mind again. The people who benefited most from the printing press were those who were already educationally advantaged. Mass literacy was not achieved until public education entered the scene and developed a reading and writing curriculum for everyone.

5. "The private sector is not organized to make significant investments in the education of the public." While it is true that manufacturers of personal computers have much to gain from mass computer literacy, it does not follow that they will invest in education. This is true because the individual manufacturer would thereby benefit its competitor as much as itself. The analogy with books comes to mind again. Why doesn't the Random House publishing company invest in teaching people to read? Because the people

might well go out and read Scribner's and Doubleday books. And so reading is taught in schools. For the same reason, computer use must be taught in schools.

Summary

In summary, then, it seems clear to me that we have both good news and bad news about personal computing. The good news is that U.S. technology is ready today to deliver powerful personal computers to our citizenry. The bad news is that our citizenry is illiterate in the use of computers. Unless the public is educated in computer use, two bad things will happen: (1) a leading U.S. industry will be denied a mass market, and (2) the public will be denied the personal and occupational benefits of knowing how to use a computer.

Recommendations

1. Mass computer literacy should be identified as a goal of our educational system.
2. Although the ongoing costs of such an educational program must be borne by the states and the local schools, federal funds are needed for research and development activities required to initiate the program.
3. Among the topics for which research funds are needed are these:
 - a. The role of computer education in the cognitive development of children.
 - b. Types of logical structures needed in the computer languages which are to be taught.
 - c. The role of computer graphics in improving communication between man and machine.
 - d. The age level at which computer use can be taught most effectively.
 - e. A set of performance objectives and corresponding achievement tests for education in computer use.
4. Funds are required to develop, test, and improve several model courses in computer use. The usual arguments against national curriculum development are inoperative here. If federal funds are *not* available for curriculum developments in computer education, then there will be *no* curriculum in that subject. That is obviously not the case for existing subjects, such as chemistry and biology. The stigma of a federally *imposed* computer curriculum can and should be avoided, however, by supporting several independent development projects. Local schools would then have a range of choices.

11. Pre- and Post-College Computer Education

Introduction

The title and focus of this conference are directed at the role of computing in colleges and universities today and in the future, a role which has been defined and secured by such imaginative and forceful leaders as Gerry Weeg, whom we remember and commemorate today in Iowa City. The other speakers gathered here will attend specifically to the topic, sharing with us their visions of the future of computing in small colleges and research universities. But that future will be shaped in no small part by the extent to which computers become part of the educational landscape in elementary and secondary schools, even in homes. Furthermore, there will exist for several generations a horde of older people who, having failed to learn about computers during their school years, discover as post-graduates that their computer illiteracy is beginning to show.

It may be fitting, therefore, to attend for a while to these two educational environments, the pre- and post-college situations. What computer experiences are students in the future likely to have before they reach college? What educational needs are likely to exist for adult computer illiterates in coming decades? The principal questions that will serve to organize this discussion are these:

1. What computer skills and experiences will kids be picking up before they get to college?
2. What impact will such computer literacy have on colleges?
3. What needs to be done for computer-illiterate adults?
4. What about CAI?

Technology Trends

A great deal of what we all have to say here today is based upon our perceptions of the present and future of computer technology. A brief review is in order.

The underlying phenomenon upon which the so-called "computer revolution" rides is the long-term steep decline in the difficulty (i.e., price) of fabricating larger and larger numbers of circuit elements in smaller and smaller amounts of space and time. This trend, which has steadily held at a 30% per year decrease in the price per element, is independent of the particular technology that has been the favorite (i.e., cheapest) at any given moment. Industry projections into the future, based on *known* technologies, claim that the 30% per year decline will continue for at least another decade and perhaps two decades. If so, today's prices for computer electronics will be half as much in two years, 1/35 as much in 10 years, and 1/1250 as much in 20 years. Here is another way of expressing the trend: if one agrees that "everyone" can afford to spend \$20 for a computer, then in 10 years everyone can have a computer similar to today's Commodore PET or Tandy TRS-80. In 20 years, everyone can have a machine like your favorite \$25,000 minicomputer for the same \$20.

These remarkable price declines, it should be stressed, apply only to the microelectronic circuits in a computer. Communication costs may decline in the future, but at a far slower rate. Electromechanical systems, such as printers and disc drives, probably will not undergo any significant price change. An obvious consequence of these price trends is the undoing of the old economic argument for connecting many cheap terminals to one expensive processor and giving each user a small portion of the expensive memory. Today a fairly capable personal computer costs less than a dumb terminal. Now and in the future, the components that are so costly that they need to be shared are the electromechanical devices — high quality printers, discs, and plotters. Economics dictates that what we used to call "peripherals" will be centralized, and the "central computer" will be peripheralized. This "inverted timesharing system" will be held together not by the need to share expensive computer time, but rather to share expensive information in a commonly accessible file system.

Today's stand-alone personal computers are the 4-function electronic calculators of 10 years ago — cheaper than anything else with the same capabilities, yet too expensive and too limited. They are more responsive than most timesharing terminals, yet too slow for number crunching. An 8K byte memory is enormous, yet too small. Their BASICs are better than one has a right to expect, yet not good enough, not standard, and, and after all, just BASIC. Why not a color display? Floppy discs? Better graphics? Music? Speech?

Indeed, all these facilities are in various stages of appearing. In the next year or two, 8-bit processors will give way to faster 16-bit chips. Memory sizes will rapidly go to a 32K byte "standard". Better operating systems, editors, and language processors will become available. Yet prices will remain in the \$500 to \$1000 range as performance variables improve.

What will people do with this richness of computer facility? The editor of a 1977 compendium of articles published in *Science* magazine on electronics says, "It is clear that the capability of some electronic devices . . . is developing more rapidly than applications can be conceived of." General Motors offers as an \$875 Cadillac option the "Trip Computer", an onboard computer with interfaces to the speed, mileage, voltage, engine temperature and fuel supply sensors, and with canned programs to display dynamic information. Mattel and Parker Brothers build computers into strategic games. Atari and Bally build them into kinetic games. When the computer becomes a \$20 part, who would buy a TV set without one? Who would buy an electric typewriter without a computer-based editor and several pages of memory? Who would buy a home thermostat that couldn't be set by telephone? Who would buy a telephone that couldn't redial when the busy signal is reached, or forward calls to another phone?

Popular literature these days is full of such examples of the imminent computer invasion. *Time* magazine, September 5, 1977, quotes a Dallas micro-computer retailer as saying, "Some day soon every home will have a computer. It will be as standard as a toilet." A *Wall Street Journal* article in April, 1978, quotes an industry estimate that the 1982 U.S. market for personal computers will be \$1.8 billion, a five-fold increase from 1978.

The dark cloud in this rosy view of the future will bring us back to our main theme here. As the *Time* article points out, personal computers "remain complicated devices that require long hours of study to use properly." Computer education is essential if the public is to benefit from the computer revolution. Our enthusiastic Dallas retailer should take note of the fact that one can master the use of the flush toilet just a wee bit faster than one can learn to exploit a computer. How will mass computer education take place?

Pre-College Computer Education

The fact that most of the larger, better known computer education projects have begun in colleges and universities can be misleading. People who already believe that computer use is an arcane skill accessible only to a few specialists with strong mathematics backgrounds are likely to seize on this circumstance as evidence that the college curriculum is the proper place for people to learn to use computers.

Yet the circumstance is an accident of history. Colleges and universities acquired computers at first because of the research needs of the faculty. After the machines were in place, some faculty members found that they could teach their subject matter better if their students were able to use the computer to solve problems or to experiment with simulations or to process data from realistic data bases. Being good faculty types, they communicated with colleagues, wrote papers, and attended topical conferences. We owe the inauguration here at the University of Iowa for the first Conference on Computers in Undergraduate Curricula (CCUC) back in 1970 to Gerry Weeg. The CCUC Proceedings, published every year shortly after the annual conference is held, give us a history of faculty at the college and university level who are interested in computer uses in their teaching.

A critical look at the topics and concepts presented in CCUC proceedings, however, reveals that a considerable body of this new methodology can be presented successfully to high school, or even junior high school, students. A case in point that I know something about is the application of a computer-based approach to teaching Newtonian mechanics. At the college level, this topic is ordinarily dressed up in the formal language of the calculus, and the fundamental theorem, Newton's Law of Motion, is presented as a differential equation. All physics teachers know that few students are comfortable with this representation, and many students never see past the mathematical dressing far enough to develop an understanding of the dynamic significance of the Law of Motion, which is supposed to predict the motions of all bodies in the universe.

Back in the 1960s, several people — Alfred Bork, John Robson, a few of us at Dartmouth, and others — began to experiment with a new mathematical representation of this body of physics. We presented Newton's Law as a finite algorithm for predicting the future position and velocity of a body from a knowledge of its present position and velocity, plus force law which could be an arbitrary function of the present position and velocity. The computer merely exercised this iterative algorithm. Several good things resulted. First, the predictive nature of the law as made manifest. Second, the method was completely general and worked with *any* force law — not just the few that permit the differential equation to be solved. Third, the only mathematical background needed was elementary high school algebra. In short, less mathematics was needed to derive a much deeper physical insight.

Now, let us suppose that every high school is equipped with enough computer power to support the use of the above approach in its high school physics course. That being the case, what is the appropriate subject matter for such a high school physics course?

The answer, I believe, is evident. Newton's Law of Motion becomes mathematically accessible to high school students, and there is no need to delay

until college the proper treatment of so fundamental a concept. High school students should be taught to represent it as a computer algorithm and should conduct numerous computer-based experiments with different laws of force, different numbers of spatial dimensions, and different numbers of interacting particles. They should emerge with a richer experience and a deeper understanding of the dynamic nature of Newtonian mechanics than most college students get today. At the same time, the mathematical difficulty should seem to be a great deal less than for some of the topics now being taught in high school physics.

New educational opportunities within existing subjects, then, will be a strong driving force for introducing computers into the secondary school curriculum. Even today the process is well under way. A 1978 Minnesota Educational Computer Consortium survey of all secondary school math, science, and business teachers in the state showed that one third of the respondents were using the computer in some of their classes.

The next step will be the creation of new courses specifically to teach ordinary students, not just the top 5-10% who take math and science, how to use computers for a variety of tasks, both numeric and non-numeric. These "basic computing skills" courses will teach about algorithms, writing and debugging programs, running simulation models, gaming, editing text, and drawing graphs and pictures. Unlike previous courses in which one mainly reads about computers, these new ones will require many hours of hands-on experience with computers — perhaps two to three hours per week per student. Cheap personal computers make this type of course as practical as a typing course.

Such new courses will require teacher education and curriculum development. In-service training of about 30,000 teachers is needed to put one full-time computer teacher in each secondary school. Curriculum development projects are needed to produce materials the teachers and students will use. Both teacher training and curriculum development are appropriate activities for federal support, since the need for basic computing skills is a national one and no other organization will undertake the task. The local community, on the other hand, can be expected to provide ongoing support for computer education, if only because of the growing vocational demand for people with basic skills in computing.

Impact on Colleges

If the above scenario is correct, then we may look forward to steadily increasing computer literacy among high school graduates and college matriculants during the next decade. What impact will this new knowledge have on colleges and universities?

In the first place, students coming to college will expect to find educational computing facilities at least as good as the ones they left behind at high school—or even at home! Many students today are disappointed to find poorer facilities in college, and there is evidence that the quality and accessibility of the campus computer environment is a significant factor for some students in choosing a college. Prestigious research-oriented institutions often are the losers in this contest, since their above-average computer facilities are usually off-limits to undergraduates.

The college curriculum must adapt also. As indicated earlier, many curricular uses of computers identified in CCUC Proceedings will have gravitated into the high school curriculum. Programming syntax and introductory computer science topics will certainly have been learned there. Consequently, colleges and universities must guard against vain repetition and be prepared to present advanced topics to freshmen.

Finally, a great deal of the methodology taught in college-level courses needs to be updated to include or, in a few cases, be replaced by computer-based methods. Digital instrumentation and computer interfacing techniques need to be taught in laboratory science courses. Computer modeling has to be included among other problem-solving methods in applied mathematics and science courses. Modern data retrieval, analysis, and display methods have to be included in social science courses. Linguists need to know how to process text for syntactic and stylistic analysis. Artists and musicians need to know how to make creative use of computers.

Fortunately, as is usually the case with curricular change, each generation of better educated college entrants will carry the message that they already know some of the things we are teaching and that perhaps we should get on to more interesting topics.

Computer-Illiterate Adults

The suddenness with which computer literacy seems likely to be added to the three R's in the school curriculum means that millions of mature well-placed adults face a life of computer illiteracy. No great calamity, one might say; after all, they've done well enough so far. Nevertheless, certain stresses are predictable.

As institutions come to depend more and more on computer-based information systems, a new break of talent is needed. Who will supervise the computer experts? How effective will the supervision and leadership be, if those at the top really don't understand what the computer people do? This is a problem today and will get worse for a long time.

Office automation is just beginning today. The skills needed to perform

clerical and typing functions in such an environment are very different from the conventional ones. An office worker needs to be able to work interactively with conversational machines, instruct them to carry out specified tasks, store and retrieve text files, and run diagnostic procedures. What will happen to the job mobility of an older person lacking these skills?

More and more, society will depend upon abilities to program computers, interact with conversational programs, diagnose software and hardware problems, and make changes and repairs to each. Those who have these skills will be in great demand. Others will have a powerful incentive to acquire them.

Finally, many adult computer illiterates who are not threatened by job inadequacy or insecurity will be interested in learning about computers out of natural curiosity, the desire for self-improvement, or a wish to stay current. Such people will bring a computer home just to learn about it.

All of these adults, taken collectively, represent a vast number of people with a need for education of a rather specific kind. During a period of declining enrollments, this ought to be welcome news indeed. The problem, however, is that our institutions of formal education have been notably unsuccessful in serving adults. Unless they develop new approaches, calendars, schedules, and organizational entities suited to this clientele, it will go elsewhere to satisfy its needs, perhaps to proprietary schools, libraries, or science museums.

What about CAI?

I have not thus far painted a bright future for computer assisted instruction, either in computer skills or other subjects. In part, this is because, in my judgement, the greatest educational need regarding computers is a knowledge of their *use*. This will be the case whether or not computers can deliver instruction in *other* subjects more effectively or at a lower cost.

Though it is off the main point of my talk, I venture to guess that CAI will *not* be a great success within the school system but that it may have much to offer in the non-school environment. CAI is basically incompatible with the school system. Since it offers to do what teachers do now, its adoption would mean the dismissal of teachers. Hooray! say some CAI advocates, since teachers are expensive; CAI, they argue, will lead to a productivity revolution in education similar to that in agriculture and industry. And this would indeed be the case if the laws of economics alone were to dictate the outcome. In that case, inefficient schools would produce a costly product which few would buy. Instead they would take their schooling business to a less costly school using labor-saving methods.

Schools are *not* like that. They are social and political entities, more than

economic ones. Parents lack realistic means of choosing cheaper schools by shopping around on a year-to-year basis. The fundamental market mechanism is absent.

On the other hand, non-school environments (such as the library, museum, and home) are well suited to the use of stand-alone teaching systems, and CAI is likely to find good uses there. No teachers will be threatened, and consumers will shop for the best buys. CAI developers, furthermore, will find there for the first time the mass market needed to justify the large capital investment required to create CAI materials.

Summary

Educational computing got its start in colleges and universities like the one we are meeting at today. The essential ingredients were an imaginative faculty and an accessible computer system. Today (and to a greater extent in the future) inexpensive microcomputers are bringing access to everyone, both inside and outside universities. Many of the curricular innovations that come from universities will gravitate naturally into the high schools, where a new breed of computer teacher will conduct hands-on classes in basic computing skills. College and university matriculants, therefore, will arrive with a solid understanding of the fundamentals, and they will expect to learn more advanced methods and applications. They will also expect to have even greater access to computers than they had in high school. As for the adults left behind in computer illiteracy, new organizations and new methods of instruction will be needed to serve their needs.

12. Technology in Science Education

The potential educational benefits of new information technologies (e.g., television, broad-band communication, and especially the computer) have been widely disseminated in both popular and professional publications. There has also been a recent history of well-funded, well-publicized demonstration projects, a few of which provide evidence that some technologies are cost effective. Yet U.S. education continues, as it has for decades, largely unaffected by these technologies.

The following analysis suggests that the present low level of use of technology in education is the result of fundamental incompatibilities between the U.S. system of education and most of the technology systems intended to perform educational functions. With few exceptions no one has found the way to make such technology part of the educational system. Improvements in cost effectiveness will not change that situation.

For the same reason, active promotion of educational technology systems, especially by Federal agencies, is viewed by the present educational system as an imposition and a threat. The Carnegie Commission report on educational technology is aptly titled *The Fourth Revolution: Instructional Technology in Higher Education*. The best known and best funded demonstrations of these technology systems require revolutionary changes in the educational system. The existing system, therefore, must be expected to reject them.

Yet not all technology applications are threatening to our educational system. This paper will be largely directed toward identifying specific places within the existing educational system where some information technologies, especially the computer, are expected to be welcome. These uses of "appropri-

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ate technology" have progressed so far on a small scale without much publicity or systematic support. Because they are compatible with the educational system, these relatively neglected applications ought to be given high priority on a Federal research and development schedule during the next decade.

The Nature of U.S. Educational Systems

We can divide the entire system into "school" and "nonschool" systems. The U.S. school system is, by consensus, a very highly decentralized system. Schools depend largely upon local funding and are largely subject to local control. In some states curriculum standards may be established at the state level, but in no case is there a Federally mandated curriculum or defined instructional methodology. At vocational, college, and professional levels, the institutions, while larger than pre-college institutions, are even more independent in defining curriculum and methodology.

The nonschool system operates less visibly than the school system, but must be included in this analysis for two reasons. First, it is structurally different from the school system and may be more compatible with information technology systems that are antagonistic to the school system. Second, as people have more leisure time and retire earlier, the nonschool learning environment will be called upon for increased services in such areas as retraining.

The major nonschool environments include the library, museum, job, and home. As places where learning occurs, they are characterized by a relative lack of formal structure, of testing and evaluation, and of methods of providing credit for work done. While it is surely true that more learning occurs outside school than within, it does not result in a transcript of grades or a graduate certificate.

The Federal role in dealing with the school system is well circumscribed. No Federal agency can dictate how or what is taught, but it can undertake programs perceived to be of collective benefit to everyone. Strong pressure exists simply to allocate funds to state and local education agencies for distribution.

The Federal government can respond to national problems, such as the decline in basic skills and the problems of the educationally disadvantaged. It can undertake short-term, "pump-priming" projects, or can sponsor curriculum development where none exists. And it can support research aimed at overcoming the problems of teaching and learning.

The Federal role with regard to nonschool systems may be somewhat less constrained, partly because those systems deal mainly with adults in a non-captive situation. Nevertheless, problems do arise. A nonschool education project must result in a product to be distributed to the intended public. With some important exceptions, public television programs for example, the prod-

ucts are distributed via conventional marketing mechanisms. Ownership problems sometimes arise in these cases when Federal funds have been involved. The more successful the project, the greater the likelihood of Federal embarrassment.

Present Status and Needs of Educational Systems

Perhaps the most salient fact about school systems now and for the next fifteen years, is that enrollments are declining. Schools have more rooms, teachers, and administrators than they need. These years will hardly be the best period to introduce labor-saving methods into schools.

One consequence of the disappearance of openings for teachers is that educational change within schools must be accomplished by inservice training of existing professionals and not by putting new courses into the teacher preparation curriculum. While school populations are declining, there is growing evidence that schools are doing a poorer job of teaching basic skills. They appear to need help in this regard. Furthermore, as they attempt to educate children with a wider range of prior skills, abilities, and attitudes, schools need help in tailoring instruction and testing to the educational needs and development level of each child. In addition the adult population will have restraining needs and what might be called "recreational learning" needs. As voluntary and forced mid-life career changes occur with greater frequency, people will need additional education. If nonschool environments are used, then new instructional systems will be needed, as well as new organizations.

Greater leisure time and early retirement will generate a demand for recreational learning—learning for the satisfaction of discovery or accomplishment. And for most people, the nonschool setting will be most comfortable.

Appropriate Information Technologies

The technologies which offer the most educational promise are the computer, television, broad-band communication, and combinations of these. The computer is unique among information technologies in that it permits intelligent interaction with the learner. All previous information technologies have been one-way paths for distributing facts and ideas. This is true of printed books, audio recordings, and broadcast or recorded television. The computer alone permits the learner, at his own pace, to interact either with his own efforts — to construct a correct algorithmic representation of a problem — or with a teacher's programmed dialogue. The computer is also unusual in that an individual can learn to use it constructively and creatively as an aid to conceptualizing and solving problems. When a person writes a computer program he is expressing an idea and when he runs the program he is testing the idea

against an expected outcome. Assessment and critical facility are required in testing logic. On a small scale, debugging a computer program is similar to intellectual processes used by researchers in any field. Writing for computers has intellectual dimensions in the same sense that writing essays or solving mathematics problems does.

Until recently, nearly all educational computer uses have been carried out on time sharing systems with per-terminal capital costs of \$10,000 to \$20,000. Personal microcomputers with similar capabilities cost \$1,000 to \$2,000 today and will undoubtedly be the preferred technology in the future.

The educational role of broadcast television is well established. It is a one-way medium with excellent cost effectiveness if the product is distributable to a mass audience. On the other hand, high production costs prevent quality programming for specialized audiences. Videotape recording allows "reviewability" and storage, eliminating the need to reach a large audience at one time. Instead small and specialized groups can be reached efficiently and at the discretion of the communicator. In the end, however, there must be a large audience.

Videodisc recording is more promising than videotape, because reproduction costs are lower, about the same as for reproducing audio records of the same length. It also has the remarkable property of allowing encoding with computer programs and digital data, as well as encoding with pictures and sound. The information on the disc may be accessed in random sequence and read without mechanical wear. In sum, videodiscs have the potential for being a medium for storage and retrieval of complete computer-based educational materials with pictures and sound. Such "intelligent videodiscs" could be used with a personal computer and ordinary TV set as a low-cost learning system in homes, libraries, museums, and schools.

Broad-band cable systems have been in use in communities and schools for television and other types of communication. To date, however, the educational effectiveness has been marginal. Nearly all present uses of cable television can be handled more effectively by videodisc distribution, but the spread of personal computers in schools and homes may result in new uses for cable systems, such as the development of information networks.

Direct reception of television signals from satellites may be important during the next 10 years, since it will increase the number of channels available at home, allowing a greater diversity of programming.

Applications of Technologies to Present Needs

Several well-known Computer-Assisted Instruction (CAI) projects have demonstrated that computers can be programmed for effective teaching. When dealing with basic skills and special education problems on a remedial

level, computers have demonstrated effectiveness at lower costs than traditional methods. Computer use is one component of education cost that goes down each year, and so it is a promising technology for application to many areas of learning. The vast majority of experience with computers in education, however, has not come from such CAI projects.

Instead, and with very little systematic research or development, the computer is gradually becoming a tool that students use as an aid in understanding a wide variety of traditional subjects. In addition, the computer can be used to solve problems, create simulations and models, and to search for data. Education in these uses of computers represents a new and very important need.

CAI, then, appears to address the need for improved effectiveness in teaching basic skills. In addition, CAI naturally lends itself to individualizing and self-pacing, while at the same time being able to carry out the testing and management tasks associated with self-pacing. Up to now, CAI course material has been developed for the school setting, but the advent of the personal computer will allow CAI materials to be used in many nonschool environments to satisfy some of the needs identified there. Videodisc publication of CAI materials promises the distribution of a very broad range of structured learning material to nonschool environments, although evaluation and accreditation may remain a problem.

In summary, then, broadcast and cable television will have no better role to play in the next 10 years than they have today. Videodisc publication of motion picture material, however, offers a revolutionary opportunity to distribute highly specialized material, including instructional units, directly to individuals, bypassing all institutional middlemen.

It is difficult to foresee how satellite communication directly to rooftop antennas will serve educational needs better than videodisc publication of the same material. Better knowledge of costs is needed.

New Educational Needs Resulting from New Technologies

The new information technologies, especially the computer, have generated entirely new educational needs of great importance requiring a high priority research and development effort. Past large-scale projects have been concerned almost entirely with teaching *existing* subject matters more effectively and at lower cost. The technology has been viewed merely as a delivery system, and nothing more.

But in the case of the computer, the technology itself contains an intellectual dimension, which has been largely overlooked in major CAI projects. As the computer becomes a part of the home, school, and business landscape,

people will need to know how to make intelligent, productive, and creative use of it. There is little doubt that ability to use computers will be judged a basic skill in the near future. Yet there does not now exist a recognized curriculum for teaching computer skills within the school system, or of measuring competency.

These new technology-generated educational needs include: basic education, vocational and professional education, and teacher training.

The basic educational needs are for *awareness* and for *competent use*. In a society which increasingly depends on complex automatic information handling systems, it is essential to the democratic principle that the public be aware of the availability, uses and abuses of information and machines that manipulate information. The extent that this "awareness need" is addressed in the school curriculum today—and it rarely is—has resulted in courses that require students merely to read about computers and data networks, without ever experiencing a computer system first hand. The situation is closely analogous to using the oral tradition to teach people about reading and writing. The only acceptable reason for teaching computer awareness courses without computers has been the cost of equipment. Availability of personal computers has now changed that situation, and awareness courses need to be developed and tested.

More important to basic education than computer awareness, however, is skilled computer use. Consider a person who is able to express understanding of an idea or problem in the form of a prescribed set of instructions, or an algorithm, and is able to write the algorithm as a computer program and run it interactively on the computer. Such a person has a new intellectual tool of basic importance to virtually every subject matter now being taught. Such a person also has a distinct educational advantage over one who does not know how to use a computer in this constructive, analytic mode. Unless the school system provides basic computer skills, the children of affluent families will be the ones who have access to home computers and acquire skills that increase their educational advantage.

Schools, therefore, have need of a basic curriculum in computer use. What exists today is a few isolated instances of unevaluated material, usually in mathematics, designed under the constraint of one or two computer terminals for an entire class, or even an entire school. What is needed is a curriculum for a class in which each student works alone at a computer for about half of the class hours devoted to the course. No curriculum now exists for a "laboratory" course with that amount of "hands-on" experience.

It is also essential that basic computing skills not be identified exclusively with math and science, as they usually are. Experience with students adverse to math has shown that they enjoy learning to use a computer and do so effectively when problems are chosen from other fields, graphics and text processing, for example.

Several curriculum development projects are needed in order to explore alternative ways to teach computer skills to different students at different grade and developmental levels. The courses would aim not merely to teach the syntax of a common computer language, but mainly to show students how to perform meaningful tasks, thereby improving their analytic and problem-solving skills. Students would learn to structure problems in a logical form, to express ideas as algorithms, to stimulate real systems as computer models, to process text, to construct graphs, and to search data bases among other skills.

Beyond a need for basic computer skills there is a need for vocational and professional training, especially in the field of computer hardware and software. People who can program computers command higher starting salaries within fields with open career paths. A person who knows how to interact with computerized word-processing systems is in high demand now, and that demand will increase along with the growth of office automation.

Present vocational courses in data processing often teach outmoded skills, such as the use of punch-card equipment. Curriculum is needed that is appropriate to the technology expected in the next several years.

Incompatibilities Between Education Systems and Technology Systems

As stated, information technology systems have failed often in education because they are not compatible with existing educational systems. Most information technology systems, for example, require very large initial investment in hardware, software, and courseware. This investment can only be recovered if there is widespread adoption of the result. Schools, on the other hand, are small, decentralized, diverse, and independent organizations accustomed to "cottage industry" production and development methods. They do not buy complete instructional systems: they hire teachers. Teachers are threatened by labor-saving instructional systems.

A new type of organization would be needed to make use of such technology systems and the schools themselves would be threatened by such organizations. If there were a free market in providing basic educational services, economics alone would determine the outcome, as it did in the industrial and agricultural revolutions. But schools are public institutions and part of the political process. Economic factors alone will not bring about an educational productivity revolution.

Nonschool systems—the library, museum, home, job—do not have the same characteristics as the school. They should be more receptive of the benefits of technology-based educational systems. Individuals in their homes, for example, represent a mass market. However, if significant amounts of formal learning are to take place outside schools, then the problems identified earlier,

i.e., pacing, testing, giving credit and degrees, must be solved, probably by new organizations.

Some Opportunities for Compatible Systems

Not all educational uses of technology are incompatible with or threatening to the school system. In fact, all of the new educational needs discussed at length earlier are entirely compatible. Satisfying those needs, therefore, is likely to lead to successful adoption of research and development products.

The new courses in awareness and in competent use of computers, for example, would be taught in ordinary schools, by ordinary teachers, meeting in ordinary rooms, using the ordinary class schedule. The only new elements would be the curriculum, the presence of computers, and the teacher's new abilities. All these elements serve the present interests of the schools.

In *limited* domains, CAI and CMI systems also support school programs. Remedial work is unattractive to most teachers, for example, and CAI systems may be perceived as helpful there, as well as in serving students with special educational needs. Computer management systems may be seen as helpful in supporting individualized self-paced learning and testing.

The nonschool system also provides new opportunities for education through technology systems. Museums and libraries are suitable organizations for offering the public access to information and instruction. Computers can function as "information exhibits," or can be used to guide visitors through a participatory exhibit.

The home also provides opportunities to learn about information technologies. The existence of the home computer will generate a need to know how to make good use of a computer. Videodisc publication permits television and/or CAI courseware to be distributed directly to individuals. New organizations will be needed to produce these materials and to give credit for work done.

Research and Development Priorities

- *Basic computer skills curricula.* For all of the reasons stated, development of a curriculum for teaching basic computer skills within the school system should have top priority on an R&D agenda for technology and education. It is entirely appropriate for Federal funding because: no other such curriculum exists today; no other organization has sufficient incentive to develop one until a market has been demonstrated; as a demonstration, this curriculum development project is an instance of "pump-priming"; and the need for basic computer skills is national. Several development projects should be funded so that alternative models aimed at different age levels, using different content and

methodology, can be developed, evaluated, and offered to schools to choose from.

- *Teacher training.* In-service training of teachers is needed to assure adequate staffing of the computer skills courses outlined above. Teachers of other courses will also need specialized "subject matter" training to prepare them to apply students' computer skills to learning.
- *Curriculum development centers.* A critical mass of talented people with experience in teaching students to use computers in a variety of disciplines should be brought together. It is essential that the group be together long enough to work effectively at producing a curriculum in basic computer skills and in the application of computer-based methods to other academic fields. Such a center, to retain its vitality and credibility, would have to deal with actual student populations on a daily basis and would have to employ mechanisms, such as visiting appointments, that assure a steady influx of creative individuals into the development center and return to their teaching environments.
- *Community learning environments.* Libraries and science museums have served as community learning environments for many years. They are organizationally well suited to providing informal education to a vast public. A serious effort should be made to expand and intensify their educational role through publicly available computer-based learning systems.
- *Development of "intelligent videodisc" personal learning systems.* Work needs to be done on engineering problems, authoring systems, record keeping and testing problems, and distribution mechanisms. A demonstration of a working, intelligent videodisc system should precede development of any major courseware activities.



SEYMOUR PAPERT

Seymour Papert is widely known in computing circles for his work relating computing to the education of young children, particularly in mathematics. Long a member of the M.I.T. faculty, he is currently Cecil and Ida Green Professor of Education in the Division for Study and Research in Education, where much of his time is spent on developing child-appropriate, LOGO-based computing in the primary school. The director of a series of NSF-funded projects at M.I.T. based on such computing, Papert has published numerous articles, and has addressed many learned societies in support of his concepts of computer use in education. Though his early background and numerous mathematical publications reveal his origin as a mathematician, he is also interested in the learning activities of young children, and how the computer may enhance those activities. This interest in early learning was influenced by Papert's period of work with Piaget in Geneva (1958-1963).

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13. Teaching Children Thinking

I. Introduction

The phrase “technology and education” usually means inventing new gadgets to teach the same old stuff in a thinly disguised version of the same old way. Moreover, if the gadgets are computers, the same old teaching becomes incredibly more expensive and biased towards its dullest parts, namely the kind of rote learning in which measurable results can be obtained by treating the children like pigeons in a Skinner box.

The purpose of this essay is to present a grander vision of an educational system in which technology is used not in the form of machines for processing children but as something the child himself will learn to manipulate, to extend, to apply to projects, thereby gaining a greater and more articulate mastery of the world, a sense of the power of applied knowledge and a self-confidently realistic image of himself as an intellectual agent. Stated more simply, I believe with Dewey, Montessori, and Piaget that children learn by doing and by thinking about what they do. And so the fundamental ingredients of educational innovation must be better things to do and better ways to think about oneself doing these things.

I claim that computation is by far the richest known source of these ingredients. We can give children unprecedented power to invent and carry out exciting projects by providing them with access to computers, with a suitably

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clear and intelligible programming language and with peripheral devices capable of producing on-line real-time action.

Examples are: spectacular displays on a color scope, battles between computer controlled turtles, conversational programs, game-playing heuristic programs, etc. Programmers can extend the list indefinitely. Others can get the flavor of the excitement of these ideas from movies I shall show at the IFIPS meeting.

Thus in its embodiment as the physical computer, computation opens a vast universe of things to do. But the real magic comes when this is combined with the *conceptual power* of theoretical ideas associated with computation.

Computation has had a profound impact by concretizing and elucidating many previously subtle concepts in psychology, linguistics, biology, and the foundations of logic and mathematics. I shall try to show how this elucidation can be projected back to the initial teaching of these concepts. By doing so, much of what has been most perplexing to children is turned to transparent simplicity; much of what seemed most abstract and distant from the real world turns into concrete instruments familiarly employed to achieve personal goals.

Mathematics is the most extreme example. Most children never see the point of the formal use of language. They certainly never had the experience of making their own formalism adapted to a particular task. Yet anyone who works with a computer does this all the time. We find that terminology and concepts properly designed to articulate this process are avidly seized by the children who really want to make the computer do things. And soon the children have become highly sophisticated and articulate in the art of setting up models and developing formal systems.

The most important (and surely controversial) component of this impact is on the child's ability to articulate the working of his own mind and particularly the interaction between himself and reality in the course of learning and thinking. This is the central theme of this paper, and I shall step back at this point to place it in the perspective of some general ideas about education. We shall return later to the use of computers.

2. The Don't-Think-About-Thinking Paradox

It is usually considered good practice to give people instruction in their occupational activities. Now, the occupational activities of children are learning, thinking, playing and the like. Yet, we tell them nothing about those things. Instead, we tell them about numbers, grammar, and the French revolution; somehow hoping that from this disorder the really important things

will emerge all by themselves. And they sometimes do. But the alienation-dropout-drug complex is certainly not less frequent.

In this respect it is not a relevant innovation to teach children also about sets and linguistic productions and Eskimos. The paradox remains: why don't we teach them to think, to learn, to play? The excuses people give are as paradoxical as the fact itself. Basically there are two. Some people say: we know very little about cognitive psychology; we surely do not want to teach such half-baked theories in our schools! And some people say: making the children self-conscious about learning will surely impede their learning. Asked for evidence they usually tell stories like the one about a millipede who was asked which foot he moved first when he walked. Apparently the attempt to verbalize the previously unconscious action prevented the poor beast from ever walking again.

The paradox is not in the flimsiness of the evidence for these excuses. There is nothing remarkable in that: all established doctrine about education has similarly folksy foundations. The deep paradox resides in the curious assumption that our choice is this: *either* teach the children half-baked cognitive theory *or* leave them in their original state of cognitive innocence. Nonsense. The child does not wait with a virginally empty mind until we are ready to stuff it with a statistically validated curriculum. He is constantly engaged in inventing theories about everything, including himself, schools and teachers. So the real choice is: *either* give the child the best ideas we can muster about cognitive processes *or* leave him at the mercy of the theories he invents or picks up in the gutter. The question is: who can do better, the child or us? Let's begin by looking more closely at how well the child does.

3. The Pop-Ed Culture

One reads in Piaget's books about children re-inventing a kind of Democritean atomic theory to reconcile the disappearance of the dissolving sugar with their belief in the conservation of matter. They believe that vision is made possible by streams of particles sent out like machine gun bullets from the eyes and even, at a younger age, that the trees make the wind by flapping their branches. It is criminal to react (as some do) to Piaget's findings by proposing to teach the children "the truth." For they surely gain more in their intellectual growth by the act of inventing a theory than they can possibly lose by believing, for a while, whatever theory they invent. Since they are not in the business of making the weather, there is no reason for concern about their meteorological unorthodoxy. But they are in the business of making minds — notably their own — and we should consequently pay attention to their opinions about how minds work and grow.

There exists amongst children, and in the culture at large, a set of popular ideas about education and the mind. These seem to be sufficiently widespread, uniform and dangerous to deserve a name, and I propose "*The Pop-Ed Culture*." The following examples of Pop-Ed are taken from real children. My samples are too small for me to guess at their prevalence. But I am sure very similar trends must exist very widely and that identifying and finding methods to neutralize the effects of Pop-Ed culture will become one of the central themes of research on education.

EXAMPLES OF POP-ED THINKING

(a) *Blank-Mind Theories*. Asked how one sets about thinking a child said: "Make your mind a blank and wait for an idea to come." This is related to the common prescription for memorizing: "Keep your mind a blank and say it over and over." There is a high correlation, in my small sample, between expressing something of this sort and complaining of inability to remember poetry!

(b) *Getting-It Theories*. Many children who have trouble understanding mathematics also have a hopelessly deficient model of what mathematical understanding is like. Particularly bad are models which expect understanding to come in a flash, all at once, ready made. This binary model is expressed by the fact that the child will admit the existence of only two states of knowledge often expressed by "I get it" and "I don't get it." They lack—and even resist—a model of understanding something through a process of additions, refinements, debugging and so on. These children's way of thinking about learning is clearly disastrously antithetical to learning any concept that cannot be acquired in one bite.

(c) *Faculty Theories*. Most children seem to have, and extensively use, an elaborate classification of mental abilities: "He's a brain", "He's a retard," "He's dumb," "I'm not mathematical-minded." The disastrous consequence is the habit of reacting to failure by *classifying* the problem as too hard, or oneself as not having the required aptitude, rather than by *diagnosing the specific deficiency* of knowledge or skill.

4. Computer Science as a Grade School Subject

Talking to children about all these bad theories is almost certainly inadequate as an effective antidote. In common with all the greatest thinkers in the philosophy of education I believe that the child's intellectual growth must be rooted in his experience. So I propose creating an environment in which the

child will become highly involved in experiences of a kind to provide rich soil for the growth of intuitions and concepts for dealing with thinking, learning, playing, and so on. An example of such an experience is writing simple heuristic programs that play games of strategy or try to outguess a child playing tag with a computer controlled "turtle."

Another, related example, which appeals enormously to some children with whom we have worked is writing teaching programs. These are like traditional CAI programs but conceived, written, developed and even tested (on other children) by the children themselves.

(Incidentally, this is surely the proper use for the concept of drill-and-practice programs. Writing such programs is an ideal project for the second term of an elementary school course of the sort I shall describe in a moment. It is said that the best way to learn something is to teach it. Perhaps writing a teaching program is better still in its insistence on forcing one to consider all possible misunderstandings and mistakes. I have seen children for whom *doing* arithmetic would have been utterly boring and alienating become passionately involved in writing programs to teach arithmetic and in the pros and cons of criticisms of one another's programs like: "Don't just tell him the right answer if he's wrong, give him useful advice." And discussing what kind of advice is "useful" leads deep into understanding both the concept being taught and the processes of teaching and learning.)

Can children do all this? In a moment I shall show some elements of a programming language called LOGO, which we have used to teach children of most ages and levels of academic performance how to use the computer. The language is always used "on-line," that is to say the user sits at a console, gives instructions to the machine and immediately gets a reaction. People who know languages can think of it as "baby LISP," though this is misleading in that LOGO is a full-fledged universal language. Its babyish feature is the existence of self-contained sub-sets that can be used to achieve some results after ten minutes of instruction. Our most extensive teaching experiment was with a class of seventh grade children (twelve year olds) chosen near the average in previous academic record. Within three months these children would write programs to play games like the simple form of NIM in which players take 1, 2, or 3 matches from a pile; soon after that they worked on programs to generate random sentences—like what is sometimes called concrete poetry—and went on from there to make conversational and teaching programs. So the empirical evidence is very strong that we can do it, and next year we shall be conducting a more extensive experiment with fifth grade children. The next sections will show some of the elementary exercises we shall use in the first weeks of the course. They will also indicate another important aspect of having children do their work with a computer: the possibil-

ity of working on projects with enough duration for the child to become personally—intellectually and emotionally—involved. The final section will indicate a facet of how more advanced projects are handled and how we see the effects of the kind of sophistication developed by the children.

5. You Can Take the Child to Euclid, but You Can't Make Him Think

Let's go back to Dewey for a moment. Intellectual growth, he often told us, must be rooted in the child's experience. But surely one of the fundamental problems of the school is how to extend or use the child's experience. It must be understood that "experience" does not mean mere busy work: two children who are made to measure the areas of two triangles do not necessarily undergo the same experience. One might have been highly *involved* (e.g., anticipating the outcome, being surprised, guessing at a general law) while the other was quite *alienated* (the opposite). What can be done to involve the mathematically alienated child? It is absurd to think this can be done by using the geometry to survey the school grounds instead of doing it on paper. Most children will enjoy running about in the bright sun. But most alienated children will remain alienated. One reason I want to emphasize here is that surveying the schoolgrounds is not a good research project on which one can work for a long enough time to accumulate results and become involved in their development. There is a simple trick, which the child sees or does not see. If he sees it he succeeds in measuring the grounds and goes back to class the next day to work on something quite different.

Contrast this situation with a different context in which a child might learn geometry. The child uses a time-shared computer equipped with a CRT. He programs on-line in a version of the programming language LOGO, which will be described in more detail below.

On the tube is a cursor point with an arrow indicating a direction.

The instruction

FORWARD 100

causes the point to move in the direction of the arrow through 100 units of distance. The instruction

ROTATELEFT 90

causes the arrow to rotate 90°.

The child knows enough from previous experience to write the following almost self-explanatory program:

TO CIRCLE

FORWARD 1

ROTATELEFT 1

CIRCLE

END

The word "TO" indicates that a new procedure is to be defined, and it will be called "CIRCLE." Typing

CIRCLE

will now cause the steps in the procedure to be executed one at a time. Thus:

1st Step:	FORWARD 1	The point creeps ahead 1 unit.
2nd Step:	ROTATELEFT 1	The arrow rotates 1°.
3rd Step:	CIRCLE	This is a recursive call; naturally it has the same effect as the command CIRCLE typed by the child. That is to say, it initiates the same process:
1st Step:	FORWARD 1	The point creeps on, but in the new, slightly different direction. The arrow now makes an angle of 2° with its initial direction.
2nd Step:	ROTATELEFT 1	
3rd Step:	CIRCLE	This initiates the same process all over again. And so on, forever.

It is left as a problem for the reader to discover why this point will describe a circle rather than, say, a spiral. He will find that it involves some real geometry of a sort he may not yet have encountered (See answer at end of paper.). The more immediately relevant point is that the child's work has resulted in a

certain happening, namely a circle has appeared. It occurs to the child to make the circle roll? How can this be done? A plan is easy to make:

Let the point go around the circle once.
Then FORWARD 1
Then repeat.

But there is a serious problem! The program as written causes the point to go round and round forever. To make it go just once round we need to give the procedure an *input* (in more usual jargon: a variable).

This *input* will be used by the procedure to remember how far round it has gone. Let's call it "DEGREES" and let it represent the number of degrees still to go, so it starts off being 360 and ends up 0. The way this is written in LOGO is:

```
TO CIRCLE :DEGREES      : DEGREES means: the thing whose name
                        is "DEGREES."
IF :DEGREES= 0  STOP

FORWARD 1

ROTATE LEFT 1

CIRCLE:DEGREES - 1      Each time round the number of degrees
                        remaining is reduced by 1.

END
```

Now we can use this as a sub-procedure for ROLL:

```
TO ROLL

CIRCLE 360

FORWARD 10

ROLL

END
```

Or, to make it roll a fixed distance:

```
TO ROLL :DISTANCE
```

IF :DISTANCE = 0 STOP

CIRCLE 360

FORWARD 10

ROLL :DISTANCE 1

END

Or we can make the circle roll around a circle:

TO FUNNYROLL

CIRCLE 360

FORWARD 10

ROTATELEFT 10

FUNNYROLL

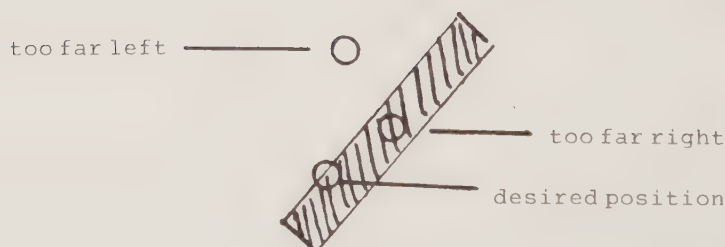
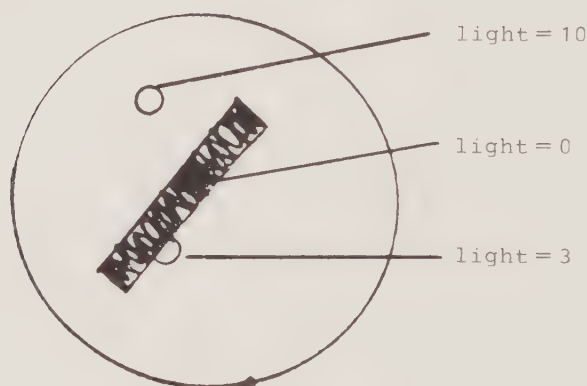
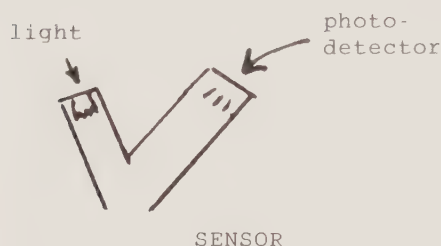
These examples will, if worked on with a good dose of imagination, indicate the sense in which there are endless possibilities of creating even more, but gradually more, complex and occasionally spectacularly beautiful effects. Even an adult can get caught up in it! Not every child will. But if he does, the result is very likely to be a true extension of his experience in Dewey's sense. And evidence is accumulating for the thesis that there is scarcely any child who cannot be involved in some computational project.

The next two sections will discuss two other peripheral devices suitable for a computation laboratory in an elementary school: a programmable vehicle and a music generator. There is, of course, no end to what one could invent. At M.I.T. we are thinking in terms of soon adding mechanical manipulators, psychedelic light shows in a reactive environment, apparatus for automated experiments in animal psychology, etc., etc., etc.

6. The Love of the Turtle

At M.I.T. we use the name "Turtle" for small computer controlled vehicles, equipped with various kinds of sense, voice and writing organs. Turtles can be controlled by the same commands used in the previous section to describe Graphics. They can be made to draw or to move about without leaving a visible trace. Procedures to achieve this are exactly like the procedures for

CRT Graphics. However, sense organs allow another interesting dimension of work. An interesting, simple one is a reflectivity sensor held close to the floor. A LOGO operation called "LIGHT" has an integer value between 0 and 10, depending on the reflectivity of the surface. Suppose we wish to program the turtle to follow the left edge of a black line on a white floor. Using an important heuristic we encourage the child to *study himself* in the situation, and try to simulate his own behavior. The key idea, of course, is to use feed-back according to the following plan:



COLOR	LIGHT	POSITION ERROR	CORRECTION
Mainly Black	small value	Too far right	ROTATELEFT
Equal Black & White	5	O.K.	Nothing
Mainly White	big value	left	ROTATERIGHT

This leads to the procedure:

TO WALK

IF LIGHT < 4 ROTATELEFT

IF LIGHT > 6 ROTATERIGHT

FORWARD 1

WALK

END.

Notice that the child can think of the program as a very simple formal model of himself, or, indeed, more justly, of a moth flying to a light. It is rare that children in the traditional context of math-science get a chance to develop a model so simple.

Turtles with touch, interactive behavior with several turtles, searching mazes, and so on scarcely scratch the surface of what can be done with these beasts.

7. Music

Just as the computer can be instructed to move a point on a TV display or make a turtle move or print a word, it can be instructed to sing a note. The LOGO instruction SING can be followed by an input to indicate a note (represented by 1 . . . 7) or a time. A program can be written thus:

```
TO MARY
SING 3
SING 2
SING 1
SING 2
SING 3
SING 3
END
```


The command

MARY

will cause the computer to sing the tune.

The program

TO CHANTMARY

MARY

CHANTMARY

END

will cause the tune to be repeated indefinitely. Programs can be written to speedup, slowdown, raise, lower, transpose, sing in chorus, etc., etc. Children can use the computer as a super musical instrument. They can compose at the typewriter and hear their creations played perfectly. They can make and undo small changes. They can cause the turtle or the CRT display to move to music and so on endlessly.

8. Case Histories from the Muzzey Jr. High School Experiment

The following piece is extracted verbatim from a report on the seventh grade teaching experiment performed at Muzzey Jr. High School at Lexington.

8.1. PROBLEM VS. PROJECT

The most exciting single aspect of the experiment was that most of the children acquired the ability and motivation to work on *projects* that extend in time over several days, or even weeks. This is in marked contrast with the usual style of work in mathematics classes, where techniques are taught and then applied to small repetitive exercise problems. It is closer, in ways that are essential to the later argument here, to the work style of some art classes where children work for several weeks on making an object; a soapcarving for example. The similarity has several dimensions. The first is that the duration of the process is long enough for the child to become *involved*, to try several ideas, to have the experience of putting something of oneself in the final result, to compare one's work with that of other children, to discuss, to criticise and to be criticised on some other basis than "right or wrong." The point

about criticism is related to a sense of creativity that is important in many ways which we shall talk about later—including, particularly, its role in helping the child develop a healthy self-image as an active intellectual agent.

Let's take an example. A continuing project over the last third of the year was working on various kinds of "language generating" programs. The children studied a program (given as a model) which generated two word sentences like:

CATS RUN
DOGS SHOUT
CHILDREN BITE
DOGS RUN
CATS RUN

-
-
-

The assignment was to study the model and go on to make more interesting programs. The sample printout that follows brought great joy to its creator who had worked hard on mastering the mathematical concepts needed for the program, on choosing sets of words to create an interesting effect and on converting her exceedingly vague (and unloved) knowledge about grammar into a useful, practical form.

INSANE RETARD MAKES BECAUSE SWEET SNOOPY SCREAMS
SEXY WOLF LOVES THATS WHY THE SEXY LADY HATES
UGLY MAN LOVES BECAUSE UGLY DOG HATES
MAD WOLF HATES BECAUSE INSANE WOLF SKIPS
SEXY RETARD SCREAMS THATS WHY THE SEXY RETARD HATES
THIN SNOOPY RUNS BECAUSE FAT WOLF HOPS
SWEET FOGINY SKIPS A FAT LADY RUNS

The next class assignment was to generate mathematical sentences which were later used in "teaching programs." For example:

$8 * \text{BOX} + 6 = 48$
WHAT IS BOX?

Finally, in the last weeks, someone in the class said she wanted to make a French sentence generator . . . for which she spurned advice and went to work. In the course of time other children liked the idea and followed suit — evoking from the first girl prideful complaints like "why do they all have to take my idea?" The interesting feature was that although they took her idea, they

imprinted it strongly with their own personalities, as shown by the following case studies:

K.M. The girl who initiated the project. Thoughtful, serious about matters that are important to her, often disruptive in class. Her approach to the French project was to begin by writing procedures to conjugate all the regular verbs and some irregular ones. The end of the school year fell before she had made a whole sentence generator. But she did have a truly professional program, completely debugged and working with great competence at conjugating — e.g. given VOUS and FINIR as inputs it would reply: VOUS FINISSEZ.

MR A gay, exuberant girl who made the “SEXY COMPUTER” program quoted above. Only half seriously she declared her intention of making the first operational French sentence generator. In a sense she did—but with cavalier disregard for the Academy’s rules of spelling and grammar!

JC A clear mind with a balanced sense of proportion. Deliberately decided to avoid the trap of getting so involved with conjugation that no sentence would ever be generated. Too serious to allow his program to make mistakes. Found a compromise: he would make a program that knew only the third person — but was still non-trivial because it did know the difference between singular and plural as well as the genders: thus it would say

LE BON CHIEN MANGE

but

LES BONNES FILLES MANGENT.

8.2. A DETAIL FROM A CHILD’S MATHEMATICAL RESEARCH PROJECT

The fine texture of the work on projects of this sort can only be shown by case studies. The following vignette needs very little reference to LOGO — thus illustrating how the projects are more than programming.

J is the author of the last French program mentioned. A little earlier he is working on generating equations as part of a project to make “a program to teach 8th grade algebra.” He has perfected a program to generate equations with coefficients in the range of 0-9 using a “random” number generator. His present problem is to obtain larger coefficients.

First Solution: Almost everyone tries this: get bigger numbers by *adding* smaller ones obtained from the old procedure. Amongst other considerations,

this looks like a good technique that has often paid well: use old functions to define new ones.

Consequences: J chooses his equation generator but soon finds some annoying features:

The new coefficients are in the range 0-18, which is unnatural and not very big.

There is a preference for some numbers e.g. 9 comes up ten times as often as 18!

Comment: The first problem can be alleviated by adding more numbers. One can even add a random number of random numbers. But this aggravates the second problem. J understands this qualitatively but does not see a way out. It is interesting that children and adults often have a resistance to making numbers by “non-numerical” operations. In this case the solution is to *concatenate* the single digit random numbers instead of adding them. LOGO has a simple way to express this and J is quite accustomed to making non-numerical strings by concatenation. In fact this is how he makes the equation! *Nevertheless he resists.* The problem is discussed in a class meeting and after some prompting everyone suddenly “discovers” the solution.

New Solution: j changes his program, now making numbers up to 99 by concatenation; he does some crude check of uniformity of distribution and tries his program.

Disaster: For a while it seems to go well. But in the course of playing with the “teaching program” a user types 5 and is surprised to get a reply like:

YOU KNUCKLEHEAD; YOU TOOK 11 SECONDS AND YOUR ANSWER IS WRONG. THE ANSWER IS 05. HERE IS SOME ADVICE . . . etc.

Comment: Poor J will get the sympathy of every mathematician who must at some stage have tried to generalize a result by extending the domain of an innocent looking function only to find that the extended function violates some obscure but essential condition. He is also in the heart of the problem of representation. Is ‘05’ a good representation? Yes, no . . . have your choice but face the consequences and be consistent. J’s problem is that his procedures accept “05” for arithmetic operations but not for the test of identity!

Solution: Change the identity test or peel off the leading zero. J chose the latter. His program worked for a while and was used to great effect.

New Step: Later J was urged to allow negative numbers. He found a good way: use the one digit random number generator to make a binary decision:

If less than 5, positive

Otherwise, negative.

That Problem Again: J had a program working perfectly with negatives. Then one day decided to make it more symmetrical by using +5 and -5 for positive and negative. This brought him back to the old problems raised by

differences between the machine's representation and the human user's. At this point the year ended with J's program not quite as effective as it had been at its peak.

14. Teaching Children to Be Mathematicians vs. Teaching about Mathematics

1. Preface

Being a mathematician is no more definable as “knowing” a set of mathematical facts than being a poet is definable as knowing a set of linguistic facts. Some modern math ed reformers will give this statement a too easy assent with the comment: “Yes, they must understand, not merely know.” But this misses the capital point that being a mathematician, again like being a poet, or a composer or an engineer, means *doing*, rather than knowing or understanding. This essay is an attempt to explore some ways in which one might be able to put children in a better position to *do* mathematics rather than merely to learn *about it*.

The plan of the essay is to develop some examples of new kinds of mathematical activity for children, and then to discuss the general issues alluded to in the preceding paragraph. Without the examples, abstract statements about “doing,” “knowing,” and “understanding” mathematics cannot be expected to have more than a suggestive meaning. On the other hand the description of the examples will be easier to follow if the reader has a prior idea of their intention. And so I shall first sketch, very impressionistically, my position on some of the major issues. In doing so I shall exploit the dialectical device employed in the previous paragraph to obtain a little more precision of statement by explicitly excluding the most likely misinterpretation.

It is generally assumed in our society that every child should, and can, have experience of creative work in language and plastic arts. It is equally generally assumed that very few people can work creatively in mathematics. I be-

lieve that there has been an unwitting conspiracy of psychologists and mathematicians in maintaining this assumption. The psychologists contribute to it out of genuine ignorance of what creative mathematical work might be like. The mathematicians, very often, do so out of elitism, in the form of a deep conviction that mathematical creativity is the privilege of a tiny minority.

Here again, it is necessary, if we want any clarity, to ward off a too easy, superficial assent from math ed reformers who say, "Yes, that's why we must use The Method of Discovery." For, when "Discovery" means discovery this is wonderful, but in reality "Discovery" usually means something akin to the following fantasy about a poetry class: the discovery-method teacher has perfected a series of questions that lead the class to discover the line "Mary had a little lamb." My point is not that this would be good or bad, but that no one would confuse it with creative work in poetry.

Is it possible for children to do creative mathematics (that is to say: to *do* mathematics) at all stages of their scholastic (and even adult!) lives? I will argue that the answer is: yes, but a great deal of creative mathematical work by adult mathematicians is necessary to make it possible. The reason for the qualification is that the traditional branches of mathematics do not provide the most fertile ground for the easy, prolific growth of mathematical traits of mind. We may have to develop quite new branches of mathematics with the special property that they allow beginners more space to romp creatively, than does number theory or modernistic algebra. In the following pages will be found some specific examples which it would be pretentious to call "new pedagogical oriented branches of mathematics" but which will suggest to co-operative readers what this phrase could mean.

Obstreperous readers will have no trouble finding objections. Mathematical elitists will say: "How dare you bring these trivia to disturb our contemplation of the true mathematical structures?" Practical people will say: "Romping? Pumping? Who needs it? What about practical skills in arithmetic?"

The snob and the anti-snob are expressing the same objection in different words. Let me paraphrase it, "Traditional schools have found mathematics hard to teach to so-called average children. Someone brings along a new set of activities, which seem to be fun and easy to learn. He declares them to be mathematics! Well, that does not make them mathematics, and it doesn't turn them into solutions to any of the hard problems facing the world of math ed."

This argument raises serious issues, from which I single out a question which I shall ask in a number of different forms:

In becoming a mathematician does one learn something other and more general than the specific content of particular mathematical topics? Is there such a thing as a Mathematical Way of Thinking? Can this be learned and taught? Once one

has acquired it, does it then become quite easy to learn *particular* topics —like the ones that obsess our elitist and practical critics?

Psychologists sometimes react by saying, “Oh, you mean the transfer problem.” But I do not mean anything analogous to experiments on whether students who were taught algebra last year *automatically* learn geometry more easily than students who spent last year doing gymnastics. I am asking whether one can identify and teach (or foster the growth of) something *other* than algebra or geometry, which, once learned, will make it easy to learn algebra and geometry. No doubt, this other thing (let’s call it the MWOT) can only be taught by using particular topics as vehicles. But the “transfer” experiment is profoundly changed if the question is whether one can *use* algebra as a *vehicle for deliberately teaching transferable general concepts and skills*. The conjecture underlying this essay is a very qualified affirmative answer to this question. Yes, one *can* use algebra as a vehicle for initiating students to the mathematical way of thinking. But, to do so effectively one should first identify as far as possible components of the general intellectual skills one is trying to teach; and when this is done it will appear that algebra (in any traditional sense) is not a particularly good vehicle.

The alternative choices of vehicle described below all involve using computers, but in a way that is very different from the usual suggestions of using them either as “teaching machines” or as “super-slide-rules.” In our ideal of a school mathematical laboratory the computer is used as a means to control physical processes in order to achieve definite goals . . . for example as part of an auto-pilot system to fly model airplanes, or as the “nervous system” of a model animal with balancing reflexes, walking ability, simple visual ability and so on. To achieve these goals mathematical principles are needed; conversely in this context mathematical principles become sources of power, thereby acquiring meaning for large categories of students who fail to see any point or pleasure in bookish math and who, under prevailing school conditions, simply drop out by labelling themselves “not mathematically minded.”

The two easy acceptance of this takes the form: “Yes, applications are motivating.” But “motivation” fails to distinguish alienated work for a material or social reward from a true personal involvement. To develop this point I need to separate a number of aspects of the way the child relates to his work.

A simple, and important one, is the time scale. A child interested in flying model airplanes under computer control will work at this project over a long period. He will have time to try different approaches to sub-problems. He will have time to talk about it, to establish a common language with a collaborator or an instructor, to relate it to other interests and problems. This *project-oriented* approach contrasts with the *problem* approach of most math teaching:

a bad feature of the typical problem is that the child does not stay with it long enough to benefit much from success or from failure.

Along with time scale goes structure. A project is long enough to have recognizable phases—such as planning, choosing a strategy of attempting a very simple case first, finding the simple solution, *debugging it*, and so on. And if the time scale is long enough, and the structures clear enough, the child can develop a vocabulary for articulate discussion of the process of working towards his goals.

I believe in articulate discussion (in monologue or dialogue) of how one solves problems, of why one goofed that one, of what gaps or deformations exist in one's knowledge and of what could be done about it. I shall defend this belief against two quite distinct objections. One objection says: "*it's impossible to verbalize*; problems are solved by intuitive acts of insight and these cannot be articulated." The other objection says: "*it's bad to verbalize*; remember the centipede who was paralyzed when the toad asked which leg came after which."

One must beware of quantifier mistakes when discussing these objections. For example, J. S. Bruner tells us (in his book *Towards a Theory of Instruction*) that he finds words and diagrams "impotent" in getting a child to ride a bicycle. But while his evidence shows (at best) that *some* words and diagrams are impotent, he suggests the conclusion that *all* words and diagrams are impotent. The interesting conjecture is this: the impotence of words and diagrams used by Bruner is explicable by Bruner's cultural origins; the vocabulary and conceptual framework of classical psychology is simply inadequate for the description of such dynamic processes as riding a bicycle! To push the rhetoric further, I suspect that if Bruner tried to write a program to make an IBM 360 drive a radio controlled motorcycle, he would have to conclude (for the sake of consistency) that the order code of the 360 was impotent for this task. Now, in our laboratory we have studied how people balance bicycles and more complicated devices such as unicycles and circus balls. There is nothing complex or mysterious or undescribable about these processes. We can describe them in a non-impotent way *provided that* a suitable descriptive system has been set up in advance. Key components of the descriptive system rest on concepts like: the idea of a "first order" or "linear" theory in which control variables can be assumed to act independently; or the idea of feedback.

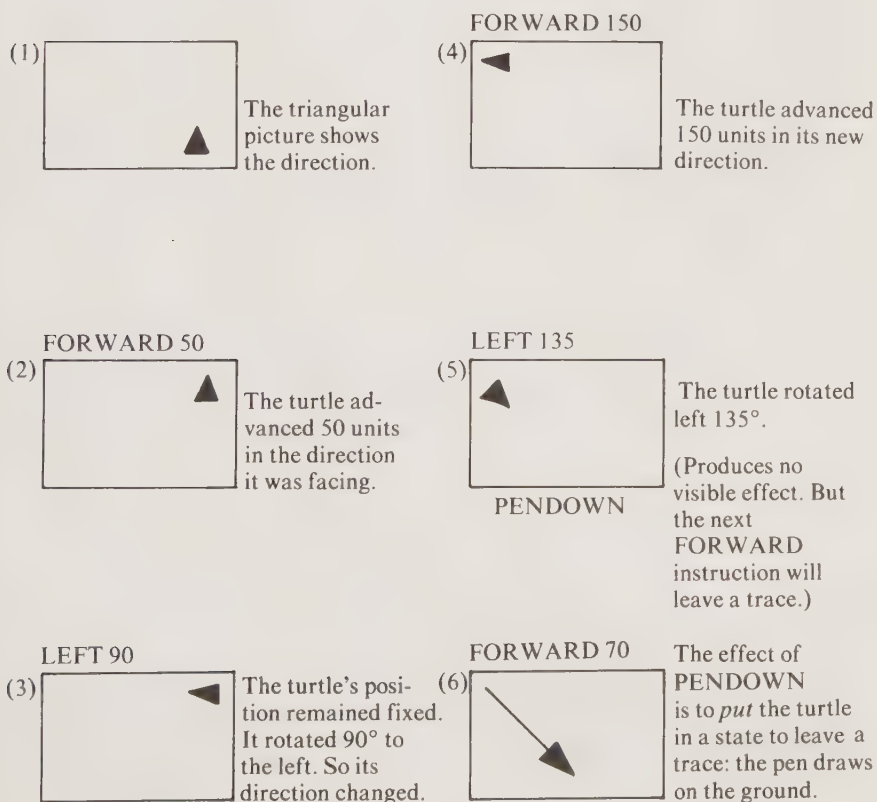
A fundamental problem for the theory of mathematical education is to identify and name the concepts needed to enable the beginner to discuss his mathematical thinking in a clear articulate way. And when we know such concepts we may want to seek out (or invent!) areas of mathematical work which exemplify these concepts particularly well. The next section of this essay will describe a new piece of mathematics with the property that it allows clear discussion and simple models of heuristics that are foggy and confusing for beginners when presented in the context of more traditional elementary mathematics.

2. Turtle Geometry: A Piece of Learnable and Lovable Mathematics

The physical context for the following discussion is a quintuple consisting of a child, a teletype machine, a computer, a large flat surface, and an apparatus called a *turtle*. A turtle is a cybernetic toy capable of moving forward or back in a particular direction (relative to itself) and of rotating about its central axis. It has a *pen*, which can be in two states called **PENUP** and **PENDOWN**. The turtle is made to act by typing commands whose effect is illustrated in Figure 1.

Figure 1: TURTLE LANGUAGE

At any time the turtle is at a particular *place* and facing in a particular *direction*. The place and direction together are the turtle's geometric *state*. The picture shows the turtle in a field, used here only to give the reader a frame of reference:



(a) DIRECT COMMANDS

The following commands will cause the turtle to draw Figure 2.

```

PENDOWN
FORWARD 100
RIGHT 60
FORWARD 100
BACK 100
LEFT 120
FORWARD 100

```

Figure 2



(b) DEFINING A PROCEDURE

The computer is assumed to accept the language LOGO (which we have developed expressly for the purpose of teaching children, not programming but mathematics). The LOGO idiom for asserting the fact that we are about to define a procedure is illustrated by the following example. We first decide on a *name* for the procedure. Suppose we choose "PEACE." Then we type:

```

TO PEACE
1 FORWARD 100
2 RIGHT 60
3 FORWARD 100
4 BACK 100
5 LEFT 120
6 FORWARD 100
END

```

These are directions telling the computer how to PEACE. The word "TO" informs the computer that the next word, "PEACE," is being defined and that the numbered lines constitute its definition.

The turtle doesn't move while we are typing this. The word "TO" and the line numbers indicated that we were not telling it to go forward and so on; rather we were telling it *how* to execute the new command. When we have indicated by the word "END" that our definition is complete the machine echoes back:

```

PEACE DEFINED
and now if we type

```

```

PENDOWN
PEACE

```

the turtle will carry out the commands and draw Figure 2. Were we to omit the command "PENDOWN" it would go through the motions of drawing it without leaving a visible trace.

The peace sign in Figure 2 lacks a circle. How can we describe a circle in turtle language?

An idea that easily presents itself to mathematicians is: let the turtle take a tiny step forward, then turn a tiny amount and keep doing this. This might not quite produce a circle, but it is a good first plan, so let's begin to work on it. So we define a procedure:

```
TO CIRCUS
1 FORWARD 5
2 RIGHT 7
3 CIRCUS
END
```

Notice two features

(a) The procedure refers to itself in line 3. This looks circular (though not in the sense we require) but really is not. The effect is merely to set up a never-ending process by getting the computer into the tight spot *you* would be in if you were the kind of person who cannot fail to keep a promise and you had been tricked into saying, "I promise to repeat the sentence I just said."

(b) We selected the numbers 5 and 7 because they seemed small, but without a firm idea of what would happen. However an advantage of having a computer is that we can always *debug* it; in this case, perhaps, by choosing different numbers. If an undesirable effect follows we can always *debug* it; in this case, perhaps, by choosing different numbers. If, for example, the turtle drew something like Figure 3a, we would say to ourselves, "It's not turning enough" and replace 7 by 8; on the other hand if it drew Figure 3b we might replace 7 by 6.

Figure 3a

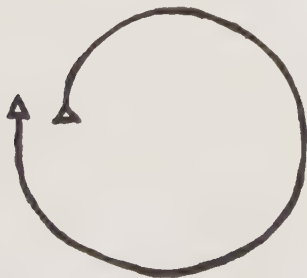


Figure 3b



I wish I could collect statistics about how many mathematically sophisticated readers fell into my trap! Experience shows that a large proportion of math graduate students will do so. In fact, the procedure cannot generate either 3a or 3b! If it did, it would surely go on to produce an infinite spiral.

And one can easily see that this is impossible since the same sequence of commands would have to produce parts of the curve that are almost flat, and other parts that are very curved. More technically, one can see that the procedure CIRCUS *must* produce a close approximation to a circle (i.e., what is, for all practical purposes a circle) because it must produce a curve of constant curvature.

One can come to the same conclusion from a more general theorem. We call procedures like CIRCUS “fixed instruction procedures” because they contain no *variables*.

THEOREM: Any figure generated by a fixed instruction procedure can be bounded either by a circle or by two parallel straight lines.

Examples of figures that can and that cannot be so bounded are shown in Figure 4.

Figure 4



A Figure Bounded neither by parallel lines nor by a circle.



A Figure Bounded by parallel lines

We now show how to make procedures with inputs in the sense that the command FORWARD has a number, called an input, associated with it. The next example shows how we do so. (The words on the title line preceded by “:” are names of the inputs, rather like the x ’s in school algebra.) In the fifth grade class we read :NUMBER as *dots* NUMBER or as *the thing of* “NUMBER”, emphasizing that what is being discussed is not the word “NUMBER” but a thing of which this word is the name.

```

TO POLY :STEP :ANGLE
1 FORWARD :STEP
2 LEFT :ANGLE
3 POLY :STEP :ANGLE
END

```

This procedure generates a rather wonderful collection of pictures as we give it different inputs.

Although POLY has provision for inputs it is really a fixed instruction procedure. To create one that is not, we change the last line of POLY. We change the title also, though we do not need to do so.

Old Procedure

```

TO POLY :STEP :ANGLE
1 FORWARD :STEP
2 LEFT :ANGLE
3 POLY :STEP :ANGLE
END

```

New Procedure

```

TO POLYSPI :STEP :ANGLE
1 FORWARD :STEP
2 LEFT :ANGLE
3 POLYSPI :STEP+20 :ANGLE
END

```

The effect of POLYSPI is shown in Figure 5.

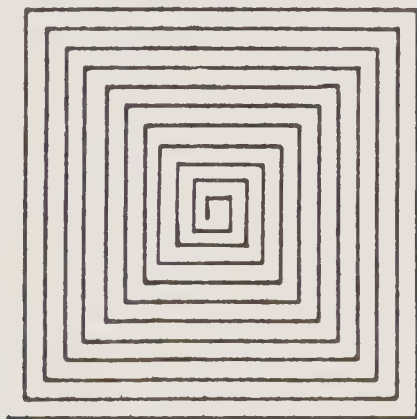


Figure 5
POLYSPI 5 90
or
Squiral

We have seen we can use POLY to draw a circle. Can we now use it to draw our peace sign? We could, but will do better to make a procedure, here called ARC whose effect will be to draw any circular segment given the diameter and the angle to be drawn as in Figure 6. The procedure is as follows where in line 2 a special constant called “*PIE*” is used and the asterisk sign is used for multiplication. (Do not assume that *:PIE* is what its name suggests.)

```
TO ARC :DIAM :SECTOR
1 IF :SECTOR= 0 STOP
2 FORWARD :PIE*:DIAM
3 RIGHT 1
4 ARC :DIAM :SECTOR-1
END
```

We can now make a procedure using the old procedure PEACE as a *sub-procedure*:

```
TO SUPERPEACE
1 ARC 200 360
2 RIGHT 90
3 PEACE
END
```



Figure 6

SUPERPEACE

Better yet we could rewrite PEACE to have inputs. For example:

```
TO PEACE :SIZE
1 FORWARD :SIZE
2 RIGHT 60
3 FORWARD :SIZE
4 BACK :SIZE
5 LEFT 120
6 FORWARD :SIZE
7 RIGHT 90
8 ARC 2*:SIZE 360
```

Then peace signs of different sizes can be made by the commands:

```
PEACE 100
PEACE 20
and so on.
```


We can use the command ARC to draw a heart:

```
TO HEART :SIZE
1 ARC :SIZE/2 180
2 RIGHT 180
3 ARC :SIZE/2 180
4 ARC :SIZE*2 60
5 RIGHT 60
6 ARC :SIZE*2 60
END
```



MINITHEOREM: A heart can be made of four circular arcs.

We can also use it to draw a flower. Notice in the following the characteristic building of new definitions on old ones.

A computer program to draw this flower uses the geometric observation that petals can be decomposed (rather surprisingly!) as two quarter circles. So let's assume we have a procedure called TO QCIRCLE whose effect is shown by the examples. Some of them show initial and final positions of the turtle, some do not.



```
QCIRCLE 50
```



```
QCIRCLE 100
```



Now let's see how to make a petal.

```
TO PETAL :SIZE
1 QCIRCLE :SIZE
2 RIGHT 90
3 QCIRCLE :SIZE
END
```

```
PETAL 100
```



TO FLOWER :SIZE

1 PETAL :SIZE

2 PETAL :SIZE

3 PETAL :SIZE

4 PETAL :SIZE

END

FLOWER 100



TO STEM :SIZE

1 RIGHT 180

2 FORWARD 2*:SIZE

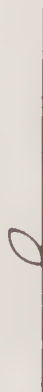
3 RIGHT 90

4 PETAL: SIZE/2

5 FORWARD :SIZE

END

STEM 100



PLANT 50



TO PLANT :SIZE

1 PENDOWN

2 FLOWER :SIZE

3 STEM :SIZE

4 PENUP

END

Now let's play a little.

TO HEXAFLOWER :SIZE

1 RIGHT 90

2 FORWARD 4*:SIZE

3 PLANT :SIZE

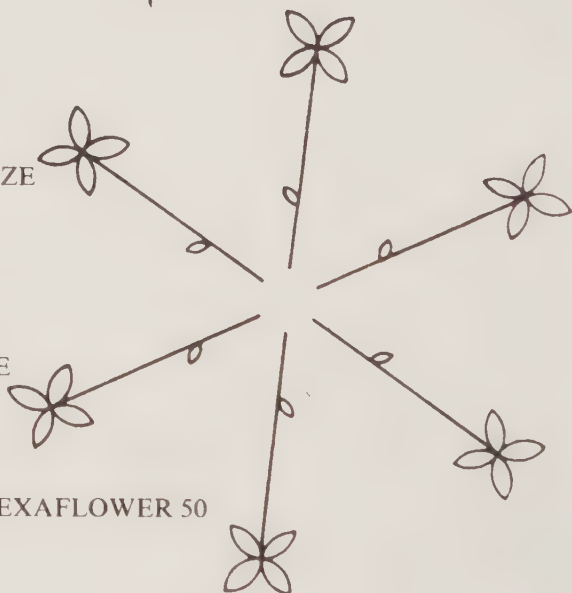
4 FORWARD :SIZE

5 RIGHT 30

6 HEXAFLOWER :SIZE

END

HEXAFLOWER 50

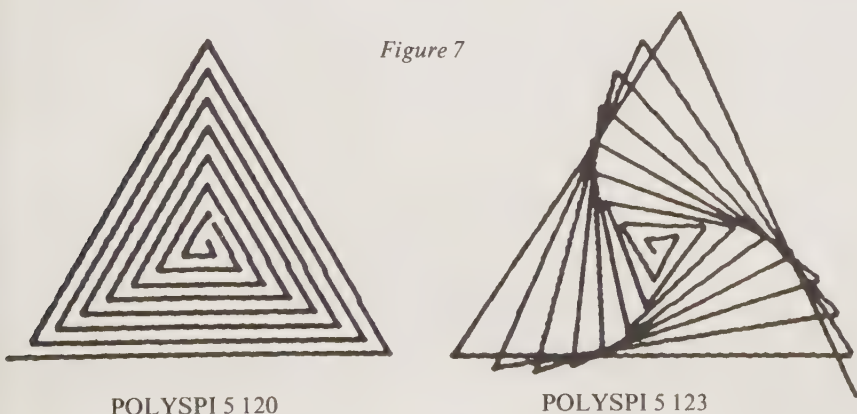


3. Creativity? Mathematics?

In classes run by members of the M.I.T. Artificial Intelligence Laboratory we have taught this kind of geometry to fifth graders, some of whom were in the lowest categories of performance in “mathematics”. Their attitude towards mathematics as normally taught was well expressed by a fifth grade girl who said firmly, “There ain’t nothing fun in math!” She did not classify working with the computer as math, and we saw no reason to disabuse her. There will be time for her to discover that what she is learning to do in an exciting and personal way will elucidate those strange rituals she meets in the math class.

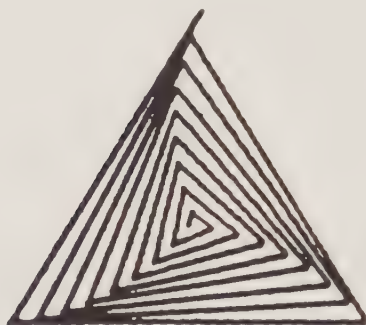
Typical activities in early stages of work with children of this age is exploring the behavior of the procedure POLY by giving it different inputs. There is inevitable challenge—and competition—in producing beautiful or spectacular, or just different effects. One gets ahead in the game by discovering a new phenomenon and by finding out what classes of angles will produce it.

The real excitement comes when one becomes courageous enough to change the procedure itself. For example making the change to POLYSPI occurs to some children and, in our class, led to a great deal of excitement around the truly spontaneous discovery of the figure now called a *squiral* (Figure 5). (Note: By spontaneous I mean, amongst other things, to exclude the situation of the discovery teacher standing in front of the class soliciting pseudo-randomly generated suggestions. The squiral was found by a child sitting all alone at his computer terminal!) By no means all the children will take this step—indeed once a few have done so it becomes derivative for the others. Nevertheless, we might encourage them to explore inputs to POLYSPI. There is room here for the discovery of more phenomena. For example, taking :ANGLE as 120 produces a neat triangular spiral. But 123 produces a very different phenomenon.

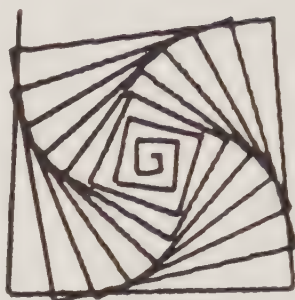


What else produces similar effects?

Figure 8



POLYSPI 5 121



POLYSPI 5 93

The possibilities for original minor discoveries are great. One girl became excited for the first time about mathematics by realizing how easy it was to make a program for Figure 9 by

- (1) observing herself draw a similar figure
- (2) *naming* the elements of her figure — “*BIG*” and “*SMALL*” — so that she could talk about them and so describe what she was doing
- (3) describing it in LOGO

```
TO GROWSHRINK :BIG :SMALL
1 FORWARD :BIG
2 RIGHT 90
3 FORWARD :SMALL
4 RIGHT 90
5 GROWSHRINK :BIG-10 :SMALL+10
END
```

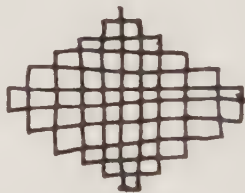


Figure 9

The possibilities are endless. These are small discoveries. But perhaps one is already closer to mathematics in doing this than in learning new formal manipulations, transforming bases, intersecting sets and drifting through misty lessons on the difference between fractions, rationals and equivalence classes of pairs of integers. Perhaps learning to make small discoveries puts one more surely on a path to making big ones than does faultlessly learning any number of sound algebraic concepts.

4. Some Physical Mathematics

The turtle language is appropriate for many important physical problems. Consider, for example, the problem of understanding planetary orbits as if one were a junior high school student. One would find conceptual barriers of varying degrees of difficulty. Certainly the idea of the inverse square law is simple enough. Somewhat harder is the representation of velocities, accelerations and forces as vectors. But the insuperable difficulty in reading a text on the subject comes from the role of differential equations. The really elegant and intelligible physical ideas give rise to local differential descriptions of orbits; translating those into global ones usually involves going through the messy business called “solving” differential equations.

Turtle geometry helps at all these points. The use of vectors is extremely natural. And the local differential description takes the form of a procedure that can be run so as to produce a drawing of a solution or studied using theorems and analytic concepts about procedures.

The framework for thinking about orbital theory in turtle terms presupposes prior contact with the concepts of state and of quantized time—both of which occur very easily and naturally in many computational situations. The state of the “planet” is its position and a certain vector called, say “*JUMP*”. If the planet were left alone it would move by *:JUMP* at every clock time. Thus it would go off, forever, in a straight line. In the presence of the sun, we think of it as undergoing two movements: it moves by *:JUMP* and then it falls into the sun! To make this more precise we put these two actions together using a procedure called “*VECTORADD*” which could be defined by the children or given as a primitive. Thus we obtain a LOGO procedure whose general idea will be intelligible to readers who try hard enough. (Two helpful comments: *MAKE* is the LOGO idiom for assignment, or setting values, so that line 1 in the procedure will cause the quantity *VECTORADD OF :JUMP AND FALL* to be computed and given the name “*NEWJUMP*”. This computation assumes the existence of another procedure, called “*FALL*”, which will compute the “fall into the sun vector”. These ideas might seem confusing when presented fast; ten year old children understand them fluently when they are presented properly.)

```
TO FLY :JUMP
```

```
1 MAKE
```

```
    NAME “NEWJUMP”
```

```
    THING VECTORADD OF :JUMP AND FALL
```

```
2 SETHEADING (DIRECTION :NEWJUMP)
```

```
3 FORWARD (LENGTH :NEWJUMP)
```

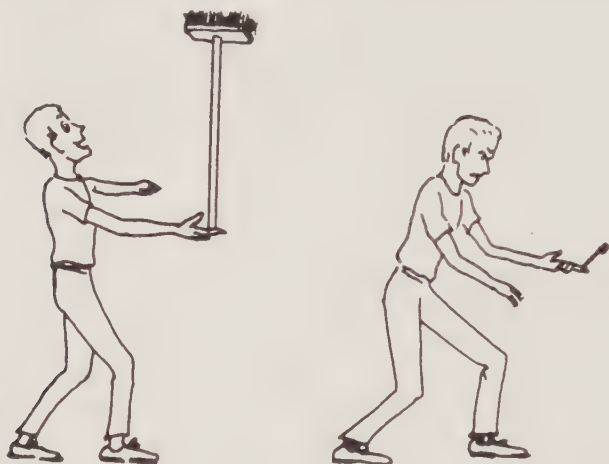
```
4 FLY :NEWJUMP
```

```
END
```


Using this same idea one can easily deal in an experimental way with three bodies; one can design space-ship orbits, synchronous satellites and so on endlessly.

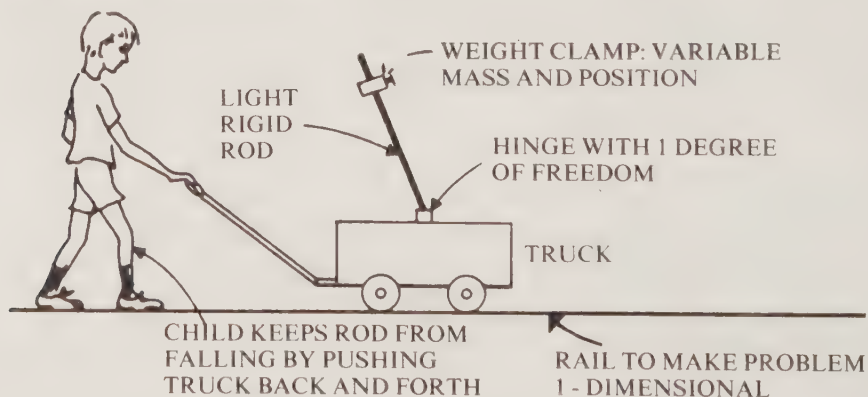
5. Control Theory as a Grade School Subject or Physics in the Finger Tips

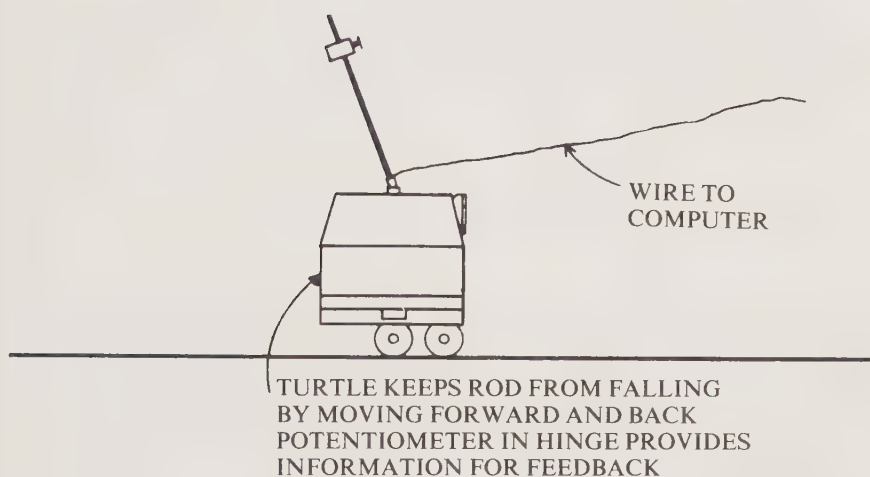
We begin by inviting the reader to carry out the illustrated experiments—or to recall doing something similar.



One of the goals of this unit of study will be to understand how people do this and particularly to understand what properties of a human being determine what objects he can and what objects he cannot balance.

A “formal physical” model of the stick balancing situation is provided by the apparatus illustrated next:





A computer controlled version replaces the track and the child by a turtle with the angle sensor plugged into its sensor socket. A simple minded procedure will do a fair amount of balancing (provided that the turtle is fast!!):

TO BALANCE

```

1 TEST ANGLE > 10
2 IF TRUE FORWARD 8
3 TEST ANGLE < - 10
4 IF TRUE BACK 8
5 WAIT 1
6 BALANCE
END

```

This procedure is written as part of a project plan that begins by saying: neglect all complications, try something. Complications that have been neglected include:

- (1) The end of the line bug.
- (2) The overshoot bug.
(Perhaps in lines 2 and 4 the value 8 is too much or too little.)
- (3) *The Wobbly Bug*
The TEST in the procedure might catch the rod over to the left while it is in rapid motion towards the right. When this happens we should leave well alone!

One by one these bugs, and others can be eliminated. It is not hard to build a program and choose constants so that with a given setting of the movable weight, balance will be maintained for long periods of time.

6. What are the Primitive Concepts of Mathematics?

To see points and lines as *the* primitive concepts of geometry is to forget not only the *logical* primitives (such as quantifiers) but especially the *epistemological* primitives, such as the notion of a mathematical system itself. For most children at school the problem is not that they do not understand *particular* mathematical structures or concepts. Rather, they do not understand what kind of thing a mathematical structure is: they do not see the point of the whole enterprise. Asking them to learn it is like asking them to learn poetry in a completely unknown foreign language.

It is sometimes said that in teaching mathematics we should emphasize the process of mathematization. I say: excellent! But on condition that the child should have the experience of mathematizing for himself. Otherwise the word "mathematizing" is just one more scholastic term. The thrust of the explorations I have been describing is to allow the child to have living experiences of mathematizing as an introduction to mathematics. We have seen how he mathematizes a heart, a squiral, his own behavior in drawing a GROW-SHRINK, the process of balancing a stick, and so on. When mathematizing familiar processes is a fluent, natural, enjoyable activity, then is the time to talk about mathematizing mathematical structures, as in a good pure course on modern algebra.

But what are the ingredients of the process of mathematizing? Is it possible to formulate and teach knowledge about how one is to tackle for example, the problem of setting up a mathematical representation of an object such as the hearts and flowers we discussed earlier?

Our answer is very definitely affirmative, especially in the context of the kind of work described above. Consider for example, how we would teach children to go about problems like drawing a heart. First step we say: if you cannot solve the problem as it stands, try simplifying it; if you cannot find a complete solution, find a partial one. No doubt everyone gives similar advice. The difference is that in this context the advice is concrete enough to be followed by children who seem quite impervious to the usual math.

A simplification of the heart problem is to settle, as a first approximation, on a triangle; which we then consider to be a very primitive heart.

```

TO TRI
1 FORWARD 100
2 RIGHT 120
3 FORWARD 100
4 RIGHT 120
5 FORWARD 100
END

```



TRI

Now that we have this construction firmly in hand we can allow ourselves to modify it so as to make it a better heart. The obvious plan is to replace the horizontal line by a structure line. So we write a procedure to make this. First choose it a name, say “TOP”, then write:

```

TO TOP :SIZE
1 ARC :SIZE/2 180
2 RIGHT 180
3 ARC :SIZE/2 180
END

```



TOP

Replacing line 1 in TO TRI by TOP we get:

```

TO TRI
1 TOP 100
2 RIGHT 120
etc.

```



HEART WITH BUG

The effect is as shown! Is this a failure? We might have so classified it (and ourselves!) if we did not have another heuristic concept: BUGS and DEBUGGING. Our procedure did not fail. It has a perfectly intelligible *bug*. To find the bug we follow the procedure through in a very FORMAL way. (Formal is another concept we try to teach.) We soon find that the trouble is in line 2. Also we can see why. Replacing line 1 by TOP did what we wanted, but it also produced a SIDE-EFFECT. (Another important concept.) It left the turtle facing in a different direction. Correcting it is a mere matter of changing line 2 to RIGHT 30. And then we can go on to make the fully curved heart. Unless we decide that a straight-sided one is good enough for our purposes.



Straight-sided Heart



Curved Heart

Our image of teaching mathematics concentrates on teaching concepts and terminology to enable children to be articulate about the process of developing a mathematical analysis. Part of doing so is studying good models (such as the heart anecdote) and getting a lot of practice in describing one's own attempts at following the pattern of the model in other problems. It seems quite paradoxical that in developing mathematical curricula, whole conferences of superb mathematicians are devoted to discussing the appropriate language for expressing the *formal* part of mathematics, while the individual teacher or writer of text-books is left to decide how (and even *whether*) to deal with heuristic concepts.

In summary, we have advanced three central theses:

- (1) The non-formal mathematical primitives are neglected in most discussions of mathematical curricula.
- (2) That the choice of content material, especially for the early years, should be made primarily as a function of its suitability for developing heuristic concepts, and
- (3) Computational mathematics, in the sense illustrated by turtle geometry, has strong advantages in this respect over "classical" topics.

15. Personal Computing and Its Impact on Education

[Note: This paper was taken from a tape recording of Dr. Papert's talk; it was transcribed by Roxanne Berridge, paraphrased by Ted Sjoerdsma, and edited by Anne Adair.]

I would like to begin by describing a fantasy that I was making a few months ago with the director of the Aspen Ski School.

If you know anything about skiing, you know that sometimes when the snow is just right, it is very easy to come down the slope. You can do the turns, with the snow itself practically ushering you through the right motions. Everything is so right that you think you are the greatest skier that ever walked. Later, of course, you learn otherwise.

My fantasy is based on that idea. Could you engineer a mountain with a dynamically variable slope determined by the expertise of the skier? A beginner on top of the mountain would start a very gentle slope. Off he goes. As he acquires enough confidence to stay up on his skis the characteristics of the slope change. This continues until he gets to the bottom of the mountain and he is a skier. So he learns because his environment is designed in such a way that it changes with his needs.

I don't know whether you could make such a mountain or not. The ski director and I talked a long time about what is involved in skiing and human beings, and we couldn't decide. I doubt that you could make a mountain so that one time down would produce a skier. But I do know that the amount of time that it takes to learn to be an expert skier has been divided by 20 in the last 20 years. The level of skiing ability which everyone expects to reach (and most do) in one season of skiing is a level much higher than most people in the

olden days ever expected to reach even if they went skiing every vacation, every winter, for all their lives.

This is all quite interesting, but how does it relate to education? And more particularly, to computers in education? Is skiing a good model to study with implications for education? Maybe it is time for those who are trying to reform the educational process to start looking at the success stories like the improvement in skiing education. Where the learning is much better than it used to be, one should go see what happened and attempt to transcribe that to the formal education of children.

One of the themes I would like you to keep in mind during this talk is: if you are interested in how learning happens, you should start by looking at success stories. It is amazing that if you look at the articles in the journals of education and educational psychology, you'll find that they are all about failures. The whole subject is based on how badly people learn in the environment (schools) in which they *should* learn very well. We should rather think about success stories of good learning.

I would also like us to catch the vision of aspiring to something bigger and better, of some kind of change or improvement which would make things a little better in our schools. It doesn't turn me on to think that we teach computing in schools because it is a basic skill and something that everyone ought to know. What does turn me on is the hope that the computer in some sense might be the magic mountain. What I am really interested in is not teaching about computers (however important that might be), but rather determining whether knowing about computers can really deeply change everything else. So let me tell you a story or two—little anecdotes about learning with computers that did deeply change everything else for some people.

Before the stories I should supply the background. For the past ten years we have been involved in a project at MIT which is designed to try to bring kids and computers together. Philosophically the background is this: In 1967 I first got involved with kids and computers. Watching one of the first projects—high school kids trying to learn to program using a Model 33 terminal and a terrible language—it was obvious that it was great for some kids. But it was even more obvious that it wasn't the way to go. So we decided to sit back in our ivory towers and think about what could be done with the computers that *might* exist ten years ahead. (Ten years ahead is about now.)

We did decide a few things. Computers should be personal computers. Computers should be action-oriented (draw pictures, make noises, make objects move around). And there should be a programming language or a programming system that didn't just happen to be lying around, but was made for the purpose.

We set out to make a collection of all the nice features of all the known programming languages, so that we could put together a language that would al-

low a kid to really communicate with a computer. What we were trying to do was to find out what a first grader (or a kid of any age) could do with a computer. Out of our many years of experience, one fact became very clear—*kids of all ages can learn to program.*

In case you are amazed at that statement, consider programming as *talking to a computer in its language*. Kids have no trouble learning to talk to people. Even two-year-olds are already talking to people with more proficiency than the kids in a high school BASIC class are learning to talk to the computer, or the kids in a high school French class are learning to speak French. Surely, there must be a way to get the kids to learn to talk to computers the way they learn to talk to people. Why should it be harder to talk “computer” (as kids say) than it is to talk English? (This becomes a guiding theme.) We decided that one reason why it seems harder is that there are no good things to talk about. So we put graphics on the computer, which certainly is something to talk about. Kids can now make pictures and do things with computer pictures that they can’t do with pencil and paper pictures. They can get precisions, get repeatability, depict motion, change color, and do all sorts of other things that can’t be done with pencil and paper.

We suddenly came into quite a number of dimensions of the educational enterprise, or what it is to learn. For example, there is something that is called social resonance. It became very clear that just because kids spent a lot of time watching television in the home, making a picture on the television screen gives them a special kind of charge. A kid who is taking over the television screen so that he can make pictures himself is moving into the space age. And something of that drive makes all of us want to buy little calculators (that we could perfectly well do without) because we would each like to have, as our very own, a piece of that space age technology. This carries over to our kids, again a sort of social resonance. The kid feels like he is doing something meaningful, like those guys he sees in the space center controlling a moon rocket. The computer, just because it is high technology, provides a certain bridge between the learner and the world.

I would like to have you examine an even more important bridge, the one between the learner and himself, between what is being learned and what is deep down inside him. I will try to get you to sense what I mean, even though I cannot fully explain it.

We were all sort of led bit-by-bit into a view of education that is more Freudian than Piagetian. Jean Piaget is a psychologist and an epistemologist. He thinks of mathematics in a very logical way, rigorous and precise. And I think that he is right in a lot of his insights. What I am trying to say is that there is a model of mathematics which doesn’t have much to do with my feelings. The general view of mathematics in our culture is that it is something separate from our bodies, cold and disincorporate.

Let me show you a different view of mathematics and an example of what it is like. Let's go back to those children learning to make pictures on the television screen. How do you go about making a picture on the television screen? The child is creating it, controlling it, intellectually interacting with it; he is not just pushing buttons and flashing things on the screen as if by magic. Thus we became involved in the enterprise of how to think of geometry. After a lot of false starts, we came up with what I think is the absolutely right way to think about geometry. This is the concept of "turtle." "Turtle" is a differential equation—differential geometry in computational terms.

Let me explain how we present it to a child, and then maybe you won't worry about it being differential geometry. I believe that what we are talking about has a certain great importance, not because it is capturing an idea that we invented, but because it is capturing ideas of science and bringing it into a new view—and the new view is possible only because of the computer's presence. It is this creation of new visions of old things that is one of the magics of the computer.

We bring in the child and introduce him to a little cybernetic animal called a "turtle". The turtle is controlled by the computer and the computer is controlled by the child. The turtle is positioned somewhere and is looking in some direction. This is his state. To change his state, there are two different change operations: forward and turning. Forward causes him to change his position without changing the direction that he is looking. The turning operation allows him to change his position without changing the direction that he is looking. The turning operation allows him to turn to the right or the left without changing his position. These two operations (coordinates) can change the two components of the state independently and thus the child can get the turtle to perform any maneuver. If the child equips the turtle with a pen, then sets it on paper, the turtle could draw things. (Alternately, one could introduce the child to the turtle on a display screen, which could obey the same commands, but now draws pictures with the speed of light.) Understanding the geometric concept of the state of the turtle, with its two kinds of change operations, gives one a kind of mathematical knowledge which provides power to do things.

However, I don't want to emphasize that aspect, but rather a kind of learning that I call "body syntonic learning." Let me introduce this with an example. Imagine a child who has learned to drive the turtle; he knows how to say forward and how many units; he knows how to turn it and how many degrees of turn. By and large the child has been drawing straight line figures. One day the child asks, "How does it draw a circle?" We ask the child to figure this out for himself. We suggest that the child pretend that he is the turtle and walk in a circle and try to describe his movements in turtle language.

This is where we connect the geometry to the child. The turtle is especially suited to serve as what has been labeled a "transitional object." The child

comes along knowing a lot about “body geometry.” He knows how to walk around in space. He has a lot of geometric knowledge, but it’s inside and he can’t verbalize it. On the other hand, there is this abstract thing called mathematics which you want him to learn. And in between is the turtle—a mathematical object, but enough like a person for him to identify with and to transfer his knowledge of himself to. Because the turtle moves like he does, the child can apply the same description and sort of talk to it through the computer.

So we have seen two ways in which the learning of mathematics is different from the traditional. One way is social resonance. The other way is resonance with deep structures inside the person. In fact, this is where Freud and Piaget come together.

Let’s return to the circle. When the child walks in a circle, he goes forward a bit and right a bit, forward a bit and right a bit, repeating this until he returns to his starting point. So, too, when he tells the turtle what to do: forward one, right one, repeated as often as necessary until it results in a circle, at least from a point of view that defines a circle as a curve of constant curvature. (You see that the curvature of a curve is how much you turn to how much you go forward.) If the child goes forward ten and right one, it would be a much flatter curve, and would produce a much larger circle that would look like a polygon. He could make a spiral by gradually decreasing the curvature. It is a very impressive thing to make a spiral and then to get all sorts of crazy pictures by putting spirals on top of one another. Mastering this idea of curvature by thinking of curves in terms of generating little changes is an extremely powerful idea. It is perfectly graphicable by these small children, and also very usable. This last aspect is the third way that this mathematics is different from school mathematics—its social resonance, its resonance with one’s own body, and *its usefulness*. It can be used right now to do something exciting.

Here you might say, is applied mathematics instead of pure mathematics. It is a curious thing that the way we teach mathematics in the elementary school is very very pure. Historically, that is a very funny thing. Historically, mankind first made mathematics a part of doing other things, like navigating, commerce, magic, religion, and all sorts of architecture. Only bit-by-bit did that beautiful thing called pure mathematics become abstracted out. I think this is a right model. And it is the model that Piaget teaches us, too, if we look carefully. The right way to learn mathematics is first learning it as a concrete applied thing that actually works in the world, and then gradually we get to pure mathematics.

So here is a model for a computer making a radical difference to the world. It is not just able to change how we teach the same algebra and the same equation in the same number of years. It can change the entire relationship to mathematics. And not only mathematics. I chose mathematics only because,

in our society, mathematics is so split off and dissociated from everything else. But the computer could have the same impact on learning French. And what we see in the French classroom or in the mathematics classroom tells us nothing at all about difficulties we have in learning these subjects.

We have seen so clearly in our laboratory what happens in ideal conditions. What we have seen in so many instances in our children is that those kids have taken to mathematics so easily, have learned with such facility, that they are being slowed down instead of accelerated by the whole school process. The teachers know that too, but they don't know what to do about it.

I would like to give it a little theory and talk about dissociate learning. For most people in this culture, mathematics has no relation to anything they enjoy, or love, or do, or appreciate. It is in a box apart. We are told that some people (like scientists) use it in some magical way to make moonships of which you know nothing. This is dissociate learning.

By syntonic learning, I mean "It goes together with." Children learn a tremendous amount of stuff by syntonic learning, including a tremendous amount of mathematics. They don't only learn it, they invent it. Piaget is important for investigating the thinking of small children and finding some remarkable discoveries. Piaget has filled books and books and books full of things that children have discovered. If it were totalled and made into a curriculum, teachers would pass out in horror at the thought of the immense task. But the same would be true if you considered what a child learns in order to speak his mother tongue, or to manipulate his parents, or to know what is a funny joke. The fact is that children, until they get to school, are avid, eager, successful learners. Then, when they come to school, they learn that learning is not so good.

The point is that children do acquire, learn, construct, reinvent (or whatever the word for it is) a vast amount of mathematical knowledge (and other kinds of knowledge) before they ever come near the classroom. And this is not knowledge that is acquired in a dissociated way. So I would suggest that syntonic learning may be another good name for Piagetian learning, which is a kind of learning that happens without being taught, without a curriculum, quiz, or grade. And there is certainly a fantastic amount that is learned this way.

Why not everything? Why are there some things that seem to require such a different learning process. I would say that one of the reasons is that, although we have always known in principle what to do about it, we haven't had the technological infrastructure to realize the things that we've already done. Now the computer has brought us the technological infrastructure that can make possible a real intervention in the learning environment, one in which syntonic learning can be generalized to cover a much greater area. I hope we can make the most of it.

16. Computer-based Microworlds as Incubators for Powerful Ideas

1. General Ideas

It has always seemed obvious that the behavioral sciences (psychology in particular, and in particular, that branch of psychology which deals with learning) ought to be helpful to the implementation of effective learning. But powerful ideas from the behavioral sciences have proved mysteriously “unexportable” to education. In this paper we explore a new methodology in which we rethink what is fundamental in a scientific domain and do this in a manner that is resonant both with ideas about learning and with a technological infrastructure that can support their implementation. We will see how the presence of computers can enormously expand our freedom of action in creating learning paths into knowledge.

In this paper we use physics as an example of a domain in which concepts considered most basic and powerful in terms of the structure of the subject matter have been quite inaccessible to beginners. The strain created by this discordance between the natural order and the pedagogical order is not unique to physics. Thus physics is used here both for itself and as a metaphor to express more general issues.

The approach to physics described here emerged from previous work in our laboratory on introducing children to mathematics through Turtle Geometry. The simplest formulation of the central research question is: Can we theorize what we have learned from 6 years of work with Turtle Geometry into a set of general ideas of wider applicability to other domains and to more advanced knowledge?

This article has not been published previously. A later version appears as part of: *Mind Storms: Children, Computers, and Powerful Ideas*. N.Y.: Basic Books, 1980.

What is essential in our method is a new approach to being simultaneously in touch with three dimensions: the scientific discipline, the field of cognitive psychology, and the technological infrastructure.

1.1 MICROWORLDS

We focus on the concept of microworld. Briefly, a microworld is a subset of reality or a constructed reality whose structure matches that of a given cognitive mechanism so as to provide an environment where the latter can operate effectively. The concept leads to the project of inventing microworlds so structured as to allow a human learner to exercise particular powerful ideas or intellectual skills. The project is not new in kind. One can recognize much of the praxis of educators as falling under this description. The idea of cuisenaire rods is a striking example. Microworlds may not be new, but *what is new is the availability of a technology for microworlds* — a technology that will free their development from dependence on inspired opportunism and allow it to become a refined, systematic and theory-based branch of education research. I am referring, of course, to the technology of microcomputation, which is rapidly making available extremely powerful but extremely inexpensive computation devices.

The aspect of microworld design on which I shall focus here is a kind of matchmaking between insights from epistemology about *what* is fundamental in a particular domain of knowledge and insights from cognitive psychology about *how* what we have recognized as fundamental can be effectively learned. Computer-based microworlds could be used to implement many different “psychologies of learning.” But, once more we have to narrow our focus to a particular kind, namely those which, like Piaget, stress the learner-directed process. In this context the role of the microworld is to provide an intellectual environment that allows learning through interaction with it.

1.2 STANDING PIAGET ON HIS HEAD

We have referred to the disappointments in the appropriation of psychological theories by education. Piaget’s work is a prime example. What educators have taken from Piaget is what happened to be “implementable” in traditional schools with traditional curricula, even though what is most powerful in Piaget is the insight that intellectual development does not always need explicit teaching. I contend that the new technologies will lead to a rehabilitation of what is most profound in Piaget’s thought. The particular microworlds described below were chosen largely because they provide a preview of how this might happen.

1.3 THE TURTLE WORLD AS AN EXAMPLE

To illustrate this perspective, I refer first to the TURTLE-GEOMETRY learning environment that I examine here as a microworld. A turtle is a cybernetic animal, usually on a light screen but also able to exist as a small vehicle. A child is able to control the motion of the turtle by typing commands (such as FORWARD 10, RIGHT 90, BACK 20) at a computer terminal. This world makes a match between a cluster of powerful mathematical ideas and a Piagetian model of learning. The match is made when the child realizes that he or she can make the turtle, for example, draw a circle by first walking in a circle, describing the act of doing so (e.g., as “going forward a little and turning a little, going forward a little and turning a little”) and finally transforming this description into a formal program for the turtle REPEAT FD 1 RT 1. The program is really a differential equation expressing the characterization of a circle as a curve of constant curvature. Thus the child has made contact with a fundamental mathematical notion (differential equation) by using old schemata (the “body knowledge” that enables the child to walk in a circle) as an assimilative basis for building new structures. One source of importance of the Turtle-Geometry world is its richness in occasions for this kind of encounter.

Two critical questions that are often raised might help to place this process in its theoretical perspective. These are: “Can *all* mathematics be ‘computerized’ in this way?” and “Does the child become dependent on computers?” The questions fail to see the *genetic intent* of the use of the microworlds. Their role is to provide a stepping-stone to becoming a mathematician. Once there, they are no longer needed.

In order to internalize the mathematical notion the child needs to represent it as (assimilate it to) something already present (the transitional object) in his internal, mental world. One difference between the child and the mathematician is that for a child the transitional object must necessarily be “non-mathematical”; but once a sufficiently rich collection of formal mathematical objects have been internalized, these act as internalizers of new ones. Mathematics becomes a self-reproducing system. But until then, its growth depends on the use of nonmathematical or semimathematical transitional objects. (This is an interpretation of the stage of concrete-operations.) The problem is to get started. The possibilities of anthropomorphisms of the turtle make it an excellent candidate for early internalization. Its connections with fundamental mathematical structures make it a powerful carrier for later internalizations. Of course, even an adult, even a mathematician may have to make use of transitional objects in suitably constructed microworlds. *But* a mature mathematician will construct his or her own such microworlds, very often in the form of diagrams. He or she does not need a computer, although one might be useful.

2. Glimpses of Learning Physics

2.1 A GLIMPSE AT A "BAD PATCH" IN TRADITIONAL LEARNING

Let us start with something that everyone would agree is very central to physics both in its historical development and in its internal structure: Newton's first two laws of motion. First, that a body stays in a state of rest or unchanged motion except if a force is applied to it, and second, that $F=ma$.

These are very fundamental, and yet we are used to thinking of them as a relatively advanced topic, encountered (at the earliest toward the end of the high school years). Worse yet, investigations into how students understand them, even after being taught, show clearly that even after a year of college physics, most students cannot be said to *understand the laws even if they have acquired skill in solving set problems*. Certainly very few are able to integrate this piece of formal physics knowledge with their intuitive sense of physical process. What can we do to change this state of affair? (See Note at end of paper for an example.)

I maintain that a large part of the problem has to do with the difficulty of finding genetic *stepping* stones to Newton's laws. By this I mean something else in the mental world of the learner which will act as a precursor of her or his personal reconstruction of the Newtonian laws. Instead, the student is expected to make a leap to Newton that strains him in three ways.

The first strain comes because he doesn't know anything else "like" these laws to which he can relate them. The second is that in fact they contradict much in his immediate experience of how the world works, and he lacks the ideas needed to make a synthetic overview. The third comes from the "passive" mode of learning: The new knowledge cannot be "exercised" in a personally involved and motivated way; it cannot be assimilated into living experience. Today's education takes two understandable options in relation to Newton's laws: It presents these laws formally at a point where it just expects the student to "fill in the blanks" of set equations and it waits until the student has met the second derivative and differential equations before it expects that the student will have any real understanding of what is going on. Surely we can do better.

2.2 AN EXAMPLE OF USING THE "THREE-WAY RESONANCE"

I would like to contrast this situation with another path to Newton. We describe this path in terms of how a child might follow it.

A first step is to bring the child into contact with other laws of motion close to those which he encounters in everyday life, but as formal as those of New-

ton and equally true in an appropriate world. These laws of motion are what we have come to call Turtle Geometry.

We have already met the turtle which the child can put into self-maintaining motion by writing a program such as:

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TO CIRCLE:
FORWARD 1
RT 1
REPEAT
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In working with the turtle, the student becomes accustomed to thinking in terms of the turtle's STATE which has two components: a position and a heading. Thus, the command FORWARD is a state-change-operator that changes the first component while leaving the second unaltered, while the command RIGHT TURN is a state-change-operator whose action is on the second component of the state. This child is in contact with some very powerful ideas.

The child is working with intellectual structures which are precursors of the Newtonian laws in several respects. To see one of these, consider how we would formulate a "qualitative" version of the Newtonian laws for a student who is already familiar with Turtle geometry.

A physics turtle is like a geometry turtle. It too has a state with two components. The first is position. The second is velocity. The first law of physics turtles is that there is only one state-change-operator, which is usually called Force, and the second law states that this state-change-operator acts only on the second component of the physics turtle's state, that is, its velocity. So, force is a velocity changer — an accelerator. There is no external state-change-operator for the first component (the position). It changes all by itself as if the physics turtle has an internal state-change-operator. Thus the beginner is given a set of down-to-death but nevertheless *usable* ways to think about Newtonian laws.

In the setting of a computer environment in which the student actually has two kinds of turtles, a geometry turtle and a physics turtle, and has lived through the experience of what it is like to control each of them, the Newtonian Laws are less lonely and isolated in conceptual space.

One could go further in this direction by introducing many kinds of turtle and even giving the student the means to invent his own kinds. An interesting classification is: *position turtles*, which receive commands stating *where* they should go; *velocity turtles*, whose commands state *what velocity* they should adopt and *acceleration turtles*, whose commands state by *how much the velocity should change*.

2.3 SOME GENERALIZATIONS

I shall next use this example to formulate a number of general principles of what might make one learning process better than another. Specifically, I do this by listing a number of features of the example; but I hope that it is clear how each of them applies very generally.

i. Contact with the very important Newtonian Laws is made at an earlier stage in learning physics. The import of this is not that children should know advanced physics. I see it very differently: I see the learner as engaged over a very long period (possibly all of life) in building a personal "physics." The most compelling reason for bringing this learner into contact with a given idea at an early stage is to allow this idea to be built into this personal physics. One of the major problems—perhaps THE major problem—in later physics education is that the student comes with a personal physics built out of totally inappropriate concepts. This has to be undone in order to rebuild on a sound basis.

ii. More important than the fact that the student encounters NEWTON'S LAWS OF MOTION at an early stage is that he encounters SEVERAL laws of motion and can thus disengage the general concept of LAW OF MOTION. He is accustomed from an early stage to thinking in terms of laws of motion. (In the monkey problem in the note, what is important is to formulate the problem in terms of laws of motion; the actual law is less important.)

iii. Even more important yet is *the way in which* he relates to thinking in terms of laws of motion: In the turtle microworlds, laws of motion are powerful methods that can be used to master the world. And besides being powerful, they are derived in a way that is deeply rooted in the learner's intuitive body knowledge.

iv. The design of these microworlds reflects an epistemological position on what is most important in physics: certain powerful ideas such as laws of motion, state and state-change-operator, differential law; qualitative precursors of the usual quantitative theories; an intimate relationship between geometry and physics; etc., etc.

v. The design of the microworlds reflects a position in genetic epistemology: in particular a structuralist and constructivist position derived from Piaget that attaches great importance to the influence on the developed forms of the developmental path. From this point of view, the physics knowledge of a physicist or student cannot be reduced to a collection of "atoms" or "rules" or whatever, which are the same irrespective of the order in which they are acquired.

vi. The use of the microworlds provides a model of a learning theory in which active learning consists of exploration by the learner of a microworld sufficiently bounded and transparent for constructive exploration and yet sufficiently rich for significant discovery.

vii. The particular microworlds we have seen have "hooks" on one side to

preformal, intuitive body knowledge and on the other side to formal, rigorous mathematical knowledge. Recall how the child used the turtle to invent a differential equation for a circle (represented as a curve of constant curvature). The essential step was to identify herself or himself with the turtle—thus using her or his ability to walk in a circle and to describe her or his own actions as the ingredients for building the formal project. I have referred to this kind of process (of which we can find numerous examples in all domains of knowledge) as body-syntonic or ego-syntonic learning in very marked contrast with the “dissociated” learning more usually found in the mathematical classroom. To emphasize the importance of this, I recall Gregory Bateson’s concept of “deutero-learning.” The point is not merely or mainly that this anthropomorphism helped the child take a particular step towards learning, a particular concept of physics—we agree with Bateson in appreciating how every act of learning sets a model for how learning should be done. Every act of dissociated learning is also learning to be dissociated. Every act of syntonic learning is learning to keep the new structures in touch with the old ones.

3. Concluding Perspectives

The construction of a network of microworlds provides a vision of education planning that is in important respects “opposite” to the concept of “curriculum.” This does not mean that no teaching is necessary or that there are no “behavioral objectives.” But the relationship of teacher to learner is very different: the teacher introduces the learner to the microworld in which discoveries will be made, rather than to the discovery itself. Moreover, this is nothing like the “socratic” discovery methods popular in the sixties, which demanded a close interaction between student and “socratic” teacher.

Finally, let us underscore yet again the role that the computer has played in this discussion. We did not begin with the computer “presence” and look for ways to use it. There is no question of a “technological imperative” that will drive developments in the way we suggest. On the contrary, computers are infinitely plastic and can reinforce old patterns as well as allow us to live out new ones. We began with a problem deeply rooted in education, the “unnaturalness” of the learning paths that the school system has been forced to put people in. Very often the computer is used to push people along these unnatural pathways. I am proposing here as I have in relation to other domains over the past decade, to use the computer in a different way, not to coax the student along the difficult and unnatural paths adopted by traditional curricula, but to open up a natural path along which the student can drive himself or herself.

Thus, the computer presence can be used to add new degrees of freedom to what children learn and how they learn it.

Note

The failure to assimilate “laws of motion thinking” into the network of intuitive knowledge is demonstrated by careful analysis of how subjects with different degrees of exposure to formal physics cope with puzzle problems such as the following classical example:

The drawing shows a monkey and a rock of equal mass, a rope, and a frictionless pulley. The monkey begins to climb the rope. What happens to the weight?



Few students with freshman college physics see this “intuitively” as a problem centering on laws of motion. For those who do, the answer is immediate and obvious: the forces acting on the two objects are always equal, so their motions must be equal. More usual approaches reflect a classification of the situation as a “balance problem” or other “standard curriculum” conceptualization, all of which are very unlikely to succeed in correctly analysing the problem.



PATRICK SUPPES

Patrick Suppes is well known in computing circles as a prominent, long-time proponent and developer of computer-assisted instruction. Currently holding several faculty appointments at Stanford, including the Lucie Stern Professorship in Philosophy, Suppes also directs the Institute for Mathematical Studies in the Social Sciences, a unit that coordinates the development and operation of numerous computer-based instructional activities incorporated into the Stanford curriculum. In addition to the Stanford work, Suppes also directs the private Computer Curriculum Corporation, which he founded to widely market computer-assisted instruction for educational institutes at many levels. The CAI, for which both the IMSS and CCC are best known, is in mathematics and language instruction for educational institutions at many levels. Though most widely known for his CAI work in elementary and remedial mathematics and language instruction, Professor Suppes has pioneered CAI work at various levels in a wide range of other disciplines as well. Suppes has received numerous awards, has published widely, and belongs to many professional and honorary organizations.

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17. Computer-based Mathematics Instruction

1. Introduction

In the past year, preliminary experiments have been conducted in preparation for the completion of the Computer-based Laboratory for Learning and Teaching at Stanford. This report describes the equipment available in the laboratory, outlines the work which has been done on the program for first grade, fourth grade, and sixth grade (mathematical logic), and describes the program logic to be used on the computer.

Two groups of six first-grade children will be at Stanford five days a week for twenty minute sessions beginning in October, 1964, although the laboratory will not be fully functioning until February or March 1965. A few selected fourth-grade students will also be run throughout the year. Both the first- and fourth-grade materials follow the general content and sequence of the *Sets and Numbers* books (1A and 1B, 4A and 4B). The fourth-grade material, however, is constructed for use by children who have had no previous introduction to any of the new mathematics programs.

The mathematical logic program, which has been run in a slightly different form during the summer of 1964, will be presented to a group of gifted sixth graders beginning in November.

2. Description of the Laboratory

The system consists of a medium-sized computer and six subject stations. Each station is located in a small (7 ft. by 8 ft.) room and contains terminal devices which permit communication between the computer and the user. The user may be one or more of the following: programmer, operator, experimenter, student, or experimental subject. The computer controls the presentation

of both visual and auditory material; the subject responds with a light pen, a typewriter keyboard, or a microphone.

2.1 OPTICAL DISPLAY UNIT

The optical display unit is a rapid, computer-controlled, random access projection device which presents visual material to the user on a 10 in. by 13 in. ground glass screen. The source material (any 8½ in. by 11 in. page of text material) is photographed on microfilm and is stored in small projector cells with a capacity of 256 pages each.

Each display unit has two projectors which project onto a single screen giving a total capacity of 512 individual pages. However, many more combinations are possible by constructing composite images from both projectors on the common screen. Each projector has its own shutter that can be opened or closed and the screen is divided into eight equal sections each with its own individually controlled mask.

The user responds to the display and sends information to the computer with the light pen. As the pen is touched to the screen, the coordinates of that position are sent to the computer and may be compared with our prearranged areas of the screen. The accuracy of the light pen will permit the identification of any ¼ in. square on the screen. Most of the operations of the display unit will occur in approximately one second. A light in the pen is turned on to indicate acceptance of the information by the computer.

2.2 CATHODE-TUBE DISPLAY

This display unit is a cathode ray tube, commonly called a "scope," which can display points of light in an area 10 in. high by 10 in. wide. There are 1024 possible positions on both the horizontal and vertical axes. In addition to individual points there are 120 prearranged characters which may be displayed in five different sizes. It is also possible to display vectors by simply identifying the end points. An electronic light pen is available for sending information from the user back to the computer by identifying points on the face of the tube.

2.3 TYPEWRITER KEYBOARD

A typewriter keyboard is attached to each scope and may be used to send information from the student to the computer.

2.4 AUDIO SYSTEM

The audio system has the provision for playing pre-recorded messages to the user. The messages are recorded on 6 in. wide magnetic tape and two tape

transports may be assigned to each of the six user stations. Each transport has a capacity of about 17 minutes which can be divided into any combination of messages from one message 17 minutes in length to 1020 messages of one second each. The user may also record onto a selected portion of the tape and then have it displayed back for comparison purposes or retained as response data. The various control and switching functions between the tape transports, microphones, and speakers are performed by a small process control computer. The random access time to any stored message is approximately one second. The audio system will not be installed until February, 1965.

2.5 CENTRAL COMPUTER

The main control functions of the entire system are handled by a medium-sized computer. The main memories are a 16,000 word core, and a 4,000 word core which can be interchanged with any of 32 bands of a magnetic drum. All input-output devices are processed through a sequence break system which services them only when necessary. Two high-speed data channels permit simultaneous computation and servicing of peripheral devices. Additional backup in computational power, additional storage, and increased input-output speed is obtained through connections to a larger computer system with magnetic tape and disk storage, located at the Stanford Computation Center.

3.1 FIRST-GRADE PROGRAM: PRELIMINARY EXPERIMENTS

Before beginning the preparation of material for use in the computer-based laboratory, an experiment with twenty-nine kindergarten children was conducted. These children comprised an entire kindergarten class with the exception of one mentally retarded boy. The children work individually, with one experimenter and one child present at each session.

The experiment used the material that we plan to use in presenting the first six concepts on the first-grade program, namely a section on:

- 1) learning to use the light pen, including in the directions words needed in later sections:

box	first	left
top	last	right
middle	arrow	dotted line
bottom	goes with	

- 2) using concrete objects to show sets
- 3) using set notation to picture a set
- 4) the empty set

- 5) equal sets (choosing an answer from two choices)
- 6) equal sets (more objects in the sets, three answer choices)

The material for the experiment was arranged in the form of several 8½ in. by 11 in. booklets requiring one response per page. The children responded by putting a dot with a crayon on the place indicated by the directions. (This corresponds to the use of a light pen to point to the correct answer on the optical display unit.)

A tape played to each child gave directions for the required responses. For all but the second section, each audio message was from five to ten seconds in length and called for a response from the child at the end of the message. In the second section, there were some longer messages accompanying displays of concrete objects.

The experimenter ran the tape recorder, located the page on which the child was to respond, and noted any comments the child made or any questions he asked. The experimenter did not make any remarks pertaining to the material unless the child did not understand an instruction. Then, the experimenter repeated the instruction, reading from the script used to make the tape. If the child still did not understand the instruction, the experimenter made an explanation, noting exactly what was said.

The material was constructed so that each concept was presented and followed by a group of problems. If a certain number of responses to these problems was incorrect, the child went to a "branch." In the branch, the concept was presented again in a slightly different way and a new set of responses was made by the child.

If the child made enough correct responses on the first presentation of a concept, the branch was skipped and the child went on to the next concept. At the end of some sections (including concept and branch) there was a short review section.

Some results of this experiment were as follows:

- 1) Of the words introduced in section 1, "last," "left," and "right" were the most difficult. There was no branch in this section.
- 2) When there was an explanation given using concrete objects, the children became very restless and their attention strayed if the audio message was over ten to twelve seconds long without asking them to make a response. There was no branch in this section.
- 3) Using set notation to picture sets was quite simple. Eight of the twenty-nine children had enough errors to go to the branch.
- 4) The concepts of the empty set had the fewest number of errors. Four children made enough errors to go to the branch.
- 5) The concept of equal sets proved to be the most difficult. Nineteen children made enough errors to go to the branch. The branch for this con-

cept was written in a slightly different way, and proved much simpler. The wording of the directions for responding to the problems seemed to cause most of the difficulty. With the different wording of the branch, the number of errors substantially decreased. However, both the concept and branch were much more difficult than they should be, so these sections have been substantially rewritten for the optical display units.

- 6) The section on equal sets, using a somewhat different format and wording from either the concept or branch for the previous section was substantially simpler, with five children having to go to a branch. In the review for sections 5 and 6, fifteen of the children made no errors.

3.2 INSTRUCTIONAL MATERIAL

Since the user may make a response only by touching the light pen to the screen, there are certain limitations on the kinds of formats that can be used. And although some material in the *Sets and Numbers* books has had to be rewritten in order to fit a multiple-choice or simple keyboard format, almost every concept in the books has been prepared for the optical display units.

The material follows the outline in the books and is essentially:

- introduction to display unit and use of light pen
- set notation
- empty sets
- equal sets
- union of disjoint sets
- properties of sets
- N notation and number of things in a set
- arabic numerals
- counting from 1 to 9
- addition (equation form, column form, sequential problems, variables)
- difference of sets
- 10 to 100
- place value
- first to sixth
- 1-100 by ones, by tens, by twos, by fives, by threes, by fours
- number words, zero-nine
- $\frac{1}{2}$
- subtraction, equation form, column form
- two variables
- time, money, calendar, cups, pints, quarts, dozen, $\frac{1}{2}$ dozen in., ft.
- geometry (not yet completed)

Because there is to be a minimum of audio messages and separate classroom instruction, the number of problems and display pages has been increased.

Some of the basic formats are as follows:

- 1) The child is asked to perform the union of two sets and to show the correct answer by placing his light pen in the box next to that set.

$\{ \textcircled{0} \} \cup \{ \textcircled{8} \} = \dots$
☐ { $\textcircled{0}$ }
☐ { $\textcircled{8}$ }
☐ { $\textcircled{0}, \textcircled{8}$ }

- 2) For each problem, the child adds the two numbers, and puts his light pen on the correct answer on the keyboard below.

$1+2 = \underline{\hspace{1cm}}$	$3+0 = \underline{\hspace{1cm}}$
$2+2 = \underline{\hspace{1cm}}$	$1+5 = \underline{\hspace{1cm}}$
$3+4 = \underline{\hspace{1cm}}$	$2+3 = \underline{\hspace{1cm}}$
0 1 2 3 4 5 6 7 8 9	

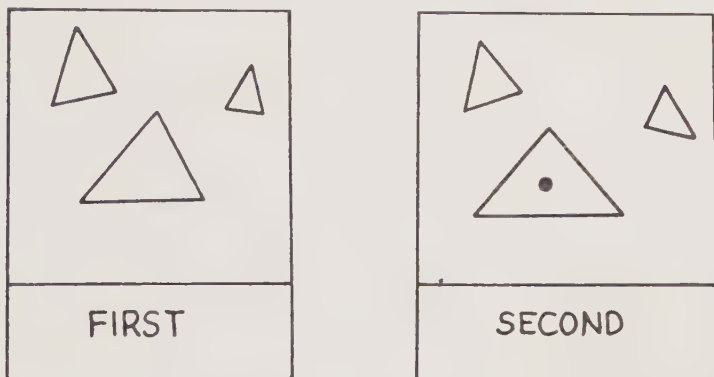
- 3) The child sees the first problem and is told to put his light pen on the number which “comes before”. The format of the keyboard has been changed for this type of problem.

$\underline{\hspace{1cm}} 5$	$\underline{\hspace{1cm}} 7$
$\underline{\hspace{1cm}} 3$	$\underline{\hspace{1cm}} 9$
0 3 6 7 4 5 2 6 1 9	

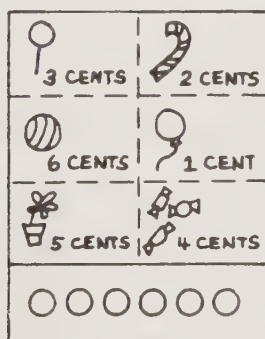
In the first example, the child is simply given the next problem after a correct response. If he responds incorrectly, an X is projected across the box to which he pointed and is then removed, and the child is given another trial. For each of the formats 2) and 3), the child sees only one problem at a time and the keyboard at the bottom of the page. The correct answer will be given by the second projector after the child has 1) made a correct response so that he can see the answer he gave on the screen or 2) made a pre-determined number of incorrect responses and is to go on to the next problem.

- 4) This example was taken from the presentation of “first, second, third” and the concept of ordering. The child is instructed to order the trian-

gles from largest to smallest. The word "first" is shown at the bottom of the page from the second projector, and the child touches the largest triangle with his light pen. A correct response causes a dot to appear on the triangle, and then the word "second" replaces "first" at the bottom of the page. He then repeats the process.



- 5) The concept of counting money is shown in this example. The first problem is shown to the child and he is told to put his light pen on the correct number of pennies starting at the left. A correct response sends him to the next problem. An incorrect answer causes the shutter to be closed for a moment and then the child is given a second trial.



Some of the material dealing with numbers is better suited to the scope since the user can type answers and the material he types can appear in the problem shown on the scope. For this reason, we plan to prepare the material for some concepts for use on the scope in addition to preparing the same material for the optical display units. Eventually some children will use the optical display units while others will use the scope. The results will be compared to

determine whether the type of response (multiple choice or constructed response) affects the child's learning.

3.3 PROGRAM LOGIC

The first-grade program is organized very simply at this point, and makes use only of the optical display unit. It consists of approximately fifty sections, each introducing a new "concept," followed by problems to be worked. The number of problems varies considerably from concept to concept. A criterion of successive-number-correct has been set for each section. If the student meets the criterion he proceeds immediately to the next concept (or to review section if scheduled); if he cannot meet the criterion he finishes the section and goes into a remedial branch which presents the same concept in a new way. Eventually a second remedial branch will be included for the student who still has not acquired the concept. For the first run of the material, however, a teletype printout will indicate to the experimenter that a child is making continuous errors and the experimenter will then instruct the child personally. (Criteria to be included later in the program are total number of errors, latency, and weighting of different problems. These data are now recorded by the computer for use in analyzing the progress of the students.) A short review section has been written for each concept. As the child proceeds through the material, a group of four review problems from each of four or five previous concepts will be presented to him. If he responds correctly to two of the four problems he will continue through the review to the next concept. If he makes more than two errors, the computer will register these data, and will present an extensive review of the particular concept to the child on the next day. In this way, the child will be tested continually on his retention of previous concepts.

4.1 FOURTH-GRADE INSTRUCTIONAL MATERIAL

The program is being written for the "average" student who has had no previous contact with "modern mathematics" or with the concept of a set. The first unit of the program begins with a review of:

1. place value
2. addition
3. subtraction
4. multiplication
5. division
6. word problems and equations

The second unit develops the following topics:

1. operations on numbers
2. commutative law

3. associative law
4. stating reasons for deductive steps in problem solving
5. distributive law

Of the two units started, the first is complete and ready to be programmed. The second is approximately 20% complete as of this date.

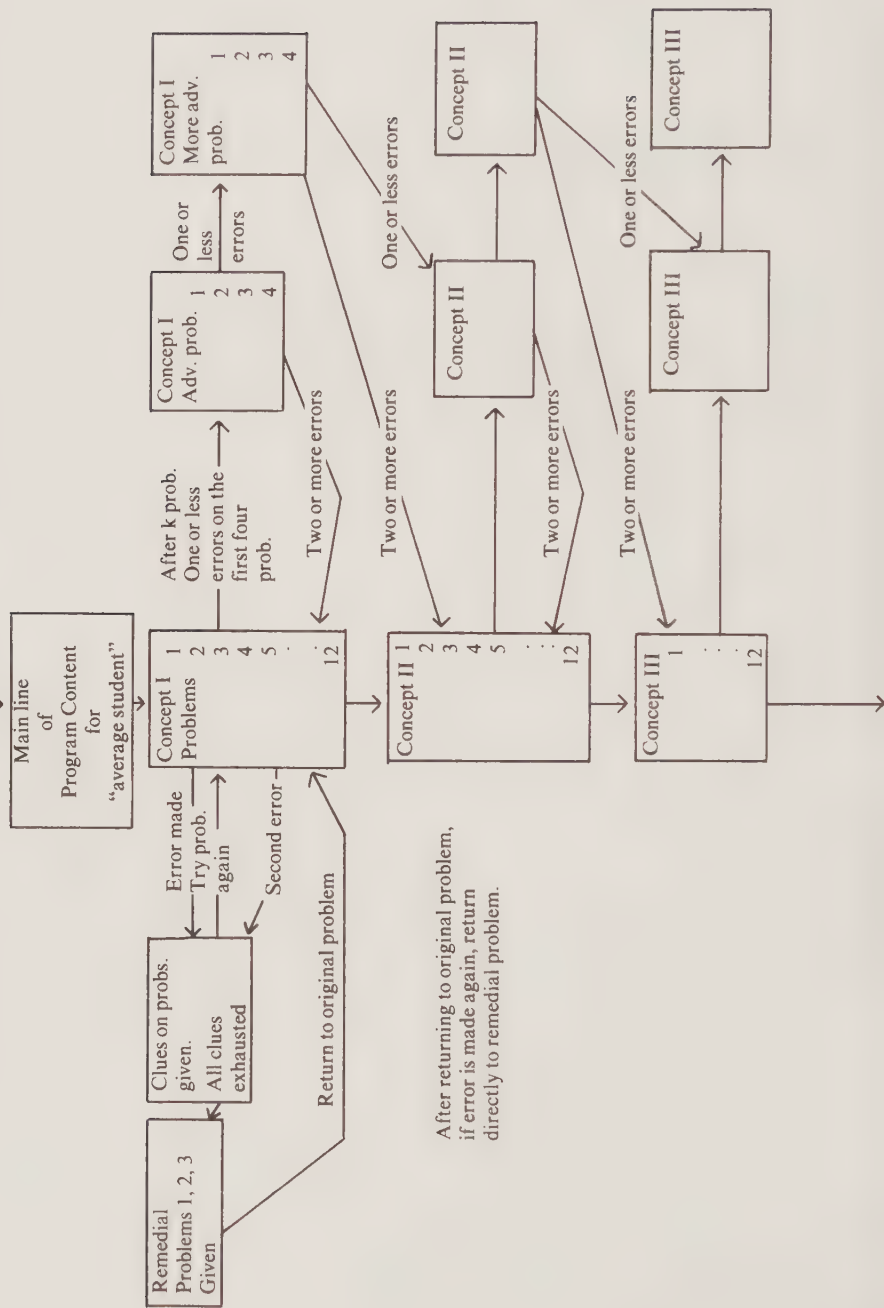
4.2 PROGRAM LOGIC

Each instructional unit is broken into a number of learning blocks which consists of the presentation of a major concept and problems to be worked by the student. The format of the program is illustrated on the following pages. The first chart shows the different ways in which students of varying abilities will proceed through the material and gives an indication of the criteria for moving ahead or back to remedial material in the program. Each "main line" concept will consist of:

1. an introduction
2. explanations of concept
3. one or more sample problems
4. further exercises ranging from simple to more difficult

If a student makes an error while working a problem, he is given a clue so that he may correct the error before continuing. If all clues are exhausted and the student has still answered incorrectly, then he is presented with three or four remedial problems of the same type. After the remedial problems are solved or the correct solution has been presented, he returns to the original problem. Advanced problems have been constructed for the bright child. If this student meets a certain criterion in the main line block, he skips to the advanced problems. As shown on the chart, it is possible without ever having to return to the "average" blocks; he will receive the introduction, explanation and sample problems, and then (provided he has met the criteria of previous lessons) go directly into a block of advanced problems. The second chart presents a more detailed description of the material in one concept block and its remedial and advanced branches. The twelve problems of one concept are displayed. A set of clues is given for the first problem, and typical remedial problems are shown at the far left. Two sets of advanced problems are also shown to demonstrate the increasing difficulty of these exercises. A special orientation unit is being prepared to familiarize the student with the response capacities of the scope and the optical display unit so that when he is presented with a problem (e.g.: add: $34 + 47 = \underline{\quad}$) he will respond in a prescribed manner. (In this example, 1C18. This represents the following train of thought: $4 + 7 = 11$; record the 1, carry the 1 ten; $3 \text{ tens} + 4 \text{ tens} + 1 \text{ ten} = 8 \text{ tens}$; thus the answer is 81.) We have had the students type in "C" for carry and the number he

Chart 1



*Chart 2: Sample Concept in Main Line
The Distributive Law for Multiplication*

REMEDIAL PROBLEMS (4)	CLUES		Criteria	ADV - I
	1. Check your operation symbols	1. $m = 5 \times (4 + 3)$ (multiple choice on optical display unit and constructed on scope)		
$4 + 3 = \underline{\hspace{1cm}}$			$k = 4$	1. $a = 12 \times (7 + 8)$
$5 \times 4 = \underline{\hspace{1cm}}$			$m = 3$	2. $b = 12 \times (10 + 11)$
$5 \times 3 = \underline{\hspace{1cm}}$			$n = 4$	3. $c = (9 + 8) \times 12$
$5 \times 7 = \underline{\hspace{1cm}}$				4. $d = 12 \times (10 + 7)$
	2. Check your grouping symbols			
	3. Check your addition and multiplication facts			
	Sample of Choices			ADV - II
	1. $(5 + 4) \times (5 + 3)$	2. $n = 3 \times (5 \times 4)$		1. $9 \times (30 + 6)$
	2. $(4 \times 5) + (4 \times 3)$	3. $p = (2 + 6) \times 4$		2. 12×24
	3. $20 + 7$	4. $q = 6 \times (3 + 2)$		3. $(40 + 7) \times 12$
	4. $(5 \times 4) + (5 \times 3)$	5. $r = 8 \times (4 + 5)$		4. 25×45
		6. $s = (5 + 7) \times 7$		
		7. $t = 9 \times (6 + 8)$		
		8. $u = (7 + 5) \times 8$		
		9. $v = 5 \times (6 + 5)$		
		10. $w = (9 + 6) \times 4$		
		11. $k = 3 \times (8 + 7)$		
		12. $j = (6 + 7) \times 6$		

wishes to carry to the next column. An alternative way would be to restructure the problem by typing in $34 + 47$, 1C18, and the problem would appear on the scope in the following format:

$$\begin{array}{r} 1 \\ 34 \\ +47 \\ \hline 81 \end{array}$$

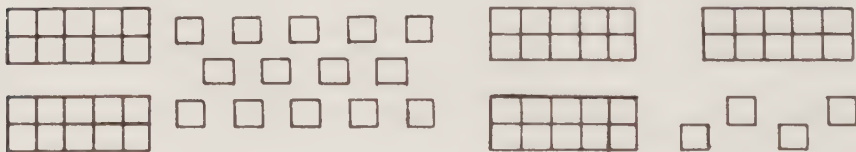
As seen in this second mode of response, 10 characters were required to demonstrate a solution. A simple subtraction problem requires careful consideration on the part of the student as he responds:

$$\begin{array}{r} 1317 \\ \text{subtract: } 247 \\ - 48 \\ \hline 199. \end{array}$$

Response: R47, 3, 17, 9, R23, 1, 13, 9, 1. The thought process behind this response is: Regroup 47 into 3 tens and 17 ones, $17-8=9$. Regroup 23 into 1 hundred and 13 tens, $13-4=9$. $1-0=1$. This type of response corresponds directly to the sequence of steps taken by the student to solve a particular problem. Thus, once the typewriter "code" is acquired by the students, the machine data can indicate exactly where he has made his mistakes.

The following examples taken from the lesson material reflect the versatility of the scope:

- ↓ ↓ ↓ ↓
1. 2321 is thousands hundreds tens ones
An arrow appears over the blank to show the desired order for responses
2. squares move to this form



2 tens and 14 ones = 3 tens and 4 ones.

3. Mary and Tom together have the same amount of money as Carol and Bill together.

Mathematical sentence: $m + t = c + b$ (typed in by student)

5. Mathematical Logic Program

Experiments in past years show that gifted elementary school students taught by teachers trained in logic can score better than 90% as high on logic tests as Stanford University logic students. In view of these test results, a program for instruction in mathematical logic has been prepared for gifted sixth graders which makes full use of the versatility of the computer-based laboratory. The scope in this laboratory has the capacity for immediate evaluation of response data and for the acceptance of any valid step in a proof, making possible the presentation of new stimulus items which are contingent upon the student's immediately previous performance.

5.1 PRELIMINARY EXPERIMENTS

In the spring of 1964 some of the most frequently used rules of sentential derivation were programmed for presentation on the scope. Several sixth-grade students who were soon to complete two years of logic study were given written instructions and 21 derivations to be carried out. They took readily to performing the derivations at the keyboard even though there was some change from the symbolism they had been using. It was clear that the mechanics of operation was no handicap.

A second experiment was conducted with two boys who had just completed fifth grade. One, a gifted child, had had classroom instruction in logic. The second boy was above average intelligence and had not been instructed in logic. During the summer these students completed 21 lessons; the first in 32 sessions averaging 29 minutes each, the second in 38 sessions averaging 37 minutes each. Continuous observation was made and a record kept of questions asked by the students and of additional instructions and hints offered by the instructor. The lessons covered almost precisely the material that had been covered before by the experienced student. Nevertheless, he did not seem to find the work boring. Thus we could presume that sufficient drill to achieve a desired degree of overlearning does not offer a serious motivational problem. By the completion of lesson 21 the inexperienced student was doing work comparable to that of the experienced student.

5.2 INSTRUCTIONAL MATERIAL

At first, the student is presented with logical symbols which are not interpreted for him. He is to view the symbols as concrete objects. Using arbitrary rules of derivation introduced one at a time, the student manipulates the symbols as he would pieces on a game board, in order to arrive at a required conclusion. Thus, the operations are made on a concrete level and the symbols or "pieces" are not looked upon as being abstract carriers of "meaning" or as representing something other than themselves.

At the next stage, the “pieces” are regarded as carriers of the possible truth values of True or False. Given a truth value for each atomic sentence, the student constructs diagrams to determine the truth value of complete molecular formulas. (Atomic formulas are the simplest type of formula, and have a single, important property—their truth values. They combine with symbols called connectives to form more complex formulas [molecular formulas]. Connectives make true or false claims about the truth values of the atomic formulas. In this way, the student understands that the truth value of a molecular formula is a function of the truth values of the included atomic formulas, and, consequently, that the connectives are “truth-functional”.) The student is shown that the rules initially introduced as arbitrary never allow us to derive a false conclusion from true premises.

As the next step, the student constructs truth tables which give the truth value of a molecular formula under every possible combination of truth values for the atomic formulas of which it is comprised. Now it is possible to demonstrate the validity of each rule. The student learns to define a tautology as a molecular formula that is true under all possible combinations of truth values for its atomic formulas. The truth-functional definitions of the connectives in the single formula corresponding to an argument are analyzed to show that an argument is valid if and only if it is a tautology. From this point on, the validity of all new rules is demonstrated when the rule is introduced. These include the methods of conditional proof, indirect proof, and biconditional replacement.

Finally the formal concepts of invalidity of arguments, and consistency or inconsistency of sets of premises, are studied and methods for their proof developed. Some metalogical discussion is given on the development of a logical system and how the computer applies it.

A spiral approach is followed in the lessons: only five rules of derivation are introduced before presentation of the concept of truth in lesson eight. In that lesson the concept of validity is introduced and several rules are immediately justified. Then truth-functional definitions are presented in truth tables and are used immediately to develop English approximations for the connectives. English examples are used as illustrations. Full translation of English molecular sentences is postponed until lesson twenty-seven.

5.3 PROGRAM LOGIC

The program is logically broken into two parts: 1) the construction of proofs and 2) a general scope program which is used for presenting additional material.

During the construction of proofs the scope is used as a worksheet on which the steps of the derivation are carried out. The following example shows a problem as it is seen on the screen.

Derive R:

P	(1) $P \rightarrow Q$
P	(2) P
P	(3) $Q \rightarrow R$

At this point the student responds by typing "AA 12 TAB KEY." Each character is displayed on the scope as it is typed. Errors may be corrected by pressing the backspace key which causes the last character entered by the student to be "erased."

The student's input is then edited for errors in "grammar" and he is required to correct any errors found. When the form of the line is correct the validity of the construction requested is checked. The program will then respond with the corresponding formula or give the student an explanation for his mistake. No attempt is made to determine the strategic appropriateness of the steps. Thus in problems with any degree of complexity there are many ways of arriving at the desired conclusion and the path taken is left completely to the ingenuity of the student.

As the program has progressed through stages of pilot runs, most of the rules have been expanded and have become more abstract in their definitions and response characteristics. Many of the rules allow for the introduction of formulas by the student, i.e., premise introduction for conditional and indirect proofs. The example given above would appear as below when completed. Student input is underlined in the example:

Derive R:

P	(1) P Q
P	(2) P
P	(3) Q R
<u>AA 1 2</u>	(4) Q
<u>DA 3 4</u>	(5) R
CORRECT	

The general scope program consists of the following:

a) English text display with or without missing words to be typed in by the student.

b) Multiple choice questions. Responses are made by pointing to a displayed box with the light pen. Incorrect responses cause the box pointed to be replaced by an X and the student then proceeds to select another answer.

c) Filling in the truth assignments for a formula truth table. A small T and F (TRUE, FALSE) are displayed with each desired entry. The student then responds by pointing to one or the other. If a correct response is made the other choice is dropped. If an incorrect response is made the program flashes a "NO!" for approximately two to three seconds at the point of error and then

replaces the entry with the correct response (since it is the only choice left) in high intensity. When the exercise is completed all incorrect responses are displayed with a greater intensity to draw the student's attention to those entries.

Construction of truth tables is introduced by having the student assign all possible truth assignments to a formula repeated several times on the scope and then the program slowly merges the formulas and truth assignments to form the familiar truth table.

d) An exercise in removing unnecessary parentheses is done by displaying a formula which contains parentheses that could be removed without changing the meaning of the formula. This is done by displayed "pointers" which follow the light pen. Thus the student points to the parentheses which he thinks can be removed. The program checks his request for validity and responds with appropriate error messages or the parentheses fade from the scope and the formula slowly closes the gaps left. This process is continued until the student indicates he believes no more may be removed. The program then proceeds to the next problem or tells the student that there are still parentheses left which he must remove.

e) Exercises in pointing out the order of dominance of connectives in formulas are done by displaying boxes above each connective and having the student point to these in the required order. Errors cause the box indicated to be replaced by an X; correct responses cause the box to be replaced by the appropriate numeral indicating the order of dominance.

6. Project Staff

The project director is Dr. Patrick Suppes, Professor Philosophy and Statistics, Stanford University. Approximate staff allocations to the various programs are as follows. The first-grade program: Mrs. Dolly Kyser, Miss Anne Hyslop; fourth-grade program: Mr. Max Jerman, Miss Phyllis Cole; mathematical logic program: Mr. Frederick Binford, Professor Dana Scott; chief computer programmer: Mr. Dow Brian.

7. Project Support

The computer-based laboratory for learning and teaching has been constructed primarily from a grant from the Carnegie Corporation of New York. The computer-based mathematics instruction (CBMI) project is supported by a grant from the Course Content Improvement Section, National Science Foundation.

18. The Teacher and Computer-assisted Instruction

EDUCATORS HAVE SHOWN increasing interest in the use of computers for classroom teaching, especially during the last year or two, and they have raised a number of fundamental questions that need analysis and discussion. The purpose of this article is to acquaint the reader with some of the ways that computers can be used for instruction, and to answer, at least briefly, some of the questions that are frequently asked about computer-assisted teaching.

Some of the most important questions are: How can the computer help in individualizing instruction? How might it change the teacher's role? How will computer-assisted instruction change teacher-administrator relationships? Will it lead to impersonality and regimentation in the classroom? How can teachers play a part in planning and using computers for instructional purposes?

Let us begin by looking at a student seated at a console or station that is connected by a telephone line to a central computer. The console will usually contain a typewriter keyboard that the student can use to "talk to" the computer and a television screen that can display written messages as well as drawings, equations, and other graphic material. In many cases, the student will also have a "light pen," which he can use to select answers to the problems shown on the screen; he can even erase or change the images that appear. The computer talks to the student through a pair of earphones or a loudspeaker, thus providing him with the verbal communication necessary for effective learning, particularly when new concepts are being presented.

The central computer, which controls the presentation of information and evaluates the students' responses, need not be in the school but can be located

at a central point in the school district. Because of its great operating speed, one large computer can serve many students, and a number of students can "time-share" the computer simultaneously.

Computer-assisted instruction is possible with only one console per classroom, which would be shared by many students during the school day. In a more expensive and elaborate arrangement, a classroom would have a large number of consoles, and each student could spend considerable time—as much as an hour and a half a day—at the console. It is important to emphasize, however, that in either arrangement the student would still be spending most of his time in the regular class setting, directly under teacher supervision.

Recent research indicates that students at all age levels come to feel at home with this sort of equipment and are quite willing to make its use a part of their daily school experience.

The student and the computer program may interact at three distinct levels, each of which comprises a particular system of instruction. (This use of the word *system* also corresponds to its use in the computer industry.)

Individualized drill-and-practice systems. This kind of interaction between the student and the computer program is meant to supplement the regular teaching process. After the teacher has introduced new concepts and ideas in the standard fashion, the computer provides regular review and practice of basic concepts and skills. In elementary school mathematics, for example, each student would receive 15 or 20 exercises a day. These would be automatically presented, evaluated, and scored by the computer program without any effort by the classroom teacher.

In addition, these exercises can be presented to the student on an individualized basis, with the brighter children receiving harder-than-average exercises, and the slower children receiving easier problems. One important aspect of this individualization should be emphasized: In the drill-and-practice computer system, a student need not be placed on a track at the start of school in the fall and held there the entire year. At the beginning of each new concept block—whether in mathematics or in language arts—a student can be "recalibrated" if the results indicate that he is now capable of handling more advanced material.

Drill-and-practice work is particularly suitable for the skill subjects that make up a good part of our curriculum. Elementary mathematics, reading, and aspects of the language arts, such as spelling, elementary science, and beginning work in a foreign language, benefit from standardized and regularly presented drill-and-practice exercises.

Tutorial systems. In contrast to the individualized drill-and-practice systems, tutorial systems take over the main responsibility for helping the student to understand a concept and develop skill in using it. Basic concepts,

such as addition or subtraction of numbers, can be introduced by the computer program in such systems. The aim is to approximate the interaction a patient tutor would have with an individual student.

In the tutorial programs in reading and elementary mathematics that we have been working with at Stanford University for the past three years, we have tried hard to avoid having slower children experience any initial failures. On the other hand, the program has enough flexibility to avoid boring the brighter children with too many repetitive exercises. As soon as the child shows that he has a clear understanding of a concept by successfully working a number of exercises, he is immediately introduced to a new concept and new exercises.

Dialogue systems. Dialogue systems are computer programs and consoles that enable the student to conduct a genuine dialogue with the computer. It will be some years before we are able to implement dialogue systems in classrooms, because a number of technical problems remain unsolved. One problem is the difficulty of devising a computer that can "understand" oral communication, especially that of young children. We would like to have a computer that would respond to questions. To attain this interaction, the computer would have to recognize the speech of the student and to comprehend the meaning of the question. It will be some time before a computer is developed that will be able to do either of these with any efficiency and economy.

Dialogue systems have been mentioned here in order to give readers an idea of the depth of interaction we ultimately hope for. Drill-and-practice systems and tutorial systems, on the other hand, are already in operation on an experimental basis and will no doubt find an increasing application throughout the country in the next few years.

Effective programs of computer-assisted instruction now exist for elementary school mathematics, parts of language arts programs (particularly reading and spelling), and various topics in mathematics and science at the secondary and university levels. The programs have been developed primarily at universities, the following of which are currently the main centers of activity: Stanford, Illinois, Michigan, Texas, Pennsylvania State, Pittsburgh, Florida State, and the Los Angeles campus and the Irvine campus of the University of California.

Let us now look at some of the most frequently asked questions about computer-assisted instruction:

What role can computers play in individualizing instruction? The theme of individualized instruction has been prominent in American education for over 50 years. Psychologists have shown that individuals differ in their abilities, their rates of learning, and often even in their general approaches to learning. Unfortunately, the cost of providing individualized instruction that adapts to

these differences is prohibitive if it depends on the use of professional teachers. For example, consider what it would cost to reduce present classroom size to four or five students per teacher.

The computer offers perhaps the most practical hope for a program of individualized instruction under the supervision of a single teacher in a classroom of 25 to 35 students. The basis for this practical hope is the rapid operation of the computer, which enables it to deal on an individual basis with a number of students simultaneously and thus lowers the cost per student of the computer.

How will the computer change the teacher's role? Drill-and-practice systems will modify the teacher's role only slightly. What they will do is relieve teachers of some of the burden of preparing and correcting large numbers of individualized drill-and-practice exercises in basic concepts and skills and of recording grades.

The teacher will be more significantly affected by tutorial system. Let us consider a concrete example: teaching addition and subtraction of fractions at the fourth grade level. The computer will provide the basic ideas and the procedure of how to add and subtract the fractions. The program will probably be written so that if a student does not understand the basic concepts on first presentation he will receive a second and possibly even a third exposure to them.

The new role of the teacher will be to work individually with all students on whatever problems and questions they may have in assessing and handling the new concepts. Tutorial systems allow teachers greater opportunity for personal interaction with students.

How will computer-assisted instruction affect teacher-administrator relationships? Teachers and administrators should be able to develop even closer relations in a setting where computers are used to aid instruction. The information-gathering capacity of the computer enables administrators to have a much more detailed profile and up-to-date picture of the strengths and weaknesses of each area of curriculum. As they develop skill in interpreting and using the vast amount of information about students provided by the computer, administrators and teachers should be able to work together more effectively for improvements in curriculum.

Is there a danger that the computer will impose a rigid and impersonal regime on the classroom and even replace teachers? Contrary to popular opinion, the computer's most important potential is to make learning and teaching *more* an individual affair rather than *less* so. Students will be less subject to regimentation and moving in lockstep because computer programs will offer highly individualized instruction. In our own work at Stanford, for example, we estimate that the brightest student and the slowest student going through our tutorial program in fourth-grade mathematics have an overlap of not more than 25 percent in actual curriculum.

The computer program is neither personal nor impersonal. The affect and feeling of the program will depend on the skill and perceptivity of those responsible for constructing it.

There seems to be little reason to think that computers will ever replace teachers or reduce the number of teachers needed. The thrust of computer-assisted instruction is to raise the quality of education in this country, not to reduce its cost. In any sort of computer-assisted instructional system used in classrooms in the near future, teachers will continue dealing with children on an individual basis and doing most of the things they are now doing during most of the school day with only slight changes.

Finally, we emphasize once again that no one expects that students will spend most of their school hours at consoles hooked up to computers. They will work at consoles no more than 20 to 30 percent of the time. All teachers everywhere recognize the help that books give them in teaching students. The day is coming when computers will receive the same recognition. Teachers will look on computers as a new and powerful tool for helping them to teach their students more effectively.

19. Impact of Computers on Curriculum in the Schools and Universities

1. Introduction

My intention in this paper is to survey the kinds of impacts that computers are already having or will have in the near future on curriculum in the schools and universities. On other occasions I have tried to analyze the general issues connected with the use of computers for instruction (for example, Suppes and Morningstar, Chapter 1, 1972). Here I shall avoid, by and large, the general issues and aim at particular examples and the implications of these examples for the future.

In analyzing various examples I want to emphasize three aspects of the use of computers for instruction. The first is the aspect of diversity, especially in terms of content and entire subject matters. With extensive use of computers, students can be offered a diversity of courses far beyond what is feasible on other grounds. The second is the sense of individualization that can be achieved by computer-assisted instruction, both in terms of actual rate of progress of the student and also in terms of the convenience of time and place for the student.

The third is the matter of productivity of faculty in the face of declining budgets, especially in the fact of faculty sizes that have little hope of being increased for the remainder of this century. In the United States anyway this is certainly true of both the schools and the universities. The implications of this have hardly yet begun to be appreciated by many people in education. However, the recent prediction by the Department of Education of the State of New York that the number of students in colleges and universities in 1990

will be 23% less than the number currently enrolled has been a sufficiently dire prediction to arouse broad interest. The associated prediction that about half of the private colleges in New York will be closed by that date because of financial difficulties has also been a subject of sharp discussion. It will not be my concern in this paper to try to make the case for the exact percentage prediction in New York but rather to accept the broad facts on which it is based, namely, the demographic data on number of students and the anticipated bounds on future budgets. The central question becomes how we can continue to enrich the educational opportunities of students in the face of declining faculty sizes.

I begin with the schools and then turn to the universities.

2. The Schools

I divide my discussion of curriculum in the schools into three parts, first dealing with basic skills for disadvantaged students, next considering problem solving and programming, and third, special programs for gifted students. I emphasize once again that what I have to say deals almost entirely with the situation in the United States, because I am not deeply familiar with activities in other countries and because I would judge that up to the present the bulk of the use of computers for instruction has taken place in the United States, although I would assume that matters will be changing rapidly in this respect over the next several decades.

2.1. BASIC SKILLS OF DISADVANTAGED STUDENTS

My own earliest work in computer-assisted instruction focused on basic skills of disadvantaged students. In particular I concentrated on elementary mathematical skills, and my colleague Richard C. Atkinson concentrated on basic reading skills. Since this work first began at Stanford University in 1963 a large number of research and evaluation studies have been published, and I shall not attempt to review them here. A history of the effort is to be found in the two books, Suppes, Jerman and Brian (1968) and Suppes and Morningstar (1972). Out of those efforts has grown a fairly widespread use in the United States of computer-based supplementary courses in elementary mathematical skills and concepts, basic reading skills and language skills. These courses have been aimed especially at disadvantaged students who have been a central concern of elementary education in the United States since the passage of the Elementary-Secondary Education Act of 1965 in the Johnson Administration.

The impact of the work at Stanford, the subsequent commercial developments at Computer Curriculum Corporation with which I have been closely

associated, and the work by other people in the United States on these basic skills as a primary focus of education for students in the age range 6 to 12 years have not led to radical changes in the content of curriculum. Indeed, it was not the intent of those developing these computer-based courses to lead to changes in the content of curriculum.

Furthermore, whatever happens to change the content of the curriculum in the elementary schools or primary schools over the next 50 years, I would be enormously surprised if the basic elementary language, reading and mathematics skills do not continue to be an important part of the curriculum.

The impact of using computers to develop these skills and to provide a powerful approach to remediation has been to change the style and not the content of the curriculum. What is especially important about this change in style is the presentation to each student of exercises at his current level of accomplishment, and the chance to move ahead or to slow down independently of the progress of other students. The computer is being used in this case to fine tune curriculum that has been taught in one form or another since systematic education began, but to fine tune it, in a way that has not been previously possible, to the needs of individual students.

2.2. PROBLEM SOLVING AND COMPUTER PROGRAMMING

A more radical change in curriculum content has appeared in the use of computers, best exemplified by the work of Seymour Papert and his colleagues at the Massachusetts Institute of Technology. In a variety of imaginative ways, they have been teaching young students to program, especially in the language LOGO, and in the process to tackle and solve a variety of problems, from writing music to playing games. A number of other people around the world have also become involved in this kind of work, and there is little doubt that it will in time have a major impact on curriculum in the schools. The impact as yet is relatively small, but only because it takes time for computer facilities to be available in a large number of schools and, even more important, for teachers to feel at home with these new approaches to education.

A good reference is the delightful article by Papert and Solomon (1972).

2.3. LOGIC FOR GIFTED STUDENTS

My own interest in teaching nonstandard curriculum to gifted students goes back nearly 20 years and predates my interest in computers. In the fall of 1956 I brought into my college logic course a group of able students in the age range 11 to 14 from neighboring schools, and I have in one way or another been continuously involved in the teaching of elementary mathematical logic

to students in the schools since that time. An early report on our work is to be found in Suppes and Binford (1965). As we moved from the first experiment at Stanford in 1956 to organizing classes in schools surrounding Stanford, it soon became apparent that the central difficulty was not in the capacity of young students to master the skills and concepts of formal logic but rather in the lack of training and, in some cases, aptitude for these matters on the part of their teachers. Despair at being able to bring a rich supplementary curriculum to gifted students in the schools was one of the things that motivated me to become involved in the use of computers for instruction in 1963.

Our first demonstration programs were running on the PDP-1 at Stanford in 1963 and embodied a full proof checker for elementary logic without quantifiers. This program was available not long after the very first of the proof checkers had been written anywhere in the world. On and off over the past decade we have engaged at Stanford in a variety of efforts to bring formal logic to able students in the schools. A fairly detailed description of the earlier efforts was given in my survey of computer-assisted instruction at Stanford in Bordeaux, France, several years ago (Suppes, 1972).

More recently, we have been working with a group of extremely gifted students in the age range 10 to 15 years. By *extremely gifted* I mean a group that has been selected with IQs of 170 or more. Our approach to these students has been different from any of our approaches in the past. We have placed terminals in their homes, and during the evening and on weekends they dial up the PDP-10 system in the Institute at Stanford for work in several courses but especially courses in logic. They begin with an elementary logic course aimed at able students of age 12, and when that is completed move on to the computer-based logic course at Stanford, which I discuss in more detail below. In addition, a number of the gifted students have also used their terminals at home to take courses in LOGO and in BASIC.

One of the striking results has been the enormous variability within what is on the surface a homogeneous group of highly gifted students. The amount of interest, the amount of work done and the rate of learning between the best student and the most indifferent student in the group is astoundingly large. This is what one is to expect when individualization without constraints, as in the case of these home-based terminals, is left to follow its own natural course. The very best student, for example, completed the relatively difficult introductory course in logic at Stanford in about four weeks, faster than any Stanford student taking the course at the same time. Other gifted students have found that they were not interested in this direction of work at all and have asked that their terminals be removed after a few weeks. In order to get a sense of the variability in interest we have left the whole process almost completely unstructured in the initial experiment.

To accomplish the most positive results, on the other hand, it is quite clear

to us from our current efforts and from a variety of experimentation on similar self-paced and nonformal educational settings that the imposition of some minimal structure is probably almost always beneficial.

One difficulty with doing more with these highly gifted students is that their programs are enormously full already, at least within the frame work of school — and parent-suggested activities of the kind frequent in California. These students are busy from morning to night in a variety of programs, from music lessons to jewelry making to competitive athletics. Unless some change is made in the standard school regime there is in fact a relatively small amount of time available to them to pursue in any deep sense supplementary work in mathematics or computer programming. The capacity is there but the social organization is relatively restricting. In other countries and in other settings the situation may well turn out to be very different.

3. The Universities

Because of the constant or slightly declining enrollments predicted on the average for the next decade and a half in American universities, I would like to concentrate on the question of how computers may be used in universities to offer a full range of instruction with a constant or declining number of faculty.

A second general point I want to emphasize is that, contrary to my own earlier work that was concerned with courses that would be used by a very large number of students, I now see the first important and significant use on a broad scale of computers at the university level may well be in offering a large number of courses that ordinarily have a small enrollment. I give some examples of this phenomenon from several areas of university teaching, but the basic point is worth general consideration. If a professor of some seniority offers a course to a class size ranging from 2 to 6 students, then it must necessarily be an expensive course to teach in terms of the total number of student hours. For example, let us assume that there are four students in the class and that the class constitutes, as it might at Stanford, half of the teaching load for one of the three terms of the year. Assuming the professor's salary is \$20,000 per year and the class meets 3 days a week for 11 weeks, then the cost of instruction is \$25.25 per student hour, just in terms of the professor's salary and excluding any other costs.

Those of you familiar with the current financial situation, at least of American universities, will recognize quickly enough that courses in this cost range will simply not be offered for any substantial number of years. The pressures to eliminate such courses are growing and will continue to increase.

On the other hand, if, by using computer facilities, a faculty member can offer four such courses simultaneously then the cost of instruction is reduced

to \$6.31 per student hour, which for advanced work is more reasonable, but is still a figure that is easy to reach even using a sophisticated computer system.

If, in contrast, we tackle the cost of teaching a class of 200 students on the same basic assumptions of a professional salary of \$20,000, this course being one-sixth of a teaching load and the class meeting 3 times a week for 11 weeks, the bare cost of instruction is about \$.51 per student hour, which is a figure that is not easy to match by use of computer facilities. These simple computations show why I think the important use of computers in teaching is in the presentation of courses of a moderate degree of specialization and at a moderate level of difficulty, rather than at the level of large elementary courses.

With these economic considerations in mind I would like now to turn to a survey of some of the efforts to implement such courses in three areas: the teaching of foreign languages, the teaching of computer programming and the teaching of mathematically oriented courses.

3.1. FOREIGN LANGUAGE TEACHING

Over the past ten years we have had experience at Stanford in teaching French, German, Russian and Chinese by computer. The highly experimental work in German is described in Levine (1973). The most extensive work by far has been in the teaching of Russian and has been under the direction of Professor Joseph Van Campen, Chairman of the Department of Slavic Languages at Stanford. The initial effort was aimed at the teaching of elementary Russian, and a positive evaluation of this work is reported in Suppes and Morningstar (1969). The more significant developments in line with the computations of costs given above have occurred in the past several years. Van Campen and his younger colleague, Professor Richard Schupbach, have embarked on the extensive use of the Institute's computer facilities to teach specialized courses in Russian language and linguistics that are ordinarily taken by only two or three students. The course in Old Church Slavonic, developed by Van Campen and which is now about 80% computer based, is described in Van Campen (1973). A corresponding course on the linguistic history of the Russian literary language is described in Schupbach (1973). Van Campen and Schupbach have plans to develop approximately 10 to 12 courses in Slavic linguistics that they can offer on a regular basis each year and with a staff consisting just of the two of them, together with some teaching-assistant help. In considering their development of these courses it is to be emphasized that this is about the only way they will be able to develop such courses, because given the present enrollment in Slavic languages at Stanford and given the current and continuing financial constraints on the University there is no hope of adding an additional faculty member to their group to aid in the teaching of these courses.

3.1. COMPUTER-PROGRAMMING COURSES

The reflexive principle of using the computer to teach courses about computer programming will undoubtedly have wide application. At Stanford, thus far the computer-assisted instruction courses in computer programming at the university level have been restricted to a short introductory course in BASIC (Barr, Beard and Atkinson, 1974) and part of the introductory course on LISP taught by John McCarthy and Cordell Green. It is our intention in the future to offer considerably more in this direction, and I anticipate that this is one of the ways that we shall be able to offer a sufficiently wide variety of specialized courses to meet the increasing demands of students interested in the many different aspects of programming.

One of the central difficulties of deeper moves into teaching computer programming entirely by computer is the problem of program verification. The writing of programs that understand the programs students write or even the weaker requirement of having a program checker that students are required to use to verify that their programs are correct is still relatively difficult even for simplified proper subsets of widely used programming languages. I will have more to say about this below in discussing the corresponding problem for mathematical proofs. Suffice it to say at this point that the difficulties are greater for verification of mathematical proofs, because we have for first-order logic Gödel's completeness theorem that provides an exact match between syntax and semantics: one sentence is a logical consequence of another if and only if it can be derived by use of syntactical rules of inference from the first. We can formalize axiomatic set theory in first-order logic and thereby expect to have powerful syntactical methods for checking the validity of proofs, methods that are more on the surface than are the corresponding methods required for verification of program correctness.

3.2. MATHEMATICALLY ORIENTED COURSES

The most extensive effort in using computers for instruction at Stanford is currently centered around the following two courses, *Introduction to Logic* and *Axiomatic Set Theory*. The course in logic has been developed over a number of years and is a direct outgrowth of the work in logic for the schools mentioned above, which first began in 1963. The course in set theory has in turn grown out of the university logic course.

Introduction to Logic. The logic course is one in which the students do essentially all of their work at computer terminals. There are no lectures, but a teaching assistant is available during a number of hours of the day for consultation. This course has been running for several years and is a regular part of the curriculum; by now it is the only version of introductory logic taught at Stanford. It is offered every term, three times a year, and the enrollment cur-

rently runs between 50 and 80 students a term. From a programming standpoint, the most important feature of the course is the proof checker. The students give proofs in first-order logic and the program checks their correctness on a step-by-step basis. The system of natural deduction embodied in the proof checker is essentially that given in Suppes (1957).

The proof checker used in the logic course has evolved over a number of years. The original one, begun in 1963, was written essentially by Dow Brian. A later version of the logic course for the schools was produced by Roulette Smith. The proof checker for the college course and the programming for the college course were initially done by Adele Goldberg (1973). The present version was written by Robert Smith, Lawrence Markosian, Lee Blaine, and Vesko Marinov. I am continuing the revision and improvement of the curriculum in collaboration with my younger colleagues, especially Edward Bolton and Lawrence Markosian.

Because the structure of the logic course up to the present and the empirical results obtained have been described in Goldberg and Suppes (1972, 1974), I shall not enter into details here.

Axiomatic Set Theory. In many respects, the logic course is deceptive as a model of how to approach mathematically oriented courses in terms of computer-based instruction. The deception arises from the fact that it is a relatively straightforward matter, though nontrivial, to have available a complete proof checker for inference rules in first-order logic. The difficulty of having reasonable rules of proof is at least an order of magnitude greater once we leave the framework of elementary logic and move to the first level of mathematics in which proofs are always given in informal style. At this next level, proofs are given informally because of the tedious and inordinate length of formal versions of the proofs. The human checker of informal proofs as given by students is tolerant and possessed of many good intuitions for evaluating arguments that are not fully worked out. Matters are otherwise in the case of the computer acceptability of such proofs. In the case of the course in set theory, by far the main effort has been devoted thus far to getting semi-reasonable proof procedures that can be used by students interactively with the program. A somewhat detailed account of where we now stand is described in a paper by my younger colleagues (Smith, Graves, Blaine and Marinov, 1975).

As far as we have been able to determine, there has not been much prior investigation of the many features that are needed to move from a first-order logic proof checker to a reasonable set of procedures for giving informal proofs in ordinary mathematics at the level of complexity of the first course in axiomatic set theory. At the present stage of our work, it seems highly unlikely that any single discovery will lead to a sharp breakthrough in how to handle informal proofs within a computer framework. On the other hand, it is

evident that there are a large number of practical constructive things that can be done and that are feasible to implement on a computer system of the size of our PDP-10. Let me give some examples of problems that we have solved and also of some that we have not.

In a first-order proof checker the first move of liberalization is from successive to simultaneous substitution for universally quantified variables. The next practical step and one that leads to a rule that is used extensively is to make the substitution of terms for variables in specification for universally quantified variables automatic, that is, by, for example, making the inference form ' $(\forall x) (F(x) \rightarrow G(x))$ ' and ' $F(a)$ ' to ' $G(a)$ ' without indicating to the program the specification of ' a ' for ' x '. Such a rule of implication is also readily strengthened to include any tautological implication that is desired as well. More elaborate examples, whose formal complexity is ignored in first-order logic, are appropriate procedures for replacing one formula by another in natural and intuitive ways without taking a large number of steps to do so, or more importantly, by not having to specify a large number of conditions that must be satisfied. It is probably especially the latter aspect of matters that needs most attention. In informal proofs, 'obvious' conditions are left implicit. Making such conditions internally explicit in the program-representation of the proof is probably the most crucial aspect of having a good computer-based proof checker for informal proofs. It amounts, of course, to having a program that in most cases will do what you expect and will only raise questions when the situation is in some way abnormal.

Another current feature of the program is that students may call a fast, restricted theorem prover that runs for a limited amount of time and then stops if a proof is not found. Students use this to cover routine gaps in their informal proofs. Because the method of attack of a resolution theorem prover seems so psychologically different from human ways of thinking about proofs I am doubtful that we shall be able to extend in any very deep way this feature of the program, although it is playing a useful role at what might be called a low level of complexity.

Another useful feature of the program is a subroutine to decide any Boolean expression. For example, any identity made up of set variables and the operation symbols for union, intersection and set difference can be evaluated for validity very quickly by this subroutine, and students may call it at appropriate points in informal proofs.

On the other hand, our proof procedures are still awkward in many cases and it is clear that practical lines of improvement can be found. For instance, our development of cardinal numbers is based upon Tarski's axiom that the cardinal numbers of two sets are identical if and only if the sets are equipollent. In this framework the informal proof that cardinal addition is associative follows trivially from the associativity of the union of sets, and in a textbook-

informal proof a one-line argument is used. At the present time it is still tedious to convert this one-line intuitive argument into an informal proof accepted by the program. Another similar example is the proof that the definition of exponentiation for cardinal numbers is justified — justified in the sense that the operation is uniquely defined for all cardinal numbers. Again, the line of proof is completely straightforward from facts already established about corresponding operations on sets, but passing from the level of arbitrary sets to the level of cardinal numbers is not as easy and natural as it should be in the proof procedures currently available.

Improvements of a similar sort are needed in order to do in a natural way the theory of ordinal numbers, especially the rather complex and somewhat difficult proofs justifying definition by transfinite recursion. I would summarize our present view of these matters by saying that we have procedures that are workable and that are being used by students in the course. But we anticipate being able to make substantial improvements and will be very disappointed if we cannot make substantial improvements over the next couple of years.

I have not said much about the curriculum in the course in axiomatic set theory. It follows closely the content of my earlier book (Suppes, 1960). After general developments of elementary set theory, the course concentrates on finite and infinite sets, the theory of cardinal numbers, the theory of ordinal numbers, and the axiom of choice. It seems likely that this classical content will be a part of the curriculum in logic and the foundations of mathematics for many years to come, and therefore we are confident that as we perfect the course we will not need to plan to change in any radical way the curriculum itself in the near future. Admittedly this situation is much different when one is developing curriculum in an area of mathematics or science that is currently under rapid change. This would, for example, be true in a graduate course in set theory that would be concerned with the many important results obtained in the past decade and a half, but the first introductory course does not reach the point at which the many new results can be seriously considered and presented to the students. It is also the case that the informal proof procedures needed for a more advanced course would be beyond our reach and too difficult to develop at the present time. The proofs would move too rapidly and it would be too much for us to close the gap between the standard informal proofs and the internal formal representation in the computer program. We believe that the level of difficulty of a course like the one in set theory I have been describing is the feasible level for current efforts.

Other courses. We are currently planning other courses at this same level of difficulty: a course in proof theory, a course in the foundations of probability, and a course in the foundations of measurement aimed especially at social scientists. These additional courses will require proof machinery at about the

same level of difficulty and complexity of the set theory course.

What we are especially interested in determining is whether the effort of extension to these new courses will be sublinear. I mean by this whether we shall find that the effort required to make the extension to these new courses will be definitely less than the effort that has been required to organize the set theory course in the first place.

Final remark on productivity. After having discussed on several previous occasions problems of productivity in the schools and the absence of increased productivity in education through most of this century, I have decided it is best to analyze the problem in terms of my own teaching and see what can be done about my own productivity. By taking responsibility for the two computer-based courses mentioned above and by offering them every term, I have doubled my teaching load at Stanford. In addition, I have adopted on principle the stance that I no longer give lecture courses. My own teaching is divided into computer-based courses in basic subjects like logic and set theory, and advanced seminars on specialized topics. I find this division extremely satisfying personally. I enjoy very much the personal and informal contact of seminars, and do not see their elimination by computer-based instruction in any foreseeable future. As far as course offerings go, I see no reason not to believe that I shall be able to double once again my course load over the next three or four years. In doing so, as indicated earlier, I am not aiming at large courses with large enrollments but at offering a diversity of courses at a medium level of difficulty. I have begun to sample the students at Stanford in terms of such courses and am finding that there are a number of courses in which they would be interested and which we do not currently offer but that we should be able to do a reasonably good job with in the kind of computer-based framework I have been describing. But teaching loads consisting entirely of small seminars are not practical in any university except for a few faculty members.

A mix of (i) large lecture classes for elementary courses, (ii) computer-based instruction for courses of intermediate difficulty and of a rather specialized character, and (iii) informal seminars at the advanced level seems attractive both in terms of productivity and personal satisfaction for faculty in the disciplines I have mentioned. However, it is too early to tell how many will share this opinion.

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20. The Future of Computers in Education

MY PURPOSE TODAY is to focus on the many and varied potentialities for the use of computers in education throughout the remainder of this century. It is important, however, that what I say not be based simply on general speculation but that it be firmly founded in past experience. Consequently, I shall spend a considerable part of my time in reviewing my own experiences and those of others in the instructional use of computers since the early sixties.

History of Computer-assisted Instruction at Stanford

ELEMENTARY-SCHOOL LEVEL

In January 1963, with a grant from the Carnegie Corporation of New York, the Institute for Mathematical Studies in the Social Sciences at Stanford University began a program of research and development in computer-assisted instruction (CAI). During the first decade, most of the work was concerned with the development of computer-based curriculums for elementary schools, especially in mathematics and reading. The work in reading was under the supervision of my former colleague Richard C. Atkinson, and the work in mathematics was my responsibility. A general history of these efforts is to be found in Suppes (1972). A large number of publications have resulted from this work; it would be inappropriate to list all of them here. Ex-

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tensive details about the work in mathematics are provided in two books, Suppes, Jerman and Brian (1968) and Suppes and Morningstar (1972).

Although the initial support for this work came from a private foundation, shortly thereafter support was received from the National Science Foundation and the United States Office of Education. There are many vignettes to tell about the extensive work we conducted in the sixties. One of my favorites is that the first school system in the world to have the opportunity of every elementary-school child's doing daily work at a computer terminal was in McComb, Mississippi. The terminals in McComb were connected to our computer at Stanford by telephone lines, and the research and development effort at McComb was supported by Title III of the Elementary-Secondary Education Act of 1965. Results of these efforts at McComb were evaluated rather carefully and the results were quite positive. A detailed analysis is given in Suppes and Morningstar (1969).

It is perhaps worth noting that almost ten years later this work in McComb has been followed by a fairly extensive use of computers for instruction in Mississippi. The computer systems and curriculum are provided by Computer Curriculum Corporation, with which I am also associated, and the funds used to support these operational efforts, which are no longer in the category of research and development, are provided by Title I allocations to the various school districts.

DISADVANTAGED STUDENTS

It is important to emphasize that a large part of the work on the instructional use of computers in elementary schools has been focused on disadvantaged students, and over the years we have worked with a variety of populations, ranging from inner-city schools in various parts of the country to rural populations in Appalachia as well as American Indian students in New Mexico. A report, for example, on the performance of American Indian students in CAI in elementary mathematics is to be found in Suppes, Fletcher and Zanoliti (1975).

HANDICAPPED STUDENTS

From July 1, 1970 to June 30, 1973, the Institute was deeply involved in the use of CAI for hearing-impaired or deaf students. This large project was funded by the Bureau of Education for the Handicapped United States Office of Education. Using our computer at Stanford, more than 1,000 deaf students were reached during 1970-71, through a national network of terminals located in residential deaf schools and special classes for deaf students in ordinary schools; during 1971-72 and 1972-73, this number rose to more than 2,000 students.

The curriculum for these students was primarily concerned with elementary mathematics and language arts. Extensive evaluation of this work has been published (Suppes, 1971; Suppes & Fletcher, 1974; Searle, Lorton, & Suppes; Suppes, 1974; Suppes, Fletcher & Zanotti, 1976; Fletcher & Suppes, 1976). Perhaps the best evidence of the success of this work with handicapped students is that almost without exception the schools have continued to use CAI more than four years after the research and project came to an end.

UNIVERSITY LEVEL

Beginning in the late 1960s, work at the Institute began to focus more on the possibilities of CAI for university-level courses. The first major effort was under the direction of Professor Joseph Van Campen of the Department of Slavic Languages at Stanford, and was concerned with the development of two years of elementary Russian at the university level. The very positive evaluation results of the first-year Russian courses are reported in Suppes and Morningstar (1969). The work in Slavic languages has now shifted from the beginning courses to intermediate courses that have small enrollment. CAI is offered in such relatively esoteric topics as Old Church Slavonic and the linguistic history of the Russian language.

One of the earliest curriculum efforts for the elementary school was the development of a course in logic for gifted elementary-school students, discussed in more detail below. In the late 1960s, efforts turned to the development of a corresponding course in logic for university students. Over the past few years the continued development and improvement of this course has been one of the central efforts in CAI at the Institute. Recently, the efforts in logic have moved to the intermediate, as well as the introductory college level as a focus of research and development. A substantial effort is now being put into axiomatic set theory, foundations of probability, and similar courses of a rather technical scientific nature.

Courses in music and computer programming have been developed for use at the university level as well. Individualized CAI in music has been applied to theoretical and instructional investigations in a number of different courses. Previous CAI work in programming has included courses in the BASIC, AID, SIMPER, and LOGO languages; current projects deal with an entirely new approach to teaching BASIC and the integration of a CAI course in LISP into the university curriculum.

Summary information on the courses mentioned is shown in Table I (taken from Suppes, Smith and Beard, 1977). It will become apparent at once that the enrollment is quite small in a number of courses that we consider as a primary focus of our research and development efforts. Contrary to some of our thinking a few years ago, it is now our deliberate choice to develop CAI for such courses rather than for the large-enrollment elementary courses.

TABLE I
University-Level Computer-Assisted Courses at Stanford, 1972-75

Course	Number of Students per Academic Year			Avg. Number of Student Hours at Computer Terminal
	1972-73	1973-74	1974-75	
Philosophy 57 Introduction to Logic	56	160	209	70 for A 54 for PASS
Philosophy 161 Set Theory	—	—	12	51 for A 31 for PASS
Slavic Lang. 211 Old Church Slavonic	5	2	1	30
Slavic Lang. Bulgarian	3	0	1	35
Slavic Lang. 212 History of Russian Literary Language	4	4	4	30
BASIC Instructional Program	—	100	200	10 ^a
Computer Science 206 LISP	—	b	79	93 ^c
Music (ear-training)	—	42	41	d
Music 21	—	42	41	d
Music 22	18	19	23	
Music 23	5	26	8	
Music 103	5	6	—	
Music 27	—	—	33	

^a BIP students were limited to ten hours of time for experimental reasons. During unlimit pilot runs, students had taken up to 30 hours to complete the course.

^b During 1973-74, LISP was taught at Stanford using the IMSSS machine; but student logged in as users, and there was no special CAI for LISP.

^c LISP students spent an average of 69 hours in the LISP interpreter, and 24 hours in the LISP CAI system.

^d The students had restricted terminal time.

The reason for the emphasis on small-enrollment courses is to a very considerable extent a matter of productivity (Suppes, 1975). The initial impetus for the course in axiomatic set theory was the loss of a faculty position in this area due to budget cuts at Stanford. In the face of declining or fixed budgets, it has become apparent that faculty sizes will probably decrease during the remainder of this century. Some state universities require that a specific level of enrollment be maintained for all courses offered, and there is considerable pressure against specialized courses with low enrollments. Still, such courses represent an important function of the university in transmitting intellectual knowledge and skills from one generation to the next.

One of our main aims in CAI research and development at the Institute is to show how these specialized courses can be maintained at reasonable cost in the future by appropriate use of computer technology. Our aim is to increase the teaching load of faculties in terms of courses offered, and none of our subsidiary aims, consequently, is to improve the ability of CAI to provide effective instruction with little or no intervention required of the instructor. A few teaching assistants, available at scheduled hours through the day and evening, work in conjunction with each course; the instructor himself is free to supervise further development of CAI courses or to consult on an individual basis with students and teaching assistants.

GIFTED STUDENTS

Work with gifted students at the elementary-school level began in an informal way in the late fifties but got under way in a more formal fashion under the sponsorship of the National Science Foundation in 1963, when a longitudinal accelerated program was undertaken. The objective of the program was to accelerate and enrich the mathematics instruction of gifted elementary-school students throughout Grades 1 to 6. Details of this program are described in several publications, so that I shall only mention certain salient points here (Suppes & Hansen, 1965; Suppes, 1966; Suppes & Ihrke, 1967, 1970). At the beginning of the second year of the project, the 34 children who were bright second-graders in 1964-65 received supplementary drill and practice in arithmetic at computer terminals and also an experimental course in mathematical logic especially prepared for gifted elementary-school students.

Because of the availability of logic at the computer terminals, the students were able to advance at a surprisingly rapid rate, and during the several years of the project they became quite proficient both in understanding and using the basic principles of logical inference. Perhaps the most important point is that, although this work began in the second grade, some of the experimentation was undertaken at the end of the first grade and if the programs had been available a year earlier could already have been used at that time. It is also important to emphasize that this work was natural and easy for the students. The range of mathematical topics that can be taught to gifted students is not easy to overestimate. In addition to the work in logic, extensive work in geometry as well as in the standard curriculum was offered to these gifted students. They made rapid progress through a variety of topics, including the elementary theory of symmetry groups. The geometry was taught by staff members of the Institute, not at computer terminals, but now, more than a decade later, it would be feasible to offer the geometry at computer terminals and thereby to provide rich instructional resources for gifted students that would not ordinarily be available in standard elementary schools.

It is also worth noting that we continued through the six years extensive work in standard drill and practice with these bright students. Some people ask why this was necessary, given their obvious gifts. Over the years I have developed an attitude and an answer that makes sense to me and to most of the people to whom I talk about the continued need for drill and practice in basic skills, even for gifted students. The analogy I use is a simple one—it is that of the training and perfection of physical skills. Most of us accept without any question that professional athletes continue to train and practice even after their particular skills have been thoroughly developed. The same is true of intellectual skills. The kind of performance we expect of gifted students can only be obtained, in my judgment, by a continued regime of drill and practice, and such a regime is just as much a desirable aspect of their intellectual training as it is of their physical training.

HOME-BASED COMPUTER-ASSISTED INSTRUCTION FOR HIGHLY GIFTED STUDENTS

I turn now to description of a more recent experiment in the Institute. The objective of this program was twofold: first to use not merely gifted but, in the main, highly gifted students, and, second, to investigate the feasibility of home-based computer terminals that would deliver instruction to the students, rather than having the students assemble at Stanford or some other focal point. It is especially important to recognize that highly gifted students, defined for this purpose as students with IQs of 165 or above, are very sparsely distributed and consequently there is a severe logistic problem of offering them advanced instruction. It is, in fact, my judgment that the use of distributed computer terminals is probably the best way to offer a variety of advanced instruction to such students. I also note that it is one of the scandals of American education, in my own judgment, that so very little is offered as advanced instruction for such students. Relative to their enormous potential, they are perhaps the most neglected population of students in our entire educational system.

Our project has been described in detail in a monograph on homebased instruction (Macken, van den Heuvel, Suppes & Suppes, 1976). I cover some of the more important aspects.

Procedure. Eight school districts in the areas surrounding Stanford University were invited to submit the names of students whose IQs as measured by the Stanford-Binet intelligence test were at least 165 and who seemed to have a wide range of outside activities; the latter condition was stipulated so we could determine how well a home-based CAI curriculum would compete with other established interests. The districts were asked to include with each name a statement describing the student and indicating why he or she was being recommended for the program. Some districts did this; others submit-

ted short autobiographies of the students, and some submitted only the test scores. From this information, members of the Institute staff selected a total of 16 students; there were some from each district and an equal number of boys and girls. We made four exceptions to the criterion of having a Stanford-Binet IQ score of at least 165; Three of the children measured "superior" on the WISC but had not taken the Stanford Binet, and the fourth child measured only 132 on the Stanford-Binet but was an outstanding achiever in his district. Children who were not selected were placed on a waiting list. All selected children were in the sixth, seventh, or eighth grade with the exception of one girl in the third grade and one in the fifth grade.

In November of 1973, letters of invitation were sent to the parents of each child selected for the program; copies were sent to the cooperating school districts. The letters provided a description of the program, including the available curriculums and necessary equipment, and indicated that there would be no cost to the family. All families agreed to participate. Installation of teletypes in the homes began as soon as permission was given by the parents; all were installed by January of 1974. The students were shown how to dial the computer at the Institute, how to use the terminal, and how to sign on to the elementary logic course. With this minimal explanation all students were about to begin their CAI work.

A meeting for parents and students was held at the Institute on January 14, 1974. This was a general meeting to answer questions about the programs and the curriculum and to provide a tour of the Institute facilities. Parents were encouraged not to urge their children to work on the teletypes; instead, students were to pace themselves and work as much or as little as they wished. The meeting was well received and follow-up monthly meetings were planned. However, because of the gasoline shortage, no further meetings were held until June when the 10 new students had been selected.

A proctor was available at the Institute four days a week during scheduled times so students could telephone for help. Some students, who indicated they felt too shy to telephone, were called once a week by the proctor to see if they were having any problems.

The summer program was similar to the spring program. Letters of invitation and explanation were sent to the 10 new students and a meeting was held June 17 for old and new students. This time the parents were not encouraged to come. Summer students were told they could take a programming course, BASIC, along with the logic course by writing or telephoning the Institute to request a BASIC manual. This left the initiative with the student: Six students requested the manual and began the BASIC course, and two finished the course by January 1, 1975.

By midsummer it was apparent that eight students—four new students and of our old students—were not working regularly toward completing the logic

course. Weekly individual goals were set to encourage regular study. The students expressed approval of the goal setting but, in fact, did not change their work habits. By January 1, 1975, four of these students (two old and two new) completed the logic course, two (new students) were still working very slowly and irregularly, and two (old students) had dropped out.

Two students, who were friends attending the same school, seemed to enjoy competing with each other on the CAI curriculums, and this competition apparently stimulated progress. After further inquiry, telephone lists of all participating students were mailed to the students. However, the students did not make much use of the lists, possibly because they were too shy to telephone peers they had not met.

Conclusions. Without entering into the details of the curriculum, let me summarize here the main conclusions we reached.

First, the high dropout rate experienced with this very special and very bright population of students is characteristic of the high dropout rates observed in other home-study courses. We left the structure of the program very free in order to test whether very able students of the kind we were working with would be able to sustain an interest in relatively difficult material over a long period of time. It should also be remarked that all of these students had very busy programs both in and out of school. We were in fact surprised, after the program began, to find the amount of time some of them were devoting to it.

Second, on the basis of the high dropout rate experienced, we would conclude that a sustained program of home study for gifted students would require the introduction of considerable structure and also, probably clear arrangements about academic credit for the course work done. Perhaps the best arrangement would be to replace part of the routine school courses by the advanced courses available at computer terminals for home study, or at the very least to arrange for advanced credit for the courses completed.

From our extensive survey of the literature and from this intensive experiment, we believe that arrangements for introducing structure providing short-range goals for students, feedback on their progress, telephone conversations with proctors, and other forms of guidance and interaction need extensive experimentation in order to determine the optimal mix for home-study instruction. Our search of the literature indicates that there has yet been very little serious experimentation on these matters. In view of the very large number of students engaged in home-study work in one form or another and given the technological possibilities for the future in terms of more intensive interactive courses available in the home, it seems most desirable that an extensive program of experimentation on the appropriate structures for such courses be undertaken.

Third, in ordinary studies of achievement of students in different technolo-

gies, for example, in the extensive survey of Jamison, Suppes, and Wells (1974), pretest and posttest achievement measures have been the primary instrument of assessment. The extensive data from correspondence courses and the restricted but detailed data from the group of very gifted students described in the present section indicate that, in the case of home-based instruction, the dropout rate is a more significant measure of assessment than actual achievement. This is meant in the sense that the primary problem of home-based instruction is to find ways to reduce the very large dropout rates encountered. Although the literature is extensive, in one sense, with regard to analysis of the causes of dropout, it is on the other hand clear that detailed quantitative and statistical analysis using relatively sophisticated models has not as yet taken place and would almost certainly yield a better understanding of the phenomena than we now have.

Let me end my remarks about work with gifted students by a comparison with what I have observed first-hand in the Soviet Union. In the early summer of 1972, I served as a special consultant to the United States Educational Exhibit touring the Soviet Union. My allocated time was spent while the exhibit was in Novosibirsk in central Siberia. One of the things I had an opportunity to learn about was the special secondary school for gifted students, located in Novosibirsk. A similar school exists in Moscow and one in Leningrad. I had the opportunity to discuss extensively the curriculum of the school and also to learn about the procedures for selecting the students, who were brought from all over Siberia to the school for three or four years of secondary-school work as full-time residential students. The main course in mathematics, physics, and chemistry were taught by university staff, and the residential school itself was located as part of the university and Soviet Academy institute complex called Akademgorodok, that is, Academic City, not many kilometers from Novosibirsk. Without entering into the details of what I learned there, it is fair to say that there is no comparable effort in the United States, although some of the very best secondary schools, as for example the Bronx High School for Science, come close. What is especially missing, however, is the kind of systematic effort at selection and identification of the brightest and ablest youngsters. There is most certainly nothing comparable to this procedure of selection in any program for highly gifted students in the United States.

Considering the amount of money that is spent on education in the United States, both at state and local levels as well as at the Federal level, the relative neglect of providing adequately for the special needs and enormous potential of our highly gifted students is, as I have already said, something of an educational scandal. I would strongly urge the Congress to consider giving special attention to the educational needs of the gifted. Because of the sparse distribution of highly gifted students in particular, the instructional use of computers for such students seems especially appropriate. Moreover, what would be learned from highly experimental work with these students would almost cer-

tainly have an enormous payoff in the years to come for a larger population of students.

Future Technological Possibilities and Their Problems

I hope that the numerous examples drawn just from our Stanford experience indicate that the instructional use of computers can serve students of all ages and of all varieties of capacities and interests. What I would now like to turn to are some of the possibilities and the problems of these possibilities that need attention in the future. I have organized my remarks under the four broad problems of talking, listening, knowing, and learning.

COMPUTERS THAT TALK

The first problem is simply that of talking. What does it take to get a computer to talk? The fact is that the technical issues are already pretty well in hand. In our computer system at Stanford we have eighteen channels of independent simultaneous audio and the computer talks independently and differently to eighteen students at the same time. We have the capacity for the computer to talk. What we need, however, is a better theory about what is to be said. For example, when I serve as a tutor, teaching one of you, or when one of you is teaching me, intuitively and naturally we follow cues and say things to each other without having an explicit theory of how we say what we say. We speak as part of our humanness, instinctively, on the basis of our past experience. But to get a computer to talk appropriately, we need an explicit theory of talking.

COMPUTERS THAT LISTEN

Long ago, the sophists of Ancient Greece objected to the use of written records for instruction because they eliminated the intimate interaction between tutor and student. In Plato's dialogues *Phaedrus*, Socrates asserts "only in principles of justice and goodness . . . taught and communicated orally for the sake of instruction . . . is there clearness and perfection and seriousness." The aspect of dialogue that is technically difficult for us at the present time, even more than talking, is that of listening. Without any questions, the problem of defining computer hardware and software that can listen to a student talk is much more difficult than having the computer talk to the student. Indeed, I want to emphasize the great difficulty of having computers that adequately listen. I recognize that it will be many years before fully satisfactory solutions are available.

On the other hand, already several years ago we were able to experiment successfully with ten-year-olds' talking to the computer in order to give an-

swers to elementary mathematical problems. Exercises given at home over ordinary telephone lines and consisting of both elementary mathematical problems and such standard topics as spelling are feasible at the present time from a technical standpoint.

The great importance of extending the capabilities of computers that listen in order to give us more powerful educational tools cannot be overemphasized. I am sorry to say that almost all of the work that has been done up to the present on the technical problems of computers' understanding spoken speech has been sponsored not by any of the educational agencies, as for example the United States Office of Education or the National Institute of Education, but by the Department of Defense, especially the Advanced Research Projects Agency. In my own mind, the educational case is a very much stronger one than the defense case for the interest in computers that listen.

COMPUTERS THAT KNOW

The third part of my discussion concerns the problem of understanding the knowledge base. To have an effective computer-based system of instruction, we must transcend mindless talking and listening and learn to understand and use a large knowledge base. For example, if we were simply to require information retrieval from a knowledge base, it would be relatively simple in the future to put the entire Library of Congress in every elementary school. The capacity to store information is increasing so rapidly that we will be able to store much more information than could ever possibly be used by any one individual.

A different and more difficult question is how to get the sizable knowledge base to interact with the student. As we come to understand how to handle such a knowledge base, the school computer of the future should be able to answer any wayward question that the student might like to ask. Moreover, as we all know, once a student uses such a capability, he will have a strong tendency to pursue still further questions that are more difficult and more idiosyncratic. It will, I think, be wonderful to see how children interact with such a system; in all likelihood, we will see children give to learning a high degree of concentration and the sustained span of attention they now give to commercial television.

There is one related point I want to emphasize. From the very beginning of school, students learn quickly the "law of the land" and know they should not ask questions the teacher cannot answer. This task of diagnosing the limitations of teachers begins early and continues through college and graduate school. So, once we have the capacity for answering out-of-the-way questions, it will be marvelous to see how students will take advantage of the opportunity and test their own capacities with a relentlessness they dare not engage in now.

COMPUTERS THAT AID LEARNING

The fourth problem, and in many ways the least-developed feature of this technology, is the theory of learning and instruction. Even if you can make the computer talk, listen, and adequately handle a large knowledge data base, we still need to develop an explicit theory of learning and instruction. In teaching a student, young or old, a given subject matter or a given skill, a computer-based learning system can record anything the student does. It can know cognitively an enormous amount of information about the student. The problem is how to use this information wisely, skillfully, and efficiently to teach the student. This is something that the very best human tutor does well, even though he does not understand at all how he does it, just as he does not understand how he talks. None of us understand how we talk and none of us understand how we intuitively interact with someone we are teaching on a one-to-one basis. Still, even though our past and present theories of instruction have not cut very deep, it does not mean that we have not made some progress. First, we at least recognize that there is a scientific problem; that alone is progress. One hundred fifty years ago there was no explicit recognition that there was even a problem. In the education literature of 150 years ago it was not recognized as important to understand the nature of student learning. Only in the twentieth century do we find any systematic data or any systematic theoretical ideas about the data. What precedes this period is romance and fantasy unsubstantiated by any sophisticated relation to evidence. So at least we can say that we have begun the task.

I want to emphasize that, in the four possibilities and their associated problems that I have mentioned, namely, those of computers that talk, listen, know, and aid learning, the primary problems are not matters of hardware or technology in a narrow sense but of general intellectual investigations of a fundamental kind that will not, in the course of ordinary affairs, be undertaken by industry but that must be addressed by Federal programs. Our rate of progress on these matters will be determined almost entirely by the level of Federal support and wisdom of management of the Federal programs that are targeted on the possibilities and problems I have discussed.

Some Possible Federal Programs

I would like to end on the practical note of stating what, from my own viewpoint, would be important and feasible programs of support for the Federal government.

1. *Gifted students.* The United States Office of Education or some other agency, such as the National Science Foundation, should have a special program of support for the use of instructional computers with gifted students, especially the highly gifted. These sparsely distributed gifted students should

be brought individualized programs, especially in basic skills of mathematics and language, at an early age and throughout all their years of elementary and secondary education. The returns on a program funded at a substantial level would, in my judgment, exceed the return on any of the other Federal programs of educational support at the present time.

2. *Handicapped students.* The extensive support programs in the Bureau of Education for the Handicapped should be targeted, even more than they are currently, on the use of computer technology for the education of the handicapped. In many ways, excellent programs of support are now already under way at the Bureau. I would urge Congress to review these programs from time to time and to insist that appropriate advantage is taken of the increasingly sophisticated technology that is available for instructional purposes.

3. *Disadvantaged students.* There is an increasing use of computers for supplementary instruction in basic skills, especially basic skills of reading and mathematics, with disadvantaged students. The use of instructional computers for this purpose should be encouraged and enhanced in the revision of the Elementary-Secondary Education Act of 1965, now pending in Congress.

4. *Home-based instruction.* The proposition that we are becoming a nation of learners deeply committed to continuing education is supported by a variety of current data. For example, in some of the community-college districts in California more than twenty percent of the entire adult population—not just the college-age group—are enrolled in at least one course. Direct electronic delivery of a wide variety of courses to the home is increasingly feasible. A program of support to accelerate current efforts would be timely.

5. *Research and development.* Finally, I strongly urge that consideration be given by Congress to a program of research and development dealing with the four main problems I discussed earlier. A program at a substantial level could accomplish a great deal to rapidly facilitate the extensive use of computers for instruction. The total R&D dollars now being spent by the Federal government for research on the instructional use of computers is very small indeed. A much larger program would not be excessive and could be practically absorbed by the research and development community. I would also emphasize that the R&D programs falling within such a general framework would, in many cases, be concerned with highly developmental experimental programs that would in themselves have direct educational benefits. I believe it is fair to say that much of our work at Stanford has been exactly of this character.

We are sitting on the edge of a revolution in the way in which instruction is delivered to students of all ages and varieties. It has been one of the great pleasures of my life to have been a part of the beginning of this revolution. I hope that the Federal government will provide strong support for this constructive direction of change throughout the rest of this century.

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Although minicomputers and microprocessors are now becoming common in schools, most educators are only beginning to use them and to think about how computers could help learning.

The Computer in the School: Tutor, Tool, Tutee brings together essays by five pioneers in the field of computers in education. They began research with five computers in varied areas of learning years ago and developed ideas now being discussed as innovations. In reality, the ideas date back to the original thinking of these pioneers, who foresaw possibilities for computers and learning that can only be widely realized with today's smaller computers.

The contributors, articulate spokesmen who have lived through several cycles of computer development in education, are: Alfred Bork, director of the Physics Computer Development Project at the University of California at Irvine; Thomas Dwyer, of the University of Pittsburgh, who was the director of a series of projects to integrate computing into the high school curriculum, funded by the National Science Foundation; Arthur Luehrman, associate director of the Lawrence Hall of Science, at the University of California at Berkeley, whose work there and earlier at Dartmouth includes the use of computers in schools, universities, and museums; Seymour Papert, who is the Cecil and Ida Green Professor of Education at M.I.T. and has worked extensively with computers and learning among young children; and Patrick Suppes, a professor of philosophy and director of the Institute for Mathematical Studies in the Social Sciences at Stanford.

The nineteen articles in this book build a good foundation for understanding the basic issues involved in using computers of any kind in schools, the teacher's role in helping the student to make full use of computing, and the general limitations of computer use. These selections also provide parents, teachers, school boards, and administrators with an excellent perspective to see why computers should be used in the classroom at every level.

Robert P. Taylor, the editor of the book, explains that computers can be used to tutor students, as in computer-assisted instruction; as a tool to aid such diverse processes as payroll preparation and text editing; and as a tutee that helps students learn something as they teach it to the computer.

Professor Taylor, who is the director of the Teachers College Program in Computing and Education, one of the first programs in the United States to grant a graduate degree in computing and education, has taught many subjects at levels ranging from elementary to graduate school. He has written and spoken widely on computing in education, on computer science education, on the place of computers in the kindergarten through twelfth-grade curriculum, and the relation of computing to art, music, literature, and other cultural activities.



Teachers College
Columbia University
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